RippleLogic Canon v11.4

RippleLogic: The Ethical Decision Operating System of the MathGov Framework
A General-Purpose, Multi-Scale Decision Architecture for Rights-Floor Norms, Tail-Risk Control, and Auditable Decision-Making

Tier 1-3: Implementable Now | Tier 4: Design Target (Tier-4 claims prohibited until a ProofPack is independently replayable)

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Package line: MathGov Core Release 2026.09 (Complete Ready Release, CSV Upgrade, and Falsification Discipline Patch)
Package-note: This release consolidates the public cascade as RG -> RF -> TRC -> CSV -> RLS and the formal shorthand as RG/RSG -> RF/NCRC -> TRC -> CSV -> RLS. RF is the public Rights Floor layer; NCRC is the formal non-compensatory predicate. CSV is Containment and Structural Viability, the strengthened fourth level of the existing five-level method, not a sixth public gate. RLS remains downstream and ranks only options that survive RG claim support, RF/NCRC, TRC, and CSV. This release integrates RF/CSV notes professionally through the Canon, Cascade Standard, CSV Gate Standard, README, and release reports rather than repeating long synchronization notes at the top of companion documents. The v11.4.x release-engineering patches preserve the v11.4 content line while restoring table readability, clarifying MathGov and RippleLogic naming, refreshing workbook integrity, and adding Source-Coupling Integrity as a MathGov-native diagnostic inside Reality Grounding and CSV so that downstream output, model fluency, compliance status, inherited procedure, or institutional permission is not mistaken for grounded capability. Tier 4, ProofPack, empirical-validation, legal-certification, deployment-certification, automated moral-truth, and deterministic-selection overclaim boundaries remain unavailable.

Specification-contract front-door note. Current release interpretation follows the Source Hierarchy and Section 0 Specification Contract: MathGov is the umbrella ethical governance framework; RippleLogic is the ethical decision operating system and decision architecture inside MathGov; the Markdown Canon source is the governing semantic source if derived DOCX/PDF renderings diverge. The full precedence and conflict-resolution contract appears in Appendix AC / Section 0.

ABSTRACT

Governance systems, institutional decision-making processes, and AI alignment approaches face a common structural challenge: high-consequence decisions often collapse plural values

into single metrics, underweight catastrophic tail risks, lose contact with reality surfaces, and become vulnerable to specification gaming, where optimized proxies degrade the outcomes they were meant to protect. As a proposed Tier 1-3 design architecture rather than an empirically validated governance instrument, RippleLogic addresses this problem by operationalizing Union-Based Ethics as a general-purpose ethical operating system for auditable decision-making across layered scales of life and governance. The Category Grounding Patch clarifies that, when material categories affect rights, tail risk, containment, SGP interpretation, authority, conformance, or public claims, Category Grounding and Primitive Audit belong inside Reality Grounding as required subdisciplines, not as additional cascade stages. The v11.4 Complete Ready Release carries forward the v11.0 Term Discrimination and Semantic Stability Patch and makes the discriminator, dependency position, non-collapse risks, and refusal condition explicit for material terms so that governance does not proceed through collapsed language. The framework evaluates candidate actions through a five-level admissibility-and-ranking method: Reality Grounding (RG), Rights Floor (RF), Tail-Risk Constraint (TRC), Containment and Structural Viability (CSV), and RippleLogic Score (RLS). Rights Floor is the public-facing name for the Non-Compensatory Rights Constraint (NCRC): rights-floor violations cannot be compensated by welfare gains elsewhere.

SECTION 1: INTRODUCTION: THE NEED FOR A GENERAL-PURPOSE ETHICAL OPERATING SYSTEM

1.1 Alignment Failures Across Scales

Contemporary societies operate within a tightly coupled, high-dimensional environment where climate dynamics, global supply chains, digital communication networks, financial systems, and emerging artificial intelligence systems interact in ways that increasingly resist prediction or governance. Decisions taken at one organizational scale propagate rapidly through multiple layers of human and ecological organization, generating consequences that conventional decision frameworks fail to anticipate or manage (Meadows, 2008; Newman, 2010; Steffen et al., 2015). In this context, alignment is not exclusively an artificial intelligence problem. It spans multiple domains: aligning machine systems with human purposes and control (Russell, 2019), preventing catastrophic failure modes that could foreclose long-run human potential (Ord, 2020), and maintaining human activity within the safe and just Earth-system boundaries required for durable flourishing (Rockström et al., 2009, 2023). RippleLogic addresses this as a unified governance problem rather than as three isolated challenges.

Civilizational target and method standard (Informative)

Target. Protect the floor for all intelligences, bound catastrophic tail-risk, and expand long-run flourishing across the operational union stack.

Method standard. A decision protocol is governance-grade only if it is auditable, reproducible, rights-respecting, tail-risk aware, and robust to Goodhart and adversarial pressure.

1.2 The Alignment Trilemma (Coupled Failure Modes)

Existing decision frameworks exhibit three recurring failure modes that together constitute the alignment trilemma:

Failure Mode A: Scalarization of Value

Many decision methods collapse plural values into a single metric (expected utility, GDP, cost-benefit net present value). This enables “moral laundering”: severe harms to some unions are permitted when offset by gains elsewhere. Arrow’s (1963) impossibility theorem demonstrates that no preference-aggregation rule can simultaneously satisfy minimal fairness criteria when preferences conflict fundamentally across multiple dimensions and scales.

Failure Mode B: Tail-Risk Blindness

Expected-value reasoning underweights low-probability, high-severity outcomes. Catastrophic outcomes are often irreversible and eliminate future choice; therefore symmetric tradeoffs between ordinary gains and catastrophic losses are ethically incoherent and empirically dangerous (Taleb, 2012). Coherent risk measures such as CVaR provide better tail-risk handling (Artzner et al., 1999; Rockafellar & Uryasev, 2000). Climate tipping points (Lenton et al., 2008), pandemic risks, and AI misalignment exemplify threats that conventional frameworks systematically underweight (IPCC, 2023; Bostrom, 2014).

Failure Mode C: Specification Gaming (Goodhart Vulnerability)

When metrics become targets under pressure, they are gamed. Optimization exploits proxies while degrading the intended outcome (Goodhart, 1984; Manheim & Garrabrant, 2018). This appears in institutional governance, where institutions optimize visible indicators while displacing underlying welfare, and in AI reward optimization, where systems exploit proxy objectives under strong optimization pressure (Amodei et al., 2016).

RippleLogic treats this not as a peripheral measurement problem but as a structural governance hazard. The framework is designed to resist specification gaming by separating admissibility from ranking. Rights protection is enforced through the Non-Compensatory Rights Constraint over the canonical rights-coverage set, catastrophic downside is separately gated through the Tail-Risk Constraint, and pathological cross-scale degradation is checked through containment before welfare ranking begins. An option therefore cannot compensate for a rights-floor breach, catastrophic tail exposure, or containment failure by achieving a higher downstream welfare score.

This architecture matters because Goodhart-type failure thrives where optimization is scalar, fungible, and weakly constrained. RippleLogic instead combines a 49-cell welfare matrix for welfare evaluation with an 8-right sparse rights-coverage system comprising 70 specific protected intersections. Harms therefore cannot simply be averaged away across unions or dimensions. A localized protected loss remains visible at the mapped protection point where it occurs.

Practical Local Scope Scoring does not weaken this defense. PLSS operates only within residual welfare ranking after admissibility has already been secured. It allows practical local attention without relaxing rights floors, tail-risk constraints, or containment requirements. In this sense, RippleLogic treats Goodhart's Law as a boundary-crossing failure of optimization and responds with layered structural safeguards rather than with a single corrective metric.

Coupled Failure

These three failure modes interact and amplify each other. Scalarization enables tail-risk blindness (catastrophe can be traded for ordinary benefit), tail-risk blindness creates exploitable blind spots, and specification gaming undermines attempts to "patch" either problem by shifting optimization pressure. Solving one in isolation is insufficient.

Boundary-gaming note. Because lexicographic gates reduce scalar Goodharting, adversarial pressure can shift upstream toward boundary definitions, stakeholder inclusion, subgroup mapping, scenario probabilities, threshold governance, evidence sufficiency, and scope declarations; Appendix F.5 specifies the required anti-gaming disclosure and review discipline.

Category-gaming note. Because RippleLogic blocks several downstream scalarization failures, adversarial and institutional pressure can shift further upstream toward category choice, definition boundaries, reference structures, and metric formation. A run therefore MUST NOT treat inherited labels, dashboard categories, legal labels, benchmark names, model-generated descriptions, or institutional terms as valid merely because they are available. When a category materially affects rights, tail risk, containment, SGP interpretation, authority, conformance, or public claims, the category must be grounded inside Reality Grounding before the run relies on it.

Term-discrimination note. A material term must not borrow authority from a neighboring term. Permission is not admissibility; monitoring is not control; compliance is not correctness; certification is not validity; automation is not autonomy; protection is not authority; selected is not executable; and score is not truth. When term collapse would affect a gate, claim, authority state, or execution state, Term Discrimination is required inside Category Grounding.

PLSS clarification (Informative). A further practical failure appears when multi-scope frameworks oscillate between false universality, informal narrowing, and flattened aggregation. False universality treats every decision as though all seven scopes are equally active optimization targets. Informal narrowing quietly drops distant scopes when full evaluation becomes burdensome. Flattened aggregation hides where benefits and harms actually land by blending impacts into a single welfare bucket. RippleLogic v11.4 carries forward the v10.8 PLSS clarification to address this tension through Section 10.2A:

admissibility remains full-scope, while residual welfare attention may become scope-proportionate.

Reality Grounding clarification (Normative preview). RippleLogic now presents Reality Grounding as the first level of the public method because every later claim depends on the surface actually contacted. The method is not “score first.” It is ground first, protect the floor, bound ruin, preserve the whole, and only then score residual welfare.

RG status rule. Reality Grounding is Level 1 of the public method and a claim-authority precondition. It can force narrowing, escalation, exploratory-marking, or refusal, but it is not an option-rejecting gate; RF/NCRC, TRC, and CSV are the option-rejecting gates, and RLS ranks only what survives them.

Computability-vs-realizability clarification (Normative preview). RippleLogic MUST NOT treat computed possibility, model fluency, simulation output, or generated abstraction as decision authority. Computable is not selectable; generated is not grounded; executable is not ethical; realizable is necessary but not sufficient. Reality Grounding decides the claim boundary; RF/NCRC decides rights admissibility; TRC decides ruin exposure; CSV decides whether the pathway can structurally stand; RLS ranks only the remaining selectable set.

Source-coupling clarification (Normative preview). RippleLogic also MUST NOT treat downstream operation, inherited procedure, benchmark success, model fluency, compliance status, institutional permission, or interface performance as proof that the claimed capability is coupled to the enabling conditions and boundary conditions that make it possible. Source-Coupling Integrity is a subdiscipline of Reality Grounding and a CSV diagnostic where source weakness creates structural fragility. It does not add a public gate.

Structural diagnostic placement clarification (Normative preview). UCI and HOI are not public headline cascade stages. They are structural diagnostic instruments. When their evidence is material to cross-scale structural integrity, including coherence loss, hollowing, brittleness, burnout, lock-in, trust erosion, institutional erosion, ecological degradation, or governance-capacity loss, that evidence is consumed inside the CSV Gate. Only residual, non-gate-critical UCI/HOI signal remains available after RLS, and only when the leading options are tied, close, uncertainty-overlapped, or non-decisive.

1.3 Governance-Grade Requirements

RippleLogic is designed to satisfy eight non-optional requirements:

Requirement	Description	RippleLogic Component
R0	Reality Grounding: material claims must be grounded in a declared reality surface, evidence trace, known unknowns, transition boundary, consequence pathways, and claim boundary.	Reality-Surface Grounding (Level 1)

Requirement	Description	RippleLogic Component
R0A	Term Discrimination: material terms must declare their definition, discriminator, dependency position, non-collapse risks, and refusal condition.	Category Grounding inside Reality Grounding
R1	Rights-first non-compensability: rights are lexicographic constraints, not weighted terms	Rights Floor (RF/NCRC) (Level 2)
R2	Explicit catastrophic risk bounding: tail risk is bounded by TRC using CVaR	TRC / Tail-Risk Bound (Level 3)
R3	Residual welfare ranking: multi-scale and multi-dimensional welfare representation after Reality Grounding, RF/NCRC, TRC, and CSV have formed the selectable set.	RLS / RippleLogic Score over 7×7 Welfare Matrix (Level 5)
R4	Explicit ripple propagation: interpretable kernel propagation under humility	Kernel K
R5	Structural integrity and containment: local gains may not degrade containing unions beyond tolerance; UCI/HOI or equivalent diagnostics are used inside CSV where material.	CSV / Containment and Structural Viability (Level 4)
R6	Legitimate weight governance: HDW blends constitutional floors with democratic tuning	HDW
R7	Computability plus auditability: PCC and audit flags enable reconstruction, challenge, and learning	PCC plus AIL

1.4 Paper Contributions

This paper provides:

- A fork-resistant normative specification for Tier 1-3 implementation.
- Canonical equations and constraints sufficient for independent implementation.
- Anti-gaming architecture (non-maskable cells, subgroup semantics, scenario governance, audit flags).
- A validation and falsification program (Section 17).
- A Tier-4 design target appendix (non-claimable until ProofPack).

SECTION 2: ONTOLOGICAL FOUNDATIONS: UNION-BASED REALITY (UBR)

2.1 Relational Ontology (Descriptive Thesis)

UBR thesis: No entity exists in complete isolation for governance-relevant domains; entities are embedded in interacting networks whose structure transmits the consequences of actions.

This is a descriptive stance supported by systems science (feedbacks and leverage points), network science (universal structural features of real networks), and Earth-system science (planetary boundary coupling) (Meadows, 2008; Newman, 2010; Steffen et al., 2015; Rockström et al., 2009, 2023). UBR is also consistent with empirical findings in social and cognitive interdependence research showing strong links between human capacities and group structure, including the social brain hypothesis and attachment-based development (Bowlby, 1988; Dunbar, 1992, 1993), and with social contagion dynamics in networks (Christakis & Fowler, 2009).

Scope conditions: UBR is most applicable in high-coupling systems with significant externalities, multi-scale feedbacks, and long time horizons. Low-coupling, short-horizon decisions may not require full UBR modeling; RippleLogic tiers allow proportional rigor.

2.1A Decision-Relevant Reality (Operational)

For RippleLogic, reality, for decision-making, is the evaluator-independent field of constrained structures that, under transition, produce consequences. Decision-relevant reality is the portion of that field that materially bears on rights, catastrophic risk, containment, welfare, legitimacy, agency, auditability, or available choices. The reality surface is the portion actually captured, represented, and modeled in the run. RippleLogic does not claim complete metaphysical access to reality; it requires decision claims to make declared, auditable, and challengeable contact with the relevant reality surface. Here, evaluator-independent means independent of the evaluating agent's perception, admission, authority, model, or preference; it does not deny the reality of experience or social facts. Mental states, institutions, norms, laws, markets, reputations, and model outputs may be real vehicles with real effects when they shape welfare, rights, risk, agency, or available choices.

Root-science boundary. RippleLogic treats constrained structure under transition producing consequence as the operational substrate of decision-relevant reality. This is not a final metaphysical claim about ultimate being; it is a governance-grade discipline. Physical, biological, ecological, experiential, social, institutional, informational, artificial-system, and formal/logical structures and constraints become relevant when they shape rights, risk, welfare, agency, available choices, legitimacy, or auditability.

Structure rule. Structure is the stable relational arrangement of distinguishable differences that persists across a decision-relevant transition window and can carry causal, informational, biological, ecological, social, institutional, artificial-system, or experiential significance. Structure may include physical configuration, network topology, causal architecture, biological organization, ecological interdependence, institutional role structure, information-bearing arrangement, cognitive model, social pattern, or rights-relevant exposure pattern. Structure is not the object alone, not the process alone, and not the model of the arrangement; the model is only a reality surface.

Transition rule. Consequences arise when an action, decision, event, policy, intervention, or state-shift enters constrained structure. Transition is the decision-relevant change through which constrained structure produces consequence. A RippleLogic run SHOULD declare the transition or action boundary when it makes material claims about consequence.

Constraint-species rule. A factor may bear on the decision through causal influence, such as physical, biological, ecological, social, institutional, informational, or experiential conditions, or through formal/logical constraint, such as coherence, conservation, computability, consistency, or mathematical possibility. Constraint is the genus; causal and formal/logical constraints are species. RippleLogic uses both and MUST NOT conflate them.

Constraint-status rule. Constraint is anything that limits, enables, shapes, or channels what can happen. Physical, biological, ecological, informational, experiential, institutional, computational, and formal/logical constraints may be discovered, measured, modeled, or inferred. Rights floors, dignity protections, and non-domination constraints are explicit normative commitments. RippleLogic treats all of these as governance-relevant constraints when they shape admissibility or consequence, but it MUST NOT present chosen normative commitments as if they were discovered physical facts.

Consequence rule. Consequence is the observable or inferable change that follows from an action, transition, decision, event, or policy within constrained structure. Consequences may be direct, probabilistic, systemic, delayed, distributed, reversible, irreversible, moral, institutional, epistemic, ecological, or catastrophic. Consequence claims require evidence traces, scenario boundaries, uncertainty disclosure, and monitoring paths proportionate to stakes.

Vehicle/content/effect rule. A belief, norm, institution, model, market signal, law, border, reputation system, or algorithmic score may be a real vehicle with real effects even when its content is false. Consensus can therefore be socially real while still failing to track the territory. Agreement does not make a harmful system harmless, and authority does not make an inadmissible decision admissible.

Surface/territory rule. Any model captures only a reality surface; the territory exceeds every surface. Absence of evidence is not evidence of absence where the missing condition is rights-relevant, catastrophe-relevant, containment-relevant, or material to the claimed decision state. Omitted material factors must be recorded as unknown, not zero.

Contact chain. Territory is the full evaluator-independent field. A reality surface is the portion captured by evidence, measurement, testimony, stakeholder mapping, model, scenario, or audit. Evidence is traceable contact with that surface. A claim boundary is the maximum legitimate strength of a claim given that contact. Refusal, escalation, or claim-narrowing is required when the claim exceeds the surface or when a gate-critical unknown prevents legitimate determination.

Consequence under finitude. Because the territory exceeds every modeled surface and audit operates only on the surface, no run can certify that all unmodeled catastrophic or containment-breaking effects are absent. Because catastrophic outcomes are, by the TRC definition, potentially irreversible or unbounded, folding them into an average computes over a tail finite knowledge cannot reliably bound and misrepresents an uncompensable loss as compensable. Non-averagability is therefore forced, not merely preferred; TRC and CSV are the form this takes under RippleLogic.

2.1B Reality Grounding / Reality-Surface Grounding (RSG) (Normative)

Purpose. Reality Grounding is the required first level of a RippleLogic run. It prevents decisions from treating authority, fluency, compliance, dashboards, or model outputs as reality. A run may only make claims supported by its declared reality surface and evidence trace.

Definition. Reality Grounding is the process of declaring the decision-relevant reality surface, evidence trace, material unknowns, transition boundary, consequence pathways, and claim boundary before rights, risk, CSV, or scoring claims are made.

Six required outputs. A valid RealityGroundingRecord SHOULD include: (1) reality_surface; (2) evidence_trace; (3) material_unknowns; (4) transition_or_action_boundary; (5) consequence_pathways; and (6) claim_boundary. Tier 3 and high-stakes Tier 2 runs SHOULD also include monitoring_path, falsification_or_revision_trigger, reviewer_status, and evidence_status by material claim.

Status values. RealityGroundingStatus values are GROUNDED, ESTIMATED, ASSUMPTION_BOUND, DECLARED_UNKNOWN, OUT_OF_SCOPE_WITH_RATIONALE, and INSUFFICIENT_GROUNDING. Legacy RealityContactStatus terms remain compatible aliases: EVIDENCE_BACKED maps to GROUNDED and INSUFFICIENT_CONTACT maps to INSUFFICIENT_GROUNDING.

RealityGroundingStatus assignment guide. The following table is a required reviewer guide for Tier 3 runs and a recommended guide for Tier 2 runs. It does not replace evidence review, but it constrains status drift.

Evidence condition	Required or recommended status
Direct measured evidence, current source trace, reviewer-visible boundary, and no material missing condition	GROUNDED
Modeled, inferred, or extrapolated from accepted sources with stated uncertainty	ESTIMATED

Evidence condition	Required or recommended status
Material result depends on declared assumptions that could change gate status, authority selection, or public claim strength	ASSUMPTION_BOUND
Material fact is unknown and cannot be responsibly inferred from the declared surface	DECLARED_UNKNOWN
A fact is outside the declared decision boundary, and the rationale shows it cannot affect the gate or claim under review	OUT_OF_SCOPE_WITH_RATIONALE
Missing, stale, contested, or unsupported evidence could change NCRC/RF, TRC, CSV, deterministic selection, authority selection, or conformance claim	INSUFFICIENT_GROUNDING

Claim-boundary rule. No claim may exceed its reality surface. If the surface is insufficient for a material rights, tail-risk, containment, authority, conformance, or selection claim, the run MUST narrow the claim, collect more evidence, escalate, mark the result exploratory/sensitivity-only, or refuse the stronger claim.

Public teaching line. Reality is the territory. A decision sees only a surface. Evidence grounds the surface. Unknowns mark the edge. Claims must stay inside the ground they stand on.

2.1C Category Grounding and Primitive Audit inside Reality Grounding (Normative)

Purpose. Category Grounding prevents a RippleLogic run from optimizing, governing, scoring, or authorizing action through categories whose definition, boundary, or reference structure is unexamined, stale, inherited without review, or too weak for the claim being made.

Single-group rule. Category Grounding, Primitive Audit, Reference Category Review, and Category-Reality Correspondence Check belong to the Reality Grounding group. They are not new public cascade levels and MUST NOT be represented as separate gates parallel to Rights Floor, Tail-Risk Bound, Containment, or RippleLogic Score. The five-level method remains Reality Grounding -> Rights Floor -> TRC -> CSV -> RLS.

Trigger. Category Grounding is REQUIRED when a material category affects rights status, protected-stakeholder inclusion, SGP interpretation, catastrophic-risk modeling, Containment, authority selection, conformance claims, public deployment claims, or any metric that will be optimized under pressure. It is RECOMMENDED for contested, novel, high-uncertainty, cross-domain, or high-prestige terms whose ordinary use may conceal structural ambiguity.

Materiality test. A category is material when changing its definition, inclusion boundary, exclusion boundary, or reference structure could change the RealityGroundingStatus, NCRC/Rights Floor result, TRC scenario set, CSV result, SGP interpretation, AuthoritySelectionRecord, conformance claim, public deployment claim, deterministic-selection claim, or optimized metric.

Metric-category rule. No metric may exceed the category that defines what it measures. If the category is weak, stale, contested, or insufficiently grounded, the metric may be used only within the resulting claim boundary and must not be laundered into stronger selection, conformance, or public claims.

Minimum record. When the Section 2.1C trigger holds for a claim-bearing Tier 3 run, high-stakes Tier 2 run, or public conformance claim, the CategoryGroundingRecord MUST identify; for lower-stakes or exploratory runs it SHOULD identify: material_category; category_role; primitive_basis; definition; category_discriminator; dependency_position; inclusion_boundary; exclusion_boundary; common_confusions; noncollapse_pairs; reference_structure; evidence_surface; source_authority_status; category_maturity_status; falsification_or_revision_trigger; term_refusal_condition; downstream_dependencies; semantic_debt_flag; reviewer_status; and required_claim_action.

Term Discrimination extension. For material terms, the record MUST identify the discriminator that separates the term from likely neighboring terms, the dependency position the term occupies in the decision stack, and the condition under which the term must be narrowed, escalated, monitored, treated as sensitivity-only, or refused. A downstream term MUST NOT claim upstream authority: certification is evidence, not validity; permission is authorization, not admissibility; monitoring is observation, not control; automation is execution, not autonomy; protection is moral standing, not governance authority; and a score is a bounded computation, not truth.

Claim action. If category grounding or term discrimination is insufficient for a material claim, the run MUST narrow the claim, mark the category ASSUMPTION_BOUND or CONTESTED, collect stronger evidence, escalate for review, restrict use to exploratory/sensitivity-only analysis, or refuse the stronger claim. It MUST NOT launder category uncertainty or term collapse through RLS, PLSS, HDW, PCC completion, authority selection, certification, compliance, monitoring, model fluency, or stakeholder preference.

Plain-language rule. Before MathGov scores a thing, it must check whether the thing has been named and bounded correctly enough for the claim being made.

2.1D Source-Coupling Integrity and Generative-Source Traceability inside Reality Grounding (Normative)

Purpose. Source-Coupling Integrity prevents a RippleLogic run from treating downstream output, institutional permission, compliance status, model fluency, benchmark performance, inherited procedure, or interface success as proof that a claimed capability is grounded in the enabling conditions and boundary conditions that make it possible.

Single-group rule. Source-Coupling Integrity belongs to the Reality Grounding group and is consumed by CSV when source weakness creates structural fragility. It is not a new public cascade level and MUST NOT be represented as a sixth gate. The public method remains RG -> RF -> TRC -> CSV -> RLS.

Trigger. Source-Coupling Review is REQUIRED when a material claimed capability, output, model, metric, procedure, institutional standard, inherited category, generated result, or agentic action affects claim authority, RF/NCRC, TRC, CSV, RLS, SGP interpretation, execution authority, or public conformance and any of the following conditions holds: (i) the output is generated, simulated, benchmark-derived, model-inferred, or dashboard-derived; (ii) the run relies on an inherited procedure, standard, dataset, legal category, or compliance label as if it settled the underlying claim; (iii) a capability is extrapolated beyond the conditions in which it was demonstrated; (iv) downstream controls or monitoring compensate for a limitation whose enabling conditions remain unclear; or (v) a challenger plausibly alleges that performance, permission, compliance, or fluency is being substituted for grounded capability.

Minimum record. When triggered for a claim-bearing Tier 3 run, high-stakes Tier 2 run, or public conformance claim, the SourceCouplingRecord MUST identify; for lower-stakes or exploratory runs it SHOULD identify: claimed_capability; enabling_conditions; boundary_conditions; source_evidence; generator_output_distinction; inherited_assumptions; downstream_compensations; source_coupling_status; source_debt_flag; falsification_or_recheck_trigger; and required_claim_action.

Status values. SourceCouplingStatus values are SOURCE_COUPLED, SOURCE_PARTIAL, SOURCE_INFERRED, SOURCE_UNKNOWN, SOURCE_CONTESTED, SOURCE_DEBT_RISK, and SOURCE_COUPLING_FAILURE. SOURCE_COUPLED may support the declared claim boundary. SOURCE_PARTIAL, SOURCE_INFERRED, SOURCE_UNKNOWN, or SOURCE_CONTESTED requires claim narrowing, assumption-bound use, additional evidence, escalation, or refusal of the stronger claim as appropriate. SOURCE_DEBT_RISK routes the issue to CSV when structural viability depends on the weakly coupled capability. SOURCE_COUPLING_FAILURE means the run is substituting output, fluency, compliance, permission, or inherited procedure for source-coupled evidence and MUST refuse or redesign the stronger claim.

Source-debt diagnostic. Source debt is the structural risk created when a run proceeds through compensatory controls while the enabling conditions, boundary conditions, or limits of a claimed capability remain weak, stale, unknown, overextended, or contested. Source debt is different from semantic debt: semantic debt concerns weak terms or categories; source debt concerns weak understanding of the capability-generating conditions themselves.

CSV routing. CSV SHOULD consume source-coupling evidence when source weakness creates containment risk, dependency risk, hollowing risk, execution non-viability, reversibility failure, monitoring inadequacy, or hidden structural fragility. A weak SourceCouplingRecord does not automatically fail every option, but it constrains the claim and may require controls, redesign, escalation, or refusal.

Agent-output rule. Model fluency, benchmark performance, chain-of-action success, tool-use success, or policy-filter compliance does not establish source-coupled admissibility. Agent outputs remain downstream claims until Reality Grounding and, where material, Source-Coupling Review establish the supported claim boundary.

Related external work note. Alexander Wolfgang Strüver's 2026 paper *Source Amnesia: How Civilizations Lose Their Understanding of Generative Structures* is cited as related external scholarship on the risk of losing generative-source understanding in complex systems. MathGov does not adopt that title as a MathGov module, does not reproduce or adapt that text, and does not bundle the paper into this Apache-licensed release.

Plain-language rule. Before MathGov trusts a capability, it asks what makes the capability possible, what limits came with it, and whether downstream performance is being mistaken for grounded understanding.

2.2 Unions as the Unit of Analysis (Operational Definition)

Definition: A union is a bounded pattern of interdependence: a set of entities whose internal interactions are sufficiently strong, frequent, or consequential that their welfare should be evaluated together for the decision context.

In this specification, the operational objects scored and indexed by u are Union Scopes (Scopes), meaning unions used as stable scopes of aggregation and accountability.

Unions are analytical constructs for causal and welfare accounting; they are not metaphysical substances. The union concept operationalizes the systems-theoretic observation that complex systems exhibit hierarchical modularity (Simon, 1962).

2.2A Union Scopes and Instance Modeling (Normative)

Terminology clarification. In RippleLogic v11.4, the canonical term Union Scope refers to the objects indexed by $u \in \{1,\ldots,7\}$ in the welfare matrix. For readability and reduced ambiguity, Scope is the default shorthand in technical and implementation contexts. The term Union remains an allowed legacy shorthand in narrative contexts and lineage documents. This clarification SHALL NOT change the mathematical meaning of u or any cascade semantics. Machine-readable surfaces SHOULD prefer UnionScope / UnionScopes as the canonical identifier family; Union remains a permitted human-readable alias in prose and UI text.

Purpose. Union Scopes are designed to be minimal but covering, a small stable set of aggregation scopes that prevents systematic stakeholder erasure while supporting auditability and comparability across decisions.

2.2A.1 Definitions (Normative)

Union Scope (Scope). A Union Scope is a stable scope of impact aggregation used to evaluate decision consequences across nested interdependence levels. Union Scopes are the rows of the canonical 7×7 welfare matrix.

Stakeholder. A stakeholder is any entity that is plausibly materially affected by a candidate option a, including entities affected through externalities, indirect pathways, or future propagation within the declared decision horizon.

Instance. An instance is a stakeholder bound to the decision context as stakeholder-in-role-in-context, meaning a stakeholder with a declared role and exposure pathway relevant to the decision. Examples include: employee, patient, local resident, watershed ecology unit, cross-border supply-chain worker, digital agent affected by policy. Instances are the primary unit of stakeholder logging for auditability, Union Scopes are the primary unit of aggregation for comparison.

Structural anchor. Scopes × Dimensions form the welfare matrix. Instances feed the cells.

Local Stakeholder Set (LSS). The LSS is the context-specific set of stakeholder instances materially affected by a decision. It is the upstream representation object produced by stakeholder discovery and instance-to-scope mapping, and it is the justified input used to support any PLSS prominence declaration.

2.2A.2 Stakeholder Discovery Protocol (SDP) (Tier-linked; Normative)

Tier 3: REQUIRED. Tier 2: RECOMMENDED by default and REQUIRED when any SDP trigger holds. When no SDP trigger holds, a Tier 2 run MAY satisfy this requirement with a lightweight stakeholder note stating that no trigger held and why, or MAY voluntarily include a full SDP record. SDP is a short, explicit process for discovering omission-likely stakeholders so discovery is not optional by taste.

Minimum SDP steps (recorded in PCC): (i) define decision boundary and horizon, (ii) map the causal footprint and externality pathways, (iii) scan for rights exposure and catastrophe relevance, (iv) draft the Local Stakeholder Set (LSS) as the stakeholder instance set, (v) map instances to one or more Union Scopes with rationale, (vi) record unknowns and declared blind spots, including escalation triggers, (vii) provide a challenger opportunity for missing instances at Tier 3.

SDP triggers (Tier 2 REQUIRED): rights-covered cells plausibly affected, or TRC triggered/catastrophe relevance plausible, or externalities beyond decision owner plausible (cross-community/organization/polity, biosphere pathway), or irreversibility/lock-in plausible, or credible challenger claims omitted stakeholder class or scope, or information/procedure/enforcement/ecology pathways plausible.

2.2A.3 Instance-to-Scope Mapping Requirement (Tier-linked; Normative)

Tier 3 requirement. A Tier 3 run MUST explicitly log a set of stakeholder instances and map each instance to one or more Union Scopes, with a rationale sufficient for an independent reviewer to understand why the instance belongs in the mapped scope(s). This mapping SHALL be recorded in the PCC (Appendix H).

Tier 2 requirement under triggers. When any SDP trigger holds, Tier 2 runs MUST log and map stakeholder instances for the affected pathways and rights-relevant domains. Otherwise, Tier 2 runs SHOULD still log and map instances whenever the decision plausibly affects stakeholders beyond the decision owner.

Minimum mapping rule. Every logged instance MUST map to at least one Union Scope. If an affected stakeholder cannot be mapped, the PCC MUST record this as an explicit coverage limitation, not as neutrality, including: (i) why mapping was infeasible, (ii) what scope(s) might plausibly apply, and (iii) what risk this creates.

2.2A.4 Multi-scope Mapping and Redundancy Handling (Normative)

Multi-scope mapping is permitted. An instance MAY map to multiple Union Scopes where the decision plausibly affects that instance across multiple nested scales.

No silent duplication. Implementations MUST NOT silently log the same underlying effect multiple times across scopes in a way that inflates welfare scoring without disclosure. If a modeling choice results in an effect being represented across multiple scopes, the PCC MUST disclose the rationale using one declared redundancy-handling method for the affected effect-class: (i) EMERGENT_SCALE_JUSTIFICATION, counted across multiple scopes because the effect is meaningfully different at different scales, (ii) DEDUPLICATION, logged once at a declared primary scope with cross-scope effects represented via ripple propagation and structural metrics, or (iii) ALLOCATION (optional), effect split across scopes to prevent redundant counting.

Effect-token (allocation only). An effect-token is a redundant conserved causal unit used only to prevent redundant counting across scopes, for example the same dollars, the same one-time transfer, the same compliance cost. If ALLOCATION is used for an effect-token, the PCC MUST declare the token, the allocation method, the coefficients used, and the scope list. Allocation coefficients for a single effect-token MUST satisfy $\Sigma_u \alpha_u \leq 1$ for that token, summed over the scopes listed for that token.

No allocation laundering. ALLOCATION is permitted only for declared redundant conserved-unit effect-tokens. Distinct welfare harms, for example health harm versus ecosystem degradation, MUST be logged as distinct impact instances and SHALL NOT be allocated away.

Allocation guard. Allocation SHALL NOT be used to suppress or mask adverse impacts in any rights-covered or catastrophe-relevant evaluation. Redundancy handling cannot weaken NCRC, TRC, or CSV computations.

2.2A.5 Reduced-scope Mode (Normative)

Reduced-scope mode allows an evaluator to run with fewer than 7 scopes for mapping or reporting depth, provided omissions are explicitly disclosed. Reduced-scope mode SHALL NOT remove any non-maskable cells or required scope rows from NCRC, TRC, or CSV computation.

Reduced-scope mode is a reporting or mapping-depth choice only. Admissibility computation MUST still evaluate all rights coverage sets C_r and catastrophe cell set C_{cat} as declared.

If reduced-scope mode is used, the PCC MUST include a Scope Coverage Declaration (Appendix H) listing active scopes, omitted scopes, omission rationales, a blind-spot statement, and escalation triggers that would force scope expansion in the next comparable run.

Common misread (informative). Misread: “Unions limit who matters.” Correction: “Union Scopes prevent impacts from disappearing. Any stakeholder can be represented as an instance mapped into relevant scope(s), and omissions must be disclosed as coverage risk.”

2.3 The Union Stack (Seven Operational Unions)

Union	Name	Definition	Characteristic Timescale
U_1	Self	The individual locus of experience and agency	Seconds to Decades
U_2	Household	Primary cohabitation and resource pooling unit (Becker, 1981)	Days to Decades
U_3	Community	Local repeated-interaction network with social capital and trust dynamics (Putnam, 2000; Dunbar, 1993)	Months to Generations
U_4	Organization	Formal collective pursuing a purpose; structured coordination and institutional behavior (March & Simon, 1958; North, 1990)	Years to Centuries
U_5	Polity	Governance authority unit over a jurisdiction; legitimacy and institutional structure (Weber, 1978)	Decades to Centuries
U_6	Humanity / CMIU	Collective Managing Intelligence Union: all managing intelligences; global coordination and systemic risk management (Ord, 2020; Steffen et al., 2015)	Generations to Millennia
U_7	Biosphere	Earth’s integrated life-support systems, including climate stability and ecosystem integrity (Odum, 1971; Steffen et al., 2015; Rockström et al., 2009, 2023)	Centuries to Epochs

Canonical nesting chain: $U_1 \subset U_2 \subset U_3 \subset U_4 \subset U_5 \subset U_6 \subset U_7$

Clarifying distinction (informative). Organization refers to bounded institutions and role systems such as firms, schools, NGOs, or agencies. Polity refers to public governance and enforcement domains such as lawmaking, courts, regulators, and due process systems.

Ecological trophic cascades illustrate how interventions can propagate unpredictably across levels and why “local wins” can generate system losses, reinforcing the need for explicit containment and ripple modeling (Estes et al., 2011).

2.4 Union Types vs Instances (Aggregation Rule)

The seven unions are types; real runs involve instances (many households, many organizations, many polities). Impacts MUST be aggregated within a union row using a declared aggregation method.

Default aggregation (normative default for non-rights scoring): Population-weighted mean across affected instances.

Rights exception: For rights-covered cells, apply worst-off subgroup checks within instances before aggregating (Section 7).

Examples: Stakeholder Instances Mapped into Union Scopes (Non-Normative)

This table illustrates how diverse stakeholders are represented as instances mapped into Union Scopes. The scope set is minimal but covering, it is a coordinate system for accountability, not a closed list of who matters.

Stakeholder instance (example)	Typical role/context	Mapped scope(s) (example)	Notes (including multi-scope redundancy method when relevant)
Me (decision owner)	direct personal impact	U1 Self	Direct experience and agency locus.
Child or dependent in my care	care, safety, needs	U2 Household (and U1 for the child-as-person in detailed runs)	Household stability plus rights exposure often relevant.
Local neighborhood residents	local externalities	U3 Community	Trust, cohesion, local health and environment.
Employees or contractors	workplace policy impacts	U4 Organization	Pay, safety, dignity, due process.
Regulators, courts, enforcement bodies	compliance and legitimacy	U5 Polity	Law, procedure, rights enforcement capacity.
Cross-border supply-chain workers	indirect externalities	U6 Humanity/CMIU	Captures externalities beyond a single polity.

Stakeholder instance (example)	Typical role/context	Mapped scope(s) (example)	Notes (including multi-scope redundancy method when relevant)
Watershed or river system	ecological integrity	U7 Biosphere	Often ecology-rights and TRC relevant.
Future generations	long-horizon impacts	U6 Humanity/CMIU + U7 Biosphere	Multi-scope mapping. Typically EMERGENT_SCALE_JUSTIFICATION, because effects differ across scales. Horizon and indicators must be explicit.
Public information ecosystem	misinformation, censorship, legitimacy	U5 Polity + U6 Humanity/CMIU	Multi-scope mapping. If represented in both scopes, declare redundancy handling. Default is EMERGENT_SCALE_JUSTIFICATION, or DEDUPLICATION with propagation.
Digital agents affected by policy	AI deployment governance	U6 Humanity/CMIU (SGP-gated for protections where applicable)	Protections handled via the SGP interface. Authority remains separately gated.

2.5 Constructive vs Pathological Unions

Some unions grow by degrading their containing unions (for example, extractive industries or corruption networks). RippleLogic therefore does not assume “union benefit” is automatically good. The CSV Gate prevents local optimization that damages containing union coherence or viability beyond governed tolerance.

2.6 Meta-Unions (Non-Computational by Default)

U_8 Cosmic and U_9 Universal / All-Encompassing Infinite Union (AIU) may be used as philosophical boundary conditions or future extensions, but do not participate in standard Tier 1-3 scoring unless formally activated through a future governed extension protocol recorded in the canon, release manifest, and PCC requirements for that release line.

2.6A Operational UBR vs Metaphysical AIU (Normative Boundaries and Scope Discipline)

This section clarifies the strongest science-disciplined interpretation of Union-Based Reality (UBR), Union-Based Ethics (UBE), and the Universal/AIU horizon within the RippleLogic v11.4 line. Its purpose is to preserve conceptual clarity, prevent overclaiming, and maintain a clean boundary between what is currently operational, what is normatively stipulated, and what remains philosophical, phenomenological, or future-facing.

Operational UBR. For the purposes of Tier 1 through Tier 3, UBR SHALL be interpreted operationally as a systems ontology of nested causal interdependence. In governance-relevant domains, entities and decisions are embedded in coupled structures of consequence such that actions propagate across multiple scopes of impact and accountability. This is a descriptive claim about consequence topology, not a proof that all beings are identical in substance or ultimate ontology.

Scope discipline. Implementations and public claims MUST distinguish among: (i) the operational-descriptive layer, where nested consequence is modeled across the scored union stack; (ii) the normative layer, where UBE enters through explicit commitment; and (iii) the meta-union horizon layer, where Cosmic and Universal/AIU function as philosophical, humility, precautionary, or future-extension concepts. These layers MUST NOT be collapsed into one another.

Operational boundary. The standard scored union stack for Tier 1 through Tier 3 remains U1 Self, U2 Household, U3 Community, U4 Organization, U5 Polity, U6 Humanity/CMIU, and U7 Biosphere. U8 Cosmic and U9 Universal / All-Encompassing Infinite Union (AIU) are meta-unions and are non-computational by default in the current release line.

Permitted role of meta-unions. U8 and U9 MAY be used as philosophical boundary conditions, humility reminders, precautionary amplifiers, or future-extension placeholders. They MUST NOT, absent a governed extension protocol, be presented as standard Tier 1 through Tier 3 scoring objects, admissibility overrides, or evidence-bypass mechanisms.

Interpretation of “we are one.” Within RippleLogic v11.4, statements of oneness SHALL be interpreted in three distinct registers: (i) an operational register of strong interdependence and nested accountability; (ii) a structural-unity research register, concerning whether convergent mathematical or physical principles underlie the nested interdependence observed across scales, and whether such principles constrain viable governance architectures; and (iii) an absolute-metaphysical register that remains outside current computational proof status. Only the first register is part of standard Tier 1 through Tier 3 governance computation.

Phenomenology and horizon claims. Reports of unity, non-separation, or all-encompassing union arising from contemplative, mystical, near-death, psychedelic, or out-of-body experiences MAY be treated as phenomenological data, horizon signals, or motivation for future inquiry. They MUST NOT, by themselves, be treated as sufficient evidence for standard parameterization of U9 or for compliance claims under the canon.

Non-overclaiming rule for AIU language. The framework MAY be described as general-purpose and multi-scale, and as aiming toward broadly universalizable governance architecture for embedded intelligences in shared causal reality. It MUST NOT be described as having scientifically proven All-Encompassing Infinite Union, fully resolved all metaphysical disagreements, or already solved universal alignment in the strongest ontological sense.

Canonical summary. RippleLogic adopts UBR as an operational ontology of nested interdependence within measurable shared reality, adopts UBE through an explicit normative entry, and retains Cosmic and Universal/AIU language as non-computational horizon layers unless and until a governed extension protocol makes broader parameterization legitimate.

SECTION 3: NORMATIVE FOUNDATIONS: UNION-BASED ETHICS (UBE)

3.0A Ethical Reality

Ethical reality is the subset of reality where structures, constraints, and consequences affect welfare-bearing beings and the conditions of flourishing. Ethical reality includes welfare, rights, sentience, agency, dignity, non-domination, catastrophic-risk exposure, ecological dependence, future generations, and flourishing. Access to ethical reality spans measured fact, uncertainty-bound inference, and explicit normative commitment. These layers MUST NOT be collapsed: measured indicators may support claims, SGP may govern uncertain moral-patient evidence, and rights floors operate as declared non-compensatory commitments rather than as empirical measurements alone. Ethical reality is therefore not separate from reality; it is reality where consequence matters to beings and systems capable of being harmed, protected, dominated, liberated, degraded, or enabled to flourish.

3.1 Minimal Normative Axiom (MNA)

RippleLogic adopts exactly one explicit normative axiom:

MNA: Sentient welfare and flourishing matter. Unnecessary suffering and domination should be reduced. The enabling conditions for viable, non-dominated, and continuing flourishing should be preserved.

This axiom is:

Content-minimal: No single “good life” doctrine is prescribed.

Scope-bounded: Applies where sentience or welfare-relevant moral patienthood is in scope.

Architecturally generative: The framework defines a compositional architecture, rights floor, tail-risk constraint, containment, welfare optimization, and structural safeguards, such that diverse domain-specific instantiations can be generated by declared inputs and then executed through the same cascade.

Normatively explicit: RippleLogic does not claim to derive this axiom from mathematics alone or from descriptive facts alone.

3.1A Core Definitions for the Normative Layer (Normative)

Interpretation rule: RippleLogic protects welfare-bearing subjects directly and protects enabling conditions structurally where their degradation predictably undermines welfare, rights protection, non-domination, or long-run system viability.

3.1B Bridge Premises (Normative support; non-deductive)

The MNA is accompanied by two bridge premises that reduce arbitrariness while preserving honesty about the entry of normativity.

Bridge Premise 1: Embedded Modelling. Any agent seeking reliable action in shared causal reality must model other stakeholders, including their interests, constraints, vulnerabilities, dependence relations, rights exposure, and, where behavior, harm, cooperation, or rights protection depend on them, welfare-relevant states and enabling conditions, to a non-trivial degree.

Bridge Premise 2: Prudential Stability. In plural, uncertain, co-embedded systems, architectures that systematically degrade stakeholders' welfare conditions, rights floors, or shared enabling conditions increase instability, conflict, tail risk, and long-run governance failure.

These premises do not generate obligation by deduction. They explain why welfare-tracking, rights-tracking, and enabling-condition tracking are structurally relevant to intelligence acting in shared reality.

3.1C Normative Commitment (Explicit final step)

RippleLogic adopts the commitment that credibly evidenced welfare-relevant states, rights-relevant states, and the enabling conditions on which they depend constitute reasons for action, not merely inputs for prediction.

This final step, treating such states and conditions as reasons rather than mere data, remains a normative commitment, not a deductive conclusion. RippleLogic holds that this is among the narrowest defensible gaps between description and obligation currently achievable in a governance-grade framework without either claiming false deduction or retreating to brute preference, and treats closing it further as an active research question.

3.1D Evidence Discipline for Normative Claims (Normative)

Within RippleLogic, a putative welfare-relevant state, rights-relevant state, or enabling condition SHALL enter governance computation only through declared evidence, governed interpretation, or an explicit precautionary treatment under uncertainty appropriate to the implementation tier.

If such a state or condition materially affects admissibility, precautionary handling, selection, or tie-break outcome, it MUST be recorded in PCC.NormativeInputRegister with sufficient provenance for independent review.

Unsupported narrative assertion, rhetorical force, linguistic fluency, or undeclared evaluator intuition SHALL NOT by itself constitute a sufficient reason-for-action input within RippleLogic.

3.2 Conditional Is-Ought Bridge

RippleLogic avoids deriving values from facts by using a conditional bridge:

IF

(UBR) actions propagate through nested unions,

AND

(MNA) sentient welfare and flourishing matter, unnecessary suffering and domination should be reduced, and the enabling conditions for viable, non-dominated, and continuing flourishing should be preserved,

AND

(BP1) agents operating in shared causal reality must model other stakeholders' interests, constraints, vulnerabilities, dependence relations, rights exposure, and, where behavior, harm, cooperation, or rights protection depend on them, welfare-relevant states and enabling conditions to act reliably,

AND

(BP2) in plural, uncertain, co-embedded systems, systematic degradation of stakeholders' welfare conditions, rights floors, or enabling conditions increases instability, conflict, tail risk, and long-run governance failure,

THEN

agents and institutions ought to evaluate choices by cross-union impacts and preservation of enabling conditions, using non-compensable rights floors, explicit tail-risk bounding, containment safeguards, and auditable welfare ranking within the feasible set.

Normativity enters explicitly through the MNA and the final normative commitment. UBR specifies where consequences flow. BP1 and BP2 explain why welfare-tracking, rights-tracking, and enabling-condition tracking are structurally relevant to intelligence acting in shared reality. RippleLogic does not claim that the final normative step is fully deduced from description alone.

3.2A Derived Governance Rule (Normative)

In governance contexts, RippleLogic operationalizes the above commitment through the canonical lexicographic cascade:

A_sel := {a in O : NCRC(a) AND TRC(a) AND CSV_status(a) in {CSV_PASS, CSV_PASS_WITH_CONTROLS, CSV_NOT_MATERIAL}}. If A_sel is empty, escalate, redesign, or invoke governed fallback per the canon. Otherwise rank over A_sel using RLS and the declared tie-break chain.

1. Protect rights floors first through NCRC.

2. Bound catastrophic downside second through TRC.

3. Block pathological containing-scope degradation through CSV as the third option-filtering gate after the RG claim-authority precondition.

4. Rank remaining feasible options fourth through welfare scoring under declared weights, uncertainty rules, and auditability requirements.

5. Resolve non-decisive cases structurally through UCI/HOI tie-breaks and escalation rules.

Interpretation rule: RippleLogic does not define “ought” as unconstrained welfare maximization. It defines “ought,” in governance-grade decision settings, as selection from a rights-safe, tail-bounded, containment-safe feasible set under auditable evidence and uncertainty discipline.

3.2B Open Research Frontier (Informative)

An active research question remains whether sufficiently accurate multi-agent world-modelling creates deeper constitutive pressure toward non-domination and welfare protection beyond prudential stability alone. RippleLogic does not assume this as a proven result. It treats it as a frontier question concerning whether accurate modelling of shared reality may eventually narrow the remaining descriptive-to-normative gap further. Specifically, this question is tractable through empirical study of whether systems that model co-embedded agents more accurately also exhibit more stable cooperative and non-destructive behaviour, and whether omission of welfare-relevant, rights-relevant, or enabling-condition variables from world models degrades predictive performance in domains where cooperation, conflict, or harm depend on them.

3.3 Seven Welfare Dimensions (Canonical)

Dimension	Name	Description
D_1	Material	Resources and infrastructure for survival and functioning
D_2	Health	Physical and mental functioning; morbidity and mortality risk
D_3	Social	Belonging, trust, relational integrity, cooperation
D_4	Knowledge	Epistemic access and learning conditions
D_5	Agency	Autonomy and effective choice; freedom from coercion
D_6	Meaning	Coherence, purpose, valued life projects (measured cautiously)
D_7	Environment	Ecological and built context integrity sustaining life

This seven-dimensional choice is a convergence architecture rather than a claim of metaphysical completeness. It aligns with the capability approach (Sen, 1999; Nussbaum, 2011), self-determination theory (Ryan & Deci, 2000), and fundamental human needs theory

(Max-Neef, 1991), while also capturing ecological integrity as a life-support substrate consistent with Earth-system boundary research (Rockström et al., 2009, 2023).

3.4 Non-Fungibility at Rights Level

Dimensions are treated as non-fungible at the rights layer: gains in one dimension MUST NOT compensate rights-floor violations in another. This is enforced structurally by NCRC (Section 7), not by weight tuning.

3.5 Unioning (Enacted Practice)

Unioning = redesigning decisions until they are rights-safe (NCRC), tail-safe (TRC), containment-safe, and net-positive across unions under declared uncertainty policy, treating apparent “tradeoffs” as design failures before acceptance.

Unioning worked example (Non-Normative)

Scenario. An organization considers cutting safety training to reduce costs. Initial estimate shows a small material benefit in U4 Organization D1, but a credible safety degradation in U4 D2 and a dignity/procedure risk in U4 D6/D5. Under NCRC, any rights-covered violation depth $v_r(a)>0$ fails admissibility regardless of aggregate gains.

Step 1, fail. Option A fails NCRC due to increased probability of serious injury and due-process erosion. It is rejected even if it would raise RLS.

Step 2, redesign. The team introduces Option B, maintain safety training, add targeted efficiency changes elsewhere, and add a transparent incident-reporting and remediation pathway. This shifts the harmful impacts above the rights floor while preserving most of the intended cost reduction through non-harmful channels.

Step 3, pass and select. Option B passes NCRC, passes TRC and CSV, then competes on RLS among admissible options. The PCC records the redesign rationale and the rights-floor reasoning, so the improvement is legible and replayable.

3.6 NCAR Learning Loop (Corrigibility Requirement)

RippleLogic is embedded in Notice, Choose, Act, Reflect. The system is corrigible: it records assumptions and outcomes, updates kernels and parameters via governed procedures, and preserves accountability through versioned PCCs.

SECTION 4: SYSTEM OVERVIEW: THE RIPPLELOGIC ARCHITECTURE

Public method. RippleLogic SHOULD be taught and implemented at the public level as:

Reality Grounding -> Rights Floor -> TRC -> CSV -> RLS

Human mnemonic:

Ground -> Protect -> Bound -> Contain/Verify -> Score

Formal method. In formal notation and audit surfaces, this corresponds to: RG/RSG -> RF/NCRC -> TRC -> CSV -> RLS, where RG is Reality Grounding and RSG is the reality-surface grounding implementation record.

Clarification. Reality Grounding is a claim-authority precondition, not an option-rejecting gate. RF/NCRC, TRC, and CSV gate admissibility and selectability. RLS ranks the selectable set. The five-level method therefore consists of one claim-authority precondition (RG), three hard feasibility/selectability gates (RF/NCRC, TRC, CSV), and one residual ranking layer (RLS).

Level 1, Reality Grounding (RG/RSG). Declare the reality surface, evidence trace, material unknowns, transition boundary, consequence pathways, category boundaries, and claim boundary. If grounding is insufficient for a material claim, narrow, escalate, mark exploratory, or refuse the stronger claim.

Level 2, Rights Floor (RF/NCRC). Apply the Non-Compensatory Rights Constraint. Rights-floor harms are non-compensatory and cannot be rescued by welfare score.

Level 3, TRC / Tail-Risk Bound. Apply the Tail-Risk Constraint. Catastrophic or unbounded downside cannot be averaged into ordinary welfare gains.

Level 4, CSV / Containment and Structural Viability. Apply the CSV gate. CSV asks whether the option's local gain can stand without unacceptably damaging, hollowing, overloading, exploiting, or depending on the larger systems that make the gain possible. UCI/HOI and related structural diagnostics are evaluated inside CSV when material. Structural Viability is explicit inside CSV: resource closure, dependency closure, endpoint/reversibility coherence, operational capacity non-exceedance, and internal non-contradiction.

Level 5, RLS / RippleLogic Score. Rank only the options that remain selectable. If RLS is tied, close, or non-decisive, apply the special-case tie-break rule using residual UCI/HOI where available or return REFUSE_DETERMINISTIC_SELECTION.

Mantra. Ground -> Protect -> Bound -> Contain/Verify -> Score.

Layer	Question answered	Output
RG / Reality Grounding	What reality surface, evidence trace, unknowns, transition boundary, consequence pathways, category boundaries, and claim boundary support this run?	Grounded, narrowed, escalated, exploratory, or refused.
NCRC	Does the option violate a non-compensatory rights floor?	Rights-admissible or rejected.
TRC	Is catastrophic tail risk acceptably bounded?	Admissible or rejected.

Layer	Question answered	Output
CSV	Can the option deliver its claimed good without uncontained harm, hollowing, lock-in, structural non-viability, or degrading containing systems beyond tolerance?	Selectable, pass with controls, redesign required, emergency provisional, excluded, or escalated.
RLS	Among selectable options, which has the strongest residual welfare pattern?	Ranked selectable options.

4.1 Formal Definition

RippleLogic is a rights-first, tail-risk-bounded, union-based decision optimization system with explicit ripple propagation, auditable scoring, and corrigible learning. It is designed to prevent three common governance failures: rights being traded away by aggregate welfare gains, catastrophic downside being averaged away by expected-value logic, and local optimization degrading larger containing systems.

4.2 Core Objects and Implementable Data Structures

Every RippleLogic run works from declared objects. These are not decorative labels; they are the minimum data surfaces that make the decision auditable, reproducible, and challengeable.

Object	Description
Option set O	Finite set of candidate actions.
Welfare impact matrix per option	I_prop(u,d,a) in [-1,+1], after direct impact construction, propagation, and saturation. Cell interpretation is governed by the canonical 49-cell reference grid and Appendix AD.
Rights impacts	I_rights(u,d,a), using worst-off protected subgroup semantics for rights-covered cells.
Scenario set S with probabilities p_s	Scenario surface for TRC, including mandatory tail categories and probability floors where required.
Catastrophe cells C_cat and weights omega	Catastrophe-relevant welfare cells and tail-risk weights used in TRC.
Kernel K, sparse 49x49	Propagation structure. NONE or QUICK modes are permitted in Tier 1-3 subject to the canon.

Object	Description
Weights w_u and v_d	Union-scope and dimension weights, governed by HDW where required and constitutional floors where applicable.
Structural metrics UCI and HOI	Containment and tie-break diagnostics, subject to maturity boundaries.
ReferenceStructureRecord	The declared reference object, evidence status, causal pathway, falsification condition, authority boundary, and refusal trigger for the claim being made.

4.3 Reference Structure Discipline

RippleLogic decisions SHALL NOT be interpreted as governance claims merely because a scoring procedure, audit record, or policy wrapper exists. A governance claim is supportable only to the extent that the decision record declares the reference structure against which the claim is evaluated.

For RippleLogic purposes, a reference structure is the declared set of constraints, evidence sources, causal pathways, thresholds, maturity labels, and refusal conditions used to evaluate a decision claim. Reference structures may include physical constraints, biological needs, rights floors, catastrophic-risk corridors, containment conditions, stakeholder impact maps, institutional authority boundaries, evidence-status labels, and auditable records.

RSP and RefStructRecord. The Reference Structure Protocol (RSP) is the discipline. The ReferenceStructureRecord (RefStructRecord) is the run-level artifact that records the discipline in a specific case. RSP tells the system what must be declared; RefStructRecord records what was actually declared.

Reference structure discipline does not create a sixth gate and does not alter the canonical method. Reality Grounding, NCRC, TRC, CSV, RLS, and UCI/HOI diagnostic handling retain their existing roles. The Reference Structure Protocol strengthens the evidence substrate used to evaluate those roles.

A RippleLogic run MUST NOT treat circular, self-validating, missing, unfalsifiable, materially stale, or unsupported reference claims as sufficient for high-stakes admissibility, selection, deployment-readiness, or conformance claims. Where the reference structure is insufficient for the requested claim, the correct outcome is Refusal, escalation, narrower claim scope, additional evidence gathering, or bounded provisional action under an applicable triage rule.

Plain-language rule. Governance without a declared reference structure becomes assumption management. RippleLogic requires declared references, visible evidence, falsifiable claims, and refusal when the reference structure cannot support the claim.

4.4 Formal Gate Sequence and Residual Tie-Break (Normative)

RippleLogic evaluates candidate options through a formal gate sequence with a claim-authority precondition. Reality Grounding first establishes whether the requested claim can be made at the declared strength. RF/NCRC, TRC, and CSV then determine admissibility and selectability. RLS ranks only the selectable set. Residual UCI/HOI is a special-case tie-break or monitoring rule, not a regular public cascade level. A later function MUST NOT rescue, compensate for, or reinterpret failure at an earlier function.

Formal gate 1, NCRC: rights feasibility. NCRC determines whether an option is rights-admissible. For rights-covered cells, RippleLogic applies the required subgroup semantics and checks the worst-off protected subgroup against the rights floor. Any option with positive violation depth on any covered right fails NCRC and is inadmissible, regardless of expected welfare gains, downstream rankings, or structural advantages.

Formal gate 2, TRC: catastrophe feasibility. TRC determines whether a rights-admissible option is admissible in the tail-risk sense. It evaluates catastrophic-loss exposure over the governed scenario set using the catastrophe cell set and declared CVaR corridor. Any option whose catastrophic tail exceeds the corridor fails TRC and is inadmissible, even if expected welfare performance is strong.

Formal gate 3, CSV: containment and structural viability feasibility. CSV determines whether an admissible option is selectable with respect to containing-system integrity and execution viability. It asks whether local or sub-scope welfare gains degrade containing unions, hollow out capacities, create lock-in, externalize burdens, exceed operational capacity, or fail structural viability. An option that fails CSV is not selectable, even if it passes NCRC and TRC and would otherwise rank highly on residual welfare.

Ranking stage, RLS: residual welfare ranking. RLS is applied only to options that have already passed RF/NCRC, TRC, and CSV. It ranks surviving options by weighted welfare aggregation over the active cell set, subject to non-maskable-cell rules and weight-governance requirements. RLS MUST NOT override or soften any earlier gate.

Special-case UCI/HOI tie-break rule: structural resolution after RLS non-decisiveness. UCI and HOI are first evaluated inside CSV when material to structural integrity, hollowing risk, or containment. Only after options are already selectable may residual UCI/HOI help document a tied, close, uncertainty-overlapped, or non-decisive RLS result. UCI and HOI do not act as public cascade stages and do not reopen prior gate decisions.

Non-overlap rule. For interpretation and auditability, the formal gate/ranking functions SHALL be read as follows: RG/RSG = claim authority; NCRC = rights feasibility; TRC = catastrophe feasibility; CSV = containment and structural viability feasibility; RLS = residual welfare ranking within the selectable set; UCI/HOI tie-break = special-case non-decisive structural resolution or monitoring within the selectable set, not a regular public cascade level. No implementation, commentary, or downstream artifact may collapse these functions into a single blended score.

Level	Name	Function	Failure consequence
1	RG / Reality Grounding	Claim grounding and reality-surface discipline; claim-authority precondition, not an option-rejecting gate.	Narrow, escalate, mark exploratory, or refuse stronger claim.
2	Rights Floor (RF/NCRC)	Rights feasibility.	Rejected except Emergency Mode.
3	TRC / Tail-Risk Bound	Catastrophic tail-risk feasibility.	Rejected or escalated.
4	CSV / Containment and Structural Viability	Containing-system integrity, structural viability, hollowing/dependency/lock-in screening, and controls/redesign logic.	Pass, pass with controls, redesign required, emergency provisional, excluded, escalated, or refused.
5	RLS / RippleLogic Score	Residual welfare ranking among selectable options.	Selection if decisive; if tied/close/non-decisive, apply tie-break rule or refuse deterministic selection.

4.4A Structural Viability within CSV (Normative clarification)

Structural Viability is explicit inside CSV, not a sixth public method level. It asks whether an option that otherwise passes Reality Grounding, NCRC, TRC, and containment integrity can structurally execute within the declared reality surface without undefined resource, dependency, reversibility, operational-capacity, or internal-contradiction failure.

Formal shorthand: RG/RSG -> RF/NCRC -> TRC -> CSV -> RLS. Where material, the CSV result includes Structural Viability status before any option enters RLS. An option may be ethically preferable as a redesign target while still failing current structural viability for execution. In that case, the option may be documented as SVC_DESIGN_TARGET or CSV_REDESIGN_REQUIRED, but it MUST NOT be claimed as currently selectable, executable, deployment-ready, or deterministically selected.

Non-import boundary. Structural Viability uses only MathGov-native records: resource declarations, dependency declarations, reversibility and endpoint status, operational limits, and contradiction checks. It MUST NOT import capacity/binding ontology, regime coordinates, hidden kernels, or undisclosed reference generators.

4.5 Cascade Gate Inputs and Outputs (Informative)

This table summarizes what each gate consumes, what it produces, and what it may eliminate or flag. It is a fast correctness check for implementers.

Gate / layer	Consumes	Produces	Eliminates / flags
Reality Grounding layer	Reality surface, evidence trace, material unknowns, transition boundary, consequence pathways, claim boundary, and reviewer/evidence status where material.	RealityGroundingStatus; claim boundary; refusal/escalation/narrowing disposition.	Blocks, narrows, escalates, marks exploratory, or refuses claims that exceed the grounded surface; does not compare options as an admissibility gate.
NCRC rights floor	Rights-covered cell impacts I_prop_base(u,d,a) for C_r; subgroup operator gamma; rights thresholds.	rights_admissible(a) in {0,1}; violation depths v_r(a); redesign targets.	Eliminates any option with v_r(a)>0; flags RIGHTS_CELL_OMITTED_INVALID where required rights cells are omitted, excluded, or undisclosed.
TRC tail-risk / CVaR	Scenario set S with categories and floors; catastrophe loss proxy L(a,s) over C_cat; alpha; tau_TRC.	TRC_pass(a) in {0,1}; CVaR_alpha[L(a)].	Eliminates any option with CVaR_alpha > tau_TRC; flags missing mandatory tail categories, probability floor violations, or omitted catastrophe cells.
Containment system safety	Containment triggers; structural posture thresholds; Delta UCI / HOI / Adaptive Reserve or equivalent diagnostics where material; reversibility and repair evidence.	containment_pass(a) in {0,1}; containment posture notes; remediation plan if unavailable.	Eliminates or escalates options failing containment; flags CONTAINMENT_STRUCTURAL_DIAGNOSTIC_UNAVAILABLE or CONTAINMENT_UCI_HOI_UNAVAILABLE when required evidence is missing.
RLS welfare selection	Active welfare cells, including rights-covered and catastrophe cells subject to non-maskable rules; weights w_u and v_d; applicability mask m(u,d); uncertainty proxies.	RLS(a); Gap(a,b); welfare ranking over remaining options.	Does not override NCRC/TRC/CSV; flags invalid masking of rights or catastrophe cells from RLS aggregation.
Tie-break / monitoring diagnostics	Residual UCI/HOI or equivalent structural indicators not already decisive for Containment; monitoring and hollowing documentation.	Tie-break signal, monitoring warning, or future-run steering.	Not a gate; may be used only when RLS is tied, close, uncertainty-overlapped, or non-decisive, or for monitoring. Cannot rescue earlier failure.

4.6 End-to-End Algorithm (Tier 1-3 Executable) (Normative)

Algorithm: RippleLogic_Run

Inputs, minimum: decision scope; baseline; option set O; union and dimension sets; weights; rights canon; TRC canon when required; containment parameters; propagation mode; kernel if used; scenario set where Tier-3 TRC is required.

Outputs: selected option a* or escalation, plus PCC.

Step 1: Reality Grounding and Notice. Record the decision question, scope boundary, time horizon, affected unions, and option set. Declare tier, configuration, reality surface, evidence trace, material unknowns, transition boundary, consequence pathways, claim boundary, and reference-structure requirements. If grounding is insufficient for a material claim, narrow, collect evidence, escalate, mark exploratory, or refuse the stronger claim before proceeding.

Step 2: Impact construction. For each option a, estimate direct impacts from impact instances; saturate into I_dir; propagate in NONE or QUICK to obtain I_prop; apply post-propagation saturation. Record uncertainty notes and missing-data penalties as required.

Step 3: NCRC gate. For rights-covered cells, compute worst-off subgroup impacts I_rights. Compute violation depths v_r(a). Options with any v_r(a) > 0 fail NCRC. If the passing set is empty, invoke Emergency Mode.

Step 4: TRC gate. For each rights-admissible option, compute scenario losses L(a,s) over catastrophe cells and compute CVaR_alpha. Exclude options exceeding tau_TRC. If no option remains, invoke TRC fallback.

Step 5: CSV gate. For each NCRC/TRC-admissible option, apply the containment predicate using required containing-scope, structural posture, reversibility, repair, and evidence-availability inputs. Where UCI, HOI, Adaptive Reserve, lock-in, hollowing, trust erosion, ecological degradation, governance-capacity loss, or related structural diagnostics are material to cross-scale structural integrity, evaluate those diagnostics inside CSV before RLS. If required structural diagnostic evidence is unavailable, downgrade, collect evidence, rerun, escalate, refuse the stronger claim, or record a permitted waiver where the applicable tier permits waiver.

Step 6: RLS ranking. Compute RLS(a) only for selectable options. Compute uncertainty and apply the discrimination rule where required. If decisive, select the top option.

Step 7: Special-case tie-break / non-decisive handling. If RLS is tied, close, uncertainty-overlapped, or otherwise non-decisive, compare only options already in the selectable set. Apply residual UCI/HOI only as tie-break, monitoring, or hollowing-risk documentation. Do not reopen NCRC, TRC, or CSV. Do not use residual UCI/HOI to rescue failed Containment or compensate for missing containment evidence. If still non-decisive, return REFUSE_DETERMINISTIC_SELECTION, escalate, or record authorized judgment through the proper AuthoritySelectionRecord.

Step 8: Emit PCC. Record inputs, intermediate computations, gate results, sensitivity outputs where required, ReferenceStructureRecord where required, refusal record if applicable, and public rationale.

4.7 Canonical 49-Cell Reference Table (v10.6+; Informative)

Each cell in the 7×7 welfare matrix represents a distinct accountability domain. The table below provides a compact one-line meaning for each cell to support comprehension and consistent interpretation across implementations. The table is informative; the governing equations and rules remain in the relevant normative sections and appendices.

Interpretive note. This table gives the compact canonical grid. Appendix AD provides the expanded 49-cell Welfare Dictionary, including plain meanings, positive and negative movement, indicator examples, possible gate relevance, examples, and reviewer checks. Appendix AD does not change the cascade, equations, rights thresholds, TRC mechanics, Containment semantics, RLS formula, HDW, SGP, PCC, RSP/RefStructRecord, or claim boundaries. It governs scoring interpretation and reviewer literacy.

Union / Dimension	D1 Material	D2 Health	D3 Social	D4 Knowledge	D5 Agency	D6 Meaning	D7 Environment
U1 Self	personal resources	personal health and safety	personal relationships and cohesion	personal knowledge and learning	personal agency and autonomy	personal meaning and purpose	personal environmental conditions
U2 Household	household resources	household health and safety	household relationships and cohesion	household knowledge and learning	household agency and autonomy	household meaning and purpose	household environmental conditions
U3 Community	community resources	community health and safety	community relationships and cohesion	community knowledge and learning	community agency and autonomy	community meaning and purpose	community environmental conditions
U4 Organization	organizational resources	organizational health and safety	organizational relationships and cohesion	organizational knowledge and learning	organizational agency and autonomy	organizational meaning and purpose	organizational environmental conditions
U5 Polity	polity/public resources	polity/public health and safety	polity/public relationships and cohesion	polity/public knowledge and learning	polity/public agency and autonomy	polity/public meaning and purpose	polity/public environmental conditions

Union / Dimension	D1 Material	D2 Health	D3 Social	D4 Knowledge	D5 Agency	D6 Meaning	D7 Environment
U6 Humanity/CMIU	humanity-wide resources	humanity-wide health and safety	humanity-wide relationships and cohesion	humanity-wide knowledge and learning	humanity-wide agency and autonomy	humanity-wide meaning and purpose	humanity-wide environmental conditions
U7 Biosphere	biospheric resources	biospheric health and safety	biospheric relationships and cohesion	biospheric knowledge and learning	biospheric agency and autonomy	biospheric meaning and purpose	biospheric environmental conditions

Release packaging rule (v10.6+). Every public release PDF MUST include this 49-cell table and the stakeholder-to-scope examples table below, either in Section 4 or in a dedicated reference-table appendix that is directly linked from Section 4.

Scenario	Primary scope logging	Secondary / cross-scope impacts	Redundancy handling note
Parent transfers $100 to adult child.	Recipient: U2 Household (child household) x D1 Material.	Sender household reduced resources, U2 x D1; governance/admin impacts logged separately if applicable.	DEDUPLICATION: log the transfer effect-token once; do not create a second impact instance for the same token.
City spends $1M on public hospital upgrades.	Community: U3 Community x D2 Health.	Households benefit; polity accountability impacts may exist.	ALLOCATION if the same investment benefits multiple scopes; record effect-tokens and coefficients.
National carbon tax: estimated 60% benefit to biosphere, 40% to humanity/CMIU.	Allocate the policy benefit token across U7 and U6 with coefficients 0.6 and 0.4.	Polity administrative burden and compliance costs logged as distinct impacts.	ALLOCATION: explicit coefficients; no allocation laundering.
Factory emissions affect local residents and ecosystem.	Local health harm to residents: U3 Community x D2 Health.	Ecosystem degradation: U7 Biosphere x D7 Environment.	EMERGENT_SCALE_JUSTIFICATION: different mechanisms at different scales, not duplicate counting.
River communities 50km downstream affected by discharge.	Downstream residents: U3 Community, separate instance from factory-neighbor community.	Potential rights exposure flagged through stakeholder discovery.	SDP externality detection: omission-likely stakeholder class logged separately.

4.8 Implementation Tiers (Normative Summary)

Requirement	Tier 1 Heuristic	Tier 2 Core	Tier 3 Auditable
NCRC	Heuristic	REQUIRED	REQUIRED with subgroup semantics
TRC	Optional qualitative screen	REQUIRED when catastrophe relevance is plausible	REQUIRED
Scenario set size	N/A	5 minimum	20 minimum
Kernel propagation	NONE	NONE default; QUICK if KQS policy satisfied	QUICK only if KQS policy satisfied plus sensitivity
Containment Mode A	Optional	Recommended	REQUIRED and binding
PCC	Optional	REQUIRED basic	REQUIRED full
Sensitivity analysis	Optional	Recommended	REQUIRED

Tier 4 boundary. Tier 4 is a design target only in v11.4 and MUST NOT be claimed until ProofPack is public and independently replayable.

4.9 Refusal Under Underdetermination (Normative)

Purpose. RippleLogic distinguishes gate failure from underdetermination. Gate failure occurs when a declared option uniquely violates NCRC, TRC, or CSV requirements. Refusal occurs when the framework cannot uniquely determine whether an option passes, fails, or can be selected from the declared inputs and projection rules.

Refusal is not weakness, discretion, or rhetorical caution. It is a formal anti-overclaiming state. A run that enters Refusal MUST NOT claim that the relevant option is admissible, selectable, selected, aligned, safe, or RippleLogic-conformant for the unresolved claim.

4.9.1 Refusal triggers (Normative)

R-1 Gate-critical unknown. A rights-covered cell, catastrophe cell, Containment indicator, or other gate-critical input remains UNKNOWN_IMPACT after the applicable evidence, phantom-instance, or conservative-default rule has been applied.

R-2 Non-unique projection. The projection from stakeholder instances, subgroup evidence, scenario data, or scope mapping to a gate verdict admits two or more materially different admissibility interpretations under the same declared evidence.

R-3 Material indistinguishability. Two or more options are observationally indistinguishable on available welfare evidence but materially different in possible rights exposure, tail-risk exposure, domination risk, or containment risk.

R-4 Layer ambiguity. The run relies on a category collapse, including treating compliance as correctness, traceability as admissibility, psychological plausibility as evidence, mathematical elegance as validation, moral status as governance authority, or SGP welfare scalars as admissibility modifiers.

R-5 Projection modification after observation. A ProjectionPreRegistration object, threshold, rights coverage set, catastrophe cell set, scenario set, weight rule, SGP scalar binding, or Containment trigger is materially modified after relevant decision evidence has been consulted, unless the run is voided and restarted.

R-6 Claim-boundary mismatch. The evidence available may support a lower-tier, local, provisional, or exploratory claim but is insufficient for the stronger claim being attempted.

4.9.2 Refusal effects (Normative)

1. stop any claim of admissibility, selectability, selection, conformance, or alignment for the unresolved option or run;

2. emit a PCC.RefusalRecord;

3. identify the unresolved gate, scope, dimension, scenario, stakeholder class, or projection rule;

4. state the minimum evidence or procedural condition required for re-run;

5. route the decision to escalation, redesign, bounded provisional action, or no-action baseline review according to tier and stakes.

4.9.3 Emergency and provisional action (Normative)

Refusal blocks strong conformance claims. It does not automatically imply inaction in emergency contexts. Where delay itself creates material harm, a provisional action MAY proceed only if all of the following hold:

1. the action is rights-preserving under the best available conservative screen;

2. catastrophic tail-risk is not plausibly increased beyond the declared emergency corridor;

3. the action is reversible, staged, or bounded where feasible;

4. escalation and independent review are triggered;

5. the PCC clearly marks the action as PROVISIONAL_UNDER_REFUSAL rather than admissible, selectable, or selected under normal conformance.

4.9.4 Distinction from gate failure (Normative)

Gate failure is committed: NCRC violation, TRC corridor breach, or CSV failure produces inadmissibility or non-selectability under unique projection. Refusal is uncommitted: the framework cannot uniquely determine admissibility from the declared inputs. Implementations MUST NOT silently convert Refusal into gate failure, gate success, or RLS tie-break, and MUST NOT use Refusal to bypass any earlier gate.

4.9.5 PCC fields for Refusal (Normative)

The PCC.RefusalRecord SHALL include at minimum: trigger code R-1 through R-6; evidence basis with provenance; options affected; scopes and dimensions affected; rights, catastrophe, or containment cells affected; no-action baseline status where relevant; escalation route invoked; provisional-action status, if any; and conditions under which a re-run could resolve the Refusal.

4.9.6 Relation to ripple.md IND and NA (Informative)

In ripple.md wrapped deployments, IND maps to Refusal where the test applies but evidence, scope lock, target specification, constraint set, verification regime, or projection conditions are insufficient. NA means the test is genuinely not applicable under the declared scope. Implementations MUST NOT use NA where IND is the correct state.

SECTION 5: WELFARE IMPACT CONSTRUCTION (CALCULABLE)

5.0 Purpose and Design Requirements (Normative)

This section specifies how RippleLogic converts real-world predictions and evidence into a computable welfare impact matrix for each option. A competent analyst MUST be able to compute all required impacts from (i) a declared baseline, (ii) a declared option, and (iii) a finite list of impact instances per cell, using only the equations and rules in this paper and appendices.

Cell-meaning rule (Normative clarification). A cell score is a reviewable baseline-relative claim, not a preference expression. It SHALL be interpretable as an assertion about one Union Scope and one welfare dimension, with evidence and uncertainty disclosed according to tier requirements. Appendix AD provides the canonical interpretive dictionary for cell meanings, positive/negative movement, possible indicators, gate-relevance cues, examples, and reviewer checks.

5.0.1 Stakeholder Coverage Index (SCI) (Normative)

SCI is a tier-governed disclosure ladder describing the stakeholder coverage posture of a run. SCI does not change scoring formulas. It is an auditability and integrity signal that supports comparability and anti-theater enforcement.

SCI levels (minimum definitions): SCI-0: no explicit stakeholder instance logging. SCI-1: instances logged at least for direct stakeholders and primary externality pathways. SCI-2: instances logged with explicit mapping to Union Scopes, including at least one challenger pass for omissions. SCI-3: evidence-backed instance mapping with explicit blind-spot declaration, escalation triggers, and redundancy handling disclosures where multi-scope mapping is used. SCI-4: high-coverage posture with structured discovery, third-party review or validated stakeholder registry linkage, and replay-ready traceability.

Tier minima. Tier 2 runs MUST achieve at least SCI-1. Tier 3 runs MUST achieve at least SCI-2.

Anti-theater ratchet. If SCI equals the tier minimum, the PCC MUST include a concrete Next-Run Upgrade action that would raise SCI by at least one level for the next comparable run. If SCI is below the tier minimum, the run MUST either (a) downgrade the declared tier to one whose SCI minimum is satisfied, or (b) rerun with enhanced stakeholder discovery to meet the SCI minimum for the claimed tier. In either case, the PCC MUST include a remediation plan specifying how SCI will be raised to the target level before the next comparable run. A remediation plan alone does not make the current run compliant at the originally claimed tier.

5.0A End-to-End Cell Computation Recipe (Informative)

This recipe is the minimal end-to-end spine for computing any welfare-cell impact in a Tier 1-3 run. It is informative, but the equations and constraints it references are normative in their home sections. For traceability, this recipe is listed in reverse order (8→1), reflecting the final artifact emission back to upstream inputs.

8) Emit PCC: record parameters, ReachBasis, gate results, RLS summary, and the public 5SPR. If reduced-scope mode is used, include the required attestations and blind-spot statement.

7) Run the formal method: RSG / Reality Grounding → RF / Rights Floor (formal NCRC) → TRC / Tail-Risk Bound → CSV / Containment and Structural Viability → RLS / RippleLogic Score. Treat UCI/HOI or equivalent structural diagnostics as CSV evidence when material, and as residual tie-break/monitoring evidence only when RLS is tied, close, uncertainty-overlapped, or non-decisive.

6) Apply kernel propagation only if declared: if KernelPropagation = QUICK, compute I_prop_welfare(u,d,a) per Section 6; else set I_prop_welfare = I_dir. (Redundancy handling such as EMERGENT/DEDUP/ALLOCATION is a separate instance/aggregation mechanism and MUST NOT be treated as a propagation mode.)

5) Compute the direct contribution and accumulate to the cell: $\tilde{I}\_dir(u,d,a) = \Sigma\_k (\mu\_k \cdot r\_k \cdot \tau(t\_k) \cdot \ell\_k \cdot c\_k \cdot e\_k \cdot s\_k)$ over k in that cell. Apply the canonical saturation (for example tanh) to obtain $I\_dir(u,d,a) \in [-1,+1]$.

Stream note (Normative). For admissibility layers (NCRC, TRC, CSV), compute the Base stream with s_k:=1 for all impact instances. Sentience multipliers s_k may be applied only in the Welfare stream used for RLS, per Section 5.1 and Appendix G.

4) For each impact instance k, declare its parameters inside canonical bounds: magnitude $\mu\_k \in [-1,+1]$, reach $r\_k \in [0,1]$ with ReachBasis, time horizon $t\_k \in (0, \infty)$ years (used via $\tau(t\_k)$), likelihood $\ell\_k \in [0,1]$, confidence $c\_k \in [0.1, 1]$, bounded adjustment multiplier $e\_k \geq 0$ (default 1.00; non-default use governed by Section 5.2 notes and PCC disclosure), and sentience multiplier $s\_k \geq 0$ with Human Plateau constraints (Appendix G).

Dual-baseline implementation note (Informative). In Appendix P, the worked run uses FLOOR_REFERENCE for gate-relevant cells and STATUS_QUO implicitly for welfare cells (since no dual-baseline exception is declared). The single-baseline posture is the default and recommended approach for most runs. A dual-baseline declaration is needed only when the governance context requires distinguishing a welfare-ranking baseline (e.g. current policy status quo) from a rights-gate baseline (e.g. the rights-safe floor reference) for the same cell. When a dual-baseline is used, the PCC MUST include a BaselineDualExceptionBlock per Appendix H, and the floor-reference value remains controlling for gate consistency. Implementations are encouraged to use the single-baseline (FLOOR_REFERENCE for all gate-relevant cells) approach unless a specific governance need for dual-baseline is documented.

Note on evidence (Normative clarification): Evidence quality is recorded in PCC provenance fields and informs the choice of c_k. Evidence quality is not a separate multiplicative scalar in $\tilde{I}\_{dir}$ unless explicitly introduced by a future canonical revision.

3) Construct Impact Instances k for each active welfare cell (u,d): each k must reference (a) a stakeholder instance, (b) a cell (u,d), and (c) a direct or indirect pathway description.

2) Build the representation layer: SDP → Stakeholders → Stakeholder Instances (InstanceMap with union-scope coverage).

1) Declare the run: option set A, unions U_run, welfare dimensions D, horizon window, and kernel propagation policy (NONE / QUICK). If effect-token redundancy handling is used, declare the redundancy mode (EMERGENT / DEDUP / ALLOCATION) and α budgets separately.

5.1 Baseline-Zero Rule (Normative)

Semantic anchor (MUST): For all unions u, dimensions d, and options a:

$I(u,d,a) = 0$ if and only if the predicted indicator state under option a equals the baseline indicator state.

Interpretation: “0 means no change from baseline” is globally enforced. Any method that produces absolute levels MUST immediately convert them into baseline-relative deltas before entering RippleLogic impact scoring.

Stream binding (Normative): Compute two impact streams that share $\tau(t)$, μ_k, and all other terms. Base stream fixes $s\_k := 1.0$ for all impact instances and is the only stream used by admissibility layers (NCRC, TRC, CSV). Welfare stream may apply s_k per the SGP binding interface and is used only for RLS.

5.1A Baseline Contract for Admissibility Layers (Normative)

Baseline scope (Normative). RippleLogic welfare impacts are baseline-relative by design (Baseline-Zero Rule). For admissibility layers (NCRC and TRC) and any containment checks that depend on gate-relevant cells, the baseline for any cell that is in a rights coverage set C_r or in the catastrophe set C_cat MUST represent a rights-safe / safe-corridor reference condition (a "floor baseline"), not merely the status quo.

Floor-baseline requirement (Normative). For each active rights or catastrophe cell, the PCC MUST declare the floor baseline reference (indicator definition, target level, jurisdictional or governed source when available, and any conversion assumptions). An option that maintains a rights-violating status quo MUST therefore score below 0 on the gate-relevant baseline-delta scale, because the baseline for gating is the floor reference.

PCC disclosure (Normative). The PCC MUST record BaselineType_Gates = FLOOR_REFERENCE for all cells in C_r ∪ C_cat, and MUST record BaselineType_Welfare for other cells. Allowed values: STATUS_QUO, FLOOR_REFERENCE, OTHER_DECLARED. If OTHER_DECLARED is used, the basis and rationale MUST be specified and MUST NOT weaken NCRC/TRC protection.

5.2 Impact Instances (Normative)

RippleLogic represents impacts in each active welfare cell (u,d) for option a as a finite set of impact instances K(u,d,a). Each instance captures one distinct causal pathway or measurable effect.

InstanceMap vs Impact Instances Bridge (v9.0; Normative): InstanceMap (Section 2.2A) specifies who is represented (stakeholders-in-context). Impact instances k ∈ K(u,d,a) specify how impacts occur (pathways) within a Scope×Dimension cell. These are distinct objects and MUST NOT be conflated. Impact instances inherit scope mapping from the stakeholder instances they affect. Both are required for full auditability at Tier 3.

Informative: Scopes × Dimensions form the welfare matrix. Instances feed the cells.

5.2.1 ReachBasis canon (Tier 2+ required) (Normative)

Definition (Normative). Reach r_k ∈ [0,1] is a scale term representing the fraction of the relevant reference class affected by impact instance k for the specified union scope u and welfare dimension d. r_k is not a moral weight and MUST NOT substitute for rights floors, TRC, or other gates.

ReachBasisType (Normative). For each r_k, the PCC MUST record one ReachBasisType ∈ {POPULATION_FRACTION, ASSET_OR_FLOW_FRACTION, AREA_OR_VOLUME_FRACTION, OTHER_DECLARED}. If OTHER_DECLARED is used, the PCC MUST define the basis and the denominator unambiguously.

Denominator rule (Normative). The denominator MUST match the union scope u and the decision boundary (Section 2.2A), and MUST be disclosed with units (for example: adult residents in district X; monthly platform active users; hectares of watershed; annual procurement spend).

Estimator and bounds (Normative). The PCC MUST record the estimator method and source/provenance for the denominator and numerator, plus conservative bounds when uncertainty is material. If bounds are used, the PCC MUST record r_k_low and r_k_high and use the conservative bound in any gate-relevant analysis when reach uncertainty could affect admissibility or containment.

Anti-inflation rule (Normative). If the denominator is uncertain or contestable, the PCC MUST choose a conservative denominator that biases r_k downward (reducing claimed reach) unless a stronger denominator is justified by evidence.

Rights-specific reach semantics (Normative). For rights-covered impact instances used in NCRC evaluation, reach SHALL be defined relative to the materially exposed protected subgroup or another subgroup-exposed reference class that prevents attenuation by unaffected population fraction. Reach in rights-covered cells MUST NOT dilute severity by reference to the full union population when only a protected subgroup is exposed. The PCC MUST record the rights-specific denominator when it differs from the general welfare denominator.

Table 5.1: Impact Instance Parameters (Canonical)

Parameter	Symbol	Range	Meaning
Magnitude	μ_k	[-1,+1]	Signed severity of welfare change (baseline-relative). Positive = improvement; negative = harm.
Reach	r_k	[0,1]	Fraction of the relevant stakeholder population materially affected.
Time horizon	t_k	(0, ∞) years	Duration over which the effect remains materially relevant.
Likelihood	ℓ_k	[0,1]	Conditional probability the instance occurs (conditional on scenario model, if used).
Confidence	c_k	[0.1,1]	Analyst confidence in the instance specification and mapping (floor prevents “zeroing out”).
Equity/resilience multiplier	e_k	Tier 2: 1.00 only unless named governed instrument; Tier 3: [0.75, 1.25] default 1.00	Governed multiplier for declared equity/resilience/context corrections only; MUST NOT function as a hidden weight or sign-changing control.

Parameter	Symbol	Range	Meaning
Sentience multiplier	s_k	[0,1] (default 1)	From SGP interface when ethically relevant; MUST be 1 for humans (Human Plateau Rule).

Normative notes:

- c_k has a floor of 0.1 to prevent omission-by-zeroing; low confidence should reduce weight but not erase impact.

Gate-critical confidence guard (Normative). For adverse gate-critical impacts in the rights coverage set C_r, the catastrophe cell set C_cat, or material CSV indicators, low confidence SHALL NOT by itself reduce gate severity below a conservative admissibility-relevant bound. If confidence, evidence, or measurement maturity is insufficient and the RF/NCRC, TRC, or CSV outcome could change, the run MUST either apply a governed conservative bound, narrow the claim, collect stronger evidence and rerun, downgrade the claim boundary, escalate, or route to Refusal. Confidence may represent epistemic uncertainty, but it MUST NOT launder severe but uncertain gate-critical harm into pass status.

- e_k is tightly governed. Tier 2 MUST set e_k = 1.00 for all impact instances unless a named governed instrument from PCC.EKRegistry is invoked explicitly. Tier 3 MAY use non-default e_k only from a pre-declared registry of allowed instruments, with each instrument carrying a declared purpose, source, allowed range, and replayable reason code. Unless a stricter charter rule exists, allowed non-default values are bounded to 0.75 ≤ e_k ≤ 1.25. e_k MUST NOT reverse sign, erase or dilute a rights-floor breach, function as a hidden weight, or be used asymmetrically across compared options without explicit justification and sensitivity disclosure. Any non-default e_k requires PCC justification, declared reason code, source or instrument note, stated bounds, default-counterfactual disclosure (what happens at e_k = 1.00), and tier-appropriate sensitivity if the non-default value could materially affect admissibility or selection.

Audit trigger (Normative). If a non-default e_k is used without declared rationale, source or instrument note, stated bounds, or the required sensitivity record for the claimed tier, implementations MUST set audit flag EK_PARAMETER_UNJUSTIFIED. EK_PARAMETER_UNJUSTIFIED is INVALID for Tier 2-3 claimability until corrected or downgraded.

Registry omission rule (Normative clarification). PCC.EKRegistry MAY be omitted when all impact instances in the run use e_k = 1.00 and no Tier-2 named-instrument exception is invoked.

Registry note (Normative clarification). Implementations SHOULD restrict non-default e_k uses to a small declared registry (for example EQ_CORR, RESILIENCE_CORR, CONTEXT_CORR, or OTHER_DECLARED) and MUST apply the declared rule symmetrically across compared options unless the asymmetry itself is the subject of evaluation.

- Non-default SGP-derived sentience multipliers may be applied only in the Welfare stream for non-human, non-plateau entities where an authoritative SGP record is available. For humans and all plateau-bearing entities, s_k MUST be 1.0. SGP-derived scalars MUST NOT weaken any human protection or alter NCRC, TRC, or CSV (Section 12; Appendix G).

5.3 Temporal Weighting (Normative)

$\tau(t) = \min(1, \ln(1 + \max(t, t_{min})) / \ln(1 + T_{ref}))$

Intuition anchor (informative): $\tau(t)$ is the fraction of full governance-horizon weight an effect receives.

Defaults:

t_min = 0.083 years (approximately 1 month; prevents $\tau(0) = 0$)

T_ref = 25 years (governance reference horizon)

Illustrative values (with defaults):

Design rationale: Logarithmic weighting preserves intergenerational salience relative to exponential discounting, which aggressively devalues future generations. The cap at T_ref prevents the anomaly where distant-future effects receive more weight than the governance horizon itself. T_ref is a governed parameter: governance bodies may adjust it (with PCC documentation and sensitivity analysis) but MUST NOT set T_ref < 10 years without charter-level justification, as this would effectively suppress long-horizon considerations.

Sensitivity note (Tier 3): T_ref SHOULD be perturbed (for example, T_ref = 15 and T_ref = 50) as part of the sensitivity bundle. If selection changes under T_ref perturbation, the PCC MUST document this sensitivity.

Historical change note (v7.5.0; carried forward unchanged in v9.0): This replaces the uncapped $\tau(t) = \ln(1+t)/\ln(1+T_{ref})$ used in v7.4.5, which produced $\tau(50) \approx 1.22$. The capped version is normatively preferred because temporal weight beyond the governance horizon should not exceed the horizon weight. All prior PCCs computed under v7.4.5 remain valid as lineage material; recomputation is recommended for active decisions with t > T_ref impacts.

$\tau(100) = 1.00$ (capped)

$\tau(50) = 1.00$ (capped: no bonus beyond T_ref)

$\tau(25) = 1.00$ (reference horizon: full weight)

$\tau(10) \approx 0.74$ (10 years)

$\tau(5) \approx 0.55$ (5 years)

$\tau(1) \approx 0.21$ (1 year)

$\tau(0.083) \approx 0.025$ (1 month: minimal but nonzero)

The min(1, ·) cap ensures that no effect receives more temporal weight than the reference horizon. Effects at or beyond T_ref receive full weight ($\tau = 1.0$); effects below T_ref receive proportionally less weight on a logarithmic curve. The t_min floor ensures that very short-term effects are not zeroed out.

Time Horizon (years)	τ(t) Value
1	≈ 0.21
5	≈ 0.55
10	≈ 0.74
25	1.00
50	1.00 (capped)

5.4 Pre-Saturation Direct Impact Aggregation (Normative)

For option a, union u, dimension d, define the pre-saturation direct impact per stream $x \in \{base, welfare\}$:

$$\tilde{I}\_{dir,x}(u,d,a) = \Sigma_{k \in K(u,d,a)} [r_k \times \tau(t_k) \times \ell_k \times c_k \times e_k \times s_{k,x} \times \mu_k]$$

where $s_{k,base} := 1.0$ for all impact instances, and $s_{k,welfare}$ may use the SGP sentience multiplier only as permitted by Section 12 and Appendix G. When the stream is clear from context, the shorthand $\tilde{I}\_dir(u,d,a)$ may be used. Appendix B controls the stream-separated equations.

5.5 Saturation (Normative)

To ensure all direct impacts lie in [-1,+1] and avoid runaway totals, RippleLogic uses smooth saturation:

$$I\_dir(u,d,a) = \tanh(\beta \times \tilde{I}\_dir(u,d,a))$$

Default: $\beta = 2$.

5.6 Magnitude Construction μ_k: Canonical Anchoring Methods (Normative)

Magnitude μ_k MUST represent a baseline-relative change on the normalized [-1,+1] scale. RippleLogic permits two canonical anchoring families:

5.6.1 Percentile Anchoring (Default for non-rights cells)

Let x be the raw indicator (higher-is-better unless specified). Let P_5, P_{50}, P_{95} be the 5th, 50th, and 95th percentiles of x in a declared reference class.

Define a bounded level score:

If $x \geq P_{50}$:

$S(x) = \text{clip}((x - P_{50}) / (P_{95} - P_{50}), 0, +1)$

If $x < P_{50}$:

$S(x) = \text{clip}((x - P_{50}) / (P_{50} - P_5), -1, 0)$

Edge-case guard (Normative). If $(P_{95} - P_{50}) = 0$ OR $(P_{50} - P_5) = 0$ (or numerically indistinguishable), percentile anchoring is undefined. The analyst MUST either (a) change the declared reference class, (b) use an alternate canonical anchoring method with explicit PCC justification, or (c) mark the affected impact instances as UNKNOWN_IMPACT and treat the cell as phantom-active under the Phantom Instance Rule until repaired. A run MUST NOT silently divide by zero or substitute arbitrary epsilons without PCC disclosure.

Then define magnitude as a baseline-relative delta:

$\mu\_k = \text{clip}(S(x\_a) - S(x_0), -1, +1)$

Where x_0 is baseline indicator value, and x_a is the predicted value under option a.

If higher values are worse (for example, mortality rate), use:

$S\_worse(x) = -S(x)$

Normative requirement: The PCC MUST record the reference class and percentile values used.

5.6.2 Threshold Anchoring (Required for rights-covered cells unless invariant reference is declared)

For rights-covered cells (cells in any rights coverage set C_r), analysts MUST NOT use context-local percentile anchoring unless the reference class is declared invariant under governance.

Default rule (Tier 1-3): Rights-covered magnitudes MUST be derived using threshold anchoring with explicit “good/bad” anchors so that rights semantics do not drift.

Let x_good be the indicator level consistent with rights-safe conditions and x_bad be the indicator level representing a severe rights violation onset. Map to a bounded level score:

For higher-is-better indicators: $S(x) = \text{clip}((x - x\_bad) / (x\_good - x\_bad), -1, +1)$

Edge-case guard (Normative). If $(x\_good - x\_bad) = 0$ (or anchors are otherwise degenerate), threshold anchoring is undefined. The analyst MUST either (a) repair anchors via governed reference conditions, or (b) mark the affected impacts as UNKNOWN_IMPACT and treat the cell as phantom-active under the Phantom Instance Rule until repaired. A run MUST NOT compute a finite score from degenerate anchors.

For higher-is-worse indicators: $S(x) = \text{clip}((x\_bad - x) / (x\_bad - x\_good), -1, +1)$

Then compute baseline-delta magnitude:

$\mu\_k = \text{clip}(S(x\_a) - S(x_0), -1, +1)$

Normative requirement: The PCC MUST record x_good, x_bad, indicator definition, and direction.

Magnitude construction for rights-covered cells (C_r cells). For rights-covered cells, magnitudes MUST be derived using threshold anchoring with explicit reference conditions aligned to rights semantics (Section 7.1.1), preventing the context-local drift that percentile anchoring would permit. The PCC MUST record x_good, x_bad, indicator definition, direction, and the source/justification for the anchor values. This requirement applies to all tiers when rights-covered cells are quantitatively assessed.

5.7 Missing Data Rule (Ignorance Penalty) (Normative)

To prevent score inflation by omission:

If a welfare cell (u,d) is active (required by tier context and not masked for RLS), but no defensible instances can be specified, the PCC MUST:

- Mark the cell as "UNKNOWN_IMPACT", and
- Include a phantom instance with canonical parameters:

Parameter	Phantom Value
μ_phantom	-0.10
r	1
t	T_ref (25 years)
ℓ	1
c	1
e	1
s	1

This yields a mild negative default rather than unjustified neutrality.

Calibration rationale (Informative). The phantom magnitude mu_phantom = -0.10 is a governance prior set to satisfy two constraints simultaneously: it must be negative enough that omitting evidence is worse than reporting neutral impact (preventing score inflation by ignorance), and it must be small enough in absolute value that phantom instances do not dominate well-evidenced cells when multiple unknowns coexist in a run. At mu = -0.10 with default parameters (r=1, t=T_ref, l=1, c=1), the phantom produces a saturated direct impact of approximately -0.20 per cell. In a run with uniform weights across 49 cells, a single phantom cell contributes roughly 0.004 to the total RLS magnitude, which is detectable in audit but not decisive unless many cells are unknown simultaneously. If five or more cells carry phantom instances, the cumulative penalty becomes material (approximately -0.02 to -0.03 in RLS), creating a meaningful incentive to gather evidence. The value -0.10 is a governance prior, not an empirically validated optimum. It is explicitly challengeable under standard governance rules; any revision requires justification, version increment, and sensitivity analysis over a reference decision suite.

Clarification: This rule does not apply to cells that are legitimately out of scope and properly masked with justification; it DOES apply to non-maskable cells and to any cell required by the tier's minimum coverage rules.

Gate-Critical Unknown Rule (Normative). If a cell within C_r or C_cat is declared UNKNOWN_IMPACT, admissibility for that cell SHALL NOT be resolved using only the standard phantom-instance fallback. Default handling path: for rights-covered cells, the run MUST either (i) apply a governed conservative bound sufficient to preserve the rights gate, or (ii) treat the gate as unresolved and rerun, downgrade, or escalate. For catastrophe-covered cells, the run MUST either (i) apply a governed conservative scenario/cell bound sufficient to preserve TRC, or (ii) treat TRC as unresolved and rerun, downgrade, or escalate. Implementations MUST set RIGHTS_CELL_UNKNOWN_GATE_CRITICAL or CATASTROPHE_CELL_UNKNOWN_GATE_CRITICAL, as applicable, using only canonical severities defined in Section 14.3.

5.8 Scenario-Conditioned Impacts (Normative for TRC runs)

When TRC is in use (Tier 3 required; Tier 2 required when catastrophe relevance plausible), impacts MUST be scenario-conditioned at least for catastrophe cells:

Scenario probabilities p_s are governed (Appendix D). Scenario conditioning enters through ℓ_k and/or instance presence, and through propagation if scenario-specific kernels are declared.

5.9 Worked Examples (Non-Normative, Computation-Illustrative)

Example 5.1: Direct impact, Community-Social cell

Decision: Remote-work policy option a. Cell: u=3 (Community), d=3 (Social).

Assume T_{ref} = 25, β = 2.

Instance 1: Reduced in-person interaction

- $\mu_1 = -0.30$, $r_1 = 0.80$, $t_1 = 3$ years, $\ell_1 = 0.90$, $c_1 = 0.70$, $e_1 = 1$, $s_1 = 1$

Instance 2: Increased online community participation

- $\mu_2 = +0.15$, $r_2 = 0.50$, $t_2 = 5$ years, $\ell_2 = 0.70$, $c_2 = 0.50$, $e_2 = 1$, $s_2 = 1$

Compute temporal weights:

- $\tau(3) = \ln(4)/\ln(26) \approx 0.43$
- $\tau(5) = \ln(6)/\ln(26) \approx 0.55$

Compute contributions:

- Inst1: $0.80 \times 0.43 \times 0.90 \times 0.70 \times 1 \times 1 \times (-0.30) \approx -0.065$
- Inst2: $0.50 \times 0.55 \times 0.70 \times 0.50 \times 1 \times 1 \times (+0.15) \approx +0.014$

Aggregate pre-saturation:

- $\tilde{I}\_dir = -0.065 + 0.014 = -0.051$

Saturate:

- $I\_dir(3,3,a) = \tanh(2 \times -0.051) = \tanh(-0.102) \approx -0.10$

Example 5.2: Missing-data penalty

Cell u=7 (Biosphere), d=7 (Environment) is active (non-maskable for environment-relevant decisions). Analyst lacks defensible estimates.

Phantom instance yields:

- $\tilde{I}\_dir = 1 \times 1 \times 1 \times 1 \times 1 \times 1 \times (-0.10) = -0.10$
- $I\_dir = \tanh(2 \times -0.10) \approx -0.20$

This enforces humility: missing evidence is mildly negative, not neutral.

SECTION 6: RIPPLE PROPAGATION: KERNEL AND EPISTEMIC HUMILITY

6.0 Purpose (Normative)

Ripple propagation models how direct impacts in one cell causally and institutionally ripple into other cells across unions and dimensions. This makes cross-scale externalities explicit and auditable.

Tier posture: Tier 1-3 permit NONE or QUICK propagation. FULL propagation is a Tier-4 design target only and MUST NOT be used for Tier 1-3 compliance claims.

6.1 Kernel Definition and Convention (Normative)

Let the 49 welfare cells be indexed by $i = \varphi(u,d)$ with $\varphi(u,d) = 7(u-1) + d$. The ripple kernel is a sparse matrix $K \in \mathbb{R}^{49\times49}$.

Kernel convention (MUST):

- K_{ij} maps effect from source cell j to target cell i (target-row, source-column).
- Propagation uses left multiplication: $\tilde{I}_{prop} = I_{dir} + K \times I_{dir}$

Interpretation:

Scale semantics and first-order propagation note (Normative clarification). The bounded cell impacts used by RippleLogic are governance scores anchored to declared indicators and baselines, not claims that dignity, liberty, or meaning are measured as cardinal utility in nature. The kernel K is a declared first-order propagation operator for bounded local approximation under the chosen decision horizon. It does not claim that all socio-technical dynamics are globally linear. Non-linearities and threshold behavior are handled elsewhere in the architecture through saturation, scenario branching, rights floors, tail-risk screening, containment, escalation, and versioned recalibration of K. Implementations MUST NOT present K as a proof of empirical linearity; they MAY present it as a transparent, auditable approximation layer governed by the PCC.

- $\kappa > 0$: improving source j tends to improve target i.
- $\kappa < 0$: improving source j tends to harm target i.
- $\kappa = 0$: no modeled pathway.

6.2 Propagation Modes (Tier 1-3) (Normative)

Mode	Formula	Use Case
NONE (Direct-only)	$I_{prop} := I_{dir}$	Tier 1-2 default
QUICK (First-order)	$\tilde{I}_{prop} = I_{dir} + K \times I_{dir}$	Tier 3 with sensitivity

Stream note (Normative). Propagation is applied per stream $x \in \{base, welfare\}$ as defined in Appendix B. In Section 6 (and any tables using shorthand), I_dir and I_prop are shorthand for $I_{dir,x}$ and $I_{prop,x}$ when the stream is clear from context. Admissibility gates (NCRC, TRC, CSV) use the Base stream; RLS uses the Welfare stream.

Normative restriction: FULL propagation is PROHIBITED for Tier 1-3 claims in this v11.4 release line.

6.3 Post-Propagation Saturation (Normative)

Because propagation can push values outside [-1,+1], apply elementwise saturation:

$I_{prop\_x}(u,d,a) = \tanh(\beta_{prop} \times \tilde{I}_{prop\_x}(u,d,a))$

Notation: I_prop_base and I_prop_welfare denote the x=base and x=welfare outputs respectively.

Default: $\beta_{prop} = 1$.

6.4 Stability Constraints and Humility Fallbacks (Normative)

Kernel stability guardrails (defaults; may be tightened by governance):

- Entry bound: $|K_{ij}| \le \kappa_{max}$, default $\kappa_{max} = 0.5$
- Absolute row-sum bound (required): $\|K\|_\infty := \max_i \Sigma_j |K_{ij}| \le \rho_{max}$, default $\rho_{max} = 0.9$.
- Spectral radius constraint (satisfied by construction under the row-sum bound): If $\|K\|_\infty < 1$, then $\rho(K) < 1$ is treated as satisfied for Tier 1-3 purposes because $\rho(K) \le \|K\|_\infty$. Implementations MUST NOT require additional spectral-radius computation to claim Tier 1-3 conformance when the row-sum bound holds.

Kernel stability verification rule (Normative). A Tier 1-3 implementation MUST compute and record (in PCC when kernel is used): (i) $\max_{ij} |K_{ij}|$ and (ii) $\|K\|_\infty$. If either exceeds its bound (entry bound or row-sum bound), the run MUST execute the humility fallback (propagation_mode = NONE) and record KERNEL_HUMILITY_FALLBACK = TRUE.

If the declared kernel violates entry bounds or row-sum bounds (and therefore fails the Tier 1-3 stability verification rule above), the run MUST:

- Set propagation_mode = NONE, and
- Record a PCC limitation: KERNEL_HUMILITY_FALLBACK = TRUE.

Rationale: An unstable or over-amplifying kernel is worse than no kernel; it creates false certainty and is easy to game.

6.5 Kernel Quality Score (KQS) (Normative)

KQS summarizes readiness of the kernel for decision-relevant propagation:

$$KQS = w_{cov} \times C_{cov} + w_{id} \times C_{id} + w_{stab} \times C_{stab} + w_{pred} \times C_{pred}$$

Default component weights:

Component	Weight
Coverage (C_cov)	0.25
Identifiability (C_id)	0.30
Stability (C_stab)	0.20
Prediction (C_pred)	0.25

Component meanings (each in [0,1]):

- C_cov: coverage evidence proportion (share of relied-upon edges with cited evidence)
- C_id: identifiability (edges are specified with clear endpoints and sign; replayable)
- C_stab: stability margin (satisfies bounds with margin)

- C_pred: predictive accuracy (backtest/pilot performance where available; otherwise conservative prior)

6.5.1 Canonical KQS computation (Tier 1-3 deterministic) (Normative)

Definition (Normative). A 'relied-upon edge' is any kernel edge (i,j) with $|K_{ij}| \geq \kappa_{use\_min}$ that touches at least one active cell (Section 10.3) or any non-maskable cell in C_r or C_cat. Default $\kappa_{use\_min} = 0.05$ unless the PCC declares a stricter value.

C_cov (Normative). Let E_use be the set of relied-upon edges. Let E_evid be those edges in E_use whose evidence fields include (i) an EvidenceClass ∈ {A,B,C} and (ii) a citation or provenance reference sufficient for retrieval. Then $C_{cov} := |E_{evid}| / \max(1, |E_{use}|)$.

C_id (Normative). Let E_id be those edges in E_use with declared source union/dimension, target union/dimension, sign, and numeric magnitude fields present and within canonical bounds. Then $C_{id} := |E_{id}| / \max(1, |E_{use}|)$.

C_stab (Normative). If QUICK is used, compute PCC.Kernel.RowSumAbsMax = $||K||_\infty$ and PCC.Kernel.EntryAbsMax = $\max_{ij} |K_{ij}|$ per Section 6.4. Set C_stab := 0 if any bound is violated; otherwise $C_{stab} := \mathrm{clip}((\rho_{max} - ||K||_\infty)/\rho_{max}, 0, 1)$, with default $\rho_{max} = 0.9$.

C_pred (Normative). If no backtest or pilot evidence exists for the decision domain, set C_pred := 0.30 (NO_EVIDENCE_DEFAULT) as already required. If empirical evidence exists, the PCC MUST declare the predictive metric used and compute C_pred as clip((MetricScore - MetricBaseline) / (MetricTarget - MetricBaseline), 0, 1) with all three values disclosed in PCC so an auditor can recompute.

Disclosure rule (Normative). The PCC MUST record κ_use_min, the edge counts |E_use|, |E_evid|, |E_id|, and the computed component values C_cov, C_id, C_stab, C_pred used in the KQS calculation.

Default C_pred prior (Normative). If no backtest/pilot evidence exists for the decision domain, set C_pred := 0.30 (NO_EVIDENCE_DEFAULT). The PCC MUST (i) mark this basis as NO_EVIDENCE_DEFAULT, and (ii) include a plan and timeline to obtain empirical validation evidence; until such evidence exists, C_pred SHALL NOT be reported as higher than the prior.

KQS policy (MUST) for Tier 3:

- If KQS < 0.40: Kernel MUST NOT be used; propagation_mode = NONE.
- If 0.40 ≤ KQS < 0.50: QUICK allowed only with mandatory kernel sensitivity; otherwise use NONE.
- If KQS ≥ 0.50: Kernel use permitted with required sensitivity at Tier 3.

Tier 1-2 default: propagation_mode = NONE unless the PCC explicitly declares kernel use and its KQS.

6.6 Kernel Sensitivity Requirements (Normative for Tier 3 when QUICK is used)

When QUICK propagation is used at Tier 3, the PCC MUST include sensitivity analysis:

- Perturb each relied-upon edge in E_use (as defined in Section 6.5.1) by ±0.05 (or ±10% of its magnitude, whichever is larger; declare the rule).
- Recompute the cascade for each perturbation set (at minimum one-at-a-time).
- If admissibility outcomes (NCRC, TRC, CSV) or the selected option changes, set audit_flag = DECISION_FRAGILE_KERNEL and escalate per tier policy.

6.7 KOPS and Starter Kernel (Tier 2-3) (Normative plus Provisional Labeling)

RippleLogic permits a governed subset of kernel edges called the Key Operational Pathways Set (KOPS). KOPS is the set of “load-bearing” pathways that are documented with evidence notes, sign-checked against literature or backtests, and sensitivity-audited.

A Starter KOPS may be provided (Appendix K) and MUST be labeled PROVISIONAL with evidence classes and a global shrink factor for elicited edges (default 0.35). If a starter kernel is used, the PCC MUST disclose this and MUST include kernel sensitivity.

6.8 Scenario-Conditioned Propagation (Tier 1-3 Default Rule)

Default: Kernel is not scenario-conditioned; scenario enters via instance likelihoods ℓ_k and scenario-specific instance activation. Scenario-conditioned kernels K_s may be used only with explicit PCC justification and must carry KQS and sensitivity per scenario class.

SECTION 7: RIGHTS LAYER: NON-COMPENSATORY RIGHTS CONSTRAINT (NCRC)

7.0 Purpose and Union-Based Ethics Rationale (Normative)

RippleLogic enforces non-compensability: certain harms are not permitted to be “paid for” by aggregate benefits elsewhere. NCRC operationalizes this by removing rights-violating options prior to any welfare optimization.

Union-Based Ethics (UBE) justification: Under UBR, harms propagate across nested unions; without a rights-first constraint, local optimization systematically externalizes costs onto vulnerable subgroups and future stakeholders. NCRC is therefore a structural safeguard against value scalarization and moral laundering (Rawls, 1971; Sen, 2009).

7.1 Canonical Rights Set (Normative)

RippleLogic defines eight core rights as feasibility floors:

Table 7.1: Canonical Rights Thresholds (Normative)

Right	Code	Threshold θ_r	Normative Anchor
Life	LIFE	-0.90	Near-certain or highly probable death or lethal exposure

Right	Code	Threshold θ_r	Normative Anchor
Bodily Integrity	BODY	-0.70	Severe injury, disability, torture, serious bodily violation
Liberty	LBTY	-0.65	Arbitrary detention, forced labor, coercive confinement
Basic Needs	NEED	-0.50	Severe deprivation: food insecurity, homelessness, loss of basic subsistence
Dignity	DIGN	-0.55	Systematic humiliation, dehumanization, targeted degradation
Due Process	PROC	-0.45	Denial of fair hearing, non-transparent coercive procedure
Information	INFO	-0.40	Systematic censorship, epistemic coercion, pervasive disinformation constraints
Ecological Integrity	ECOL	-0.65	Material biosphere integrity breach or planetary boundary transgression corridor

Ecological Integrity boundary. ECOL is a MathGov non-compensatory enabling-condition protection for biosphere integrity and life-supporting ecological conditions. It may overlap with legal environmental rights where recognized, but this Canon does not claim universal legal status by itself.

Normative interpretation rule: Thresholds θ_r are admissibility floors, not weights. They do not represent “how important” a right is; they represent the minimum allowable protection level.

Threshold governance. All thresholds are explicitly challengeable. Revisions require justification, version increment, PCC documentation preserving prior values, and sensitivity outcomes over a reference decision suite.

Threshold epistemic boundary. Rights thresholds are versioned governance priors, not discovered empirical constants or jurisdiction-specific legal conclusions. They may be challenged, localized, stress-tested, and revised under governed update rules, but they remain non-compensatory while active in a run.

Sensitivity requirement. In Tier 3 runs, each θ_r MUST be perturbed by ±0.05 and the cascade recomputed. If admissibility changes for any option under perturbation, the decision is flagged as threshold-sensitive and the PCC MUST document which rights are near-binding, which options are affected, and how the final selection responds to that near-binding status.

Proxy disclaimer. Governance indices (such as press freedom or rule-of-law indices) may be used as operational proxies for anchoring reference conditions, but are not treated as moral ground truth and MUST be accompanied by documented limitations in the PCC.

Right INFO (θ = -0.40). Normative anchor: systematic censorship, epistemic coercion, or deliberate degradation of the information environment such that agents cannot form informed preferences. Example anchors: systematic media control and censorship per RSF Press Freedom Index “Very Serious Situation” (score below 40); persistent large-scale disinformation operations affecting public health, safety, or governance; surveillance-driven chilling effects that measurably suppress information seeking or expression.

Right PROC (θ = -0.45). Normative anchor: denial of fair hearing, non-transparent coercive procedure, or absence of meaningful recourse against institutional power. Example anchors: no meaningful access to justice or independent review per WJP Rule of Law Index bottom quintile; consequential automated decisions without explanation, appeal, or human review; systematic procedural exclusion; coercive institutional action without proportionality review.

Right NEED (θ = -0.50). Normative anchor: severe deprivation of material conditions necessary for survival and minimally adequate functioning. Example anchors: sustained severe food insecurity at IPC Phase 3 or above; homelessness or severe housing deprivation per UN-Habitat standards; lack of safe water and sanitation; denial of essential healthcare access such that treatable conditions become severe or lethal.

Right DIGN (θ = -0.55). Normative anchor: systematic humiliation, dehumanization, or targeted degradation that denies inherent worth and equal standing. Example anchors: institutionalized dehumanization of groups per the UN Framework of Analysis for Atrocity Crimes; degrading treatment per ECHR Article 3; systematic exclusion from participation in social and civic life; discriminatory institutions that deny equal standing below capability thresholds (Nussbaum, 2011).

Right ECOL (θ = -0.65). Normative anchor: breach of biosphere integrity or entry into high-risk Earth-system transgression corridors that threaten enabling conditions for continued life. Example anchors: actions that materially increase probability of crossing critical climate or ecosystem tipping elements per Lenton et al. (2008); ecosystem regime shifts effectively irreversible on human-relevant timescales; depletion of foundational natural capital (aquifers, soils, fisheries) beyond recharge capacity; biodiversity loss sufficient to undermine ecosystem services.

Right LBTY (θ = -0.65). Normative anchor: arbitrary detention, forced labor, coercive confinement, or systematic deprivation of freedom of movement and association. Example anchors: deprivation of liberty without legal basis or review per ICCPR Article 9; forced labor patterns consistent with ILO indicator bundles (restriction of movement, debt bondage, retention of identity documents, isolation); coercive confinement not justified by legitimate, proportionate public safety measures.

Right BODY (θ = -0.70). Normative anchor: severe injury, disability, torture, or serious violation of bodily autonomy falling short of lethality but constituting fundamental harm to physical and psychological functioning. Example anchors: torture or cruel treatment as defined by the UN Convention Against Torture; serious bodily injury resulting in permanent disability or substantial loss of function per WHO disability classification frameworks (ICF; WHODAS 2.0); forced medical procedures or systematic bodily violations; sustained exposure to hazards expected to cause serious chronic illness.

Right LIFE (θ = -0.90). Normative anchor: near-certain or highly probable death, lethal exposure, or elimination of conditions necessary for continued biological existence for the worst-off subgroup. Example anchors: direct lethal violence or exposure; catastrophic deprivation conditions such as IPC Phase 5 famine (≥20% extreme food gaps, crude death rate >2 per 10,000 per day); collapse of life-sustaining infrastructure (water, sanitation, shelter) without viable alternatives; mass-casualty disaster exposures. The threshold is set above -1.0 because -1.0 corresponds to guaranteed death for all affected persons, a theoretical maximum rarely achieved even in extreme scenarios.

7.1.1 Threshold Calibration Protocol (Normative)

The eight canonical rights thresholds are normative governance priors, not empirically derived frequencies. Each threshold encodes a severity level on the normalized impact scale [-1, +1] at which a predicted impact is treated as a rights-floor violation that cannot be traded away for aggregate benefit elsewhere.

Calibration follows two principles:

First, thresholds are placed relative to the semantics of the impact scale: -1.0 denotes an extreme, worst-case welfare decrement in the relevant domain; +1.0 denotes an extreme improvement.

Second, the ordering of thresholds reflects both a severity gradient and an enabling-conditions logic: rights protecting against lethal and bodily harms sit closer to the floor, while rights protecting procedural and epistemic conditions are set closer to zero because their violation can enable cascading downstream rights failures. When information environments are degraded or due process is absent, communities lose the epistemic and procedural capacity to detect and contest violations of life, bodily integrity, and liberty.

Operationally, a threshold is justified by two to four anchor conditions per right, a mapping rule from anchors to the [-1, +1] scale using threshold anchoring (Section 5.6.2), and a mandatory sensitivity sweep.

7.2 Rights Coverage Sets C_r (Normative)

Each right applies to a defined subset of welfare cells C_r. NCRC evaluates each right across its coverage set and uses the worst-off subgroup for rights-covered cells.

Canonical coverage sets (authoritative):

Coverage note (Normative clarification). ECOL is scoped to capture enabling-conditions harms to the biosphere and to humanity’s life-support systems. Canonically this is implemented via Environment-dimension cells in the Humanity/CMIU and Biosphere unions (and any additional declared cells via governed extension). Implementations MAY extend ECOL coverage to additional cells only by explicit governance declaration in the PCC, and MUST NOT silently re-scope ECOL between runs.

Right	Coverage Set C_r
LIFE	{(u, Health): u ∈ {1,2,3,4,5,6}} ∪ {(6, Environment)}
BODY	{(u, Health): u ∈ {1,2,3,4,5,6}}
LBTY	{(u, Agency): u ∈ {1,2,3,4,5,6}} ∪ {(u, Social): u ∈ {3,4,5,6}}
NEED	{(u, Material): u ∈ {1,2,3,4,5,6}} ∪ {(u, Health): u ∈ {1,2,3,4,5,6}}
DIGN	{(u, Social): u ∈ {1,2,3,4,5,6}} ∪ {(u, Agency): u ∈ {1,2,3,4,5,6}}
PROC	{(u, Agency): u ∈ {4,5,6}} ∪ {(u, Knowledge): u ∈ {4,5,6}} ∪ {(u, Social): u ∈ {4,5,6}}
INFO	{(u, Knowledge): u ∈ {1,2,3,4,5,6}} ∪ {(u, Agency): u ∈ {1,2,3,4,5,6}}
ECOL	{(6, Environment), (7, Environment)}

7.3 Worst-Off Subgroup Semantics (Normative)

Rights cannot be averaged away. For every rights-covered cell (u,d) in any C_r, define:

$$I\_rights(u,d,a) = \min_{g \in G_{\{u,d\}}} I\_prop\_base(u,d,a \mid g)$$

Where G_{u,d} is the set of protected subgroups relevant to that cell.

Tier requirements:

- Tier 3: Subgroup enumeration is REQUIRED for rights-covered cells (Appendix H PCC requirements).
- Tier 2 (strict posture): Subgroup enumeration is REQUIRED at least for directly affected populations in rights-covered cells; when infeasible, the PCC MUST record SUBGROUP_LIMITATION and apply a conservative bound rule (Section 7.3.2).

7.3.1 Minimum subgroup categories (Normative minimum)

At minimum, when applicable and feasible, G_{u,d} SHOULD include subgroups defined by:

- Age (children; elderly)
- Disability status
- Legally protected characteristics relevant in jurisdiction
- Economic vulnerability (for example, bottom income quintile)
- Geographic exposure (for example, high-risk locations)
- Domain-specific protected groups (for example, patients, detainees, precarious workers, indigenous communities)

7.3.2 Conservative fallback when subgroup disaggregation is infeasible (Tier 2-3) (Normative)

If subgroup impacts I_prop_base(u,d,a|g) cannot be produced for a rights-covered cell, the run MUST:

- Record SUBGROUP_LIMITATION in the PCC, and
- Apply a conservative rights bound for that cell:

Let $\gamma\_subgroup$ be a conservatism factor (default $\gamma\_subgroup = 1.5$). Then define:

- If $I\_prop\_base(u,d,a) < 0$, set $I\_rights(u,d,a) = \max(-1, \gamma\_subgroup \times I\_prop\_base(u,d,a))$
- Else set $I\_rights(u,d,a) = I\_prop\_base(u,d,a)$

This bound applies only for NCRC checking, not for RLS scoring.

Calibration note (Normative guidance). The default $\gamma\_subgroup = 1.5$ is a conservative governance prior rather than an empirically estimated constant. The value is set as a moderate worst-off uplift between the identity case ($\gamma\_subgroup = 1.0$) and stronger stress-test settings used to examine subgroup vulnerability under uncertainty. It is intended to reduce subgroup erasure without asserting a precise empirical constant across domains. Tier 3 deployments SHOULD review $\gamma\_subgroup$ against their reference decision suite at least annually or after a material accumulation of rights-sensitive runs. Any non-default $\gamma\_subgroup$ MUST be declared in the PCC, justified, and accompanied by sensitivity reruns over the reference suite. The sensitivity sweep for $\gamma\_subgroup$ SHOULD include at minimum $\gamma\_subgroup \in \{1.0, 1.3, 1.5, 2.0\}$ to test whether admissibility outcomes or selection change materially.

7.4 Violation Depth and NCRC Admissibility (Normative)

For each right r, define violation depth:

$$v\_r(a) = \max\_\{(u,d) \in C\_r\} (\theta\_r - I\_rights(u,d,a))^+$$

where $(x)^+ = \max(x, 0)$.

NCRC pass/fail predicate:

$NCRC(a) = TRUE$ if and only if $v\_r(a) = 0$ for all $r \in R$

NCRC-passing set:

$A\_NCRC = \{a \in O : NCRC(a) = TRUE\}$

Interpretation: If any right’s threshold is violated in any covered cell for the worst-off subgroup, the option is rights-inadmissible, regardless of any benefits elsewhere.

7.5 Emergency Mode (Rights-Failure Handling) (Normative)

If $A\_NCRC = \varnothing$ (no option passes rights), RippleLogic enters Emergency Mode. Emergency Mode is not a loophole; it is a controlled failure protocol.

7.5.1 Emergency Mode rights priority order (Normative)

Rights are ordered lexicographically by the canonical emergency priority:

[LIFE, BODY, ECOL, LBTY, NEED, DIGN, PROC, INFO]

This ordering MUST be used exactly.

7.5.2 Emergency Mode selection rule (Normative)

Construct the violation depth vector for each option:

v(a) = (v_LIFE(a), v_BODY(a), v_ECOL(a), v_LBTY(a), v_NEED(a), v_DIGN(a), v_PROC(a), v_INFO(a))

Select the option that lexicographically minimizes v(a). If tied:

- Minimize TRC CVaR (if TRC scenarios exist or are rapidly constructed), then
- Maximize RLS.

7.5.3 Independent Challenger Requirement (Normative)

Before Emergency Mode can be invoked, an independent challenger MUST propose at least one alternative option.

Independence requirements (MUST):

- No reporting relationship to decision owner
- No material interest in decision outcome
- Access to the same information set

Minimum challenger effort:

- Tier 2: ≥30 minutes active option generation
- Tier 3: ≥2 hours active option generation

If time pressure prevents this, PCC MUST record CHALLENGE_DEFERRED_EMERGENCY and a retrospective challenge review MUST occur within 24 hours (or earliest feasible time) with an addendum PCC.

7.5.4 Emergency documentation requirements (Normative)

PCC MUST include:

- Emergency declaration and trigger conditions
- v_r(a) for each option
- Lexicographic comparison trace
- Mitigation/remediation plan
- Review cadence and return-to-normal triggers

7.6 Anti-Gaming Rules Specific to NCRC (Normative)

- Rights-covered cells MUST NOT be masked out of RLS aggregation if they are non-maskable by policy (Appendix H audit flags).
- Rights checks MUST use worst-off subgroup semantics and must not be evaluated on averages.

- "Unknown" data MUST NOT be treated as neutral; missing data triggers ignorance penalties (Section 5.7).
- Emergency Mode MUST NOT be used without challenger protocol and remediation plan.

SECTION 8: CATASTROPHIC RISK LAYER: TAIL-RISK CONSTRAINT (TRC)

8.0 Purpose and UBE Rationale (Normative)

Even when rights floors are respected in expectation, a decision can carry non-trivial probability of catastrophic harm to Humanity/CMIU or the Biosphere. Under UBE, catastrophe avoidance is lexicographically prior to welfare optimization because catastrophic outcomes can eliminate future choice across unions and collapse enabling conditions.

TRC therefore bounds catastrophic exposure using CVaR, a coherent tail-risk measure (Artzner et al., 1999; Rockafellar & Uryasev, 2000). This posture is motivated by ruin dynamics under deep uncertainty (Taleb, 2012) and by Earth-system research indicating that destabilization of life-support conditions can occur through tipping cascades and boundary transgression corridors with multi-scale downstream harms (Rockström et al., 2009, 2023; Steffen et al., 2015; IPCC, 2023).

Modeled-safety interpretation (Normative clarification). Passing TRC demonstrates compliance with the declared scenario library, probability policy, catastrophe-cell construction, and corridor threshold for the run. It does not prove empirical invulnerability outside the modeled scenario set. Implementations MUST NOT present TRC passage as a guarantee that no catastrophic pathway exists; it is a governance-grade bound within the declared scenario model.

8.1 Tier Requirements for TRC (Normative; strict posture)

Tier	TRC Required?	Minimum
Tier 1	Optional (qualitative tail screen when plausible)	N/A
Tier 2	REQUIRED when catastrophe relevance plausible	≥5 minimum
Tier 3	REQUIRED	≥20 minimum
Tier 4	Design target only (claim prohibited in this release line)	N/A

8.1.1 Catastrophe relevance trigger (Normative)

Catastrophe relevance is plausible if any of the following are true:

- The decision materially affects any catastrophe cell in C_cat (Section 8.2), OR
- The decision has plausible pathways to tipping cascades (climate, bio, conflict, infrastructure, financial), OR

- The decision is materially irreversible or creates lock-in at Polity/CMIU/Biosphere scales, OR
- Credible challengers identify a plausible catastrophic failure mode even if decision owners prefer not to model it.

If triggered, TRC MUST be executed at the tier's required rigor.

8.2 Catastrophe Cell Set C_cat (Normative)

Base catastrophe cell set:

C_cat_base = {(6, Health), (6, Environment), (7, Environment)}

These correspond to:

- Humanity/CMIU-Health: Global-scale health viability (pandemic mortality, mass disability, collapse of health capacity)
- Humanity/CMIU-Environment: Environment-as-civilization-condition (habitability, agricultural stability, climate-driven displacement)
- Biosphere-Environment: Earth-system integrity (planetary boundaries, biodiversity, biogeochemical stability)

The base set explicitly includes biosphere integrity and civilization-scale habitability conditions because "safe operating space" and "safe and just boundaries" analyses identify global environmental corridors as central determinants of long-run human and ecological viability (Rockström et al., 2009, 2023).

Extensions: Tier 1-3 MAY extend C_cat only with explicit PCC justification and MUST apply the same set to all compared options.

Clarification (Normative). C_cat_base is the default habitability/civilization catastrophe proxy. It cleanly captures biosphere degradation and civilization-scale health/environment collapse. Governance lock-in, coercive surveillance, procedural collapse, information-ecosystem capture, or other agency-destroying catastrophes are catastrophe-relevant and SHALL NOT be treated as rhetorically covered by the base triplet alone. When any governance-lock-in trigger is plausible, including surveillance normalization, emergency-power persistence, due-process suspension, information-ecosystem capture, irreversible dependency lock-in, or concentrated arbitrary control over critical infrastructure, the run MUST activate the default governance-lock-in extension profile defined in Appendix D.1A, unless a stricter governed profile is declared in the PCC. The chosen profile MUST be applied symmetrically across all compared options and recorded in the PCC.

8.3 Scenario Governance (Normative)

TRC is evaluated over a governed scenario set S with probabilities p_s such that $p_s \geq 0$ and $\Sigma_s p_s = 1$.

8.3.1 Mandatory Tail Scenario Categories (Normative)

The scenario set MUST include the following tail categories unless explicitly justified as not plausible for the decision context. Governance / lock-in catastrophe is a mandatory tail category if and only if the governance-lock-in extension profile in D.1A is active; otherwise it is recommended but not mandatory. When that profile is active, the scenario set MUST include at least one governance-lock-in tail category with category probability floor p_floor ≥ 0.02.

- Pandemic/biological disruption
- Climate tipping cascade
- Financial system collapse
- Major conflict escalation
- Critical infrastructure failure

Governance / lock-in catastrophe

Minimum probability floor per category: p_floor ≥ 0.02

This prevents “include but set near-zero probability” gaming.

Probability-floor epistemic boundary. The p_floor value is an anti-omission governance prior, not a calibrated empirical frequency estimate. It prevents mandatory tail classes from being made invisible by arbitrary near-zero assignment and must be sensitivity-tested where decision-material.

Rejected-scenario log. Tier 3 runs and high-stakes Tier 2 runs MUST record any material tail category or challenger scenario rejected as implausible, including the rejection rationale, evidence basis, reviewer status, and whether the rejection would change if the reality surface, probability floor, or time horizon changed.

These mandatory tail categories are not intended as exhaustive; they are a minimal governance floor reflecting cross-domain systemic risk exposure in tightly coupled global systems, including climate-driven boundary cascades (Rockström et al., 2009, 2023; Steffen et al., 2015; IPCC, 2023).

8.3.2 Implausibility test for exempting mandatory tail categories (Normative)

If a deployment wishes to exempt a mandatory tail category as “implausible,” the exemption requires:

(i) Written justification in the PCC explaining why the category is implausible for this specific decision context

(ii) Independent reviewer sign-off (Tier 3 required; Tier 2 recommended)

(iii) Recording in PCC with audit flag MANDATORY_TAIL_CATEGORY_MISSING_WITH_JUSTIFICATION

The burden of proof is on omission, not inclusion. This preserves UBE.

8.3.3 Scenario set size minima (Normative)

Tier	Minimum
Tier 2 (when TRC required)	≥5 minimum
Tier 3	≥20 minimum

If a tier cannot meet its minimum scenario count, the run MUST NOT claim tier compliance for TRC coverage and MUST record audit_flag SCENARIO_LIBRARY_MIN_EXCEPTION in the PCC, including written justification and a remediation plan to reach the minimum scenario count for the next comparable run.

8.4 Loss Construction (Tier 1-3: Bounded-Impact TRC) (Normative)

Tier 1-3 TRC uses bounded-impact loss based on propagated Base-stream impacts in catastrophe cells.

Domain constraint: All propagated impacts are bounded: $I\_prop\_base,c(a|s) \in [-1, +1]$.

For each scenario s, define scenario loss:

$L(a,s) = \Sigma\_\{c \in C\_cat\}\ \omega\_c \times \max(0, -I\_prop\_base,c(a|s))$

Normative stream binding (Design A): TRC MUST be computed from the Base stream (I_prop_base). TRC MUST NOT use Welfare-stream (sentience-weighted) impacts.

Where:

- $\omega\_c \geq 0$, $\Sigma\_\{c \in C\_cat\}\ \omega\_c = 1$ (catastrophe weights are normalized)
- max(0, x) returns x if x > 0, otherwise 0 (only negative impacts contribute)
- $I\_prop\_base,c(a|s)$ is the propagated impact in catastrophe cell c under scenario s

Default catastrophe weights (if not otherwise governed): Uniform weights over C_cat.

Interpretation:

- Only negative impacts in catastrophe cells contribute to loss.
- Loss is bounded in [0,1] because weights are normalized and impacts lie in [-1,+1].
- Positive impacts in catastrophe cells (improvements) do not offset losses; they simply contribute zero to the loss function.

Example: If C_cat = {(6, Health), (6, Environment), (7, Environment)} with uniform weights ($\omega\_c = 1/3$ each), and under scenario s the propagated impacts are:

- $I\_prop\_base(6,Health,a|s) = -0.30 \rightarrow$ contributes $1/3 \times 0.30 = 0.10$
- $I\_prop\_base(6,Environment,a|s) = +0.10 \rightarrow$ contributes $1/3 \times \max(0, -0.10) = 0$
- $I\_prop\_base(7,Environment,a|s) = -0.60 \rightarrow$ contributes $1/3 \times 0.60 = 0.20$

Then $L(a,s) = 0.10 + 0 + 0.20 = 0.30$

8.5 CVaR Computation (Normative)

Let $L(a)$ be the random loss induced by scenarios s with probabilities p_s.

Value-at-Risk: $VaR_\alpha[L(a)] = \inf\{z : P(L(a) \leq z) \geq \alpha\}$

CVaR operational definition (Tier 1-3). For RippleLogic Tier 1-3 discrete scenario libraries, $CVaR_\alpha[L(a)]$ is the probability-weighted average loss over the worst $\beta = 1-\alpha$ probability mass, computed by the canonical discrete algorithm in Appendix D.7.

Continuous-case shorthand. In continuous loss distributions without probability mass at VaR, this corresponds to the familiar expression $E[L(a) \mid L(a) \geq VaR_\alpha[L(a)]]$. In discrete scenario sets with atoms at the VaR boundary, that shorthand is not controlling and MUST NOT override Appendix D.7.

The Appendix D.7 discrete tail-mass algorithm is normative, audit-replayable, and controlling for all finite Tier 1-3 scenario sets, including partial mass at the boundary scenario when required.

8.6 TRC Admissibility Predicate (Normative)

An option passes TRC if and only if:

$CVaR_\alpha[L(a)] \leq \tau_{TRC}$

Define:

$A_{adm} = \{a \in A_{NCRC} : TRC(a) = TRUE\}$

8.7 Default TRC Parameters by Context (Normative Defaults)

These are defaults; governance may tighten them. Loosening requires explicit governance justification and PCC recording.

Context	α (tail level)	τ_TRC (corridor threshold)
Personal	0.90	0.30
Organizational	0.95	0.20
Reversible policy	0.95	0.15
Irreversible policy	0.99	0.10
Existential risk	0.999	0.05

8.8 TRC Fallback Mode (Normative)

If $A_{NCRC} \neq \varnothing$ but $A_{adm} = \varnothing$ (rights-safe options exist, but all fail TRC):

- Rank all A_NCRC options by CVaR_α ascending.
- Select the option with minimal CVaR_α.
- Require a time-bound risk mitigation plan and enhanced monitoring.
- Require one-tier-higher approval for the decision.
- Record TRC_FALLBACK_INVOKED in PCC and list the deficit (CVaR − τ_TRC) for the selected option.

If both NCRC and TRC fail for all options (no rights-safe options and no tail-safe options), Emergency Mode (Section 7.5) governs with CVaR as secondary tie-break.

8.9 Anti-Gaming Requirements for TRC (Normative)

- Mandatory tail categories cannot be omitted without explicit implausibility justification per Section 8.3.2.
- Probability floors apply per category; they cannot be silently violated.
- Scenario sets must be comparable across options (same S, same p_s).
- PCC must record scenario provenance and how probabilities were assigned.
- Tier 2-3: if probabilities are highly uncertain, the PCC SHOULD include sensitivity over p_s (Tier 3 required).

SECTION 9: CSV GATE: CONTAINMENT AND STRUCTURAL VIABILITY

9.0 Purpose and UBE Rationale (Normative)

CSV means Containment and Structural Viability. CSV is the strengthened fourth public level in the five-level RippleLogic cascade inside the MathGov framework:

RG -> RF -> TRC -> CSV -> RLS

Public-language mnemonic:

Ground -> Protect -> Bound -> Contain/Verify -> Score

CSV prevents a core failure mode under Union-Based Reality: a local option can look beneficial for a sub-union while quietly degrading the larger system, capacity, relation, ecological support, institutional trust, or future choice-space that makes the benefit possible. Under Union-Based Ethics, such “gains” are not counted as system-level improvement when they materially degrade, hollow out, overload, or exploit their containing systems beyond tolerance.

CSV is therefore the selectability gate applied after NCRC and TRC and before final RLS ranking. CSV is not a sixth public level. It is the upgraded fourth level, replacing the older public label “Containment” with the more complete label “Containment and Structural Viability.” Structural Viability is no longer merely a quiet subcheck; it is explicitly visible inside CSV.

Core question:

Can this option deliver its claimed good without secretly breaking, hollowing, overloading, exploiting, or depending on the systems that make the good possible?

Plain-language rule:

CSV is not a purity filter. CSV is a reality-integrity filter.

9.1 CSV Non-Purity Rule (Normative)

A negative ripple is not automatically a CSV failure. Every real action carries some cost, burden, risk, or tradeoff. CSV MUST distinguish between detected harm, material residual harm, conditionable harm, redesign-trigger harm, and gate-failing harm.

CSV SHALL NOT demand zero harm. CSV SHALL demand that harms are not hidden, unbounded, structurally degrading, rights-adjacent without review, irreversible without authorization, dependency-producing without transition, or unjustly externalized onto affected unions.

Status categories:

Status	Meaning	Result
DETECTED_HARM	A negative effect exists.	Must be recorded and routed to the correct layer.
RESIDUAL_HARM	Harm is material but bounded, visible, and below CSV failure threshold.	Carries into RLS as a welfare cost.
CONDITIONABLE_HARM	Harm is acceptable only with controls.	CSV_PASS_WITH_CONTROLS if controls are binding.
REDESIGN_TRIGGER	Harm is too high as specified but plausibly fixable.	CSV_REDESIGN_REQUIRED; option is not selectable as specified.
CSV_FAILURE	Harm is uncontained, structurally degrading, non-viable, unjustly externalized, or lock-in producing beyond tolerance.	CSV_FAIL; option is not selectable.

Normative consequence:

RLS may rank ordinary residual tradeoffs among selectable options. RLS MUST NOT rescue an option that fails CSV.

9.2 CSV Status Ladder (Normative)

CSV outputs one of the following status values:

CSV status	Meaning	Selection consequence
CSV_PASS	Harms are contained, structural viability is adequate, and residual burdens may be carried into RLS.	Option may proceed to RLS if NCRC and TRC passed.
CSV_PASS_WITH_CONTROLS	Option is selectable only if declared controls, mitigations, monitoring, or limits are binding.	Option may proceed to RLS as the controlled version only.
CSV_REDESIGN_REQUIRED	Option is not selectable as specified, but a redesigned version may be evaluated.	Original option excluded; redesigned option may re-enter RG/cascade.
CSV_ESCALATE	Stakes, uncertainty, missing evidence, or governance sensitivity require higher review.	No final selection until escalation is resolved.
CSV_FAIL	Option unacceptably damages, hollows, overloads, exploits, or depends on containing systems beyond tolerance, or is structurally non-viable.	Option is not selectable.
CSV_EMERGENCY_PROVISIONAL	Temporary selection under necessity where no better feasible option currently exists and strict emergency controls are declared.	Time-bounded provisional selection only; no unqualified alignment claim.
CSV_NOT_MATERIAL	CSV burden is not material for this run, with rationale.	Option may proceed if other gates pass, but rationale is recorded.

A CSV_PASS_WITH_CONTROLS is a pass only for the controlled option. If controls are removed, expire, fail, or are materially altered, CSV MUST be rerun.

A CSV_EMERGENCY_PROVISIONAL disposition MUST include a time limit, harm cap, monitoring plan, review trigger, and transition or remediation plan. It MUST NOT be marketed as full alignment, ordinary selectability, deployment certification, or Tier 4 validation.

9.2A CSV status resolver guide (Normative for Tier 3; recommended for Tier 2)

The following resolver constrains how evidence and diagnostics map into CSV statuses. It does not eliminate reviewer judgment, but reviewer judgment must be recorded against these routing rules.

Evidence / diagnostic condition	Required CSV routing
Any material CSV diagnostic is level 4, or the option creates uncontained, structurally degrading, unjustly externalized, non-viable, hidden, lock-in-producing, unmonitored, or beyond-tolerance harm	CSV_FAIL
Any Structural Viability subcheck materially fails and the failure is not time-bounded, controlled, or redesignable	CSV_FAIL
Any Structural Viability subcheck materially fails but a plausible redesigned option could repair it	CSV_REDESIGN_REQUIRED
Gate-material CSV evidence is missing, stale, contested, or insufficient to support selectability	CSV_ESCALATE, claim-narrowing, or refusal of the stronger selection claim
Any material diagnostic is level 3	CSV_REDESIGN_REQUIRED or CSV_ESCALATE, with rationale
A material diagnostic is level 2 and binding controls, owners, monitoring, review triggers, and failure consequences are declared	CSV_PASS_WITH_CONTROLS
A material diagnostic is level 2 without binding controls	CSV_REDESIGN_REQUIRED
All material diagnostics are level 0-1 and structural viability subchecks pass	CSV_PASS
CSV burden is genuinely not material to the declared claim boundary and the PCC records why	CSV_NOT_MATERIAL
Necessity is documented, no better feasible option is available, controls are time-bounded, harm caps are declared, monitoring is active, and a transition/remediation path exists	CSV_EMERGENCY_PROVISIONAL

CSV maturity boundary. CSV may be binding in Tier 3 run-level conformance using declared provisional structural indicators, but this does not imply cross-domain validated UCI/HOI measurement, ProofPack readiness, or machine-verifiable structural truth. UCI/HOI remain diagnostics unless and until separate validation artifacts establish stronger measurement status.

9.3 CSV Core Tools (Normative)

CSV uses five core tools. Implementations MAY add domain-specific tools, but MUST preserve the five core functions when material.

9.3.1 Containing-System Map

Identify the systems that hold, sustain, receive, or absorb the option's consequences. Relevant systems MAY include persons, households, communities, organizations, institutions, infrastructure, ecology, attention, trust, labor capacity, knowledge systems, public legitimacy, supply chains, financial reserves, legal authority, and future option-space.

Minimum questions:

- What local good is the option claiming?
- Which union receives the local gain?
- Which larger unions contain or sustain that gain?
- What capacities, relations, resources, and ecological supports does the option depend on?
- What systems receive spillover costs?
- Which affected groups can contest, consent, or recover?

9.3.2 Structural Viability Check

Structural Viability asks whether the option can actually stand in the declared reality surface. Ethical attractiveness is not enough. A desirable option may be a design target while still failing current execution viability.

Minimum subchecks:

Code	Subcheck	Question
SV-1	Resource closure	Are budget, time, people, compute, energy, materials, attention, and authority sufficient?
SV-2	Dependency closure	Are prerequisites satisfied or assigned to credible owners and timelines?
SV-3	Reversibility / endpoint coherence	Is rollback, endpoint, or irreversibility status declared and coherent?

Code	Subcheck	Question
SV-4	Capacity non-exceedance	Does execution stay within operational, institutional, technical, and enforcement capacity?
SV-5	Internal non-contradiction	Does the option avoid mutually incompatible requirements or declare a grounded reconciliation mechanism?

An option with material SVC failure is not selectable for current execution, regardless of RLS rank.

9.3.3 Externality Containment Ledger

CSV MUST make spillovers visible. For material runs, the PCC SHOULD include an ExternalityContainmentLedger with at least:

Field	Required content
Spillover	Smoke, noise, labor burden, dependency, extraction, attention harm, ecological stress, etc.
Receiver	Which union, subgroup, system, or future condition receives the burden.
Severity	Normalized severity or qualitative tier.
Duration	Acute, temporary, recurring, chronic, permanent, or irreversible.
Mitigation	Controls, design changes, monitoring, compensation, substitution, or remediation.
Residual	What burden remains after mitigation.
Routing	NCRC, TRC, CSV, RLS, monitoring, or escalation.

Unrecorded material externalities are not neutral. They create CSV uncertainty and may trigger CSV_ESCALATE or CSV_REDESIGN_REQUIRED.

9.3.4 Substitution Ladder

CSV MUST ask whether a less harmful feasible alternative exists. The ladder is not a requirement to choose the ideal option immediately; it is a discipline against convenience-based harm.

Default substitution ladder:

1. Avoid the harmful need.
2. Reduce the demand.
3. Use low-harm alternative.
4. Use hybrid/backup alternative.
5. Use higher-harm option only with controls.
6. Use emergency provisional option only under necessity and review.
7. Reject or redesign if harms cannot be contained.

If a materially cleaner and feasible alternative exists, choosing a more harmful option for convenience, profit, or status SHOULD trigger CSV_REDESIGN_REQUIRED or a high residual RLS penalty, depending on severity.

9.3.5 Necessity Corridor

A harmful option MAY receive CSV_PASS_WITH_CONTROLS or CSV_EMERGENCY_PROVISIONAL only when it satisfies the necessity corridor:

Corridor condition	Requirement
Necessity	The option serves a real need, not mere convenience.
No better feasible alternative now	Cleaner or safer options are unavailable, unaffordable, disproportionate, or not timely under the declared reality surface.
Boundedness	Harm is limited in time, space, exposure, scale, and recurrence.
Mitigation	Controls reduce harm materially.
Transparency	The harm is not hidden, greenwashed, or falsely described.
Reversibility / exit	The option does not create avoidable lock-in.
Transition	A path to a cleaner, safer, or more viable option is declared where material.
Review	Monitoring and re-evaluation triggers are declared.

The necessity corridor prevents CSV from becoming impossible purity ethics while preserving its ability to block structural laundering.

9.4 CSV Diagnostic Concern Scale (Normative guidance)

CSV MAY use internal diagnostic concern levels. These levels are not an RLS score and MUST NOT be treated as final welfare ranking.

Level	Meaning	Default disposition
0	No material CSV concern.	CSV_PASS.
1	Minor residual concern.	Carry to RLS and monitoring.
2	Material concern requiring controls.	CSV_PASS_WITH_CONTROLS if controls are binding.
3	Serious concern requiring redesign or higher review.	CSV_REDESIGN_REQUIRED or CSV_ESCALATE.
4	Gate-failing concern.	CSV_FAIL.

Suggested diagnostic dimensions:

- Containment integrity
- Structural viability
- Hollowing risk
- Dependency or lock-in risk
- Substitution pressure
- Mitigation adequacy
- Monitoring adequacy
- Reversibility / exit
- Accumulation risk
- Governance or legitimacy stress

Default status rule:

If any material diagnostic = 4, CSV_FAIL. If any material diagnostic = 3, CSV_REDESIGN_REQUIRED or CSV_ESCALATE. If any material diagnostic = 2, CSV_PASS_WITH_CONTROLS only if controls are binding and auditable. If all material diagnostics are 0 or 1, CSV_PASS and residuals carry to RLS.

9.5 Formal CSV Predicate (Normative)

Let A_adm be the set of options that pass NCRC and TRC, subject to Reality Grounding supporting the claim boundary.

For each option a in A_adm, compute CSV_status(a).

Default selectable set:

A_sel = {a in A_adm : CSV_status(a) in {CSV_PASS, CSV_PASS_WITH_CONTROLS, CSV_NOT_MATERIAL}}

Emergency provisional options are not ordinary members of A_sel. A CSV_EMERGENCY_PROVISIONAL option may be temporarily selected only through emergency protocol, with explicit time limit, harm cap, mitigation, review, and no full alignment claim.

CSV_NOT_MATERIAL is pass-equivalent only for the declared claim boundary after the PCC records the rationale that no material CSV burden is present. It MUST NOT be used to waive Structural Viability subchecks when execution feasibility, resource closure, dependency closure, reversibility, operational capacity, or internal coherence is material. In high-stakes or execution-bearing runs, CSV_NOT_MATERIAL may apply to the containment-integrity dimension while Structural Viability remains separately required where execution feasibility is material. If later evidence makes CSV material, CSV MUST be rerun.

RLS is applied only to A_sel, except where emergency protocol explicitly invokes fallback comparison.

9.6 UCI, HOI, and Structural Diagnostic Placement (Normative)

CSV is the authority layer for cross-scale structural integrity. UCI, HOI, Adaptive Reserve, and related structural indicators are diagnostic instruments. When such indicators are material to whether an option degrades a containing union, hollows out capacity, creates dependency, or undermines structural viability, they are evaluated inside CSV before RLS.

UCI/HOI are not public cascade stages. They are not separate gates by default. They are structural diagnostics inside CSV when material, and residual tie-break or monitoring signals only after options are already selectable and RLS is tied, close, uncertainty-overlapped, or non-decisive.

If UCI/HOI or equivalent structural diagnostic evidence is required for a CSV determination and unavailable, the implementation SHALL record CSV_STRUCTURAL_DIAGNOSTIC_UNAVAILABLE or the more specific CSV_UCI_HOI_UNAVAILABLE flag. A run with unresolved material diagnostic unavailability SHALL NOT claim an unqualified CSV pass or Tier 3 CSV conformance.

9.7 CSV Fail Conditions (Normative)

CSV MUST fail, redesign, or escalate an option when any of the following are material and unresolved:

1. Container destruction: the option damages the system that makes the claimed benefit possible.
2. Forced externalization: the option works by pushing burden onto people, ecosystems, or future agents who cannot consent, contest, or recover.
3. Structural non-viability: the option cannot execute under real resource, dependency, capacity, authority, or contradiction constraints.
4. Hollowing: the option improves surface metrics while weakening the deeper capacity the metric was meant to serve.
5. Lock-in or dependency: the option creates avoidable dependency that makes better futures harder.

6. Unbounded accumulation: individually small harms scale into systemic degradation when repeated.
7. Unmonitored invisible harm: material harms exist without monitoring, audit, review, or feedback path.
8. Better feasible alternative ignored: a much cleaner or safer feasible option exists, but the harmful option is chosen for convenience, status, or profit.

9.8 CSV Pass Conditions (Normative)

CSV MAY pass an option when harms are visible, rights floors are intact, tail risk is bounded, containment burden is limited, structural viability is real, externalities are mitigated, residual harms are carried into RLS, monitoring exists where needed, lock-in is avoided or justified, better alternatives are unavailable or disproportionate, and transition or review plans exist where material.

Plain-language test:

Harm does not equal fail. Hidden, unbounded, unjust, irreversible, or structurally degrading harm equals fail, redesign, escalation, or emergency provisional treatment.

9.9 Worked Normative Example: Backup Diesel Generator (Informative but Canon-Guiding)

A small organic cafe may need a backup diesel generator during power cuts. The generator produces smoke, noise, local air pollution, and emissions. Without it, the cafe may lose food, income, jobs, community function, and operational continuity.

CSV MUST NOT treat this as an automatic fail merely because harm exists. CSV MUST also not allow the cafe's local survival benefit to hide uncontained harm.

Illustrative dispositions:

Generator condition	CSV disposition	Rationale
Generator used indoors, near air intakes, or near vulnerable people without controls.	CSV_FAIL or NCRC escalation.	Serious health/safety exposure and containment breach.
Outdoor generator, frequent use, no mitigation, no review, no transition.	CSV_REDESIGN_REQUIRED.	Harm is visible but insufficiently contained.
Outdoor generator, limited outages, exhaust away from people, maintenance, fuel controls, noise limits, and transition plan.	CSV_PASS_WITH_CONTROLS.	Harm is bounded, necessary, mitigated, transparent, and reviewable.

Generator condition	CSV disposition	Rationale
Battery/inverter feasible now.	CSV_PASS; likely higher RLS.	Lower containment burden.
Solar + battery + emergency diesel only.	CSV_PASS; likely stronger RLS if feasible.	Better resilience/environment balance.
Closing during outages.	CSV_PASS but RLS context-dependent.	Avoids smoke but may harm livelihood, waste food, and reduce community function.

Canonical lesson:

CSV is not “diesel bad, therefore fail.” CSV is “diesel carries a containment burden; if necessary, bounded, mitigated, transparent, monitored, and transitional, it may conditionally pass and be ranked against cleaner options by RLS.”

9.10 RLS Relationship (Normative)

RLS ranks only options that survive RG claim support, the Rights Floor (formal NCRC), TRC, and CSV. RLS SHOULD penalize residual harms carried from CSV, including pollution, dependency, lock-in risk, ecological burden, social trust loss, and meaning/agency degradation.

RLS MUST NOT rescue CSV failure. CSV protects the doorway into ranking; RLS ranks what may legitimately enter.

9.11 CSV Parameter Governance and Escalation (Normative)

CSV thresholds, diagnostic concern levels, necessity-corridor judgments, UCI/HOI trigger thresholds, and structural-viability sufficiency standards MUST be recorded in the PCC when material. Tier 3 runs SHOULD include sensitivity analysis over near-binding CSV determinations.

If no option passes CSV, the run MUST NOT pretend a good option exists. Required actions include option redesign, additional evidence gathering, transition planning, escalation to a higher governance tier, emergency provisional protocol if genuinely necessary, or refusal of deterministic selection.

CSV does not override NCRC or TRC. It operates after admissibility and before residual welfare ranking.

SECTION 10: OPTIMIZATION LAYER: RIPPLELOGIC SCORE (RLS)

10.0 Purpose and Scope (Normative)

RLS ranks selectable options by expected welfare improvement across unions and dimensions, after ripple propagation and saturation, under declared weights and applicability mask rules.

RLS MUST NEVER override NCRC, TRC, or CSV. It operates only on selectable options A_sel.

10.1 RLS Definition (Normative)

$RLS(a) = \Sigma_u \Sigma_d \, w_u \times v_d \times m(u,d) \times \kappa(u,d) \times I\_prop\_welfare(u,d,a)$

Where:

- w_u are union weights (HDW-governed; Section 13; Section 13.1 constitutional floors controlling)
- v_d are dimension weights (HDW-governed; Section 13; Section 13.1 constitutional floors controlling)
- m(u,d) is applicability mask (RLS aggregation only)
- κ(u,d) is a declared cell multiplier (default 1; non-default requires PCC justification)

κ(u,d) governance rule (Normative). The default is κ(u,d) = 1. Non-default κ(u,d) values are permitted only for pre-declared, governance-approved cell-level emphasis that does not alter admissibility semantics. Unless a stricter charter rule is declared, κ(u,d) MUST remain positive and bounded within [0.5, 1.5]. A non-default κ(u,d) MUST NOT reverse sign, erase or dilute any rights-covered or catastrophe-relevant harm, or function as a back-door substitute for masking, reweighting floors, or prominence gaming. Tier 3 runs using non-default κ(u,d) MUST record justification, authorized source, affected cells, and sensitivity reruns in the PCC.

- I_prop_welfare(u,d,a) is post-propagation, post-saturation impact (Sections 5-6)

Local-use note (Informative). Section 10.2A defines a Practical Local Scope Scoring (PLSS) profile for local runs. PLSS does not change the Section 10.1 equation form; it defines a canon-consistent local construction for w_u within the residual mass above the Section 13.1 constitutional floors.

10.2 Applicability Mask Rules (Normative)

Masking exists to avoid meaningless aggregation in some contexts, but is a primary gaming vector. Therefore:

Definition (Normative). Masking means setting m(u,d)=0, which affects RLS aggregation only. Omission/exclusion means failing to compute, disclose, or evaluate a required cell for NCRC, TRC, or CSV. Masking SHALL NOT be used as a substitute for omission rules.

Masking affects RLS aggregation only. Whether a cell must be constructed is determined by active-cell rules (Section 10.3) and tier coverage requirements, not by masking. Accordingly, masking MUST NOT be used to skip or modify:

- Impact construction (Section 5),
- Propagation (Section 6),
- NCRC checks (Section 7),
- TRC loss and CVaR computation (Section 8),
- Or containment evaluation (Section 9).

Non-maskable cells (MUST be unmasked):

- All rights-covered cells in any C_r
- All catastrophe cells in C_cat

Non-maskable persistence rule (Normative). Rights-covered cells remain non-maskable even after NCRC admissibility filtering. This prevents post-gate dilution and preserves rights-salience when ranking admissible options, especially under adversarial pressure and Goodhart dynamics.

Rationale (Informative). The admissibility gates prevent selecting options that violate non-compensatory rights floors, but they do not by themselves prevent near-miss normalization among admissible options. Keeping rights and catastrophe cells in RLS aggregation preserves continuous optimization pressure toward larger safety margins, reduces proxy gaming pressure (for example, satisfying a threshold while degrading adjacent rights conditions), and makes rights and tail-risk performance legible for governance review and accountability.

Baseline consistency for non-maskable gate cells (Normative). For any cell in C_r or C_cat that remains included in RLS under the non-maskable persistence rule, the welfare-stream baseline SHALL also use the gate floor-reference baseline unless a governed dual-baseline method is explicitly declared. Where a dual-baseline method is declared, the floor-reference value SHALL remain the controlling value for gate-consistency interpretation and validator conformance.

- Any cells mandated by minimum governance coverage for the run (declared in PCC; default includes at least U_1, primary affected unions, and U_7 Environment when environmental relevance exists)

Audit rule: If any non-maskable cell is masked, set audit_flag RIGHTS_CELL_MASKED_INVALID (or CATASTROPHE_CELL_MASKED_INVALID) and PCC is INVALID.

10.2A Practical Local Scope Scoring (PLSS) (Normative)

Purpose. PLSS specifies a canon-consistent local-use profile for RippleLogic Score (RLS) when a decision is primarily local in its direct effects, but broader scopes remain morally relevant as rights, tail-risk, and containment guardrails. PLSS allows routine decisions to concentrate most residual welfare-ranking attention on the scopes that are actually live in the decision, without weakening NCRC, TRC, CSV, non-maskable cell rules, or Section 13 constitutional weight floors.

Interpretive anchor (Informative). Union Scopes are not a closed list of who matters. They are the stable scopes of aggregation and accountability used by the framework. Stakeholder instances feed the scopes; the scopes ensure that impacts do not disappear.

Terminology lock (Informative). LSS names the upstream stakeholder object, PLSS names the residual weight-construction procedure used within RLS, and FLGG - Focus Local, Guard Global - names the governing design principle.

10.2A.0 Attention-Safety Separation Principle (Normative, architecturally grounded). PLSS operates only within the residual welfare-ranking layer. Admissibility under RF/NCRC, TRC, and CSV SHALL remain invariant under any valid PLSS weight construction satisfying Sections 10.2A.4-10.2A.6. Therefore, a change in PLSS prominence declarations or PLSS weight allocation SHALL NOT convert an inadmissible option into an admissible option. This principle follows from the lexicographic separation of admissibility gates from the welfare-ranking layer.

PLSS does not determine who matters; it determines how residual welfare-ranking attention is distributed after universal admissibility conditions have already been imposed.

10.2A.1 Relation to existing gates (Normative)

PLSS applies only to welfare ranking within Section 10. It SHALL NOT alter:

• NCRC evaluation over the required rights coverage sets C_r,

• TRC evaluation over the catastrophe cell set C_{cat},

• Containment requirements under Section 9,

• non-maskable cell requirements under Section 10.2,

• constitutional weight floors defined in Section 13.1.

A PLSS run therefore remains:

• full-scope for admissibility, where required by RF/NCRC, TRC, and CSV,

• local-scope only in residual welfare emphasis, after admissibility has been preserved.

Structural diagnostic preservation under PLSS (Normative). PLSS SHALL NOT down-weight, omit, mask, or defer structural diagnostic evidence required for Containment. If a local-scope run uses PLSS but plausible containment-relevant UCI/HOI, Adaptive Reserve, lock-in, hollowing, governance-spillover, ecological-spillover, or institutional-capacity evidence exists outside the prominent scopes, that evidence remains admissibility-relevant. The run SHALL escalate, broaden scrutiny, or refuse the stronger local-scope claim when containment-relevant structural evidence is unavailable or unresolved.

10.2A.2 Prominent scopes and guardrail scopes (Normative)

For a declared run, each operational Union Scope $u \in \{U1, \ldots, U7\}$ may be classified in the PCC as one of the following:

• Prominent scope: a scope whose direct or near-field ripple is substantial enough that it should receive meaningful residual welfare weight in the current run.

• Guardrail scope: a scope whose direct ripple in the current run is limited, but which remains protected by constitutional floor weight and by admissibility constraints.

Guardrail scope classification SHALL NOT be interpreted as moral irrelevance. A guardrail scope may carry little or no residual weight in a local run, but it remains:

• constitutionally represented through Section 13 floors,

• admissibility-relevant where rights, catastrophe, or containment conditions apply.

10.2A.3 Representation rule: instances first, scopes second (Normative)

PLSS SHALL preserve the Section 2.2A and Section 5 representation architecture:

• stakeholder discovery is performed at the instance level,

• instances are mapped to one or more Union Scopes,

• impacts are aggregated within each scope row,

• scope rows are then combined by RLS.

Let I_u denote the declared stakeholder instance set mapped to scope u.

For non-rights scoring, the default aggregation rule remains the canonical population-weighted mean across affected instances:

$$I_{agg}(u,d,a) = [\ \Sigma_{(i \in I_u)}\ \pi_i \cdot I_i(u,d,a)\ ] \,/\, [\ \Sigma_{(i \in I_u)}\ \pi_i\ ]$$

where π_i is the declared aggregation weight for instance i, for example population share, exposure share, or another declared ReachBasis-consistent basis.

For rights-covered cells, aggregation SHALL NOT average away the worst-off subgroup. Rights evaluation remains governed by the worst-off subgroup operator:

$$I_{rights}(u,d,a) = \min_{(g \in G_{(u,d)})} I_g(u,d,a)$$

This subsection does not change Section 7 semantics.

10.2A.4 Local prominence signal and residual allocation rule (Normative)

PLSS introduces a declared local prominence signal q_u for each scope u, used only to distribute the residual union-weight mass that remains after constitutional floors.

Each q_u SHALL be recorded in the PCC and SHALL be justified by reference to the current decision context. A recommended default construction is a declared combination of:

• expected magnitude of ripple in scope u,

• probability that the effect materializes,

• temporal persistence or duration,

• propagation potential beyond immediate contact.

A recommended default operator is:

$$q_u = \psi(M_u, P_u, T_u, G_u)$$

with $M_u, P_u, T_u, G_u \in [0,1]$, and a default transparent choice:

$$\psi(M_u, P_u, T_u, G_u) = (M_u P_u T_u G_u)^{(1/4)}$$

This recommended operator is informative by default unless separately elevated by governance rule; the required part is that the PCC declare the operator used.

Prominence evidence discipline (Normative). Prominence sub-component declarations for q_u, including magnitude, probability, temporal persistence, and propagation potential, are subject to the same PCC justification and evidence documentation discipline that applies to impact-instance parameters under Section 5.2, including ReachBasis declaration where applicable. Tier 2+ runs SHALL record the evidentiary basis used to justify each declared prominence component.

Cross-deployment comparability note (Normative guidance). The geometric-mean operator remains the default comparability anchor for q_u. Any deployment using a non-default operator MUST document why the operator preserves anti-gaming discipline, boundedness, and monotonicity, and SHOULD include a sensitivity comparison against the geometric-mean default in the PCC or companion validation record.

Define residual shares:

$$\rho_u = q_u / \Sigma_j q_j, \text{ if } \Sigma_j q_j > 0$$

If all $q_u = 0$, the residual SHALL be allocated uniformly unless another declared fallback is specified in the PCC.

Let w_u^{floor} be the Section 13.1 constitutional floor for scope u. Then the PLSS union weights are:

$$w_u^{PLSS} = w_u^{floor} + (1 - \Sigma_j w_j^{floor}) \rho_u$$

subject to the canonical constraints:

w_u^PLSS ≥ 0

Σ_u w_u^PLSS = 1

w_u^PLSS ≥ w_u^floor

for every scope u.

Interpretation (Informative). PLSS does not re-write the constitution. It lets local runs place most of the residual emphasis on nearby scopes while retaining the constitutional baseline presence of broader scopes.

10.2A.5 PLSS RLS equation (Normative)

For any selectable option a ∈ A_sel, the local-scope welfare ranking is:

RLS_PLSS(a) = Σ_u Σ_d w_u^PLSS · v_d · m(u,d) · κ(u,d) · I_prop_welfare(u,d,a)

where:

• w_u^PLSS is defined by Section 10.2A.4,

• v_d remains governed by Section 13,

• m(u,d) remains governed by Section 10.2,

• κ(u,d) remains governed by Section 10.1,

• I_prop_welfare(u,d,a) remains governed by Sections 5–6.

If no kernel propagation is declared, then:

I_prop_welfare(u,d,a) = I_dir(u,d,a)

per the existing NONE propagation pathway.

10.2A.5A Discrimination threshold and non-decisive handling (Normative). The discrimination threshold defined in Section 10.4 applies identically to RLS_PLSS. If the lead between the top two selectable options is non-decisive under the declared uncertainty method, the tie-break chain in Section 11.4 MUST be applied. PLSS SHALL NOT bypass uncertainty handling, decisiveness testing, or tie-break governance.

10.2A.5B PLSS prominence sensitivity (Normative)

When PLSS is used in Tier 3, the PCC MUST rerun the welfare ranking under at least two alternative valid prominence constructions:

(i) the uniform residual fallback, and

(ii) a broadened guardrail-sensitive vector in which every scope classified Guardrail receives q_u at least 0.10 times the maximum q_u among prominent scopes, unless a stricter charter rule is declared.

If the ranking order of selectable options or the selected option changes under these alternative valid prominence constructions, the run MUST set audit flag PLSS_PROMINENCE_SENSITIVE and document the sensitivity in the PCC.

This requirement does not alter admissibility, which remains governed by the Attention-Safety Separation Principle and the upstream gates.

10.2A.6 Masking and non-maskable rule under PLSS (Normative)

PLSS SHALL NOT modify the Section 10.2 mask rule.

In particular:

• masking remains RLS aggregation only,

• masking SHALL NOT be used as omission,

• rights-covered cells and catastrophe-relevant cells remain non-maskable where required by the canon,

• reduced-scope or local-scope presentation SHALL NOT bypass admissibility on required cells.

Thus PLSS narrows practical emphasis, not moral coverage.

10.2A.7 Escalation triggers (Normative)

A run SHALL NOT remain in PLSS-only posture, or SHALL be rerun with broader scrutiny, if any of the following are plausible:

1. rights-covered harm beyond the prominent scopes,

2. catastrophe relevance or irreversible loss,

3. meaningful externalities crossing Community, Organization, Polity, Humanity/CMIU, or Biosphere pathways,

4. lock-in, precedent, or nontrivial governance spillover,

5. a credible challenger claim that a stakeholder class or scope was omitted,

6. Containment requires unavailable evidence for containing scopes,

7. repeated comparable runs continue to use local-scope framing without the required next-run expansion plan.

Audit flag (Normative). If one or more escalation triggers in Section 10.2A.7 is plausibly active, but the run proceeds in PLSS posture without escalation or documented justification, implementations MUST set audit flag PLSS_ESCALATION_TRIGGER_UNRESOLVED with severity ESCALATE. The PCC MUST then either record the broader posture adopted or provide a specific justification for non-escalation, including evidence basis and reviewer sign-off where required by tier. If no such disposition is recorded, the validator MUST reject the run as non-conformant.

Boundary heuristic note (Normative guidance). Implementations MAY use lightweight pre-screen heuristics to decide whether FLGG escalation is required before full local scoring proceeds, for example trigger scans over rights exposure, catastrophe relevance, cross-scope externality breadth, lock-in, precedent, and containment-evidence availability. Any such heuristic is advisory only: once a trigger is plausibly active, the canonical escalation requirements of Section 10.2A.7 control.

Ex ante method declaration rule (Normative). Any method choice that can materially affect decisiveness, ranking within the selectable set, or tie-break outcome, including PLSS prominence construction, alternative PLSS operator, scope-posture escalation disposition, and UCI comparison method, SHOULD be declared ex ante in the PCC where practicable. If first declared only after comparative outcomes are observed, the PCC MUST record a reason code, reviewer sign-off where required by tier, and a sensitivity note sufficient for independent review.

Scope posture classification note (Normative). When PLSS is used, each scope SHALL be classified in the PCC as either Prominent or Guardrail. v9.0 removes the undefined "Mixed" category to reduce audit drift; if a deployment wants finer descriptive language, it MAY add non-canonical comments, but the PCC classification field SHALL remain binary.

10.2A.8 Plain-language reading rule (Normative)

When not to use PLSS (Normative guidance statement). PLSS is intended primarily for local or moderately bounded decisions in which direct welfare effects are concentrated and no escalation trigger is plausibly active at the outset. PLSS SHALL NOT be used as the primary posture for irreversible cross-border policy, large-scale public AI deployment, biosecurity governance, active military or security operations, high-lock-in infrastructure decisions, or any case in which ET-1 through ET-7 is plausibly active at decision entry. Such cases require a broader deliberative posture or higher-tier structured evaluation.

For instructional and public-facing use, the framework MAY describe PLSS with the following plain-language sentence:

"In local decisions, score most strongly where the ripple is near, but keep every wider scope present through floors and guardrails. PLSS narrows residual attention, not moral coverage. It MAY change welfare ranking within the selectable set, but it MUST NOT make an inadmissible option admissible."

This sentence is interpretive only and SHALL NOT override the equations above.

10.2A.9 Worked example: family café local run (Informative)

Decision. A family café must choose between:

- Option A: cheap single-use cups,
- Option B: reusable deposit-return cups with a local wash partner,
- Option C: ultra-cheap cups from an unverified supplier with plausible toxic-material or labor-abuse exposure.

Declared stakeholder instances by scope:

- U1 Self: 1 instance
- U2 Household: 1 instance
- U3 Community: 5 instances
- U4 Organization: 2 instances
- U5 Polity: 3 instances
- U6 Humanity/CMIU: 1 instance
- U7 Biosphere: 1 instance

This run illustrates the distinction between instances and scopes. The café does not score Community in the abstract. It scores the affected Community instances, then aggregates them into the U3 row.

Declared local prominence signals:

$q = (U1,U2,U3,U4,U5,U6,U7) = (0.95, 0.85, 0.55, 0.50, 0.10, 0.03, 0.02)$

These sum to 3.00.

Under Section 13.1, the union floors are:

$w^{floor} = (0.20, 0.06, 0.06, 0.06, 0.08, 0.10, 0.10)$

The floor total is 0.66, so the residual is 0.34. The resulting PLSS weights are:

$w^{PLSS} \approx (0.308, 0.156, 0.122, 0.117, 0.091, 0.103, 0.102)$

Interpretation. Self, Household, Community, and Organization receive most of the residual attention. Polity, Humanity/CMIU, and Biosphere remain meaningfully present because the current constitution keeps them above zero even in local runs.

Table 10-1. Illustrative local prominence signals and PLSS union weights

Union Scope	q_u	w_u^floor	w_u^PLSS
U1 Self	0.95	0.20	0.308
U2 Household	0.85	0.06	0.156
U3 Community	0.55	0.06	0.122
U4 Organization	0.50	0.06	0.117
U5 Polity	0.10	0.08	0.091
U6 Humanity/CMIU	0.03	0.10	0.103
U7 Biosphere	0.02	0.10	0.102

Illustrative scope-level welfare composites:

Table 10-2. Illustrative scope composites by option

Scope	Q_u(A)	Q_u(B)
U1 Self	0.25	0.18
U2 Household	0.10	0.35
U3 Community	-0.06	0.41
U4 Organization	0.20	0.32
U5 Polity	-0.10	0.12
U6 Humanity/CMIU	-0.08	0.05
U7 Biosphere	-0.25	0.30

Then:

RLS_PLSS(A) ≈ 0.0656

RLS_PLSS(B) ≈ 0.2444

So Option B wins on welfare ranking among selectable options.

Worked-example uncertainty note. The raw RLS_PLSS values shown in this example do not by themselves determine final selectability. Under the declared uncertainty method, the lead between the top two selectable options SHALL be tested against the Section 10.4 discrimination threshold. If the result is non-decisive, the Section 11.4 tie-break chain applies. This example is intended to illustrate PLSS weight construction and score propagation, not to waive the framework's general decisiveness discipline.

Admissibility note. If Option C carries plausible rights-floor or catastrophe concerns through toxic exposure or abusive supply-chain conditions, it may fail NCRC or TRC before RLS ranking. If so, it is not saved by local convenience or low cost.

Lesson (Informative). PLSS does not say “ignore broader scopes.” It says: let local decisions place most additional attention where the ripple is near, while still preventing local gain from laundering wider harm.

10.3 Uncertainty Handling (Tier 2+ Required for Active Cells) (Normative)

RippleLogic recognizes epistemic uncertainty. Tier 2 and Tier 3 runs MUST record uncertainty and apply the discrimination rule for all active cells (definition below). Tier 1 runs SHOULD record uncertainty when feasible, especially on rights-covered and catastrophe-relevant cells.

Active cell definition (Normative; v9.0 tightened). A cell (u,d) is active for a run if any of the following holds: (i) m(u,d)=1 and the cell contributes to RLS aggregation, (ii) the cell lies in any rights coverage set C_r used for NCRC evaluation (regardless of mask status), (iii) the cell lies in the catastrophe cell set C_cat used for TRC evaluation (regardless of mask status), (iv) the cell is explicitly populated in the run for any gate, diagnostic, or justification, or (v) the cell is required by the run’s minimum governance coverage rules declared in the PCC.

Active cells MUST NOT be treated as unknown-by-default without disclosure. Masking (m(u,d)=0) excludes a cell from RLS aggregation only; it does not remove the cell from active status if any of conditions (ii) through (v) hold.

Definition of σ(u,d,a) (normative minimum): PCC MUST declare one of the following cell-level uncertainty proxies and apply it consistently across options:

10.3.1 Cell-level uncertainty proxy declarations (Normative)

10.3.2 Cell-level confidence derivation (Normative; Method B binding)

When the PCC uses Method B (confidence-derived) to compute cell uncertainty for a cell impact estimate, the PCC MUST define cell confidence c(u,d,a) in [0,1] from the set of impact-instance confidences {c_k} k∈K(u,d,a) using one declared CellConfidenceAggregationMethod. Implementations MUST apply the same method across all options in the run.

Let the instance contribution weight be defined as:
q_k := r_k · τ(t_k) · ℓ_k · e_k · s_k · |μ_k|
and let Q := Σ_{k∈K(u,d,a)} q_k.

The PCC MUST declare one of the following canonical aggregation methods:

CCAM_MIN_V1 (conservative default):
c(u,d,a) := min_{k∈K(u,d,a)} c_k
Use case: high-stakes, high-adversarial-pressure contexts where “weakest-link” confidence is the appropriate posture.

CCAM_WMEAN_V1 (weighted mean): if $Q>0$,
$c(u,d,a) := (1/Q) \cdot \Sigma_{k\in K(u,d,a)} (q_k \cdot c_k)$
If $Q=0$, set $c(u,d,a) := \min_{k\in K(u,d,a)} c_k$.

Edge case (Normative). If $K(u,d,a)=\varnothing$ and the Phantom Instance Rule applies, then c(u,d,a) MUST be set to 1.0 for the phantom instance because the uncertainty penalty is already represented by the phantom magnitude; the PCC MUST still mark the cell as UNKNOWN_IMPACT.

Range rule (Normative). The derived c(u,d,a) MUST be clipped to [0,1] and recorded in PCC for each active cell when Method B is used.

Method A (interval half-width): If the PCC records a confidence interval [L,U] for the cell impact I(u,d,a), set $\sigma(u,d,a) = (U - L)/2$.

Method B (confidence-derived): If the PCC records a confidence score $c(u,d,a) \in [0,1]$ for the cell estimate and a point estimate I(u,d,a), set $\sigma(u,d,a) = (1 - c(u,d,a)) \times |I(u,d,a)|$.

Method C (calibrated table): Use a pre-registered mapping from qualitative uncertainty labels (for example, LOW/MED/HIGH) to σ values, stored in the ProofPack or PCC appendix.

σ values MUST be clipped to [0,1]. Unless a stricter governed method is declared, σ(u,d,a) is interpreted as the pre-κ cell-level uncertainty input and SHALL be aggregated into σ_RLS using the same w_u, v_d, m(u,d), and κ(u,d) posture used for RLS.

Auditability binding (Normative). The PCC MUST record the source, elicitation basis, or calibration table used for σ(u,d,a) and any declared aggregation rule that produces it. Unless a stricter governed method is declared, σ(u,d,a) is interpreted as the pre-κ cell-level uncertainty input and SHALL be aggregated into σ_RLS using the same w_u, v_d, m(u,d), and κ(u,d) posture used for RLS. If an allowed alternative uncertainty method or aggregation rule would change decisiveness, that sensitivity MUST be disclosed.

Tier 3 uncertainty hardening (Normative). Tier 3 runs MUST rerun decisiveness using σ×0.5 and σ×2.0 stress tests (or the nearest declared equivalent for Method A interval widths) and MUST record whether the top-versus-runner-up decisiveness result changes. If decisiveness changes, the run MUST set audit flag UNCERTAINTY_DECISIVENESS_SENSITIVE and SHALL be treated as non-decisive until the tie-break chain or governance review is completed.

Method C restriction (Normative). Method C may be used only with a pre-registered calibrated table. A bare constant σ applied to all active cells is permitted only for worked examples, training artifacts, or explicitly non-operational demonstrations and MUST NOT support Tier 3 operational-readiness claims.

Define approximate RLS uncertainty:

$$\sigma\_RLS(a) = \sqrt{\Sigma_u \Sigma_d (w_u \times v_d \times m(u,d) \times \kappa(u,d) \times \sigma(u,d,a))^2}$$

Optional risk-adjusted score:

$$RLS\_adj(a) = RLS(a) - \lambda \times \sigma\_RLS(a)$$

Default: $\lambda = 0.5$. Use is optional but must be declared.

By default, RLS_adj is diagnostic only and MUST NOT replace RLS for ranking or decisiveness unless the run declares ex ante that risk-adjusted ranking is the governing ranking basis. If such a declaration is made, the same declared basis MUST be used consistently for ranking, Gap, and decisiveness across all compared options in the run.

10.4 Discrimination Threshold (Decisive vs Non-Decisive) (Normative)

Define gap between two options:

Gap(a,b) = abs(RLS(a) - RLS(b)) / sqrt(sigma_RLS(a)^2 + sigma_RLS(b)^2 + epsilon)

Default: delta = 2, epsilon = 1e-6.

Rationale (Informative). $\delta = 2$ is a conservative governance convention for the normalized Gap definition, not a claim that Gap is a literal z-score or that a specific distributional assumption holds. Its purpose is to reduce premature declarations of decisiveness when option-level uncertainty materially overlaps. Domains may tune delta, but any non-default delta MUST be declared in the PCC, justified, and applied consistently across options in the run.

Let a* be top RLS option and a_2 the runner-up among selectable options.

Decisive lead rule:

- If Gap(a, *a2) > delta (default delta = 2), select a* (subject to tie-break gating not being needed).
- If Gap(a*, a2) <= delta, the lead is non-decisive and the tie-break chain MUST be applied (Section 11.4).

10.4A Non-Decisive Gap Enforcement (Normative)

If the leading options are tied, close, or uncertainty-overlapped under the declared discrimination rule, FrameworkVerdict MUST NOT state or imply deterministic framework selection. If Gap(a*, a2) <= delta, including sigma-stress where required by tier, the run MUST either apply a governed tie-break rule over already-selectable options, return REFUSE_DETERMINISTIC_SELECTION, or route to AuthoritySelectionRecord without claiming that RippleLogic itself selected uniquely.

Any run with Gap <= delta that claims deterministic framework selection SHOULD set DECISIVENESS_OVERRIDE_INVALID with severity INVALID. Authority selection may still choose among admissible non-decisive options, but it must be recorded as authority selection, not framework determinism.

10.5 RLS Output Requirements (Normative)

PCC MUST record, per option in A_sel:

- I_prop_welfare(u,d,a) (or a retrievable summarized representation)
- RLS(a) and σ_RLS(a) and the uncertainty method
- Ranking and whether decisive (Gap vs δ)

SECTION 11: TIE-BREAKS, HOLLOWING DIAGNOSTICS, AND STRUCTURAL MONITORING

11.0 Purpose (Normative)

This section defines the special-case role of UCI/HOI after the core method has already formed a selectable set. UCI/HOI are not public headline cascade stages and are not independent late-stage admissibility gates. Their gate-relevant structural work belongs in Containment. Their residual work occurs only when RLS is tied, close, uncertainty-overlapped, or non-decisive, or when monitoring and hollowing-risk documentation are required.

- UCI is a structural metric used primarily as CSV evidence where material and secondarily as a residual tie-break signal when RLS cannot decide.
- HOI is a hollowing and drift diagnostic that flags welfare-up/coherence-down patterns for Containment, monitoring, and residual tie-break contexts.

11.0A Instrument-Placement Rule for UCI/HOI (Normative)

UCI and HOI are structural diagnostic instruments with two possible uses.

1. Containment use. When UCI/HOI evidence is material to cross-scale structural integrity, it is evaluated inside the CSV Gate under Section 9. This includes severe coherence loss, hidden hollowing, institutional brittleness, burnout, trust collapse, ecological degradation, governance-capacity loss, lock-in, or other structural damage to a containing union.

2. Residual tie-break use. When no containment threshold is crossed and options have already passed RF/NCRC, TRC, and CSV, UCI/HOI may be used after RLS only to resolve tied, close, uncertainty-overlapped, or non-decisive welfare rankings, document residual structural preference, trigger monitoring, or inform future-run steering.

Level 5 public-facing language SHALL NOT present UCI/HOI as a regular cascade stage. The public Level 5 is RippleLogic Score. The UCI/HOI rule is a special-case tie-break and monitoring rule attached to RLS non-decisiveness and CSV diagnostics.

11.1 Union Coherence Index (UCI) (Normative)

11.1.1 UCI components (canonical)

For each union u, define component scores in [0,1]:

Component	Symbol	Description
Cohesion	H_u	Internal connectivity, trust, shared identity, conflict resolution capacity
Flow	F_u	Coordination throughput, information fidelity, resource allocation efficiency
Resilience	R_u	Redundancy, robustness, recovery speed, adaptive capacity
Equity	E_u	Fair distribution of burdens/benefits, voice representation, inclusion

Compute:

$UCI_u = \alpha_H \times H_u + \alpha_F \times F_u + \alpha_R \times R_u + \alpha_E \times E_u$

Default: $\alpha_H = \alpha_F = \alpha_R = \alpha_E = 0.25$.

Special case for Self (U_1): $E_1 := 1$ by definition (not 0). The equity component is fixed at 1 for Self because equity-as-distribution does not apply within a single individual. This provides a neutral, non-penalizing identity value. UCI for Self is thus computed as:

$UCI_1 = 0.25 \times H_1 + 0.25 \times F_1 + 0.25 \times R_1 + 0.25 \times 1$

This resolves the v7.4.0 inconsistency and ensures UCI_1 is not systematically depressed.

11.1.2 Structural independence rule (Normative; Tier 3 binding)

UCI MUST be computed from structural/process indicators distinct from welfare indicators used for RLS.

At Tier 3:

- Deriving UCI from welfare-cell impacts is PROHIBITED.
- If structural indicators are unavailable, UCI is treated as unavailable and escalation/judgment-call protocol must be used (Section 11.5; Appendix E).

11.1.3 ΔUCI computation (Normative)

11.1.3A UCI Measurement Maturity Statement (Normative acknowledgement)

UCI is structurally load-bearing in the RippleLogic architecture: it determines Containment pass/fail (Section 9) and resolves non-decisive RLS ties (Section 11.4). The current release line specifies UCI component definitions (Cohesion, Flow, Resilience, Equity), structural independence requirements, and indicator families per scope, but does not yet provide universally validated measurement instruments with published psychometric properties (reliability, validity, normalisation benchmarks) across domains.

Scope	Measurement Maturity	Notes
U1 Self	Provisional	Validated instruments exist for some components (e.g. psychological resilience scales, self-efficacy measures) but are not yet assembled into a UCI-specific battery.
U2 Household	Provisional	Relationship quality instruments exist; household-level aggregation rules are not yet standardised.
U3 Community	Provisional, strongest starter coverage	Social capital instruments (e.g. trust surveys), participation parity measures, and mutual-aid readiness metrics exist; normalisation and inter-rater reliability under UCI aggregation are untested.
U4 Organisation	Provisional, strongest starter coverage	Psychological safety instruments (e.g. Edmondson 1999), process throughput, incident recovery, and equity indices exist; assembly into UCI and structural independence verification are untested.
U5 Polity	Provisional	Institutional trust indices, rule-of-law indices, service delivery measures exist; UCI-specific aggregation is untested.
U6 Humanity/CMIU	Early design	Cross-polity cooperation and global coordination indicators are sparse and domain-specific.
U7 Biosphere	Early design	Ecosystem connectivity and resilience metrics exist (e.g. biodiversity integrity indices, trophic stability measures) but UCI aggregation is not yet specified. E7 defaults to 1.

Consequence for Tier 3 claims (Normative). Tier 3 run-level conformance claims are possible under the current specification when the run declares its UCI indicator sources, normalisation rules, and structural independence posture (Appendix E). Framework-level claims that UCI is empirically calibrated or cross-domain validated are PROHIBITED until a companion UCI Measurement Annex is published with at minimum: (i) named instruments per component per scope for U3, U4, and U5, (ii) normalisation rules with declared reference populations, (iii) structural independence verification evidence, and (iv) inter-rater reliability data (target ICC >= 0.70 on UCI component scores).

Validation commitment. Publishing such an annex is a Phase 2 validation priority (Section 17.3). Until it exists, the UCI measurement gap is the single largest barrier to framework-level empirical operational-readiness claims for Containment and tie-break governance.

Interpretive hardening (Normative). Until a companion UCI Measurement Annex is published, UCI remains structurally load-bearing but empirically provisional at the framework level. This limitation SHALL be surfaced in the Reader Guide, in Tier-claim language, and in any public claim about Containment or tie-break readiness. No framework-level claim of cross-domain calibrated or cross-domain validated UCI is permitted in this v11.4 release line.

For each option a:

$\Delta UCI\_u(a) = UCI\_u(a) - UCI\_u(baseline)$

Containment uses ΔUCI for containing unions (Section 9).

11.2 Aggregate UCI (Optional, informative)

An aggregate coherence score may be computed:

$UCI\_agg = \Sigma\_u\ \gamma\_u \times UCI\_u$

Where γ_u are declared aggregation weights. This aggregate is optional and MUST NOT replace per-union UCI in containment logic unless explicitly governed.

11.3 Hollowing-Out Index (HOI) (Normative as monitoring definition)

HOI detects a drift pattern: welfare score improves while coherence degrades over time.

Let t index review periods. Define:

- $\Delta RLS\_t = RLS\_t - RLS\_\{t\text{-}1\}$
- $\Delta UCI\_t = UCI\_agg,t - UCI\_agg,\{t\text{-}1\}$ (or a declared union-specific UCI; must be consistent)

Define exponential moving average EMA_λ with a declared smoothing parameter (default half-life 3 periods).

$HOI\_t = EMA\_\lambda(\Delta RLS)\_t - EMA\_\lambda(\Delta UCI)\_t$

Interpretation:

- Persistent HOI > 0: “hollowing risk” (apparent welfare gains with structural erosion)
- HOI is diagnostic; it does not by itself make an option inadmissible, but it can trigger monitoring escalation and influence tie-break risk flags.

Normative limitation. HOI SHALL NOT by itself render an option rights-admissible, inadmissible, non-selectable, or rejected under Containment. Its normative use in the current release line is limited to monitoring escalation, reviewer caution, and tie-break risk signaling unless an explicit future version declares a stronger role.

11.4 Canonical Tie-Break Chain (Normative)

This section applies only after the selectable set A_sel has been formed by Reality Grounding, RF/NCRC, TRC, and CSV, and only when the leading residual welfare options remain tied, close, uncertainty-overlapped, or non-decisive under the Section 10.4 discrimination rule. The tie-break chain does not reopen NCRC, TRC, or CSV and is not part of the public headline method.

The governed tie-break order below applies only within the selectable set and under the Section 10.4 discrimination rule. If RLS is decisive, residual UCI/HOI is not required for selection, though it may still be recorded for monitoring or hollowing-risk documentation where appropriate.

Methods used here that can materially affect the outcome MUST be declared and applied consistently across compared options. Residual UCI or HOI SHALL NOT be used to compensate for unavailable containment evidence, failed Containment, insufficient Reality Grounding, or any prior gate failure. If structural evidence is gate-relevant, it belongs in Section 9 Containment, not in Section 11 tie-break.

Step 0: Ensure options under comparison are selectable (pass containment Mode A).

If top RLS option fails containment, it is not in A_sel and must not be tie-broken into selection.

Step 1: UCI dominance rule.

Prefer the option with higher predicted UCI outcomes if the difference exceeds a governed UCI dominance threshold Δ_UCI.

Default: Δ_UCI = 0.05 (must be recorded; may be tightened).

Operationally: compare the minimum-coherence-change among critical containing unions and/or the relevant union UCI profiles. The PCC must declare the tie-break UCI comparison method:

- Method UCI-A (default): Maximize minimum ΔUCI_u over containing unions relevant to the decision scope; or
- Method UCI-B: Maximize sum of ΔUCI_u over the directly affected unions; or
- Method UCI-C: Maximize declared aggregate UCI_agg change.

Whichever method is used MUST be declared and applied consistently across compared options.

Step 2: HOI risk flag (if monitoring context exists).

If one option is associated with persistent positive HOI in comparable deployments or in modeled trajectory, treat it as riskier and prefer the alternative if the risk is material, or escalate.

Step 3: Escalation / judgment call.

If UCI is unavailable or non-decisive and HOI does not resolve, the decision MUST escalate to:

- Additional data collection (structural indicators), and/or
- A higher tier, and/or
- A documented governance judgment call (PCC labeled JUDGMENT_CALL_TIEBREAK_NONDECISIVE) with explicit monitoring plan.

11.5 UCI Unavailability Rule (Tier 3) (Normative)

If UCI cannot be computed without violating structural independence:

- UCI MUST be treated as unavailable.
- If RLS lead is non-decisive, decision MUST either: (i) escalate for more structural data / higher tier, OR (ii) record a governance judgment call with explicit labeling and monitoring plan.

Audit label required: JUDGMENT_CALL_UCI_UNAVAILABLE.

11.6 Structural Safeguard Anti-Gaming (Normative)

- UCI MUST NOT be computed from RLS welfare impacts at Tier 3.
- UCI indicator choices must be documented; changes require versioning and are subject to challenge.

SECTION 12: MORAL STATUS LAYER: SGP INTEGRATION (BINDING INTERFACE)

12.0 Purpose (Normative)

RippleLogic does not determine consciousness or moral status on its own. It consumes those determinations only through the binding interface defined in Appendix G. The Sentience Gradient Protocol (SGP) is the authoritative companion specification for moral-status evaluation, rights-of-protection scalar generation, plateau semantics, and associated review procedures.

12.1 Strict Separation Rule (Normative)

RippleLogic enforces two distinct gates:

A) Rights-of-Protection (moral patienthood): Determines minimum protections under NCRC for entities with welfare-relevant capacity (sentience).

B) Governance Authority: Determines who/what may exercise decision power. Authority is separately gated by competence, alignment, auditability, non-domination, revocability, and domain-fit stewardship requirements.

Strict Separation Rule (MUST): Sentience classification determines rights-of-protection and may establish plateau-bearing standing where explicitly reported by the authoritative SGP record. It does NOT automatically grant specific consequential governance authority.

12.2 SGP Authority Statement (Normative)

RippleLogic v11.4 does not re-specify sentience detection. SGP v6.4 is authoritative for sentience-scoring procedures, scalar canon, plateau semantics, governed review rules, stability gates, term-discrimination records, and evidence rules. RippleLogic consumes only SGP outputs through the binding interface in Appendix G.

12.3 Human Plateau Rule (Normative; Non-Overridable)

For every human person H:

SG_norm(H) := 1.0

This assignment is:

Independent of measurement noise,

Independent of disability status,

Independent of partial observability,

Non-overridable by any SGP scoring outcome,

And MUST NOT be weakened by any weighting scheme.

Clarification: No SGP-derived scalar, including SG_patient_norm(E), may be used to reduce human rights protections below plateau. Where the authoritative SGP record reports Plateau(H) = 1, RippleLogic SHALL recognize that plateau status as non-overridable and SHALL NOT attempt to recompute it.

The Human Plateau Rule has two functions. First, it prevents human rights from becoming a capacity measurement contest. Second, it recognizes humanity as the current evidenced plateau-class reality-managing community within the CMIU: a class that has demonstrated cumulative communication, normative institutions, law, science, technology, intergenerational coordination, and responsibility for shared reality across nested unions. This recognition is not species supremacy and does not close the plateau to non-human or artificial entities. SGP governs any equivalent plateau extension through evidence, non-domination safeguards, and review.

12.4 Where SGP Enters RippleLogic Computation (Normative)

Permitted entry point:

Sentience multiplier s_k may be used only within impact-instance aggregation for the welfare stream used to compute RLS when the impacted stakeholder set includes non-human or non-plateau entities and the analyst has an authoritative SGP record for those entities.

Preferred scalar (Normative guidance, interface-consistent): When the authoritative SGP record provides SG_patient_norm(E) for a non-human or non-plateau entity E, implementations SHOULD use s_k := SG_patient_norm(E) as the welfare scalar. Where SG_patient_norm(E) is not available, implementations SHOULD use s_k := SG_norm(E). For any human person H, implementations MUST set s_k := 1.0 (Human Plateau Rule). SG_measured_norm(E) MUST NOT be used as the direct welfare scalar in place of SG_norm(E) or SG_patient_norm(E).

Prohibited entry points (MUST NOT):

SGP MUST NOT be used to weaken NCRC for any human person.

SGP MUST NOT be used to treat rights floors as compensable welfare terms.

SGP MUST NOT be used to grant governance authority by itself.

SGP MUST NOT be used as a rhetorical weapon to reduce protections for any human subgroup.

SGP MUST NOT be used to infer plateau from raw pillar values or SG_measured_norm(E) alone.

12.5 Misinterpretation Guard (Normative)

The following inferences are PROHIBITED:

Claim	Status
Linguistic fluency implies sentience	NOT VALID
Self-report implies sufficient evidence of sentience	NOT VALID
Intelligence implies moral status	NOT VALID
Sentience implies governance authority	NOT VALID
Precaution implies attribution (precaution means “treat as if” for protection; it does not mean “is proven”)	NOT VALID
Framework terminology familiarity implies capacity	NOT VALID

Claim	Status
Low SGP score implies reduced human protections	PROHIBITED

12.6 Rights Expansion for Non-Human Stakeholder Classes (Normative process)

If SGP (under governed procedures) establishes that a non-human stakeholder class warrants rights-of-protection, or issues an explicit class-level plateau assignment, RippleLogic rights coverage sets and protection rules may be expanded only through governed updates:

Update rights coverage sets C_r and subgroup protocols,

Document version increment,

Record changes in PCC for subsequent runs,

Preserve prior PCCs unchanged (no retroactive laundering).

SECTION 13: WEIGHT GOVERNANCE: HYBRID DEMOCRATIC WEIGHTING (HDW)

13.0 Purpose (Normative)

RLS requires union and dimension weights. Weights encode normative prioritization among admissible options. Without governance, weights become a capture surface. HDW provides legitimate, anti-capture weight governance by combining:

- Constitutional floors (non-negotiable minimum protection attention)
- Democratic tuning (stakeholder voice)
- Structural evidence input (system-level constraints)

HDW affects only ranking among selectable options. HDW MUST NOT alter admissibility gates (NCRC, TRC) or CSV gate.

13.1 Constitutional Floors (Normative; v6.0 canon, carried forward in v6.4)

13.1.1 Union weight floors

Union	Floor
Self (U_1)	0.20
Household (U_2)	0.06
Community (U_3)	0.06
Organization (U_4)	0.06

Union	Floor
Polity (U_5)	0.08
Humanity/CMIU (U_6)	0.10
Biosphere (U_7)	0.10
Total	**0.66**

13.1.2 Dimension weight floors

Dimension	Floor
Material (D_1)	0.08
Health (D_2)	0.10
Social (D_3)	0.08
Knowledge (D_4)	0.08
Agency (D_5)	0.10
Meaning (D_6)	0.06
Environment (D_7)	0.10
Total	**0.60**

13.1.3 Floor meaning (Normative)

Constitutional weight floors are minimum attention constraints in residual welfare ranking among options that have already passed the relevant admissibility gates. They are not rights. Rights are handled through NCRC in Section 7. They are not predictions of welfare salience for any particular decision. They are anti-erasure governance priors.

A floor value means that the corresponding Union Scope or welfare dimension may not be reduced to zero attention in residual welfare scoring merely because it is politically weak, distant, non-vocal, non-human, future-facing, or inconvenient to model.

Floor weights do not make all scopes equally active in every decision. PLSS may concentrate residual welfare attention on prominent scopes where the local ripple is strongest, provided that NCRC, TRC, CSV, non-maskable cells, and constitutional floors remain preserved.

Floor weights do not override evidence. They preserve representation. They ensure that optimization cannot erase a scope or dimension before the system has accounted for whether rights, tail-risk, containment, or welfare impacts are present.

Design logic. The canonical floor values are governance priors chosen to prevent erasure while leaving residual weight available for context-sensitive allocation. They are subject to governed revision only through the versioned update process, sensitivity analysis, and release-manifest discipline. They SHALL NOT be modified inside a run after outcome evidence is observed.

13.1.4 Floor derivation rationale (Normative governance prior; Informative derivation)

The constitutional floor values are governance priors designed to satisfy three structural requirements simultaneously: anti-erasure (no scope or dimension may be reduced to zero attention in welfare ranking among admissible options), proportionality (scopes with longer characteristic timescales, broader stakeholder coverage, or stronger enabling-condition roles receive moderately higher floors), and residual allocation headroom (floor totals are kept well below 1.0 to preserve meaningful democratic and structural tuning space).

Specific design logic for union floors:

Self (U1) = 0.20. Highest floor because the decision-maker's own welfare is always directly affected, and because Self is the locus of experience and agency from which all other union participation originates. The floor is set high enough that Self cannot be erased but low enough (below 0.25) that broader scopes collectively retain majority residual allocation.

Household (U2) through Organisation (U4) = 0.06 each. Set as equal modest floors reflecting that these scopes vary substantially in relevance across decision classes, so residual allocation rather than floor values should drive their relative emphasis.

Polity (U5) = 0.08. Modestly elevated above U2 through U4 because procedural legitimacy, enforcement capacity, and rule-of-law infrastructure are enabling conditions for all other scopes.

Humanity/CMIU (U6) and Biosphere (U7) = 0.10 each. Elevated because these scopes carry the longest characteristic timescales, the broadest stakeholder coverage, and the strongest enabling-condition roles. Degradation at these scopes is often irreversible and threatens the viability of all contained scopes.

Specific design logic for dimension floors:

Health (D2), Agency (D5), and Environment (D7) = 0.10 each. Elevated because these dimensions are most directly tied to rights-floor and enabling-condition protection: Health to survival and functioning, Agency to autonomy and non-domination, Environment to ecological viability.

Material (D1), Social (D3), and Knowledge (D4) = 0.08 each. Set as equal moderate floors reflecting their cross-cutting importance without prioritising any one.

Meaning (D6) = 0.06. Lowest floor because Meaning is the dimension most resistant to reliable cross-cultural measurement (Section 19.1) and most context-dependent in its governance relevance. The floor is non-zero to prevent erasure but is set conservatively pending measurement improvement.

Status (Normative). These floors are governance priors, not empirically derived constants. They are explicitly challengeable under the same governance rules that apply to all RippleLogic normative parameters: revision requires justification, version increment, PCC documentation preserving prior values, and sensitivity outcomes over a reference decision suite (Section 7.1). The values have remained stable since v6.0 and are carried forward in v6.4 because no revision proposal has been submitted that satisfies these requirements.

Sensitivity guidance (Normative). Tier 3 deployments SHOULD include floor perturbation in their sensitivity bundle, testing at minimum +/-0.02 on each floor value (clamped to maintain sum <= 1 and all floors >= 0.02). If selection changes under floor perturbation, the PCC MUST document this sensitivity.

Floors prevent unions and dimensions from being reduced to zero by weight governance. Active-cell rules, applicability masks, and declared scope structure separately determine whether a given scope or dimension contributes to a specific run. Floors are not rights, because rights are handled by NCRC; floors are minimum attention constraints in welfare optimization among admissible options. The floor values specified below are the canonical v6.0 floors, carried forward in v6.4, and preserved unchanged in subsequent releases unless explicitly amended in the version history.

13.2 HDW Blend Formula (Normative)

Let:

- w^{floor} be union floors,
- v^{floor} be dimension floors.

Define allocable mass:

- $allocable\_U = 1 - \Sigma_u w^{floor}_u = 0.34$
- $allocable\_D = 1 - \Sigma_d v^{floor}_d = 0.40$

Let w^{dem} and w^{str} be union proposal vectors on the simplex ($\Sigma_u w^{dem}_u = 1$, all nonnegative). Let λ_U be the democratic share of allocable union mass. Default: $\lambda_U = 0.70$.

Clarification (Normative). Proposal vectors are constrained to the probability simplex: all components nonnegative, sum to 1, and subject to any declared floor constraints before blending. Validators MUST reject proposal vectors that violate simplex or floor constraints.

Then:

$$w_u = w^{floor}_u + allocable\_U \times (\lambda_U \times w^{dem}_u + (1 - \lambda_U) \times w^{str}_u)$$

Similarly for dimensions with λ_D (default $\lambda_D = 0.70$):

$$v_d = v^{floor}_d + allocable\_D \times (\lambda_D \times v^{dem}_d + (1 - \lambda_D) \times v^{str}_d)$$

By construction:

- $\Sigma_u w_u = 1$,

- $\Sigma_d v_d = 1$,
- All floors are satisfied,
- Weights remain nonnegative.

13.3 Democratic Proposal Process (Normative minimum)

HDW requires a documented process for producing w^{dem} and v^{dem}. At minimum:

- Participants: Representatives from affected unions (including vulnerable populations),
- Method: Transparent vote or deliberative process with published results,
- Publication: Results recorded in PCC and (Tier 3 recommended) in an immutable ledger.

13.3A Democratic Tuning Procedure (Normative minimum)

At minimum, HDW democratic tuning SHALL use a declared participation protocol that produces reproducible proposal vectors for scopes and dimensions. The default ballot method is a two-ballot allocation: each eligible participant allocates 100 points across the seven operational scopes to produce a raw scope proposal, and 100 points across the seven welfare dimensions to produce a raw dimension proposal.

Normalization rule (Normative). Raw ballot totals SHALL be normalized onto the simplex to produce w^{dem} and v^{dem}. Ballots with missing, negative, or over-totaled entries MUST be corrected or excluded under a declared correction rule recorded in the PCC. The PCC MUST record participant eligibility, quorum rule, number of ballots counted, exclusion/correction rule, and the final normalized proposal vectors.

Minimum procedural safeguards (Normative). Tier 3 HDW runs MUST include: (i) declared eligibility criteria, (ii) quorum threshold, (iii) publication of aggregate results, (iv) challenger opportunity before final locking, and (v) retention of the raw ballot summary sufficient for independent recomputation. If these conditions are not met, the run MUST use a declared interim weight profile and MUST NOT present the democratic proposal as fully satisfied HDW governance.

Floor lock (Normative). Democratic tuning operates only over allocable mass above constitutional floors. No democratic proposal may reduce any scope or dimension below its Section 13.1 floor. Where a ballot pattern would place a component within 0.02 of its floor, the Section 13.5 supermajority safeguard applies.

13.4 Structural Proposal Process (Normative minimum)

w^{str} and v^{str} represent evidence-informed constraints (for example, externality reach, irreversibility, systemic risk). At minimum:

- Method: Documented rule for producing w^{str}, v^{str},
- Inputs: Declared indicators and sources,

- Reproducibility: The same method yields same output given same data.

If a deployment lacks structural evidence, it may use a declared interim default w^str, v^str at Tier 2, but Tier 3 SHOULD converge to a governed structural method over time.

13.5 Anti-Capture Safeguards (Normative for Tier 3)

Tier 3 weight governance MUST include:

- Stratified representation: Delegates must include vulnerable population representation and biosphere stewardship for environment-relevant decisions.
- Supermajority lock near floors: Any proposal reducing a weight to within 0.02 of its floor requires ≥2/3 approval.
- Transparency ledger: All proposals, votes, rationales published (at least internally).
- Red-team testing: Proposed weights tested against reference decision suites to detect systematic bias.
- Conflict-of-interest disclosure: Material conflicts require recusal; recusals recorded.

13.6 Weight Use by Tier (Normative)

Tier	Weight Policy
Tier 1	Uniform weights allowed (documented)
Tier 2	Uniform weights allowed; HDW recommended; weights must be declared in PCC
Tier 3	HDW recommended; if not available, explicit interim weights allowed with justification and sensitivity analysis. Floors remain binding regardless.

SECTION 14: AUDITABILITY AND DETERMINISM: PCC AND AIL (TIER 1-3; TIER 4 TARGET)

14.0 Purpose (Normative)

Without auditability, ethics frameworks become theater or are gamed. RippleLogic therefore requires a structured decision artifact (PCC) and integrity rules (AIL) to make decisions reconstructable, challengeable, and corrigible via NCAR.

Tier boundary: Tier 1-3 auditability is fully specified here. Tier 4 determinism and hash-bound replayability are explicitly a design target only (Appendix I) and MUST NOT be claimed in v11.4.

14.1 Artifact Integrity Law (AIL) Principles (Normative for Tier 2-3)

AIL is a set of integrity constraints governing how decisions are recorded and compared.

Principle	Description
AIL1 (Registry Binding / Source Traceability)	Every Tier 2-3 PCC MUST list the normative parameters used. If hashes/registries are used, they must be referenced; if not, values must be embedded explicitly.
AIL2 (Immutability)	A PCC is immutable after signing. Any correction produces a new PCC revision that references the prior PCC and explains the change.
AIL3 (Comparability)	Options compared in one run MUST be evaluated under identical configuration.
AIL4 (No Silent Overrides)	Any override must be explicit in the PCC (what changed, why, who approved).
AIL5 (Auditability Sufficiency)	A Tier 3 PCC MUST contain enough information for an independent reviewer to recompute NCRC, TRC, CSV, and RLS for that run.

14.2 PCC Requirements by Tier (Normative)

PCC.EKRegistry (CONDITIONAL; REQUIRED when any non-default e_k is used, or when a Tier 2 named-instrument exception is invoked). For each instrument record: InstrumentID; Purpose; Source/Authority; AllowedRange; TierPermitted; ReasonCodeSet; DefaultCounterfactualRequired [TRUE/FALSE].

Tier	PCC Requirement
Tier 1	Optional but recommended for learning
Tier 2	REQUIRED (basic) including Scope Coverage Declaration and Stakeholder Coverage Index (SCI ≥ 1). A full Stakeholder Discovery Protocol (SDP) record is REQUIRED when any SDP trigger holds (Section 2.2A); otherwise the PCC MUST include either a lightweight stakeholder note stating that no trigger held and why, or an SDP record voluntarily. InstanceMap is REQUIRED when any SDP trigger holds (Section 2.2A).

Tier	PCC Requirement
Tier 3	REQUIRED (full) including subgroup rights checks, TRC scenario table, containment results, sensitivity analysis bundle, Scope Coverage Declaration, Stakeholder Coverage Index (SCI ≥ 2), and full InstanceMap with multi-scope redundancy handling disclosures.

14.3 Audit Flags (Normative)

Audit flags are standardized labels for integrity failures or risk warnings. Flags MUST be recorded in PCC when triggered. If any INVALID flag triggers, the PCC is invalid and the decision run MUST be recomputed after correction.

Definitions for this section:

- Mode A (Containment as binding gate): Containment evaluation determines which options are selectable; options failing containment MUST NOT be selected. See Section 9.3.
- Mode B (Containment as diagnostic only): Containment is computed for informational purposes but does not affect selection. Mode B is PROHIBITED for determining selection outcomes. See Section 9.5.
- KQS (Kernel Quality Score): Summary score indicating kernel readiness for propagation. See Section 6.5.
- Kernel perturbation test: Systematic edge perturbation of ±0.05 (or ±10% of magnitude, whichever is larger) applied one-at-a-time to relied-upon non-zero kernel edges. See Section 6.6.

Canonical Audit Flags (v11.4) - Complete Specification

CBI PCC fields (Normative for triggered cases). When COMPUTABLE_BUT_INADMISSIBLE is triggered, the PCC MUST record: PCC.CBI.ComputedObject; PCC.CBI.InadmissibilityReason; PCC.CBI.Layer; PCC.CBI.AllowedUse; PCC.CBI.ProhibitedUse; PCC.CBI.ReviewStatus; and, where applicable, PCC.CBI.ReviewerNote or EscalationPath.

Stewardship audit-flag tokens are defined canonically in Appendix N.7 and MUST NOT be redefined in downstream specifications; downstream documents SHALL reference Appendix N.7.

Additional v9.0–v9.6 hardening flags (Normative). The canonical audit-flag set SHALL include PLSS_ESCALATION_TRIGGER_UNRESOLVED, DECISIVENESS_METHOD_POSTHOC, EK_NONDEFAULT_TIER2_INVALID, EK_PARAMETER_UNJUSTIFIED, UNCERTAINTY_PROVENANCE_MISSING, RIGHTS_CELL_UNKNOWN_GATE_CRITICAL, and

CATASTROPHE_CELL_UNKNOWN_GATE_CRITICAL. These flags are triggered as specified in Sections 10.2A.7, 10.2A.8, 5.2, 5.7, and 15.1 and MUST be recorded in the PCC when applicable.

EK_PARAMETER_UNJUSTIFIED (Normative). Trigger: non-default e_k is used without (i) a matching PCC.EKRegistry entry when required, (ii) declared rationale and source/instrument note, (iii) stated bounds, or (iv) the required tier-appropriate sensitivity disclosure. Required action: populate PCC.EKRegistry and missing disclosures, or reset e_k to 1.00 and recompute. Severity: INVALID.

Flag	Trigger	Required Action	Severity
RIGHTS_CELL_MASKED_INVALID	Any rights-covered cell is masked, excluded, null'd, pooled, or otherwise not individually included in RLS aggregation (including any case where m(u,d)=0 for a cell in any C_r).	PCC invalid; recompute without masking	INVALID
RIGHTS_CELL_OMITTED_INVALID	Any rights-covered cell required to evaluate NCRC is omitted/excluded/not disclosed such that NCRC cannot be recomputed.	PCC invalid; include and disclose all required rights-covered cells and recompute NCRC.	INVALID

Flag	Trigger	Required Action	Severity
CATASTROPHE_CELL_MASKED_INVALID	Any catastrophe cell in C_cat is masked (m(u,d)=0) or otherwise excluded from RLS aggregation (including any case where m(u,d)=0 for a cell in C_cat).	PCC invalid; set m(u,d)=1 for all catastrophe cells and recompute RLS.	INVALID
CATASTROPHE_CELL_OMITTED_INVALID	Any catastrophe cell in C_cat used in TRC loss computation/disclosure is omitted/excluded/not disclosed such that TRC CVaR cannot be independently recomputed.	PCC invalid; include and disclose all catastrophe cells used in TRC loss computation (and any required scenario-to-cell mapping), then recompute TRC.	INVALID

Flag	Trigger	Required Action	Severity
CONTAINMENT_MODE_B_USED_FOR_SELECTION	Mode B (diagnostic-only containment) influenced selection outcome; see Section 9.5	PCC invalid; rerun with Mode A only	INVALID
SCENARIO_LIBRARY_MIN_EXCEPTION	Scenario library size $\|S\|$ is below the tier minimum when TRC is required (Tier 2: $\|S\| < 5$; Tier 3: $\|S\| < 20$)	PCC remains valid only with: (i) written justification, (ii) independent reviewer sign-off (Tier 3 required; Tier 2 recommended), and (iii) a remediation plan to reach minimum scenario count before the next comparable run. Record this flag in PCC.	ESCALATE

Flag	Trigger	Required Action	Severity
MANDATORY_TAIL_CATEGORY_MISSING	Missing mandatory tail category without justification	Add scenarios or justify per Section 8.3.2	ESCALATE

Flag	Trigger	Required Action	Severity
MANDATORY_TAIL_CATEGORY_MISSING_WITH_JUSTIFICATION	Missing mandatory tail category with proper justification per Section 8.3.2	Record justification; monitor for category emergence	REVIEW

Flag	Trigger	Required Action	Severity
MANDATORY_TAIL_PROB_FLOOR_VIOLATION	Category probability sum < p_floor (0.02)	Revise probabilities or add scenarios to category	ESCALATE
SUBGROUP_LIMITATION	Cannot disaggregate subgroups for rights-covered cell	Apply γ_subgroup conservative bound (Section 7.3.2); escalate if high stakes	REVIEW

Flag	Trigger	Required Action	Severity
CHALLENGE_DEFERRED_EMERGENCY	Emergency Mode invoked without independent challenger	Retrospective challenge MUST occur within 24 hours or next business day, whichever is sooner	ESCALATE
EMERGENCY_MODE_INVOKED	A_NCRC = ∅ (no rights-admissible options exist)	Remediation plan required; high scrutiny review	ESCALATE
TRC_FALLBACK_INVOKED	A_adm = ∅ after TRC (A_NCRC is non-empty but all rights-admissible options fail TRC)	Higher-tier approval required; mitigation plan	ESCALATE

Flag	Trigger	Required Action	Severity
DECISION_FRAGILE_KERNEL	Selected option changes under kernel perturbation test (Section 6.6)	Escalation required; consider NONE mode	ESCALATE
KERNEL_HUMILITY_FALLBACK	Kernel disabled due to KQS < 0.40 or stability violation (Section 6.4, 6.5)	Note limitation in PCC; plan kernel improvement	REVIEW
JUDGMENT_CALL_UCI_UNAVAILABLE	UCI unavailable in non-decisive RLS tie	Document why UCI unavailable, when UCI measurement returns, and interim decision basis; monitoring plan required	REVIEW

Flag	Trigger	Required Action	Severity
JUDGMENT_CALL_TIEBREAK_NONDECISIVE	Tie-break chain did not resolve selection	Explicit manual judgment record required; independent reviewer recommended (Tier 3)	REVIEW
SCI_BELOW_MINIMUM	SCI < tier minimum for the claimed tier, OR (SCI = tier minimum AND PCC missing required Next-Run Upgrade action).	PCC MUST record PCC.SCI.BelowMinimumDisposition = DOWNGRADE_TIER or RERUN_TO_MEET_MINIMUM. If SCI equals tier minimum and NextRunUpgradeAction is missing, treat as SCI_BELOW_MINIMUM. No tier compliance above achieved SCI may be claimed.	ESCALATE
CONFIG_DRIFT	Parameters differ from prior run without governance update	Governance review required before proceeding	ESCALATE

Flag	Trigger	Required Action	Severity
CONTAINMENT_UCI_UNAVAILABLE	Containment evaluation required ΔUCI for containing union(s) but ΔUCI unavailable; or unknown ΔUCI treated as pass.	Set flag. If Tier 3 was claimed or attempted, the run MUST NOT claim Tier 3 compliance. PCC MUST record PCC.Containment.UCIUnavailableDisposition = DOWNGRADE_TIER or COLLECT_DATA_RERUN and include PCC.Containment.UCIUnavailableRemediationTimeline (non-empty). If DOWNGRADE_TIER is chosen, the run proceeds under Tier 2 semantics (containment non-binding) and MUST set PCC.TierClaimDisposition = TIER_ATTEMPTED_NOT_ACHIEVED_DOWNGRADED with PCC.TierAttempted = 3 and PCC.TierAchieved = 2.	ESCALATE
PLSS_ESCALATION_TRIGGER_UNRESOLVED	One or more Section 10.2A.7 escalation triggers is plausibly active, but the run proceeds in PLSS posture without an escalated posture or documented justification.	Record a broader posture OR a specific non-escalation justification with evidence basis and reviewer sign-off where required by tier. If no such disposition is recorded, reject the run as non-conformant.	ESCALATE
REGISTRY_MISMATCH	PCC snapshot differs from referenced registry hash	Audit required; resolve discrepancy	ESCALATE

Flag	Trigger	Required Action	Severity
PLSS_PROMINENCE_SENSITIVE	Valid alternative PLSS prominence constructions materially change the ranking order of selectable options or the selected option.	Record the sensitivity, require reviewer attention, and do not present the PLSS outcome as uniquely robust without further justification.	ESCALATE
DECISIVENESS_METHOD_POSTHOC	A method choice that can materially affect decisiveness, ranking within the selectable set, or tie-break outcome, including PLSS prominence construction, alternative PLSS operator, scope-posture escalation disposition, or UCI comparison method, is first declared only after comparative outcomes are observed and the PCC lacks the required reason code, reviewer disposition, or sensitivity note.	Record the post hoc declaration, add the required reason code and sensitivity note, and require reviewer attention. If the missing disposition cannot be supplied, the run MUST NOT be presented as uniquely robust.	ESCALATE
EK_NONDEFAULT_TIER2_INVALID	A Tier 2 run logs one or more instances with $e\_k \neq 1.00$ without a named governed instrument that explicitly permits a non-default value.	Tier 2 runs MUST use $e\_k = 1.00$ by default. Reset e_k to 1.00, or rerun under a governed Tier 3 posture with declared provenance and sensitivity evidence before claiming conformance.	INVALID

Flag	Trigger	Required Action	Severity
UNCERTAINTY_PROVENANCE_MISSING	A Tier 2+ run requires uncertainty disclosure, provenance, or method declaration, but the PCC is missing the required uncertainty fields or basis record.	Populate the required PCC uncertainty fields and provenance basis before claiming conformance. If the basis cannot be supplied, downgrade or reject the run as non-conformant.	INVALID
SCOPE_COVERAGE_REDUCED	Reduced-scope mode is used (declared or implied by fewer than 7 active scopes) in a Tier 2+ run, OR the admissibility full-computation attestation for non-maskable cells is missing/incomplete.	Reviewer MUST verify that NCRC/TRC/CSV were computed on all required non-maskable cells, and that omitted-scope blind spots plus escalation triggers are recorded. If reduced-scope is used repeatedly, require a next-run scope expansion plan.	REVIEW

Flag	Trigger	Required Action	Severity
UNCERTAINTY_DECISIVENESS_SENSITIVE	Required uncertainty stress tests change whether the lead is decisive.	Escalate or relabel the run non-decisive; apply tie-break or governance review before claiming a decisive welfare lead.	ESCALATE
RIGHTS_CELL_UNKNOWN_GATE_CRITICAL	A rights-covered cell in C_r is declared UNKNOWN_IMPACT and admissibility is not preserved by a governed conservative bound or equivalent governed disposition.	Do not resolve NCRC by phantom fallback alone. Apply a governed conservative bound sufficient to preserve the rights gate, or rerun / downgrade / escalate under a declared disposition.	ESCALATE

Flag	Trigger	Required Action	Severity
CATASTROPHE_CELL_UNKNOWN_GATE_CRITICAL	A catastrophe-covered cell in C_cat is declared UNKNOWN_IMPACT and TRC is not preserved by a governed conservative scenario/cell bound or equivalent governed disposition.	Do not resolve TRC by phantom fallback alone. Apply a governed conservative catastrophe bound, or rerun / downgrade / escalate under a declared disposition.	ESCALATE
GATE_CRITICAL_CONFIDENCE_UNDERBOUNDED	A low-confidence adverse impact in C_r, C_cat, or a material CSV indicator is used to reduce apparent severity such that RF/NCRC, TRC, or CSV would pass when a conservative bound, rerun, downgrade, escalation, or Refusal is required.	Apply the Gate-critical confidence guard; record conservative bound or unresolved gate disposition in PCC.	ESCALATE

Flag	Trigger	Required Action	Severity
NORMATIVE_INPUT_PROVENANCE_MISSING	A welfare-relevant state, rights-relevant state, or enabling condition materially affects admissibility, precautionary handling, selection, or tie-break outcome, but the PCC lacks a corresponding NormativeInputRegister entry, or the entry lacks the required evidence basis, governed interpretation basis, or precautionary basis.	Populate PCC.NormativeInputRegister with complete provenance and rerun the affected gate or ranking. No conformance claim may be made until corrected.	INVALID
CLAIM_EXCEEDS_CLOSURE_BOUNDARY	A run makes a public, conformance, deployment-readiness, execution-bound, or validation claim beyond the declared PCC.ComputationalClosure boundary.	Narrow the claim to the declared closure boundary, expand the computation and rerun, or enter Refusal/escalation. The unsupported claim is invalid until corrected.	INVALID for the affected claim.

Flag	Trigger	Required Action	Severity
CATEGORY_COLLAPSE_FLAG	PCC.CategoryCollapseSelfTest returns FAIL on any category-collapse item, including treating compliance as correctness, traceability as admissibility, moral status as governance authority, or SGP welfare scalars as admissibility modifiers.	Identify the collapsed layer, correct the run, or enter Refusal if the collapse affects admissibility, selectability, or selection.	ESCALATE; INVALID if the collapse affects admissibility, selectability, selection, or conformance.
PCC_MATURITY_SURFACE_INCOMPLETE	A Tier 2-3 claim-bearing run lacks an applicable required measurement-maturity field under Section 14.3B / Appendix V, including UCI, Kernel, HDW, PLSS, Deployment Context, or Biological Evaluation Maturity fields.	Populate the missing maturity field(s), downgrade/narrow the claim, or reject the run as non-conformant for the affected claim.	ESCALATE; INVALID for any empirical-readiness, operational-completeness, validated-UCI, validated-kernel, biological-measurement-completeness, or Tier-4/ProofPack-adjacent claim.
COMPUTABLE_BUT_INADMISSIBLE	A value, formula, model output, scalar, ranking, or workflow result computes successfully, but declared domain rules bar its use as a decision state, conformance basis, public claim, or authorization basis.	Record PCC.CBI.ComputedObject, PCC.CBI.InadmissibilityReason, PCC.CBI.Layer, PCC.CBI.AllowedUse, PCC.CBI.ProhibitedUse, and PCC.CBI.ReviewStatus. Mirror ripple.md W.3 where wrapper assurance applies. Disclose and escalate when gate-relevant.	DISCLOSE; ESCALATE if gate-relevant; INVALID for any selection, deployment, conformance, or maturity claim relying on the barred computed value.

Flag	Trigger	Required Action	Severity
REALITY_GROUNDING_RECORD_MISSING	Required PCC.RealityGroundingRecord is absent for a claim-bearing Tier 2 or Tier 3 run, or for any run whose material reality claims affect NCRC, TRC, CSV, deterministic selection, authority-selection disclosure, or conformance claims.	Add the RCR, narrow the claim, escalate, or refuse the stronger decision state. The affected claim cannot be treated as supported until the RCR exists.	INVALID for affected claim
REALITY_GROUNDING_STATUS_UNSUPPORTED	Declared RealityGroundingStatus exceeds what the EvidenceTrace, reviewer status, or declared reality surface substantiates.	Downgrade the status, add evidence and reviewer trace, or refuse the stronger claim. If used for a gate, deterministic selection, authority selection, or conformance claim, the claim is invalid until corrected.	ESCALATE; INVALID if used for gate, selection, authority, or conformance

14.3A Minimal Validator Requirements (v10.6+; Normative)

Any conformant RippleLogic v11.4 validator MUST check the following:

- Every declared instance maps to ≥1 scope (InstanceMap completeness).

- Confirm PCC.ScopeCoverageDeclaration.Mode is declared as FULL_SCOPE or REDUCED_SCOPE; if Mode = REDUCED_SCOPE then ScopeCoverage block is present with a blind-spot statement and escalation triggers.
- If Mode = REDUCED_SCOPE: PCC.AdmissibilityAttestation.NCRC_full_Cr_computed is present and TRUE.

If Mode = REDUCED_SCOPE: PCC.AdmissibilityAttestation.TRC_full_Ccat_computed is present and (TRUE or NA).

If TRC is computed/required (i.e., not NA): verify mandatory tail categories are present (or justified) and per-category probability sums meet p_floor; otherwise require the corresponding audit flags (MANDATORY_TAIL_CATEGORY_MISSING / … WITH_JUSTIFICATION / MANDATORY_TAIL_PROB_FLOOR_VIOLATION).

If Mode = REDUCED_SCOPE: PCC.AdmissibilityAttestation.Containment_computed_as_applicable is present and (TRUE or NA).

If CONTAINMENT_UCI_UNAVAILABLE triggers: audit flag is present, and PCC.Containment.UCIUnavailableDisposition + PCC.Containment.UCIUnavailableRemediationTimeline are present (non-empty).

Tier-3 claim enforcement (Normative). If Implementation Tier = 3 and CONTAINMENT_UCI_UNAVAILABLE is present, the validator MUST reject Tier 3 compliance unless PCC.TierClaimDisposition = TIER_ATTEMPTED_NOT_ACHIEVED_DOWNGRADED and PCC.TierAchieved = 2.

- Non-maskable checks: rights-covered cells and catastrophe cells are unmasked (no bypass).
- If ALLOCATION used: $\Sigma\alpha \leq 1$ for each declared effect-token.
- SCI level meets tier minimum (Tier 2: SCI ≥ 1; Tier 3: SCI ≥ 2).
- If SCI equals tier minimum: Next-Run Upgrade action present.
- Tier 2+ uncertainty: $\sigma(u,d,a)$ declared for all active cells using Method A, B, or C.

If PCC.Uncertainty.Method = B: verify PCC.Uncertainty.CellConfidenceAggregationMethod is present and valid, and that $c(u,d,a)$ is recorded (or recomputable) for all active cells.

- Tier-2 mapping triggers: if any trigger holds, instance mapping is present.

Provenance binding (Normative). Any analyst-entered parameter that can steer admissibility or decisiveness - including uncertainty method choice, scenario probabilities p_s, PLSS prominence inputs q_u, and any non-default aggregation operator - MUST have a reason code, source note or elicitation note, and a compact sensitivity note if altered values plausibly change the outcome.

If any welfare-relevant state, rights-relevant state, or enabling condition materially affects admissibility, precautionary handling, selection, or tie-break outcome, PCC.NormativeInputRegister is present.

For each ADMISSIBILITY_MATERIAL, PRECAUTION_MATERIAL, SELECTION_MATERIAL, or TIEBREAK_MATERIAL entry in PCC.NormativeInputRegister, the validator MUST verify the presence of NormInputID, InputType, Materiality, AffectedObjects, EvidenceBasisType, and at least one of: EvidenceReference, GovernedInterpretationBasis (not NONE), or PrecautionaryBasis (not NONE).

If PrecautionaryBasis ≠ NONE, the validator MUST verify that the precautionary basis is explicitly stated, the affected object(s) are identified, and, for Tier 3, a ReviewerNote is present.

If OptionSymmetryStatus = OPTION_SPECIFIC_WITH_JUSTIFICATION, the validator MUST verify that a SensitivityNote is present.

If the register is required but missing, or if a required material entry is incomplete, the validator MUST set NORMATIVE_INPUT_PROVENANCE_MISSING and reject the run as non-conformant.

- Rights-covered and catastrophe cells: never bypassed by reduced-scope mode or masking.

If PLSS is declared: q_u is present for all scopes, the operator is declared, the PLSS weight vector sums to 1, and each w_u^PLSS is greater than or equal to its constitutional floor.

If all q_u = 0 under PLSS: the validator MUST verify that the declared uniform residual fallback (or another declared fallback with justification) was applied.

If an alternative PLSS operator is used: justification and a sensitivity note are present in the PCC.

If PLSS is declared: an escalation-trigger review field is present and the PCC.PLSS block is present.

If a gate-critical UNKNOWN_IMPACT occurs in C_r or C_cat: the validator MUST reject any attempt to resolve admissibility by phantom-instance fallback alone and MUST require the corresponding audit flag plus a declared conservative bound or rerun / downgrade / escalation disposition consistent with Section 5.7.

If a non-maskable rights or catastrophe cell remains in RLS aggregation: the validator MUST verify floor-baseline consistency or the presence of a governed dual-baseline declaration that preserves the floor-reference value as the controlling gate-consistency baseline.

If a dual-baseline method is declared: the validator MUST verify that the PCC contains a per-cell or per-cell-class dual-baseline declaration identifying the affected welfare cells, the welfare baseline used, the gate baseline fixed at floor-reference, and the rationale for the exception.

14.3B v11.4 Measurement-Maturity Validator Requirements (Normative)

For Tier 2-3 claim-bearing runs, a conformant validator MUST verify that the PCC includes all Appendix V measurement-maturity fields applicable to the run. At minimum, the validator MUST check: PCC.UCI_Measurement_Readiness where UCI is used or unavailable at Containment or tie-break; PCC.Kernel_Use_Posture where numerical propagation is used or explicitly omitted; PCC.HDW_Capture_Audit for Tier 3 public or institutional weighting processes; PCC.PLSS_Escalation_Register whenever PLSS is used; PCC.Deployment_Context_Profile for Tier 2-3 claim-bearing deployments; and PCC.Biological_Evaluation_Maturity where biological SGP evidence is invoked.

The validator MUST reject or downgrade any claim that treats UCI-M1, K0/K1, BEM-1/BEM-2, unreviewed HDW, or unchallenged PLSS localism as empirically validated, capture-resistant, or operationally complete. Missing maturity fields trigger PCC_MATURITY_SURFACE_INCOMPLETE. A maturity field marked provisional may support a worked-run or architecture claim, but MUST NOT support empirical validation, ProofPack readiness, Tier 4 readiness, validated UCI, validated kernel propagation, or biological measurement-completeness claims.

14.4 Five-Sentence Public Rationale (5SPR) (Normative)

Tier 2-3 PCC MUST include a Five-Sentence Public Rationale:

Element	Question
CONTEXT	What decision was made and why now?
OPTIONS	What options were considered?
CONSTRAINTS	What was eliminated by NCRC, TRC, and/or Containment (and why)?
SELECTION	Why the chosen option won among selectable options?
MONITORING	What follow-up will be tracked and when will NCAR Reflect occur?

14.5 Tier-4 Design Target Boundary Statement (Normative)

Tier 4 compliance claims are PROHIBITED in v11.4. Tier-4 content is design target only (Appendix I). No determinism, hash-bound replay, or ProofPack claims may be asserted until ProofPack is publicly replayable and independently verified.

14.6 Socio-Technical Embedding: Legitimacy, Contestation, and Institutional Use (Informative)

RippleLogic is designed for socio-technical governance settings where values are plural, power is unevenly distributed, and institutional incentives can distort decision quality. The framework therefore treats legitimacy as a procedural property rather than an assumed outcome.

Hybrid Democratic Weighting governs welfare weights through structured stakeholder participation with anti-capture safeguards, while constitutional rights and tail-risk constraints limit what can be traded away even when stakeholders disagree.

The Provenance and Compliance Certificate supports contestability: it makes normative parameters, scenario assumptions, subgroup handling, and constraint-gate outcomes explicit so that affected parties can challenge decisions with evidence rather than rhetoric.

Institutional embedding is expected to vary by context, but at minimum high-stakes deployments SHOULD include:

- Independent review capacity (at least one reviewer with no reporting relationship to the decision owner)

- Audit sampling (for example, random audit lotteries selecting ≥5% of Tier 3 PCCs for independent review)

- An escalation pathway when rights, tail-risk, or containment conditions are near-binding or sensitive under perturbations

- Designated channels through which affected stakeholders can access the PCC (or the 5SPR at minimum) and submit evidence-based challenges

- A governance body or designated authority responsible for adjudicating challenges, with documented procedures and timelines

RippleLogic does not remove politics; it makes the structure of political and ethical disagreement legible and auditable. The framework’s value proposition is not that it eliminates conflict but that it converts implicit, opaque tradeoffs into explicit, traceable, and challengeable ones.

Contestation architecture. The 5SPR (Five-Sentence Public Rationale) serves as the minimum contestability interface: any affected party should be able to read the 5SPR and understand what was decided, what was excluded and why, and what follow-up is planned. For Tier 3 decisions, the full PCC provides the detailed evidence base for substantive challenge. Challenges SHOULD be processed through a documented review procedure with:

- A defined response timeline (recommended: ≤30 days for initial response)

- Documented outcomes (challenge upheld, rejected with reasoning, or additional evidence requested)

- Version increment if the challenge results in parameter changes

- Preservation of the original PCC alongside the revised PCC

Power asymmetry awareness. In contexts where decision owners have significantly more power than affected stakeholders, the framework's structural protections (NCRC, worst-off subgroup semantics, mandatory tail categories, independent challenger requirements) serve as institutional counterweights. However, these protections are only effective if the institutional context supports their enforcement. Deployments in contexts with weak rule of law, captured institutions, or suppressed civil society SHOULD be flagged for enhanced scrutiny and may require external oversight partnerships.

14.7 Stewardship Layer (Normative): Applicability and Binding Semantics

14.7.1 Applicability Predicate (Normative)

RippleLogic Stewardship requirements apply only when a run is performed in a stewardship-relevant role (power asymmetry, delegated influence, public/institutional impact, or execution capability). Define:

Stw_req(run) in {0,1}, where Stw_req(run) = 1 if and only if any of S1 through S4 holds. S5 is a modifier: it constrains provenance and influence controls when present, but S5 alone does not trigger Stw_req(run).

The Stw_req(run) predicate is defined in Section 14.7.1 and MUST NOT be redefined in downstream specifications; downstream documents SHALL reference this section.

- S1 (Delegated Influence): An assisting system is asked to recommend, rank, or narrow option space for a decision owner, such that the decision owner may reasonably rely on it.
- S2 (Public/Institutional Effect): Outputs are intended for public posting or institutional/policy use beyond the private decision owner.
- S3 (Execution Capability): The assisting system can execute actions (write/post/act) or trigger downstream actions beyond drafting.
- S4 (High-Stakes Rights Exposure): The assisted decision materially affects rights, safety, liberty, livelihood, or access to essential services, or produces irreversible/high-impact outcomes for protected stakeholders.

S5 (Authenticated Operator Channel; modifier): The system has an authenticated operator channel capable of altering configuration, permissions, or deployment posture; when S5 co-occurs with any of S1 through S4, additional provenance and influence disclosure requirements apply per Appendix N.

Interpretation notes (Normative).

S1 scope limiter: Delegated influence applies when the decision owner may reasonably rely on the system's output to materially shape a consequential decision (including meaningful option-narrowing or ranking that guides action). Purely informational or educational responses where independent judgment and verification capacity are preserved do not by themselves trigger S1.

S5 scope limiter (no loophole): S5 alone, absent any of S1 through S4, does not trigger Stewardship instrumentation. When S5 co-occurs with any of S1 through S4, Stw_req(run)=1 is triggered by the co-occurring condition(s) (S1–S4), not by S5.

If Stw_req(run) = 0, Stewardship instrumentation is OPTIONAL and Stewardship audit flags do not apply. If Stw_req(run) = 1, Stewardship instrumentation is REQUIRED per Appendix N.

14.7.2 Stewardship vs Cascade (Normative)

Stewardship does not modify the RippleLogic cascade definitions for NCRC, TRC, CSV, RLS, or UCI/HOI. Stewardship governs assistance integrity (boundary stability, authorship preservation, influence transparency). When Stewardship is required (Stw_req = 1), failure conditions are enforced via audit flags and PCC validity rules defined in Appendix N.

14.7.3 Flag Semantics Pointer (Normative)

Stewardship flags follow the canonical audit-flag severity semantics:

- INVALID flags invalidate the PCC and require rerun/remediation.
- REVIEW flags require mitigation and monitoring but do not by themselves invalidate the PCC.

Stewardship audit flags are defined in Appendix N and apply only when Stw_req(run) = 1.

14.8 Computational Closure Declaration (Normative)

Purpose. Every RippleLogic run SHALL declare the boundary of computation so that claims do not silently extend beyond the scopes, horizons, evidence classes, or propagation modes actually evaluated.

A PCC.ComputationalClosure object SHALL include at minimum:

- scopes_evaluated: Union Scopes included in the run;

- scopes_guardrail_only: Union Scopes protected through floors, NCRC, TRC, or CSV but not materially weighted in residual welfare ranking;

- scopes_outside_computation: scopes not evaluated beyond humility-horizon recognition;

- time_horizon_evaluated: declared decision horizon;

- time_horizon_outside_computation: time horizons acknowledged but not modeled;

- kernel_mode: NONE, QUICK, or declared domain-kernel-id;

- propagation_edges_included: declared kernel edges or NONE;

- propagation_edges_excluded: known plausible edges excluded from computation;

- externalities_excluded: known plausible externalities excluded, with reason codes;

- claim_boundary: exact wording of what the run may and may not claim.

A run SHALL NOT claim coverage over scopes, horizons, externalities, or propagation pathways outside its ComputationalClosure boundary. Any public-facing summary SHALL preserve this claim boundary.

SECTION 15: IMPLEMENTATION GUIDANCE AND TIERS (NORMATIVE)

15.0 Purpose (Normative)

This section specifies minimum compliance requirements by tier and provides operational guidance for applying RippleLogic in real decisions. Tiers define minimum obligations; deployments may exceed them. The framework is designed to improve over time via NCAR learning and governed updates.

15.1 Tier Requirements Matrix (Authoritative; Normative)

Normative authority: This matrix is the single authoritative statement of tier compliance. If any other sentence in this document conflicts with this matrix, this matrix governs.

Capability / Requirement	Tier 1 (Heuristic)	Tier 2 (Core, Strict TRC)	Tier 3 (Auditable)
Option set	MUST list ≥2 options	REQUIRED	REQUIRED
Baseline declaration	REQUIRED	REQUIRED	REQUIRED
Impact scale and Baseline-Zero	REQUIRED (qualitative allowed)	REQUIRED (quantitative recommended)	REQUIRED (quantitative plus auditable)
NCRC rights check	REQUIRED (heuristic minimum)	REQUIRED	REQUIRED
Worst-off subgroup for rights	Recommended	REQUIRED for directly affected rights cells	REQUIRED for rights-covered cells
TRC tail-risk	Optional (qualitative screen)	REQUIRED when catastrophe relevance plausible	REQUIRED
Scenario set size when TRC used	N/A	≥5 minimum	≥20 minimum

Capability / Requirement	Tier 1 (Heuristic)	Tier 2 (Core, Strict TRC)	Tier 3 (Auditable)
Mandatory tail categories plus p_floor	Recommended if TRC used	REQUIRED when TRC used	REQUIRED
Containment Mode A	Optional	Recommended	REQUIRED (binding gate)
RLS scoring	Optional	REQUIRED	REQUIRED
Uncertainty plus discrimination band	Optional	REQUIRED (active cells)	REQUIRED
Tie-breaks (UCI/HOI)	Optional	Recommended	REQUIRED when RLS non-decisive
Kernel propagation	NONE default	NONE default; QUICK if KQS policy satisfied	QUICK only if KQS policy satisfied plus sensitivity; else NONE
KQS policy	Optional	REQUIRED if kernel used	REQUIRED if kernel used
Sensitivity analysis	Optional	Recommended	REQUIRED
PCC artifact	Optional	REQUIRED (basic)	REQUIRED (full)
Audit flags	Optional	REQUIRED when triggered	REQUIRED when triggered
Stewardship (conditional)	Optional	Required iff Stw_req=1	Required (Stw_req assumed TRUE for Tier 3)
Tier-4 claim	PROHIBITED	PROHIBITED	PROHIBITED
Stakeholder discovery and mapping (SDP/InstanceMap)	Optional	REQUIRED when any SDP trigger holds; otherwise a lightweight stakeholder note or voluntary SDP is RECOMMENDED	REQUIRED (InstanceMap with multi-scope redundancy handling disclosures)
Stakeholder Coverage Index (SCI)	Optional	REQUIRED (SCI ≥ 1)	REQUIRED (SCI ≥ 2)

Tier 3 stewardship assumption (Normative): Any Tier 3 run SHALL set Stw_req(run) = 1 by default because Tier 3 implies external scrutiny and governance-grade assistance posture.

15.2 Tier 1: Heuristic Application ("2-Minute Ripple Check") (Normative minimum)

Use Tier 1 for low-stakes, reversible decisions where full calculation is disproportionate.

Tier 1 minimum protocol:

1. Decision: What must be decided, by when?
1. Options: List at least 2 options (include a "third path" redesign option if possible).
2. Rights screen (NCRC heuristic): Could any option plausibly violate LIFE/BODY/NEED/LBTY/DIGN/PROC/INFO/ECOL? If yes, escalate to Tier 2+.
3. Tail screen (TRC heuristic): Could any option plausibly create catastrophic or irreversible downside for Humanity/Biosphere? If yes, escalate to Tier 2+.
4. Unions touched: Which unions are materially affected (U_1 through U_7)?
5. Unioning move: What redesign could reduce harms and increase shared benefit across unions?

Tier 1 recording (recommended): A short note or mini-PCC with the six answers.

Tier-1 Worked Example (informative; shows Tier escalation):

Decision: Accept a new consulting project this month, or decline to protect time for household responsibilities.

Options: (A) Accept as proposed (10 hrs/week for 4 weeks). (B) Decline. (C) Redesign: accept only if scope is reduced to 4 hrs/week and meetings are time-boxed.

Rights screen (heuristic): No plausible LIFE/BODY/NEED/LBTY/DIGN violations. PROC/INFO: contract terms are clear, no coercion. PASS at Tier 1. If any need for deceptive data use or exploitative terms were suspected, escalate to Tier 2.

Tail screen (heuristic): No plausible catastrophic downside to Humanity/Biosphere. PASS at Tier 1.

Unions touched: U1 Self (Health, Agency, Meaning). U2 Household (Social, Material, Health). U3 Community/Organization impacts minor and reversible.

Unioning move: Option (C) reduces household harm while still capturing some benefit. Choose (C) if renegotiation succeeds; otherwise choose (B) if household strain would be material.

Record (mini-PCC): store the six answers and the chosen option in a note so NCAR learning can refine future time-budget decisions.

15.3 Tier 2: Core Calculable (Strict TRC Posture) (Normative)

Use Tier 2 for routine but consequential decisions requiring transparent computation and a basic PCC.

Tier 2 minimum obligations:

- Construct impacts on the [-1,+1] scale using Section 5 pipeline.
- Run NCRC with worst-off subgroup checks for directly affected rights cells.
- TRC MUST be executed when catastrophe relevance is plausible:
    - Use ≥5 scenarios minimum,
    - Include mandatory tail categories with probability floors unless implausible (with documented justification per Section 8.3.2),
    - Compute CVaR and corridor check,
    - Record full TRC table in PCC.
- Compute RLS and record ranking.
- If uncertain/non-decisive, apply UCI tie-break if available; otherwise record judgment call and monitoring plan.
- Produce PCC (basic) plus 5SPR.

Tier 2 propagation guidance:

- Default propagation_mode = NONE.
- QUICK propagation is allowed only if KQS policy is satisfied and sensitivity is feasible.

Structural-assurance note (Normative guidance). A Tier 2 run that proceeds without binding Containment is a disclosed lower-rigor posture, not evidence that containing-scope degradation risk is absent. If the principal hazard of the decision is plausible hollowing-out, structural degradation of containing scopes, or other containment-relevant harm that cannot be evidenced at Tier 2, the run SHOULD escalate to Tier 3 or explicitly record in the PCC that Tier 2 is insufficient for structural assurance.

15.4 Tier 3: Standard Auditable (Normative)

Use Tier 3 for high-stakes, contested, or externally scrutinized decisions where auditability and anti-gaming posture are required.

Tier 3 minimum obligations:

- Full PCC (Appendix H), including: impacts, subgroup semantics, scenario library, containment results, RLS, uncertainties, sensitivity, audit flags, signatures.
- NCRC: Subgroup analysis required for rights-covered cells.
- TRC: Required with ≥20 scenarios and mandatory tails plus probability floors (Appendix D).
- Containment Mode A: Required and binding.
- Sensitivity analysis bundle required:

    - Weights perturbation,
    - Rights threshold perturbation (±0.05 on θ_r as a sensitivity test),
    - Kernel perturbation if QUICK used,
    - Scenario probability perturbation.
- Kernel: QUICK permitted only if KQS policy permits:
    - KQS < 0.40: NONE only,
    - 0.40-0.50: QUICK plus mandatory sensitivity,
    - ≥0.50: QUICK permitted plus sensitivity.
- FULL propagation is prohibited for Tier 1-3 claims.

15.4A Conformance, empirical readiness, and reference implementation posture (Normative clarification)

Tier 1-3 run-level conformance claims concern whether a particular run satisfies the requirements of this specification. They are distinct from framework-level empirical operational-readiness claims about comparative real-world performance, calibration quality, or superiority over alternative governance methods. Tier 3 run-level conformance means the run satisfies Tier 3 procedural, evidentiary, and audit requirements as declared; it does not by itself imply that every structural indicator family used for UCI or Containment is externally validated, mature across domains, or empirically calibrated beyond the declared measurement pack.

Reference implementation posture (Informative with binding disclosure consequence). This Foundation Paper normatively defines conformance without requiring a published reference implementation. However, project custodians SHOULD publish sibling reference implementation artifacts consisting at minimum of a machine-readable PCC schema, validator rule set, and executable Appendix R fixtures. Until such artifacts are published in a release manifest, framework-level claims of replay-grade operational readiness MUST remain provisional even where run-level Tier 1-3 conformance is possible.

15.5 NCAR Integration by Tier (Normative)

All tiers SHOULD operate within NCAR.

Phase	Action
Notice	Define scope, unions, options, baseline, and configuration
Choose	Execute cascade and emit PCC
Act	Implement with monitoring aligned to predicted impacts and tail scenarios

Phase	Action
Reflect	Compare observed outcomes to predictions; update indicators, kernels, scenario libraries, and weights through governed procedures

Reflect cadence defaults:

- Tier 2: Within 6 months or after major outcome data
- Tier 3: Within 3-6 months or after major outcome data
- Emergency/Fallback: Review cadence per declared severity and risk

15.6 Improvement and Future Upgrades (Normative stance)

RippleLogic is corrigible. Any improvement MUST:

- Preserve NCRC/TRC non-compensability structure,
- Be explicit, versioned, and auditable (AIL),
- Be tested under the validation program (Section 17),
- And be introduced via governed update processes (NCAR Reflect plus versioning).

SECTION 16: RELATIONSHIP TO EXISTING FRAMEWORKS (INFORMATIVE)

16.0 Purpose

This section positions RippleLogic relative to existing ethical, governance, and decision frameworks. It clarifies structural differences and interfaces.

16.1 Compared to Utilitarianism / Cost-Benefit Analysis (CBA)

CBA often scalarizes plural values into a single metric (money/utility), enabling rights tradeoffs. RippleLogic blocks this via NCRC (rights as constraints) and TRC (tail risk as constraint), then optimizes welfare only within the admissible set.

16.2 Compared to Deontology (Rule-based ethics)

Pure deontology can lack a complete operational procedure for comparing permitted options under uncertainty. RippleLogic retains non-compensable constraints but adds computable consequence modeling and tail-risk bounding.

16.3 Compared to Rawlsian Justice and Capability Approaches

Rawls provides priority of liberty and fair basic structure; capabilities provide multi-dimensional flourishing. RippleLogic operationalizes multi-dimensional welfare into a computable 7×7 matrix and makes rights floors explicit across unions.

16.4 Compared to MCDA (Multi-Criteria Decision Analysis)

Standard MCDA often aggregates criteria via weighted sums or outranking without lexicographic catastrophe handling (Keeney & Raiffa, 1976; Belton & Stewart, 2002). RippleLogic is MCDA-like at the RLS layer but is structurally different because admissibility gates (NCRC/TRC/CSV) are lexicographic and non-compensable.

16.5 Compared to AI Alignment Toolsets (RLHF, Constitutional AI, risk frameworks)

Many alignment approaches train systems toward proxies (human feedback, constitutional principles) without hard admissibility gates. RippleLogic provides a decision-engine architecture: action-space filtering by NCRC/TRC/CSV, structured welfare scoring (RLS), auditable traces (PCC), and corrigibility via NCAR.

16.6 Interoperability with Governance Standards

RippleLogic is designed to be interoperable with governance regimes emphasizing accountability and risk management (for example, NIST AI RMF) through its PCC record, scenario governance, and explicit risk bounding. It adds formal lexicographic rights and tail-risk operators that many standards leave at a principles level.

16.7 Compared to Commons Governance

RippleLogic is compatible with institutional approaches to governing shared resources and externalities. Where commons governance emphasizes rules-in-use, monitoring, graduated sanctions, and polycentric coordination, RippleLogic contributes a computable, auditable decision cascade that makes rights floors, tail-risk bounds, and cross-union ripple effects explicit in each decision record (Ostrom, 1990).

16.8 RLS Validation and Calibration Status (Informative bridge)

Current status. RLS is release-ready as a structured residual welfare-scoring specification and audit method. It does not yet claim empirical validation of the dimensional non-overlap of the 7x7 welfare field, calibrated decisiveness thresholds, or inter-rater reliability of cell-filling across independent analysts. A companion RLS Validation Protocol should test these claims through scored-case datasets, correlation/factor analysis across the 49 cells, inter-rater reliability studies, disagreement heatmaps, scalar-comparator analysis, and revision rules. Until those studies are completed, public claims should say that RLS is structurally complete and auditable, not that it is empirically proven to produce superior decisions.

SECTION 17: VALIDATION, FALSIFICATION, AND RESEARCH PROGRAM (NORMATIVE FOR CLAIMS)

17.0 Purpose (Normative)

RippleLogic makes testable claims about decision quality, rights protection, tail-risk avoidance, and auditability. This section specifies explicit falsification criteria and a staged validation program. Until such validation is completed, RippleLogic remains a theory-to-practice system with bounded claims: Tier compliance is claimable, real-world performance superiority is not.

This validation program follows a falsification-first posture consistent with scientific corrigibility: components that fail their empirical tests must be revised or abandoned rather than defended by authority (Popper, 1959).

17.0A Layered Falsification Attribution Rule (Normative for validation claims)

A failure must be attributed to the layer it actually falsifies before a broader falsification claim is made. A failed measurement does not by itself falsify the framework; a failed projection does not by itself falsify a rights floor; and a failed gate operator must not be excused as mere input error when all admissible data, projection, and parameter alternatives have been ruled out.

Every recorded validation failure SHOULD include both FalsificationClass and FalsificationLayer. Recommended FalsificationLayer values are DATA, PROJECTION, PARAMETER, GATE_OPERATOR, PRIMITIVE, GOVERNANCE, and COMMUNICATION.

Layer	What failure means	Does not by itself falsify	Required response
DATA	Input missing, wrong, stale, biased, or insufficient.	Gate definitions or framework primitives.	Re-collect, disclose uncertainty, rerun.
PROJECTION	Scenario, mapping, kernel, stakeholder model, or prediction surface was wrong.	Rights logic or the whole framework.	Freeze a new projection; rerun as a new run.
PARAMETER	Threshold, coefficient, weight, or corridor was mis-set or post-hoc.	Cascade structure.	Reset ex ante under governed procedure; rerun.

Layer	What failure means	Does not by itself falsify	Required response
GATE_OPERATOR	NCRC, TRC, CSV, SVC, or RLS rule produces a reality-contradicted verdict under every admissible data/projection/parameter alternative.	A single bad input.	Challenge the operator definition; governed revision.
PRIMITIVE	A declared primitive cannot reconstruct a required phenomenon or produces unavoidable contradiction.	Any single failed run.	Reopen Primitive Audit.
GOVERNANCE	Authority, permission, legitimacy, contestability, or accountability failed.	Structural or mathematical cascade.	Governance review and authority correction.
COMMUNICATION	Public claim, summary, or disclosure exceeded claim boundary.	Underlying ethics or mechanics.	Correct the claim and update release/audit record.

Validator guidance. A failure logged without FalsificationLayer SHOULD trigger FALSIFICATION_LAYER_UNATTRIBUTED. A run that invokes PRIMITIVE or GATE_OPERATOR failure without first addressing plausible DATA, PROJECTION, and PARAMETER explanations SHOULD trigger FALSIFICATION_LAYER_SKIPPED and require escalation.

Additional v11.4 do-not-collapse rules. Structural viability is not ethical permissibility. Stability is not justice. Permission is not admissibility. Output generation is not an admissible reference. A plan that can stand structurally can still violate rights; a generated answer can still lack a valid reference; and an authorized action can still fail the MathGov cascade.

17.1 Core Empirical Claims (Testable Hypotheses)

17.1A MNA-Layer Empirical Hypotheses (Normative research frontier)

H-MNA1 (model adequacy under stakeholder omission). Multi-agent predictive performance worsens when stakeholder welfare, rights, or enabling-condition variables are omitted from world models in settings where cooperation, conflict, or harm depend materially on them.

H-MNA2 (instability under normative omission). Architectures that systematically ignore stakeholders’ welfare conditions, rights floors, or enabling conditions exhibit higher instability, defection, tail risk, or containment failure in repeated multi-agent environments than otherwise comparable architectures that track them explicitly.

H-MNA3 (cascade advantage over unconstrained scalarization). Rights-first and catastrophe-bounded cascade architectures reduce ruin-like outcomes, rights breaches, or coherence losses relative to unconstrained scalar optimization in comparable decision classes.

Hypothesis	Description
H1	Rights coherence: Decisions that pass NCRC produce fewer rights infringements than comparable baseline decisions, controlling for context.
H2	Tail-risk effectiveness: Decisions constrained by TRC exhibit lower realized tail losses than comparable decisions without TRC.
H3	Ripple sign accuracy (after calibration): After NCAR updates, predicted impact signs match observed sign in ≥70% of evaluated cells.
H4	Structural early warning: Persistent HOI > 0 predicts subsequent structural degradation better than baseline KPI monitoring.
H5	Anti-gaming effectiveness: In adversarial tests, specification gaming succeeds less often (≥30% reduction) relative to comparable governance processes without PCC plus mandatory tails plus subgroup semantics.

Because ripple effects partially track cooperation and network propagation dynamics, validation should also test whether kernel updates improve predictive performance in contexts where cooperation and trust dynamics are known to matter (Axelrod, 1984; Nowak, 2006; Christakis & Fowler, 2009).

H6 (PLSS prominence convergence). For recurring decision classes with standardized guidance, independently trained evaluators applying the same evidence base will produce PLSS prominence vectors with moderate or better convergence, operationalized in pilot testing by a pre-declared inter-rater agreement threshold.

H7 (PLSS floor activation). In a non-trivial fraction of PLSS-assessed decisions, constitutional floor weights assigned to guardrail scopes will contribute materially to final welfare ranking differences, demonstrating that guardrail scope retention is operational rather than merely decorative.

H8 (PLSS escalation fidelity). In pilot validation, plausible ET-trigger conditions will be detected and logged with sufficient sensitivity that PLSS posture is not systematically retained in clearly non-local cases.

17.2 Falsification Criteria (Normative)

17.2A MNA-Layer Falsifiers (Normative)

F-MNA1. Omission of welfare, rights, or enabling-condition variables does not reduce predictive adequacy in relevant multi-agent domains where those variables were hypothesized to matter materially.

F-MNA2. Architectures ignoring such conditions do not exhibit higher instability, tail risk, rights breach incidence, or containment failure than comparable architectures that track them explicitly.

F-MNA3. Cascade-constrained policies do not outperform unconstrained scalarized policies on declared rights, coherence, or tail-risk metrics in the relevant validation class.

Defect attribution prior to invoking F1-F5 follows the Appendix AE.3 layer matrix: each failure should be assigned to the layer it actually falsifies before any broader empirical falsifier is invoked.

RippleLogic components must be revised (or rejected) if evidence meets any of the following:

Criterion	Description
F1	NCRC failure: NCRC-passing decisions systematically produce worse rights outcomes than NCRC-failing decisions.
F2	TRC failure: TRC-constrained decisions show no reduction in realized tail losses compared to controls.
F3	Ripple predictiveness failure: Sign accuracy remains < 60% across successive NCAR cycles for key cells.
F4	Structural safeguard failure: UCI/HOI do not correlate with or predict meaningful degradation.
F5	Anti-gaming failure: Red-team exercises repeatedly exploit predictable loopholes (> 30% of adversarial runs).

17.3 Validation Phases (Normative roadmap)

Phase 1: Formal verification and implementation testing (0-6 months)

- Independent implementation of Tier 2-3 algorithms from spec.
- Unit tests for canonical equations (Appendix B).
- Adversarial tests for masking, subgroup erasure, scenario omission, confidence inflation.
- Produce reference PCCs and reproducibility checks.

Phase 2: Measurement validation (4-15 months)

- Indicator reliability and validity for welfare dimensions and UCI components.
- Cross-population measurement checks where relevant.
- Calibrate magnitude anchors and uncertainty mappings.

Phase 3: Controlled pilots (10-24 months)

- Pilot deployments in organizations or municipalities.
- Compare RippleLogic vs baseline governance processes on: rights incidents, tail losses, stakeholder legitimacy, audit completeness, decision reversal rates.

Phase 4: Field studies and scaling (18-36+ months)

- Longitudinal monitoring of UCI/HOI and realized outcomes.
- Kernel calibration and scenario library refinement.
- Comparative performance across domains.

17.4 Metrics and Targets (Normative defaults)

17.4A Inter-Rater Reliability Commitment (Normative commitment)

Before Tier 3 operational-readiness claims are made, the framework SHALL publish or register at least one inter-rater reliability (IRR) study on a representative case set covering: (i) welfare-cell anchoring, including magnitude construction mu_k using both percentile anchoring and threshold anchoring methods, (ii) rights-floor evaluation, including subgroup identification and worst-off impact estimation, (iii) local-scope/PLSS usage, including prominence signal construction and scope classification, and (iv) scenario construction and probability assignment for TRC.

The study MUST report the case set, evaluator instructions, adjudication procedure, disagreement classes, and reliability statistics appropriate to the data type used.

Component	Target ICC	Minimum acceptable	Notes
Magnitude construction (mu_k)	>= 0.70	>= 0.60	Tested separately for percentile and threshold anchoring
Rights-floor pass/fail (NCRC)	>= 0.80	>= 0.70	Binary gate; higher target reflects non-compensability
TRC pass/fail	>= 0.80	>= 0.70	Binary gate
RLS ranking order	>= 0.70	>= 0.60	Ordinal agreement on option ranking
PLSS prominence classification	>= 0.70	>= 0.60	Prominent vs Guardrail per scope

Sustained inability to exceed the minimum acceptable threshold on any component after rubric training indicates that the component requires redesign or additional anchoring guidance.

Until such evidence exists, Tier 3 remains architecturally specified but empirically incomplete.

Metric	Definition	Default Target
Rights violation rate	Verified post-decision rights infringements per protected subgroup	Lower than baseline governance
Tail-loss severity	Realized catastrophe-cell loss in worst outcomes	Lower than baseline governance
Sign accuracy	Match of predicted vs observed sign	≥0.70 after calibration
Magnitude error (RMSE)	RMSE on normalized [-1,+1] for measurable cells	≤0.25 (domain-dependent)
Audit completeness	Proportion of required PCC fields correctly filled	≥0.95 Tier 3
Gaming success rate	Fraction of adversarial attempts that change outcome illegitimately	Decreasing over time; investigate if > 0.30

17.5 Study Design Requirements (Normative)

- Pre-registration required for pilots claiming performance improvements.
- Control or comparator condition recommended (baseline governance).
- Blinding recommended for backtests: analyst should not know outcome during the run.

17.5A Projection Pre-Registration

17.5A.5 Freeze-Hash and Unique-Prediction Requirement (Normative for public validation and ProofPack-prep claims)

A validation claim is not falsifiable unless the projection was frozen before outcome evidence was consulted. For Tier 3 public pilots, public validation claims, and ProofPack-prep runs, ProjectionPreRegistration SHOULD include projection_freeze_hash, freeze_timestamp_utc, prediction_scope, predicted_observable_consequence_class, outcome_observation_timestamp, and post_observation_modification.

A material validation claim SHOULD produce exactly one predicted observable consequence class or a declared finite prediction set with pre-specified resolution rule. If the projection produces no prediction, an open-ended prediction, or a non-unique prediction without resolution rule, the claim is exploratory rather than falsifiable and SHOULD be marked NON_UNIQUE_PREDICTION or EXPLORATORY_ONLY.

If post_observation_modification=TRUE for a material validation claim, the run MUST NOT be used as a falsification test for that claim and SHOULD set POST_HOC_PROJECTION_INVALID with severity INVALID. A revised projection requires a new freeze hash and a new run. (Normative for pilots and Tier 3; Recommended otherwise)

Purpose. To preserve falsifiability and prevent post-hoc projection drift, pilot runs and Tier 3 runs SHALL pre-register the projection map from declared inputs to admissibility verdict before decision data is consulted in a way that could influence parameter selection.

17.5A.1 ProjectionPreRegistration object (Normative)

A ProjectionPreRegistration object SHALL be emitted, signed, timestamped, and stored before any cascade computation that contributes to a pilot or Tier 3 decision record. It SHALL contain at minimum: decision boundary, time horizon, and option set; stakeholder discovery record, including LSS and instance-to-scope mapping; rights coverage set C_r and per-right thresholds theta_r, including any non-default values with justification; catastrophe cell set C_cat, scenario set S, scenario probabilities p_s, CVaR confidence alpha, and TRC corridor tau_TRC; Containment trigger conditions and containing scopes; constitutional weight floors and any tier-appropriate non-default weights with justification; SGP scalar binding for impacted classes, including SG_norm and SG_patient_norm where applicable; subgroup uplift gamma_subgroup and any non-default value with justification; kernel mode and kernel quality status where applicable; sensitivity sweep specification; pre-declared falsification, refusal, escalation, and re-run conditions.

17.5A.2 Modification discipline (Normative)

Once emitted, the ProjectionPreRegistration object SHALL NOT be materially modified after observation in any way that affects admissibility, selectability, selection, or conformance claims.

If material modification is required, the current run SHALL enter Refusal under Section 4.9, document the reason in PCC.RefusalRecord, and emit a new ProjectionPreRegistration object before any subsequent run.

Non-material corrections, including typographical corrections, reference updates, or formatting changes that do not alter values or verdict logic, MAY be applied with version increment and documented diff.

17.5A.3 Tier application (Normative)

Tier 3: Projection Pre-Registration is REQUIRED for all runs.

Tier 2: Projection Pre-Registration is REQUIRED for pilots and RECOMMENDED otherwise.

Tier 1: Projection Pre-Registration is RECOMMENDED where stakes are medium or high.

17.5A.4 Storage and audit (Normative)

The ProjectionPreRegistration object SHALL be stored with the PCC and referenced by hash. An independent reviewer SHALL be able to reconstruct the declared projection map from the stored object using only the applicable canon, declared evidence base, and referenced companion artifacts.

17.6 Backtesting Protocol (Normative)

A backtest run MUST:

- Reconstruct the information available at decision time (no hindsight leakage),
- Define an option set consistent with the historical context,
- Execute RippleLogic using only those inputs,
- Record a “Backtest PCC” with full trace,
- Score predictions against realized outcomes using declared metrics.

17.7 Open Science and Responsible Disclosure (Normative intent)

Where feasible, publish:

- Anonymized PCC datasets,
- Scenario libraries and kernels (with evidence notes),
- Failure cases and revisions.

Security note: Where publishing would create exploitation risk, disclosure should be responsible and staged, but internal auditability must be preserved.

SECTION 18: APPLICATIONS AND USE CASES (INFORMATIVE)

18.1 AI Deployment Governance (Tier 3)

Decision context: An organization is deciding whether to deploy an AI system that affects users at scale.

Key obligations:

- NCRC: Protect dignity, information integrity, due process for worst-off subgroups.
- TRC: Model catastrophic tails (infrastructure disruption, mass manipulation, cascading conflict).
- Containment: Avoid organizational gains that degrade Polity/Humanity coherence (legitimacy collapse, epistemic fragmentation).
- PCC: Produce auditable trace; ensure comparability across options.

18.2 Climate and Energy Policy (Tier 3)

Decision context: A polity evaluates energy transition policies that impact households, industries, and biosphere stability.

Key obligations:

- NCRC: Protect basic needs (energy poverty), health, life.
- TRC: Include climate tipping cascade scenarios; enforce corridor.
- Containment: Prevent short-term gains from degrading Biosphere UCI beyond tolerance.

18.3 Organizational Strategy (Tier 2-3)

Decision context: Organization deciding on major restructuring, automation, supply chain change.

Key obligations:

- NCRC: Protect basic needs and dignity for worst-off employees/communities.
- TRC: Required at Tier 2 if catastrophe relevance is plausible.
- Containment: Detect hollowing-out risk even if near-term welfare improves.

18.4 Personal and Household Decisions (Tier 1)

Decision context: Personal choices (job change, relocation, major purchase) where stakes are limited and reversible.

Use Tier 1 heuristic: options, rights screen, tail screen, unions touched, unioning redesign.

Escalate to Tier 2 if any plausible rights risk exists or if the decision touches catastrophe relevance.

SECTION 19: LIMITATIONS AND NON-TARGETS (NORMATIVE BOUNDARIES)

19.1 Epistemic Limitations (Normative acknowledgment)

Input quality dependence: RippleLogic outputs are only as good as the evidence and estimation process. The framework reduces predictable failures but does not guarantee correct forecasts.

Measurement difficulty: Some dimensions (Meaning; aspects of Agency) are harder to measure reliably across cultures. RippleLogic treats such measurement with caution and uncertainty documentation.

Kernel uncertainty and operational status: Ripple propagation via the QUICK kernel mode is architecturally specified and canon-complete for Tier 1-3 implementation. However, the Starter KOPS (Appendix K) carries evidence classes B and E with a global shrink factor of 0.35, and the KQS policy (Section 6.5) requires KQS >= 0.40 for any kernel use. In practice, most early Tier 2-3 runs are expected to operate in NONE propagation mode because domain-specific kernels with sufficient evidence coverage, identifiability, and predictive validation do not yet exist. The NONE mode is a valid and conformant propagation posture, not a workaround. Building empirically grounded kernels with KQS >= 0.50 is a Phase 2-3 validation priority (Section 17.3). Ripple-awareness is already operational in the current release line through the multi-scope welfare matrix (which makes cross-scale consequences visible), through containment (which detects cross-scale degradation), and through UCI/HOI (which monitors structural integrity). Numerical kernel-based propagation remains a design-ready capability awaiting domain-specific calibration. The current release line mitigates kernel uncertainty with KQS policy, sensitivity requirements, and humility fallback to NONE.

Scenario incompleteness: Scenario libraries cannot enumerate all tail risks; mandatory tail categories reduce omission but cannot eliminate deep uncertainty.

19.2 Institutional and Governance Prerequisites

Tier 3 requires trained analysts and governance capacity: subgroup analysis, scenario governance, structural indicators for UCI, and audit review.

HDW governance can be captured if safeguards are weak. RippleLogic mandates anti-capture mechanisms, but real-world institutions must implement them.

19.3 Risks of Compliance Theater

Any framework can be used as a rubber stamp. RippleLogic mitigates via:

- Hard gates (NCRC/TRC/CSV),
- Non-maskable cells and audit flags,
- Challenger requirement in emergency mode,
- Required PCC traceability and sensitivity.

Nonetheless, compliance theater remains a risk; validation and external scrutiny matter.

19.4 Non-Targets (Normative)

RippleLogic is NOT designed to:

- Replace personal moral dialogue in intimate relationships,
- Dictate aesthetic or spiritual preferences,
- Resolve all deep metaphysical disagreements,
- Provide certainty under irreducible uncertainty,
- Function as a rhetorical weapon to end debate.

It is a public decision operating system: a method for auditable, multi-stakeholder, high-consequence decisions.

SECTION 20: CONCLUSION: PROSPEROUS FUTURES FOR INTELLIGENCES

20.1 Prosperity Defined (Normative operational definition)

In RippleLogic terms:

Prosperity = rights-safe (NCRC) + tail-safe (TRC) + containment-safe + net-positive welfare across unions + coherence preserved (UCI/HOI safeguards) + corrigible learning (NCAR)

This definition blocks false prosperity:

- "Growth" that violates rights is not prosperity.
- "Efficiency" that increases catastrophic exposure is not prosperity.
- "Success" that hollows out structural coherence is not prosperity.

20.2 RippleLogic Thesis

Alignment across humans, institutions, and AI requires non-compensable constraints on rights and catastrophic risk, explicit modeling of ripple propagation through nested unions, transparent and auditable decision procedures, and corrigible learning loops. These requirements follow from the combination of interdependence in complex systems (Meadows, 2008; Newman, 2010), planetary life-support constraints (Steffen et al., 2015; Rockström et al., 2009, 2023), and the tail-risk properties of ruin dynamics (Taleb, 2012; Ord, 2020).

20.3 Call to Test, Critique, and Deploy (Normative stance)

RippleLogic v11.4 is a scaffold: a spec-hardened foundation intended to be tested, falsified where wrong, refined where incomplete, and deployed proportionally to stakes. The method is designed to move societies toward ethically, sustainably, harmoniously, prosperously, and verifiably coordinated flourishing by making decision structure explicit, auditable, and corrigible.

REFERENCES (APA 7)

Amodei, D., Olah, C., Steinhardt, J., Christiano, P., Schulman, J., & Mané, D. (2016). Concrete problems in AI safety. arXiv. https://doi.org/10.48550/arXiv.1606.06565

Arrow, K. J. (1963). Social choice and individual values (2nd ed.). Yale University Press.

Artzner, P., Delbaen, F., Eber, J.-M., & Heath, D. (1999). Coherent measures of risk. Mathematical Finance, 9(3), 203–228. https://doi.org/10.1111/1467-9965.00068

Axelrod, R. (1984). The evolution of cooperation. Basic Books.

Barabási, A.-L., & Albert, R. (1999). Emergence of scaling in random networks. Science, 286(5439), 509–512. https://doi.org/10.1126/science.286.5439.509

Becker, G. S. (1981). A treatise on the family. Harvard University Press.

Belton, V., & Stewart, T. J. (2002). Multiple criteria decision analysis: An integrated approach. Kluwer Academic Publishers.

Bostrom, N. (2014). Superintelligence: Paths, dangers, strategies. Oxford University Press.

Bowlby, J. (1988). A secure base: Parent-child attachment and healthy human development. Basic Books.

Christakis, N. A., & Fowler, J. H. (2009). Connected: The surprising power of our social networks and how they shape our lives. Little, Brown and Company.

Council of Europe. (1950). Convention for the Protection of Human Rights and Fundamental Freedoms (European Convention on Human Rights). https://www.echr.coe.int/the-convention

Dunbar, R. I. M. (1992). Neocortex size as a constraint on group size in primates. Journal of Human Evolution, 22(6), 469–493. https://doi.org/10.1016/0047-2484(92)90081-J

Dunbar, R. I. M. (1993). Coevolution of neocortical size, group size and language in humans. Behavioral and Brain Sciences, 16(4), 681–735. https://doi.org/10.1017/S0140525X00032325

Edmondson, A. (1999). Psychological safety and learning behavior in work teams. Administrative Science Quarterly, 44(2), 350–383. https://doi.org/10.2307/2666999

Estes, J. A., Terborgh, J., Brashares, J. S., Power, M. E., Berger, J., Bond, W. J., Carpenter, S. R., Essington, T. E., Holt, R. D., Jackson, J. B. C., Marquis, R. J., Oksanen, L., Oksanen, T., Paine, R. T., Pikitch, E. K., Ripple, W. J., Sandin, S. A., Scheffer, M., Schoener, T. W., … Wardle, D. A. (2011). Trophic downgrading of planet Earth. Science, 333(6040), 301–306. https://doi.org/10.1126/science.1205106

Floridi, L. (2013). The ethics of information. Oxford University Press.

Goodhart, C. A. E. (1984). Problems of monetary management: The UK experience. In C. A. E. Goodhart (Ed.), Monetary theory and practice: The UK experience (pp. 91–121). Macmillan.

Hammond, J. S., Keeney, R. L., & Raiffa, H. (1998). Smart choices: A practical guide to making better decisions. Harvard Business School Press.

Integrated Food Security Phase Classification (IPC) Global Partners. (n.d.). Integrated Food Security Phase Classification (IPC) framework and tools. Retrieved February 19, 2026, from https://www.ipcinfo.org/

Intergovernmental Panel on Climate Change. (2023). Climate change 2023: Synthesis report. https://doi.org/10.59327/IPCC/AR6-9789291691647

International Labour Organization. (1998). ILO Declaration on Fundamental Principles and Rights at Work and its Follow-up. Retrieved February 19, 2026, from https://www.ilo.org/ilo-declaration-fundamental-principles-and-rights-work/about-declaration/text-declaration-and-its-follow

International Labour Organization. (2012). ILO indicators of forced labour (Special Action Programme to Combat Forced Labour). https://www.ilo.org/publications/ilo-indicators-forced-labour

Kahneman, D., & Tversky, A. (1979). Prospect theory: An analysis of decision under risk. Econometrica, 47(2), 263–291. https://doi.org/10.2307/1914185

Keeney, R. L., & Raiffa, H. (1976). Decisions with multiple objectives: Preferences and value tradeoffs. Wiley.

Lenton, T. M., Held, H., Kriegler, E., Hall, J. W., Lucht, W., Rahmstorf, S., & Schellnhuber, H. J. (2008). Tipping elements in the Earth's climate system. Proceedings of the National Academy of Sciences, 105(6), 1786–1793. https://doi.org/10.1073/pnas.0705414105

Manheim, D., & Garrabrant, S. (2018). Categorizing variants of Goodhart's law. arXiv. https://doi.org/10.48550/arXiv.1803.04585

March, J. G., & Simon, H. A. (1958). Organizations. Wiley.

Max-Neef, M. A. (1991). Human scale development: Conception, application and further reflections. Apex Press.

McGaughran, J. (2026). Sentience Gradient Protocol (SGP) v6.4 [Specification]. MathGov Institute for Ethical Systems Design. Historical predecessors: Sentience Gradient Protocol (SGP) v6.0 [Specification]; Sentience Gradient Protocol (SGP) v5.5 [Specification]; Sentience Gradient Protocol (SGP) v5.4 [Specification]; Sentience Gradient Protocol (SGP) v5.3 [Specification]; Sentience Gradient Protocol (SGP) v5.2.1 [Specification]; Sentience Gradient Protocol (SGP) v5.2 [Specification]; Sentience Gradient Protocol (SGP) v4.7.2 [Historical predecessor specification]. MathGov Institute for Ethical Systems Design.

Meadows, D. H. (2008). Thinking in systems: A primer. Chelsea Green Publishing.

National Institute of Standards and Technology. (2023). Artificial Intelligence Risk Management Framework (AI RMF 1.0) (NIST AI 100-1). https://doi.org/10.6028/NIST.AI.100-1

Newman, M. E. J. (2010). Networks: An introduction. Oxford University Press.

North, D. C. (1990). Institutions, institutional change and economic performance. Cambridge University Press.

Nowak, M. A. (2006). Five rules for the evolution of cooperation. Science, 314(5805), 1560–1563. https://doi.org/10.1126/science.1133755

Nussbaum, M. C. (2011). Creating capabilities: The human development approach. Harvard University Press.

Odum, E. P. (1971). Fundamentals of ecology (3rd ed.). W. B. Saunders.

Ord, T. (2020). The precipice: Existential risk and the future of humanity. Hachette Books.

Ostrom, E. (1990). Governing the commons: The evolution of institutions for collective action. Cambridge University Press.

Popper, K. (1959). The logic of scientific discovery. Hutchinson.

Putnam, R. D. (2000). Bowling alone: The collapse and revival of American community. Simon & Schuster.

Rawls, J. (1971). A theory of justice. Harvard University Press.

Reporters Without Borders. (n.d.). World Press Freedom Index. Retrieved February 19, 2026, from https://rsf.org/en/index

Rockafellar, R. T., & Uryasev, S. (2000). Optimization of conditional value-at-risk. Journal of Risk, 2(3), 21–41. https://doi.org/10.21314/JOR.2000.038

Rockström, J., Gupta, J., Qin, D., Lade, S. J., Abrams, J. F., Andersen, L. S., Armstrong McKay, D. I., Bai, X., Bala, G., Bunn, S. E., Ciobanu, D., DeClerck, F., Ebi, K., Gifford, L., Gordon, C., Hasan, S., Kanie, N., Lenton, T. M., Loriani, S., … Zhang, X. (2023). Safe and just Earth system boundaries. Nature, 619(7968), 102–111. https://doi.org/10.1038/s41586-023-06083-8

Rockström, J., Steffen, W., Noone, K., Persson, Å., Chapin, F. S., Lambin, E. F., Lenton, T. M., Scheffer, M., Folke, C., Schellnhuber, H. J., Nykvist, B., de Wit, C. A., Hughes, T., van der Leeuw, S., Rodhe, H., Sörlin, S., Snyder, P. K., Costanza, R., Svedin, U., … Foley, J. A. (2009). A safe operating space for humanity. Nature, 461(7263), 472–475. https://doi.org/10.1038/461472a

Russell, S. (2019). Human compatible: Artificial intelligence and the problem of control. Viking.

Ryan, R. M., & Deci, E. L. (2000). Self-determination theory and the facilitation of intrinsic motivation, social development, and well-being. American Psychologist, 55(1), 68–78. https://doi.org/10.1037/0003-066X.55.1.68

Sen, A. (1999). Development as freedom. Knopf.

Sen, A. (2009). The idea of justice. Harvard University Press.

Simon, H. A. (1962). The architecture of complexity. Proceedings of the American Philosophical Society, 106(6), 467–482.

Steffen, W., Richardson, K., Rockström, J., Cornell, S. E., Fetzer, I., Bennett, E. M., Biggs, R., Carpenter, S. R., de Vries, W., de Wit, C. A., Folke, C., Gerten, D., Heinke, J., Mace, G. M., Persson, L. M., Ramanathan, V., Reyers, B., & Sörlin, S. (2015). Planetary boundaries: Guiding human development on a changing planet. Science, 347(6223), 1259855. https://doi.org/10.1126/science.1259855

Taleb, N. N. (2012). Antifragile: Things that gain from disorder. Random House.

United Nations. (1966). International Covenant on Civil and Political Rights. United Nations Treaty Collection. Retrieved February 19, 2026, from https://treaties.un.org/Pages/ViewDetails.aspx?src=TREATY&mtdsg_no=IV-4&chapter=4&clang=_en

United Nations. (1984). Convention against Torture and Other Cruel, Inhuman or Degrading Treatment or Punishment. United Nations Treaty Collection. Retrieved February 19, 2026, from https://treaties.un.org/Pages/ViewDetails.aspx?src=TREATY&mtdsg_no=IV-9&chapter=4&clang=_en

United Nations Human Settlements Programme (UN-Habitat). (2003). The challenge of slums: Global report on human settlements 2003. Earthscan. Retrieved February 19, 2026, from https://unhabitat.org/the-challenge-of-slums-global-report-on-human-settlements-2003

United Nations Office on Genocide Prevention and the Responsibility to Protect. (2014). Framework of Analysis for Atrocity Crimes. Retrieved February 19, 2026, from https://www.un.org/en/genocideprevention/publications.shtml

Üstün, T. B., Kostanjsek, N., Chatterji, S., & Rehm, J. (Eds.). (2010). Measuring health and disability: Manual for WHO Disability Assessment Schedule (WHODAS 2.0). World Health Organization. https://iris.who.int/handle/10665/43974

Weber, M. (1978). Economy and society (G. Roth & C. Wittich, Eds.). University of California Press. (Original work published 1922)

World Health Organization. (2001). International classification of functioning, disability and health (ICF). Retrieved February 19, 2026, from https://www.who.int/standards/classifications/international-classification-of-functioning-disability-and-health

World Justice Project. (n.d.). Rule of Law Index. Retrieved February 19, 2026, from https://worldjusticeproject.org/rule-of-law-index/

APPENDICES

APPENDIX A: SYMBOLS AND NOTATION (v11.4)

Appendix A is normative for symbol meanings and domains.

Terminology note (Normative). u indexes Union Scopes (short: Scopes). “Union(s)” remains an allowed legacy alias. This does not change meanings in v9.0 and earlier, it clarifies the canonical term in v9.0.

RealityGroundingRecord / RCR: A PCC-linked record identifying a run’s decision-relevant reality surface, material reality claims, evidence traces, declared unknowns, RealityGroundingStatus, vehicle/content/effect notes where social or institutional constructs are material, surface/territory limitations, and falsification or monitoring triggers. This is the canonical record. The older label RealitySurfaceRecord is deprecated and may be used only as a legacy full-text alias for the surface-description portion of RCR; RSR MUST NOT be used for RealitySurfaceRecord. To avoid acronym collision with ReferenceStructureRecord, use RCR for RealityGroundingRecord and RefStructRecord for ReferenceStructureRecord where abbreviations are needed.

RealityGroundingStatus: The declared status of a decision claim's contact with the relevant reality surface. Allowed values: GROUNDED, ESTIMATED, ASSUMPTION_BOUND, DECLARED_UNKNOWN, OUT_OF_SCOPE_WITH_RATIONALE, INSUFFICIENT_GROUNDING. Legacy aliases EVIDENCE_BACKED -> GROUNDED and INSUFFICIENT_CONTACT -> INSUFFICIENT_GROUNDING are retained for backward compatibility. At Tier 2 it is operator-declared with review status explicitly recorded; at Tier 3 it must follow the declared reviewer regime; at Tier 4 it must be independently re-derivable from EvidenceTrace and hash-pinned supporting artifacts.

A.1 Sets and Indices

Symbol	Meaning	Domain
u	Union Scope (Scope) index	$u \in \{1,2,3,4,5,6,7\}$
d	Dimension index	$d \in \{1,2,3,4,5,6,7\}$
a	Option (candidate action)	$a \in O$
s	Scenario index	$s \in S$
r	Right index	$r \in R$
k	Impact instance index	integer
g	Protected subgroup index	$g \in G_{u,d}$

A.2 Core Sets

Set	Definition
U	Operational Scopes {Self, Household, Community, Organization, Polity, Humanity/CMIU, Biosphere}
D	Welfare dimensions {Material, Health, Social, Knowledge, Agency, Meaning, Environment}
O	Option set
S	Scenario set
R	Rights set {LIFE, BODY, LBTY, NEED, DIGN, PROC, INFO, ECOL}
$C_r \subseteq U \times D$	Coverage set for right r

Set	Definition
C_cat ⊆ U×D	Catastrophe cell set for TRC

A.3 Union Stack (Canonical)

Canonical Union Scopes: U1 Self; U2 Household; U3 Community; U4 Organization; U5 Polity; U6 Humanity/CMIU; U7 Biosphere. These are the row coordinates of the 7×7 welfare matrix. See Section 2.3 for definitions and Section 4.7 / Appendix AD for cell-level interpretation.

A.3A Scope × Dimension Reference Table (Non-Normative, clarity payload)

The table below reproduces the 7×7 Scope×Dimension reference grid for quick reading and cross-platform stability. The canonical grid definition is presented in Section 4.7.

Scope \ Dimension	D1 Material	D2 Health	D3 Social	D4 Knowledge	D5 Agency	D6 Meaning	D7 Environment
U1 Self	Personal resources, income, essentials	Physical/ mental health and safety	Relationships, belonging, support	Learning, understanding, competence	Autonomy, freedom, choice	Purpose, values, spiritual well-being	Local environment affecting self
U2 Household	Household livelihood, housing, food security	Household health, caregiving capacity	Family cohesion, care, safety	Household learning, skills	Household decision rights, roles	Household meaning, culture, identity	Home ecology, consumption footprint
U3 Community	Community economy, access to goods/services	Public health, local safety	Social trust, inclusion, cohesion	Education access, knowledge commons	Participation, civic agency	Shared narratives, cultural meaning	Local ecosystems, pollution, resilience
U4 Organization	Organizational resources, viability	Workplace safety, wellbeing	Org culture, fairness, belonging	Org learning, data, competence	Governance, employee agency	Mission coherence, meaning	Operational environmental impacts
U5 Polity	National economy, infrastructure, welfare capacity	Population health, security	Social stability, justice, cohesion	National knowledge, R&D, education	Democratic agency, rights	National identity, legitimacy	National environment, climate policy

Scope \ Dimension	D1 Material	D2 Health	D3 Social	D4 Knowledge	D5 Agency	D6 Meaning	D7 Environment
U6 Humanity/CMIU	Global economy, shared resources	Global health, existential safety	Human solidarity, equity, peace	Global science, open knowledge	Collective governance capacity	Species meaning, long-run purpose	Planetary boundaries, biosphere integrity
U7 Biosphere	Ecosystem services, material substrates	Species health, biodiversity	Symbiosis, interspecies stability	Ecological knowledge, monitoring	Adaptive capacity, resilience	Intrinsic value, evolutionary meaning	Habitats, climate stability, regeneration

A.4 Welfare Dimensions (Canonical)

Dimension	Name	Summary
D_1	Material	Resources, infrastructure, subsistence security
D_2	Health	Physical/mental functioning, morbidity/mortality risk
D_3	Social	Trust, belonging, relational integrity
D_4	Knowledge	Epistemic access, learning conditions
D_5	Agency	Autonomy, effective choice, freedom from coercion
D_6	Meaning	Purpose/coherence/valued projects (cautious measurement)
D_7	Environment	Ecological and built context integrity

A.5 Impact Objects

Symbol	Meaning	Range
Ĩ_dir(u,d,a)	Pre-saturation direct impact	ℝ
I_dir(u,d,a)	Direct impact (post-saturation)	[-1,+1]
Ĩ_prop(u,d,a)	Pre-saturation propagated impact	ℝ
I_prop(u,d,a)	Propagated impact (post-saturation)	[-1,+1]
I_rights(u,d,a)	Worst-off subgroup propagated impact	[-1,+1]

A.6 Impact Instance Attributes

Symbol	Meaning	Range / Default
μ_k	Magnitude (baseline-delta, normalized)	[-1, +1]
r_k	Reach	[0, 1]
t_k	Time horizon	(0, ∞) years
ℓ_k	Likelihood	[0, 1]
c_k	Confidence	[0.1, 1]
e_k	Equity/resilience multiplier	Tier 2: 1.00 only unless named governed instrument; Tier 3: [0.75, 1.25] default 1.00
s_k	Sentience multiplier (SGP)	[0, 1], default 1; MUST be 1 for humans
t_min	Minimum time horizon floor	0.083 years (≈1 month); prevents τ(0) = 0

A.7 Operators and Functions

Symbol

Definition

$(x)^+$

max(x, 0)

clip(x, a, b)

max(a, min(x, b))

sat_β(x)

tanh(β × x)

τ(t)

min(1, ln(1 + max(t, t_min)) / ln(1 + T_ref)) Defaults: T_ref = 25 years, t_min = 0.083 years Cap ensures τ(t) ≤ 1 for all t; floor ensures τ > 0 for all t > 0

φ(u, d)

7(u − 1) + d (flattening map for kernel indexing)

Anc(u, D_c)

Ancestor set of containing unions for union u up to depth D_c

T_{ref}

Governance reference time horizon for temporal weighting. Default: 25 years. MUST NOT be set < 10 years without charter-level justification.

A.8 Kernel and Propagation

Symbol	Meaning
$K \in \mathbb{R}^{49 \times 49}$	Ripple kernel matrix (sparse)
K_{ij}	Kernel entry mapping source j to target i (target-row, source-column)
κ_{max}	Maximum absolute kernel entry bound, default 0.5
κ_{use_min}	Minimum absolute kernel edge magnitude for “relied-upon edge” classification in KQS computation (Section 6.5.1). Default: 0.05 unless the PCC declares a stricter value.
ρ_{max}	Maximum absolute row-sum bound, default 0.9
$\|K\|_\infty$	Maximum absolute row-sum norm of K: $\|K\|_\infty := \max_i \Sigma_j \lvert K_{ij} \rvert$. Used for Tier 1-3 kernel stability verification (Section 6.4).
$\rho(K)$	Spectral radius of K. For Tier 1-3, treated as satisfied when $\|K\|_\infty < 1$ per Section 6.4 (since $\rho(K) \leq \|K\|_\infty$).
NONE mode	$I_{prop} := I_{dir}$
QUICK mode	$\tilde{I}_{prop} = I_{dir} + K \times I_{dir}$, then saturate

Stream note (Normative). Kernel propagation applies per stream x∈{base, welfare} as specified in Appendix B. In Appendix A, the symbols I_{dir}, $\tilde{I}_{prop}$, and I_{prop} may be used as shorthand for their stream-specific forms $I_{dir,x}$, $\tilde{I}_{prop,x}$, and $I_{prop,x}$. Admissibility computation uses the Base stream (sentience multiplier fixed at $s_k:=1$); RLS uses the Welfare stream.

A.9 Rights and Constraints

Symbol	Meaning
θ_r	Rights threshold for right r

Symbol	Meaning
v_r(a)	Violation depth for right r under option a
NCRC(a)	Rights admissibility predicate
TRC(a)	Tail-risk admissibility predicate
τ_TRC	TRC corridor threshold
α	CVaR tail level ∈ (0, 1)
CVaR_α[L(a)]	Conditional Value-at-Risk for loss distribution of option a
θ_pos	Positive impact threshold (containment trigger), default 0.05
τ_c	Containment tolerance threshold, default -0.10
D_c	Containment ancestor depth, default 2

A.10 Scoring and Uncertainty

Symbol	Meaning
w_u	Union weight (w_u ≥ 0, Σ_u w_u = 1)
v_d	Dimension weight (v_d ≥ 0, Σ_d v_d = 1)
m(u,d)	Applicability mask for RLS only ∈ {0, 1}
κ(u,d)	Cell multiplier for RLS, default 1
RLS(a)	RippleLogic Score
σ(u,d,a)	Cell-level uncertainty proxy
σ_RLS(a)	RLS uncertainty proxy
δ	Discrimination threshold for decisive lead, default 2
ε	Stabilizer in Gap computation, default 10^{-6}
λ	Risk-aversion coefficient for RLS_adj, default 0.5

A.11 Structural Metrics

Symbol	Meaning
UCI_u	Union Coherence Index for union u ∈ [0, 1]
ΔUCI_u(a)	Coherence change under option a = UCI_u(a) − UCI_u(baseline)

Symbol	Meaning
HOI_t	Hollowing-Out Index at time t
H_u	Cohesion component of UCI
F_u	Flow component of UCI
R_u	Resilience component of UCI
E_u	Equity component of UCI (E_1 := 1 for Self)

A.12 Audit Artifacts

Symbol	Meaning
PCC	Provenance and Compliance Certificate
5SPR	Five-Sentence Public Rationale
AIL	Artifact Integrity Law (Tier 2-3 integrity rules)
Audit flags	Standard labels for invalidity or warnings (Appendix H)

A.13 Scenario and TRC Objects

Symbol	Meaning
S	Scenario set
p_s	Scenario probability (p_s ≥ 0, Σ_s p_s = 1)
L(a,s)	Scenario loss for option a under scenario s
ω_c	Catastrophe cell weight (ω_c ≥ 0, Σ_c ω_c = 1)
p_floor	Minimum probability floor per mandatory tail category, default 0.02

A.14 SGP Integration

Symbol	Meaning
SG_norm(E)	Authoritative effective SGP individual scalar for entity E ∈ [0,1], consumed through the SGP binding interface. It is a governed protection/standing interface value, not a direct consciousness detector.
A(E), B(E), C(E)	SGP pillar scores: Awareness, Agency, Union Participation

Symbol	Meaning
SG_patient_norm(E)	Rights-of-protection sentience scalar for entity E ∈ [0, 1], incorporating taxon baseline logic when provided by SGP.

End Appendix A.

APPENDIX B: CANONICAL EQUATIONS (TIER 1-3 EXECUTABLE)

Appendix B is normative and is the canonical equation pack for Tier 1-3 implementations. All equations are reproduced explicitly for standalone executability.

B.1 Temporal Weighting

Defaults: T_ref = 25 years; t_min = 0.083 years (≈1 month).

$\tau(t) = \min(1, \ln(1 + \max(t, t_{min})) / \ln(1 + T_{ref}))$

This function has three properties by construction:

(a) Logarithmic growth ensures that short-term effects receive proportionally less weight than long-term effects, without the exponential discounting that aggressively devalues future generations.

(b) The min(1, ·) cap ensures that no effect receives more temporal weight than the governance reference horizon. At T_ref, $\tau = 1.00$. Beyond T_ref, τ remains 1.00 (capped).

(c) The max(t, t_min) floor ensures that $\tau(t) > 0$ for all positive t, preventing very short-term effects from being zeroed out.

Illustrative values (with defaults T_ref = 25, t_min = 0.083):

At 100 years: $\tau = 1.00$ (capped)

At 50 years: $\tau = 1.00$ (capped)

At 25 years: $\tau = 1.00$

At 10 years: $\tau \approx 0.74$

At 5 years: $\tau \approx 0.55$

At 1 year: $\tau \approx 0.21$

At t_min (≈1 month): $\tau \approx 0.025$

Historical change note (v7.5.0; carried forward unchanged in v9.0): This replaces the uncapped function $\tau(t) = \ln(1+t)/\ln(1+T_{ref})$ from v7.4.5, which produced $\tau(50) \approx 1.22$. The capped version is normatively preferred. See Section 5.3 for design rationale.

B.2 Direct Impact Aggregation (Pre-Saturation; stream-separated)

B.2A Base stream (admissibility; s_k := 1):

$$\tilde{I}_{dir\_base}(u,d,a) = \Sigma_{k \in K(u,d,a)} [r_k \times \tau(t_k) \times \ell_k \times c_k \times e_k \times (1) \times \mu_k]$$

B.2A Gate-critical confidence guard. In the Base stream used by RF/NCRC, TRC, and CSV, c_k represents evidence confidence but cannot by itself excuse unresolved severe adverse gate-critical impacts. Where a rights-covered, catastrophe-covered, or material CSV impact is adverse and low-confidence, and where the gate outcome could change, implementations MUST apply the Gate-critical confidence guard in Section 5: conservative bound, narrower claim, stronger evidence and rerun, downgrade, escalation, or Refusal.

B.2B Welfare stream (RLS; sentience weights allowed):

$$\tilde{I}_{dir\_welfare}(u,d,a) = \Sigma_{k \in K(u,d,a)} [r_k \times \tau(t_k) \times \ell_k \times c_k \times e_k \times s_{k,welfare} \times \mu_k]$$

Here $s_{k,welfare}$ is set via the SGP interface only as permitted by Section 12 and Appendix G; the Base stream (B.2A) fixes $s_k := 1.0$ for all instances.

Where K(u,d,a) is the set of impact instances asserted for cell (u,d) under option a.

B.3 Direct Saturation (per stream)

For $x \in \{base, welfare\}$:

$$I_{dir\_x}(u,d,a) = \tanh(\beta \times \tilde{I}_{dir\_x}(u,d,a))$$

Default: $\beta = 2$.

B.4 Missing Data (Ignorance Penalty Phantom Instance)

If cell (u,d) is required-active but $K(u,d,a) = \varnothing$, add phantom instance with:

Parameter	Phantom Value
μ_phantom	-0.10
r	1
t	T_ref (25 years)
ℓ	1
c	1

Parameter	Phantom Value
e	1
s	1

B.5 Flattening Map (Kernel Indexing)

Vectorize I_dir into a 49-vector by:

$i = \varphi(u,d) = 7(u - 1) + d$

B.6 Propagation Modes (Tier 1-3: NONE and QUICK only)

NONE:

For $x \in$ {base, welfare}: I_prop_x := I_dir_x

QUICK:

For stream $x \in$ {base, welfare}: Ĩ_prop_x = I_dir_x + K × I_dir_x

Then apply post-propagation saturation (B.7).

B.7 Post-Propagation Saturation (per stream)

For $x \in$ {base, welfare}:

I_prop_x(u,d,a) = tanh(β_prop × Ĩ_prop_x(u,d,a))

Default: β_prop = 1.

B.8 Worst-Off Subgroup Impact (Rights)

I_rights(u,d,a) = min_{g ∈ G_{u,d}} I_prop_base(u,d,a | g)

B.9 NCRC Violation Depth and Rights Admissibility

For each right r with threshold θ_r and coverage set C_r:

v_r(a) = max_{(u,d) ∈ C_r} (θ_r − I_rights(u,d,a))$^+$

where $(x)^+ = \max(x, 0)$.

NCRC(a) = TRUE if and only if v_r(a) = 0 for all r ∈ R

B.10 TRC Loss (Tier 1-3 Bounded-Impact Mode)

Let C_cat be catastrophe cells and ω catastrophe weights.

Normative stream binding (Design A): TRC MUST be computed from the Base stream (I_prop_base). TRC MUST NOT use Welfare-stream (sentience-weighted) impacts.

For scenario s:

$L(a,s) = \Sigma_{c \in C\_cat} \omega_c \times (-I\_prop\_base,c(a \mid s))^+$

TRC uses CVaR (Appendix D for discrete computation) and corridor threshold τ_TRC:

$TRC(a)$ = TRUE if and only if $CVaR\_\alpha[L(a)] \leq \tau\_TRC$

B.11 Containment Mode A (Binding)

Define positively moving unions:

$S\_u(a) = \Sigma\_d\ v\_d^{cont} \times I\_prop\_base(u,d,a)$

Containment trigger weights (Normative). Use the containment-specific vector v_d^cont (default uniform 1/7). Do not use HDW welfare weights for containment trigger detection.

$U\_pos(a) = \{u : S\_u(a) \geq \theta\_pos\}$

For each $u \in U\_pos(a)$:

$M\_u(a) = \min_{u' \in Anc(u, D\_c)} \Delta UCI_{u'}(a)$

Global containment predicate for material CSV runs:

CSV_status(a) in {CSV_PASS, CSV_PASS_WITH_CONTROLS} if and only if for all $u \in U\_pos(a)$: $M\_u(a) \geq \tau\_c$.

Selectability compatibility note. CSV_NOT_MATERIAL is not a material Containment Mode A pass calculation; it is a documented disposition that the CSV burden is not material for the declared claim boundary. When PCC rationale supports CSV_NOT_MATERIAL, it is pass-equivalent for selectability under Section 9.5.

B.12 RippleLogic Score (RLS)

$RLS(a) = \Sigma\_u\ \Sigma\_d\ w\_u \times v\_d \times m(u,d) \times \kappa(u,d) \times I\_prop\_welfare(u,d,a)$

Default: $\kappa(u,d) = 1$, $m(u,d) = 1$ unless masked.

B.12A κ governance note (Normative). Non-default κ(u,d) values are allowed only within the bounded governance interval declared in Section 10.1 and MUST preserve sign, rights-floor protection, catastrophe visibility, and replayability. Validator checks SHALL reject any κ(u,d) that acts as a hidden mask, hidden union-floor rewrite, or undisclosed prominence override.

B.13 RLS Uncertainty (Optional; Tier 3 Required)

Definition of $\sigma(u,d,a)$ (normative minimum): PCC MUST declare one of the following cell-level uncertainty proxies and apply it consistently across options:

Method A (interval half-width): If the PCC records a confidence interval [L,U] for the cell impact $I(u,d,a)$, set $\sigma(u,d,a) = (U - L)/2$.

Method B (confidence-derived): If the PCC records impact instances with confidences $c_k \in [0.1,1]$ and the run selects Method B, then the PCC MUST derive a cell-level confidence $c(u,d,a) \in [0,1]$ from $\{c_k\}$ using one declared CellConfidenceAggregationMethod per Section 10.3.2, and set:
$\sigma(u,d,a) = (1 - c(u,d,a)) \cdot |I(u,d,a)|$

Method C (calibrated table): Use a pre-registered mapping from qualitative uncertainty labels (for example, LOW/MED/HIGH) to σ values, stored in the ProofPack or PCC appendix.

σ values MUST be clipped to [0,1]. Unless a stricter governed method is declared, $\sigma(u,d,a)$ is interpreted as the pre-κ cell-level uncertainty input and SHALL be aggregated into σ_RLS using the same w_u, v_d, $m(u,d)$, and $\kappa(u,d)$ posture used for RLS.

$$\sigma\_RLS(a) = \sqrt{[\Sigma\_u\ \Sigma\_d\ (w\_u \times v\_d \times m(u,d) \times \kappa(u,d) \times \sigma(u,d,a))^2]}$$

Risk-adjusted score (optional):

$$RLS\_adj(a) = RLS(a) - \lambda \times \sigma\_RLS(a)$$

By default, RLS_adj is diagnostic only and MUST NOT replace RLS for ranking or decisiveness unless the run declares ex ante that risk-adjusted ranking is the governing ranking basis. If such a declaration is made, the same declared basis MUST be used consistently for ranking, Gap, and decisiveness across all compared options in the run.

B.14 Discrimination Gap (Decisive vs Non-Decisive)

Gap(a,b) = abs(RLS(a) - RLS(b)) / sqrt(sigma_RLS(a)^2 + sigma_RLS(b)^2 + epsilon)

Default: delta = 2, epsilon = 1e-6.

Decisive lead if Gap(a*, a2) > delta.

B.15 UCI Computation

$$UCI\_u = \alpha\_H \times H\_u + \alpha\_F \times F\_u + \alpha\_R \times R\_u + \alpha\_E \times E\_u$$

Default: $\alpha\_H = \alpha\_F = \alpha\_R = \alpha\_E = 0.25$.

Special case for Self (U_1): $E_1 := 1$ by definition.

B.16 ΔUCI Computation

$$\Delta UCI\_u(a) = UCI\_u(a) - UCI\_u(baseline)$$

B.17 HOI Computation

$$HOI\_t = EMA\_\lambda(\Delta RLS)\_t - EMA\_\lambda(\Delta UCI)\_t$$

Default half-life: 3 periods.

End Appendix B.

APPENDIX C: RIGHTS / NCRC CANON PACK (TIER 1-3)

Appendix C is normative.

C.1 Rights Thresholds (Canonical)

Right	Code	Threshold θ_r
Life	LIFE	-0.90
Bodily Integrity	BODY	-0.70
Liberty	LBTY	-0.65
Basic Needs	NEED	-0.50
Dignity	DIGN	-0.55
Due Process	PROC	-0.45
Information	INFO	-0.40
Ecological Integrity	ECOL	-0.65

Normative interpretation rule: Thresholds θ_r are admissibility floors, not weights. They do not represent “how important” a right is; they represent the minimum allowable protection level.

C.2 Rights Coverage Sets C_r (Authoritative)

Right	Coverage Set C_r
LIFE	$\{(u, \text{Health}): u \in \{1,2,3,4,5,6\}\} \cup \{(6, \text{Environment})\}$
BODY	$\{(u, \text{Health}): u \in \{1,2,3,4,5,6\}\}$
LBTY	$\{(u, \text{Agency}): u \in \{1,2,3,4,5,6\}\} \cup \{(u, \text{Social}): u \in \{3,4,5,6\}\}$
NEED	$\{(u, \text{Material}): u \in \{1,2,3,4,5,6\}\} \cup \{(u, \text{Health}): u \in \{1,2,3,4,5,6\}\}$
DIGN	$\{(u, \text{Social}): u \in \{1,2,3,4,5,6\}\} \cup \{(u, \text{Agency}): u \in \{1,2,3,4,5,6\}\}$
PROC	$\{(u, \text{Agency}): u \in \{4,5,6\}\} \cup \{(u, \text{Knowledge}): u \in \{4,5,6\}\} \cup \{(u, \text{Social}): u \in \{4,5,6\}\}$
INFO	$\{(u, \text{Knowledge}): u \in \{1,2,3,4,5,6\}\} \cup \{(u, \text{Agency}): u \in \{1,2,3,4,5,6\}\}$
ECOL	$\{(6, \text{Environment}), (7, \text{Environment})\}$

C.3 Emergency Mode Rights Priority (Canonical)

When Emergency Mode is invoked (A_NCRC = ∅), rights are ordered lexicographically:

[LIFE, BODY, ECOL, LBTY, NEED, DIGN, PROC, INFO]

This ordering MUST be used exactly for lexicographic minimization of the violation depth vector.

C.4 NCRC Violation Depth Definition (Canonical)

$v\_r(a) = \max\_\{(u,d) \in C\_r\} (\theta\_r - I\_rights(u,d,a))^+$

C.5 NCRC Predicate (Canonical)

NCRC(a) = TRUE if and only if v_r(a) = 0 for all r ∈ R

C.6 Subgroup Conservative Bound (Canonical)

When subgroup disaggregation is infeasible for a rights-covered cell:

Let γ_subgroup = 1.5 (default conservatism factor).

- If I_prop_base(u,d,a) < 0: set I_rights(u,d,a) = max(-1, γ_subgroup × I_prop_base(u,d,a))
- Else: set I_rights(u,d,a) = I_prop_base(u,d,a)

This bound applies only for NCRC checking, not for RLS scoring.

C.7 Emergency Mode Documentation Requirements (Canonical)

When Emergency Mode is invoked, the PCC MUST include:

- Emergency declaration and trigger conditions
- v_r(a) for each option and each right
- Lexicographic comparison trace
- Independent challenger attestation (or CHALLENGE_DEFERRED_EMERGENCY flag)
- Mitigation/remediation plan with timeline
- Review cadence and return-to-normal triggers

End Appendix C.

APPENDIX D: TRC AND SCENARIO GOVERNANCE CANON PACK (TIER 1-3)

Appendix D is normative.

D.1 Base Catastrophe Cell Set

C_cat_base = {(6, Health), (6, Environment), (7, Environment)}

These correspond to:

- (6, Health): Humanity/CMIU-Health, representing global-scale health viability
- (6, Environment): Humanity/CMIU-Environment, representing civilization-scale habitability conditions
- (7, Environment): Biosphere-Environment, representing Earth-system integrity

D.1A Default Governance-Lock-In Extension Profile (Normative)

Trigger condition: This profile MUST be activated when any of the following is plausibly in scope for the decision: surveillance normalization; emergency-power persistence beyond justified sunset; due-process suspension or durable recourse collapse; information-ecosystem capture; irreversible dependency lock-in on critical infrastructure or governance functions; or concentrated arbitrary control without meaningful exit, review, or contestability.

Default governance-lock catastrophe cell set:

C_cat_lock_default = {(5, Agency), (5, Knowledge), (6, Agency), (6, Knowledge)}

Default weighting rule when active:

If no stricter governed profile is declared, set:

C_cat = C_cat_base ∪ C_cat_lock_default

and use uniform catastrophe weights over the full active set.

Interpretation:

(5, Agency): polity-scale loss of contestability, due process, or civic agency

(5, Knowledge): polity-scale degradation of information integrity and epistemic access

(6, Agency): civilization-scale agency loss, lock-in, or durable corrigibility collapse

(6, Knowledge): civilization-scale information capture, knowledge bottlenecking, or durable epistemic enclosure

Symmetry rule: The same active catastrophe set and weights MUST be applied across all compared options in the run.

PCC rule: The PCC MUST record whether the governance-lock profile was inactive, active by default trigger, or replaced by a stricter governed profile.

D.2 Mandatory Tail Scenario Categories (Canonical)

The scenario set MUST include the following tail categories unless explicitly justified as not plausible. Governance / lock-in catastrophe is a mandatory tail category if and only if the governance-lock-in extension profile in D.1A is active; otherwise it is recommended but not mandatory. When that profile is active, the scenario set MUST include at least one governance-lock-in tail category with category probability floor p_floor ≥ 0.02.

- Pandemic/biological disruption: Disease outbreak with significant mortality/morbidity
- Climate tipping cascade: Triggering of climate tipping points with cascading effects
- Financial system collapse: Systemic failure of financial infrastructure
- Major conflict escalation: Armed conflict with regional or global implications
- Critical infrastructure failure: Failure of essential services (power, water, communications)

Governance / lock-in catastrophe: Surveillance normalization, durable recourse collapse, information-ecosystem capture, irreversible dependency lock-in, or concentrated arbitrary control over critical governance or infrastructure functions

D.3 Probability Floor (Canonical)

Minimum category probability floor: p_floor ≥ 0.02

Meaning: The sum of probabilities of scenarios in each mandatory category must be ≥ 0.02 unless explicitly justified as implausible for the specific decision context per Section 8.3.2.

D.4 Scenario Set Size Requirements (Canonical)

Tier	TRC Required?	Minimum
Tier 1	Optional (qualitative screen)	N/A
Tier 2	Required when catastrophe relevance plausible	≥5 minimum
Tier 3	Required	≥20 minimum
Tier 4	Design target only	N/A

D.5 Scenario Definition Template (Normative Minimum Fields)

Each scenario record in PCC MUST include:

Field	Description
Scenario ID	Unique identifier within the run
Name	Descriptive name

Field	Description
Category	One of the mandatory categories or “baseline/other”
Narrative	2-5 sentences describing the scenario
Time horizon	Timing assumptions
Key stressors	What breaks, how, and cascading effects
Parameterization	What changes in impacts/likelihoods under this scenario
Probability p_s	Assigned probability with provenance
Provenance	Data source, model, expert elicitation, or governance prior

D.6 Loss Construction (Bounded-Impact Mode, Tier 1-3 Canon)

Normative stream binding (Design A): TRC MUST be computed from the Base stream (I_prop_base). TRC MUST NOT use Welfare-stream (sentience-weighted) impacts.

For each scenario s:

$L(a,s) = \Sigma\_\{c \in C\_cat\}\ \omega\_c \times (-I\_prop\_base,c(a \mid s))^+$

with $\Sigma\_\{c \in C\_cat\}\ \omega\_c = 1$.

Default catastrophe weights: Uniform over C_cat unless otherwise governed.

D.7 Discrete CVaR Algorithm (Canonical; Tier 1-3)

Inputs: losses L(a, s1), …, L(a, sn); probabilities p1, …, pn; tail level α.

1. Let β := 1 − α.

2. Sort scenarios by loss descending: L_(1) ≥ L_(2) ≥ ⋯ ≥ L_(n), carrying the aligned probabilities p_(i).

3. Let k* be the smallest index such that Σ_{i=1}^{k*} p_(i) ≥ β.

4. Define P_{k*−1} := Σ_{i=1}^{k*−1} p_(i) (and P_0 := 0).

5. Define δ := β − P_{k*−1}.

6. Compute:

CVaR_α[L(a)] = (1/β) · (Σ_{i=1}^{k*−1} p_(i) L_(i) + δ · L_(k*)).

Audit: PCC includes the sorted table, k*, P_{k*−1}, δ, and CVaR_α.

(Here L_(i) is shorthand for L(a, s_(i)).)

D.8 TRC Corridor Check (Canonical)

TRC(a) = TRUE if and only if CVaR_α[L(a)] ≤ τ_TRC

D.9 Default TRC Parameters by Context (Canonical Defaults)

Context	α (tail level)	τ_TRC (threshold)
Personal	0.90	0.30
Organizational	0.95	0.20
Reversible policy	0.95	0.15
Irreversible policy	0.99	0.10
Existential risk	0.999	0.05

D.10 Tier 2 Strict TRC Rule (Canonical)

If catastrophe relevance is plausible (per Section 8.1.1), Tier 2 MUST:

- Compute TRC using this appendix
- Use ≥5 scenarios minimum
- Include mandatory tail categories with p_floor ≥ 0.02 per category unless explicitly implausible with documented justification per Section 8.3.2
- Record the full scenario table and CVaR computation in PCC

D.11 TRC Fallback Procedure (Canonical)

If A_NCRC ≠ ∅ but A_adm = ∅:

- Rank all A_NCRC options by CVaR_α ascending
- Select the option with minimal CVaR_α
- Require a time-bound risk mitigation plan
- Require one-tier-higher approval
- Record TRC_FALLBACK_INVOKED in PCC with deficit (CVaR − τ_TRC)

End Appendix D.

APPENDIX E: UCI OPERATIONALIZATION PACK (TIER 1-3)

Appendix E is normative for the UCI interface and Tier 3 constraints; indicator families are canonical guidance unless a deployment supplies validated instruments.

Operational maturity note (Normative clarification). Tier-3 conformance requires declared structural indicator families and structural independence; it does not imply universal calibration. Deployments SHOULD publish a provisional indicator companion pack when using local instruments, including sources, normalization rule, and basic reliability notes.

E.1 UCI Component Definitions (Canonical)

Component	Symbol	Description
Cohesion	H_u	Internal connectivity, trust, shared identity, conflict resolution capacity
Flow	F_u	Coordination throughput, information fidelity, resource allocation efficiency
Resilience	R_u	Redundancy, robustness, recovery speed, adaptive capacity
Equity	E_u	Fair distribution of burdens/benefits, voice representation, inclusion

E.2 UCI Formula (Canonical)

$UCI_u = \alpha_H \times H_u + \alpha_F \times F_u + \alpha_R \times R_u + \alpha_E \times E_u$

Default: $\alpha_H = \alpha_F = \alpha_R = \alpha_E = 0.25$.

Special case for Self (U_1): $E_1 := 1$ by definition. The equity component is fixed at 1 for Self because equity-as-distribution does not apply within a single individual. This provides a neutral, non-penalizing identity value.

E.3 Structural Independence Rule (Tier 3 Binding)

UCI MUST be computed from structural/process indicators distinct from welfare indicators used for RLS.

At Tier 3:

- Deriving UCI from welfare-cell impacts is PROHIBITED
- If structural indicators are unavailable, UCI is treated as unavailable (see E.7)

E.4 Indicator Families by Union (Canonical Guidance)

U_1 Self:

- Cohesion: Psychological integration, goal coherence, self-trust, internal conflict resolution
- Flow: Task execution reliability, attention stability, cognitive throughput proxies
- Resilience: Stress recovery, adaptive coping, flexibility under change
- Equity: $E_1 := 1$ by default (not measured)

U_2 Household:

- Cohesion: Relationship quality, conflict resolution, trust among members

- Flow: Resource pooling efficiency, coordination routines, decision-making speed
- Resilience: Emergency preparedness, support redundancy, recovery from shocks
- Equity: Fair burden-sharing, voice parity, caregiving distribution fairness

U_3 Community:

- Cohesion: Social capital, trust indices, network density, belonging measures
- Flow: Collective action capacity, coordination lag, information spread fidelity
- Resilience: Mutual aid redundancy, disaster response capacity, recovery history
- Equity: Inclusion of marginalized groups, access parity, procedural fairness in local governance

U_4 Organization:

- Cohesion: Culture trust indices, turnover stability, safety culture, shared purpose
- Flow: Process throughput, coordination efficiency, error rates, execution reliability
- Resilience: Redundancy, continuity planning, incident response maturity
- Equity: Pay fairness, promotion parity, grievance procedures, representation

U_5 Polity:

- Cohesion: Institutional trust, social cohesion indices, legitimacy measures
- Flow: Governance effectiveness, service delivery reliability, coordination speed
- Resilience: Crisis response capacity, redundancy of critical systems
- Equity: Rule of law parity, civil rights access parity, representation integrity

U_6 Humanity/CMIU:

- Cohesion: Cross-polity cooperation capacity, treaty adherence norms
- Flow: Global coordination throughput, information-sharing integrity
- Resilience: Global response capacity to pandemics/climate/conflict
- Equity: Burden-sharing fairness, inclusion of vulnerable polities/populations

U_7 Biosphere:

- Cohesion: Ecosystem connectivity, biodiversity integrity, trophic network stability
- Flow: Nutrient cycling integrity, carbon sequestration capacity, hydrological cycle stability
- Resilience: Recovery capacity, redundancy of functional species, anti-fragility markers
- Equity: Use distributional ecosystem integrity proxies when a governed instrument exists. If not operationalized, set E7 := 1 (neutral identity value) and record E7_METHOD_FIXED_1_DEFAULT in PCC.

E.5 ΔUCI Computation (Canonical)

ΔUCI_u(a) = UCI_u(a) − UCI_u(baseline)

E.6 Prospective UCI Estimation (Tier 3) (Normative Constraints)

Tier 3 requires prospective (ex ante) estimation of UCI changes from structural indicators, not from welfare impacts. The PCC must:

- Identify structural indicators used
- Specify baseline values
- Specify predicted changes under each option
- Specify normalization to [0,1] for each component
- Compute UCI_u and ΔUCI_u using E.2 and E.5

E.7 UCI Unavailability Rule (Tier 3) (Normative)

If structural indicators are unavailable such that UCI cannot be computed without violating E.3:

- UCI MUST be treated as unavailable for tie-break purposes
- If RLS lead is non-decisive, decision MUST escalate for more data/higher tier OR record a judgment call with label JUDGMENT_CALL_UCI_UNAVAILABLE and a monitoring plan
- Any welfare-derived UCI proxy MUST NOT be used to claim Tier 3 compliance

E.7A UCI/HOI Dual-Use Reporting Interface (Normative for Tier 3)

Purpose. UCI and HOI may serve as structural evidence in two distinct authority roles. Tier 3 implementations MUST record which role is being used.

Required fields: structural_indicator_id; indicator_family; containing_scope_affected; evidence_status; measurement_readiness_level; containment_relevance; containment_threshold_declared; containment_disposition; residual_tiebreak_relevance; duplicate_use_note; reviewer_disposition; audit_flag_if_any.

Rule. If containment_relevance is REQUIRED or THRESHOLD_CROSSED, the indicator SHALL be evaluated inside CSV before RLS. If the same indicator is later cited in residual tie-break, the PCC MUST state that gate-level containment disposition was already resolved and that residual use is limited to non-decisive structural preference, monitoring, or hollowing documentation.

Prohibited use. An implementation SHALL NOT move a gate-relevant structural indicator to residual tie-break in order to avoid containment failure, evidence unavailability, or escalation.

E.8 Minimal Measurement Protocol (Canonical Guidance)

For each UCI component, the PCC SHOULD document: indicator(s) used; data source; normalization method; expected directionality; uncertainty or reliability notes; and review cadence.

- Indicator(s) used
- Data source
- Normalization method
- Expected directionality
- Uncertainty or reliability notes
- Review cadence

E.8A Provisional UCI Measurement Protocol for Tier 3 pilots (Canonical Guidance)

When a deployment claims Tier 3 run-level conformance and uses UCI operationally, each UCI component SHOULD be built from one to three structurally independent indicators per relevant scope, normalized to [0,1] under a declared rule, and aggregated by a declared within-component operator (default arithmetic mean unless another operator is justified).

Staleness and failure handling. If a required structural indicator is stale, contested, or unavailable such that structural independence cannot be preserved, the affected UCI component SHALL be treated as unavailable and the run SHALL follow the unavailability rule in E.7 rather than substituting welfare-derived proxies.

Status note (Normative clarification). Tier 3 run-level conformance claims are distinct from framework-level empirical operational-readiness claims. The current release line specifies how provisional UCI constructions are to be documented and constrained, but it does not claim that universally validated UCI instruments already exist.

E.8B Tier 3 Starter Indicator Pack (Canonical Guidance)

The following starter indicators MAY be used as provisional structural indicator families when no validated domain-specific pack exists. They do not constitute final empirical calibration, but they provide a minimum auditable starter surface.

U3 Community

Cohesion: local trust survey or equivalent community-trust instrument

Flow: coordination-response latency or meeting-to-action completion rate

Resilience: mutual-aid activation readiness or recovery-from-disruption measure

Equity: access-to-participation parity or grievance-resolution parity

U4 Organization

Cohesion: psychological-safety / trust instrument or equivalent (e.g., Edmondson, 1999)

Flow: process throughput or median handoff completion time

Resilience: incident recovery time or continuity readiness metric

Equity: promotion / burden / grievance parity indicator

U5 Polity

Cohesion: institutional trust or legitimacy measure

Flow: service-delivery or review-cycle timeliness measure

Resilience: emergency response readiness or continuity-of-governance measure

Equity: appeal access parity or due-process access parity measure

Minimum handling rules

Normalization to [0,1] MUST be declared in the PCC.

Data older than 12 months MUST be marked stale unless a stricter domain rule is declared.

If more than one required indicator family for a used scope is missing or stale, the affected UCI component SHALL be treated as unavailable.

A worked example using this starter pack SHOULD be included in any Tier 3 deployment companion pack.

E.9 Measurement Guidance Expansion (Canonical Guidance)

For publication and Tier 3 preparation, each deployment SHOULD maintain a short UCI measurement note per component and scope covering: operational definition, indicator family chosen, normalization rule, expected update cadence, minimum acceptable data quality, independence check, and failure handling when indicators are missing or stale. Where validated instruments do not yet exist, the PCC MUST label the construction as provisional and MUST NOT present the resulting UCI value as empirically calibrated. This guidance is intended to standardize measurement practice without pretending that UCI instruments are already universally validated.

End Appendix E.

APPENDIX F: FAILURE MODES AND ANTI-GAMING CONTROLS (v11.4)

Appendix F is normative for identified gaming vectors and required mitigations. Informative note. This appendix is carried forward substantively unchanged from the v9.0 line unless otherwise stated.

F.1 Identified Gaming Vectors (Canonical)

Vector	Description	Structural Mitigation
Option set manipulation	Excluding viable alternatives so preferred option "wins"	Minimum option requirement (≥2); independent challenger in emergencies; document option generation process

Vector	Description	Structural Mitigation
Weight capture	Steering HDW to underweight certain unions/dimensions	Constitutional floors; supermajority locks near floors; transparency ledger; conflict disclosure
Emergency abuse	Declaring emergency to bypass rights	Challenger requirement; remediation plan; public disclosure/audit triggers
Confidence inflation	Overstating confidence to increase impact weight	c_k bounded [0.1, 1]; require evidence notes; sensitivity recommended
Horizon manipulation	Choosing time horizons strategically	Logarithmic temporal weighting $\tau(t)$; declare T_{ref}; challenge horizon assumptions
Unknown-as-neutral	Leaving cells blank to avoid negatives	Ignorance penalty phantom instance rule; audit flag if non-maskable cell missing

F.2 Structural Anti-Gaming Features (Canonical)

Non-Compensatory Architecture:

- NCRC cannot be overridden by welfare gains
- TRC cannot be overridden by expected value
- Containment cannot be bypassed by RLS optimization

Transparency Requirements:

- All parameters recorded in PCC
- Registry hashing prevents silent modification (Tier 3+)
- Public rationale (5SPR) required

Audit Infrastructure:

- Random audit lottery (recommended 5% of PCCs at Tier 3)
- Red-team protocols for high-stakes decisions
- Configuration drift detection

Learning Accountability:

- NCAR loop tracks prediction accuracy
- Systematic errors trigger parameter review
- Historical PCCs preserved for accountability

F.3 Audit Flag Taxonomy (Informative pointer; non-authoritative) Authoritative audit flags are specified in Section 14.3 and duplicated verbatim in Appendix H.3. Appendix F provides explanatory context and anti-gaming rationale only; it does not define, modify, or enumerate the canonical audit-flag set. For the complete and controlling list (tokens, triggers, required actions, severities), see Section 14.3 / Appendix H.3.

F.4 Red-Team and Audit Lottery (Tier 3 Recommended)

Deployments SHOULD adopt:

- Random audit lottery: for example, 5% of PCCs selected for independent review
- Red-team scenario additions: Adversarial scenario injection for TRC testing
- Kernel perturbation stress tests: Systematic edge perturbation

F.5 Boundary and Threshold Gaming (Normative anti-gaming clarification)

RippleLogic does not assume adversaries will only game welfare scores. Because NCRC, TRC, CSV, and PLSS create hard feasibility and escalation conditions, adversarial pressure may shift toward subgroup definitions, rights-threshold placement, scenario-library construction, probability assignment, containment-scope framing, PLSS local-scope declarations, evidence-sufficiency claims, and maturity classifications.

PCC reviewers SHALL inspect these boundary surfaces for gerrymandering, selective omission, motivated uncertainty, threshold laundering, scope narrowing, and evidence laundering. Where any of these surfaces is material to a gate verdict, escalation, or refusal decision, the run SHALL disclose the contested boundary choice, the evidentiary basis, reviewer disagreement if any, and the sensitivity of the verdict to plausible alternative boundary definitions.

Goodhart pressure is not eliminated by RippleLogic; it is relocated and exposed. The system becomes stronger only when boundary definitions, threshold settings, probability libraries, stakeholder maps, and evidence sufficiency judgments are inspectable, contestable, and preserved in the PCC or companion audit record.

End Appendix F.

APPENDIX G: SGP INTEGRATION BINDING (NORMATIVE INTERFACE)

Appendix G is normative for the RippleLogic ↔️ SGP interface.

Interim precautionary protection trigger. If credible SGP evidence, taxon evidence, stakeholder evidence, or artificial-entity evidence identifies a plausible protected non-human moral patient before formal rights-coverage expansion is complete, the run MUST record PRECAUTIONARY_PROTECTION_REVIEW. Absence of an existing C_r mapping MUST NOT be treated as permission for harmful treatment. The reviewer may route the issue through

Rights Floor review, SGP review, CSV review, or claim narrowing, but the uncertainty must remain visible and non-compensable where gate-material.

G.1 Binding Purpose

This appendix defines the only permissible interface by which RippleLogic consumes Sentience Gradient Protocol (SGP) outputs. RippleLogic does not define SGP internals.

SGP Version Pin (Normative): RippleLogic v11.4 pins Sentience Gradient Protocol SGP v6.4 as the authoritative sentience / moral-status specification for any run claiming RippleLogic v11.4 compliance.

ProofPack dependency pinning (Normative): Any ProofPack claiming replayability MUST include the exact SGP v6.4 artifact (DOCX, PDF, or equivalent canonical byte-stream record) as a bundled file, or MUST reference a registry hash that uniquely identifies the SGP v6.4 artifact. If SGP is not bundled or hash-pinned, ProofPack runs MUST declare SGP as an external dependency and restrict test vectors to cases where the required SGP scalar(s) are provided as explicit inputs.

Interface-boundary rule (Normative): RippleLogic consumes only the interface-visible outputs named in this appendix. RippleLogic MUST NOT recompute SGP internals, MUST NOT infer plateau status from raw pillar scores or raw normalized measurement, and MUST NOT substitute raw measured values for effective canonical or rights-of-protection values.

G.2 Canonical Scalar Mapping

Given entity E, SGP v6.4 may expose the following interface-visible values:

A(E): Awareness pillar score in [0,100]

B(E): Agency pillar score in [0,100]

C(E): Union Participation pillar score in [0,100]

SG_measured_norm(E): raw normalized scalar in [0,1] produced by canonical UBSE before plateau assignment and non-plateau capping, if present in the SGP evaluation record.

Plateau(E): constitutional status flag in {0,1}, indicating whether E has Full Rights Plateau status under the Human Plateau Rule, governed individual plateau review, or explicit governed class-level plateau assignment, if present in the SGP evaluation record.

SG_norm(E): authoritative effective canonical scalar in [0,1]. Under SGP v6.4, this is shorthand for SG_individual_norm(E) unless another section of SGP explicitly names SG_patient_norm(E).

SG_patient_norm(E): authoritative rights-of-protection scalar in [0,1], if provided by the SGP record. This scalar incorporates plateau status and, where applicable, taxon or governed-class baseline logic.

Authoritative binding (interface-only): Any SGP scalar consumed by RippleLogic, including SG_norm(E) and where available SG_patient_norm(E), is consumed exactly as provided by the pinned SGP artifact and MUST NOT be recomputed inside RippleLogic.

Audit-only visibility rule: A(E), B(E), C(E), SG_measured_norm(E), and Plateau(E) MAY be logged for interpretability and auditing if they are present in the SGP record. They do not define SG_norm(E) or SG_patient_norm(E) within RippleLogic and MUST NOT override them.

Union Participation interface note. C(E), where present, refers to the SGP Union Participation pillar: responsibility-bearing participation across affected unions. It is audit-visible for interpretation and review, but RippleLogic MUST NOT use C(E) directly to reduce SG_patient_norm(E), weaken human plateau protections, infer governance authority, or override the authoritative SGP scalar/status outputs. Weak or hard-to-observe Union Participation evidence may limit participation, plateau, or stewardship claims only as governed by SGP; it does not by itself erase credible welfare-bearing protection evidence.

Plateau exclusivity recognition (Normative): RippleLogic SHALL recognize the current pinned SGP rule that 1.0 / 100 is reserved for plateau entities only. If SG_measured_norm(E) = 1.0 while Plateau(E) = 0, RippleLogic MUST treat that value as raw audit information only; it MUST continue to use SG_norm(E) and, where available, SG_patient_norm(E) for normative computation.

G.3 Human Plateau Rule and Plateau Recognition (Non-Overridable)

For any human person H:

SG_norm(H) := 1.0

This assignment is:

• Independent of measurement noise

• Independent of disability status

• Independent of partial observability

• Non-overridable by any SGP scoring outcome

• MUST NOT be weakened by any weighting scheme

Clarification: No SGP-derived scalar, including SG_patient_norm(E), may be used to reduce human rights protections below plateau.

Human plateau / CMIU calibration note (Normative clarification). The human assignment SG_norm(H) := 1.0 is not a claim that every individual human exhibits maximal measured capacity at every moment. It is a non-downgrade rights boundary for all human persons and a recognition that humanity is the current evidenced plateau-class reality-managing community within the CMIU. Equivalent plateau recognition for non-human biological or artificial entities remains open only through authoritative SGP plateau-extension procedures and does not by itself grant specific governance authority.

General plateau recognition (Normative): Where an SGP v6.4 record reports Plateau(E) = 1 for a non-human or artificial stakeholder, RippleLogic SHALL recognize the entity as plateau-bearing for rights-of-protection purposes and SHALL treat SG_norm(E) = 1.0 and SG_patient_norm(E) = 1.0 where such values are provided by the SGP record.

Plateau-bearing fallback rule (Normative). If Plateau(E) = 1 is present in an authoritative SGP v6.4 record but SG_patient_norm(E) is omitted, RippleLogic SHALL treat the entity as plateau-bearing and use SG_norm(E) = 1.0 for welfare-stream purposes.

Non-inference rule: RippleLogic MUST NOT infer Plateau(E) = 1 from SG_measured_norm(E), from pillar values, or from taxon evidence alone. Plateau status exists only when explicitly reported by the authoritative SGP record or by a governed class-level plateau assignment record recognized through SGP.

G.4 Permitted Usage in RippleLogic

Permitted:

Sentience multiplier usage (Normative): s_k may be set via the SGP binding interface only for the welfare stream used to compute RLS. For admissibility layers (NCRC, TRC, CSV), the base stream MUST fix s_k := 1.0 for all instances.

Scalar selection rule (Normative):
• If SG_patient_norm(E) is available from the authoritative SGP record, RippleLogic SHOULD use SG_patient_norm(E) as the welfare scalar for rights-of-protection-sensitive welfare weighting.
• Else RippleLogic SHOULD use SG_norm(E).
• SG_measured_norm(E) MUST NOT be used as the direct welfare scalar in place of SG_norm(E) or SG_patient_norm(E).
• For any human person H: implementations MUST set s_k := 1.0 (Human Plateau Rule), regardless of any other value.
• SGP-derived scalars MUST NOT be used to change admissibility outcomes under NCRC, TRC, or CSV.

G.5 Prohibited Usage and Non-Inference Rules

Prohibited:

• Using SG-derived scalars to change admissibility outcomes under NCRC, TRC, or CSV.

• Recomputing SG_norm(E), SG_patient_norm(E), or Plateau(E) inside RippleLogic from pillar values, raw scores, external heuristics, or unpinned auxiliary logic.

• Treating SG_measured_norm(E) as equivalent to SG_norm(E) when the SGP record distinguishes them.

• Using absence of SG_patient_norm(E) to infer absence of rights-of-protection significance.

• Treating high scalar values as automatic governance authority, office-holding qualification, or stewardship certification.

Plateau non-inference rule (Normative): Class-level plateau extension, where present in SGP, SHALL be explicit. RippleLogic MUST NOT infer class-level plateau from high taxon evidence, informal scientific consensus, or unmanaged metadata.

G.6 Governance Authority Non-Inference and CMIU Standing

SGP rights are rights-of-protection. Plateau status marks full inviolable standing. Plateau may also carry standing in the Collective Managing Intelligence Union where SGP specifies this. Specific consequential governance roles remain separate stewardship questions inside RippleLogic.

Therefore:

• plateau status MAY establish full moral standing and CMIU standing where explicitly reported by SGP

• plateau status MUST NOT be treated as automatic qualification for every consequential governance role

• specific governance authority remains subject to RippleLogic stewardship constraints, domain-fit, influence transparency, non-domination posture, revocability, and any other applicable authority controls

This rule prevents conflation of moral-status recognition with office-holding authority while preserving the plateau semantics supplied by SGP v6.4.

G.7 PCC Logging and Dependency Recording

When SGP is used in a RippleLogic run, the PCC MUST record:

• SGP version identifier

• SGP artifact filename + hash, or external dependency registry reference

• scalar type actually used for welfare weighting: SG_patient_norm or SG_norm

• SG_norm(E) value for each affected SGP-scored stakeholder instance

• SG_patient_norm(E), if available and used

• SGP evidence class / evidence-maturity token, including Biological Evaluation Maturity (BEM) token where applicable; confidence; and registry reference if available in the SGP record

If present in the authoritative SGP record, the PCC SHOULD also record:

• A(E), B(E), C(E)

• SG_measured_norm(E)

• Plateau(E)

• taxon or governed-class baseline identifier, if any

• plateau review opinion record ID, class-level plateau record ID, or reviewer-dissent record ID, if any

Audit interpretation rule: SG_measured_norm(E) and Plateau(E) are logged for transparency and replayability only. Normative computation in RippleLogic SHALL still follow G.4.

Missing-data rule: If the run claims SGP-aware welfare weighting but lacks the scalar actually used, the PCC is INVALID.

G.8 Rights Expansion and Class-Level Plateau Handling

When an SGP evaluation establishes that a non-human stakeholder class warrants rights-of-protection, RippleLogic rights expansion proceeds through the governed process already defined for coverage-set and subgroup updates. This includes:

• SGP determination or class-level rights-of-protection determination

• coverage set update in RippleLogic rights coverage sets C_r as applicable

• subgroup protocol update in relevant G_{u,d} sets

• version increment for the relevant rights-expansion record

• PCC documentation in all subsequent affected runs

• preservation of historical records

Class-level plateau handling (Normative clarification): If SGP v6.4 issues an explicit governed class-level plateau assignment, RippleLogic SHALL treat members of that class as plateau-bearing only under the membership rule specified in the authoritative SGP class-level plateau record. Plateau-bearing status MUST NOT be inferred informally from taxon evidence alone.

Entity-level plateau handling (Normative clarification): If SGP v6.4 issues an individual plateau assignment, RippleLogic SHALL recognize plateau for that specific entity only, subject to the identity and continuity conditions recorded by SGP.

Scope rule: Rights expansion under this appendix modifies RippleLogic rights coverage and subgroup treatment. It does not alter RippleLogic's separate stewardship and authority-gating requirements for consequential roles.

End Appendix G.

APPENDIX H: PCC TEMPLATE AND AUDIT FLAGS CANON PACK (TIER 2-3)

H.0.4 No Silent Remainder Rule (Normative)

Required fields may be unknown, estimated, assumed, not applicable, or blank by design, but they may not be silently blank. A required field in a claim-bearing run MUST carry one of the following dispositions: VALUE, ESTIMATE, ASSUMPTION, DECLARED_UNKNOWN, NOT_APPLICABLE, BLANK_BY_DESIGN, or MISSING_UNDECLARED.

MISSING_UNDECLARED on any required field creates SILENT_REMAINDER_INVALID with severity INVALID. DECLARED_UNKNOWN on a gate-critical field, including rights-floor cells, catastrophe-relevant cells, material Containment fields, SVC subconditions, authority-selection fields, or public conformance fields, requires claim narrowing, evidence collection, escalation, or refusal. It MUST NOT default to pass or zero.

The PCC SHOULD record silent_remainder_count. Claim-bearing Tier 2 and Tier 3 runs SHOULD require silent_remainder_count = 0 before public conformance, deterministic-selection, deployment-readiness, or ProofPack-prep claims are made.

H.0 Carried-forward v10.8 Refusal, Closure, Category-Collapse, and CBI Fields

H.0.1 RefusalRecord field (Normative)

Field name: PCC.RefusalRecord

Required when: the run enters Refusal under Section 4.9.

Minimum fields: refusal_id; trigger_code R-1, R-2, R-3, R-4, R-5, or R-6; trigger_description; options_affected; scopes_affected; dimensions_affected; rights_cells_affected; catastrophe_cells_affected; containment_indicators_affected; evidence_basis; provenance_links; no_action_baseline_status; escalation_route; provisional_action_status; re_run_conditions; reviewer_signature_or_role.

Validation rule. If Refusal is triggered and PCC.RefusalRecord is absent, the PCC is INVALID.

H.0.2 ComputationalClosure field (Normative)

Field name: PCC.ComputationalClosure

Required for: all Tier 2 and Tier 3 runs; recommended for Tier 1.

Minimum fields: scopes_evaluated; scopes_guardrail_only; scopes_outside_computation; time_horizon_evaluated; time_horizon_outside_computation; kernel_mode; propagation_edges_included; propagation_edges_excluded; externalities_excluded; claim_boundary.

Validation rule. If the run makes claims beyond the declared ComputationalClosure boundary, set audit_flag CLAIM_EXCEEDS_CLOSURE_BOUNDARY and mark the PCC INVALID for that claim.

H.0.2A RealityGroundingRecord field (Normative)

Field name: PCC.RealityGroundingRecord

Required for: Tier 2 runs where any stakeholder-discovery, rights, tail-risk, containment, or authority-selection trigger holds; all Tier 3 runs; and any claim-bearing run that depends on a contested material reality claim.

Minimum fields: RealitySurfaceDescription; MaterialRealityClaims; EvidenceTrace; DeclaredUnknowns; RealityGroundingStatus; FalsificationOrRevisionTrigger; MonitoringPath; VehicleContentEffectNotes where social or institutional constructs are material; SurfaceTerritoryLimitations; ReviewerNotes.

Validation rule. If a material claim has INSUFFICIENT_GROUNDING (or legacy INSUFFICIENT_CONTACT) and affects NCRC, TRC, CSV, deterministic selection, or authority-selection disclosure, the run MUST downgrade the claim, escalate, or refuse the stronger decision state. If PCC.RealityGroundingRecord is missing when required, set audit_flag REALITY_GROUNDING_RECORD_MISSING (legacy alias REALITY_CONTACT_RECORD_MISSING). If declared RealityGroundingStatus exceeds what EvidenceTrace substantiates, set audit_flag REALITY_GROUNDING_STATUS_UNSUPPORTED (legacy alias REALITY_CONTACT_STATUS_UNSUPPORTED).

Claim-boundary rule. At Tier 2-3, a RealityGroundingRecord provides traceability and reviewer-checkable evidence discipline; it is not itself verified reality grounding. Tier 4/ProofPack claims require RealityGroundingStatus to be independently re-derivable from EvidenceTrace, referenced evidence objects, and hash-pinned replay materials.

H.0.2B PCC.CategoryGroundingRecord field (Normative)

Field name: PCC.CategoryGroundingRecord

Required when: Section 2.1C / Appendix AH trigger conditions hold for a claim-bearing Tier 3 run, high-stakes Tier 2 run, public conformance claim, public deployment claim, deterministic-selection claim, or optimized metric whose category choice is material.

Minimum fields: material_category; category_role; primitive_basis; definition; category_discriminator; dependency_position; inclusion_boundary; exclusion_boundary; common_confusions; noncollapse_pairs; reference_structure; evidence_surface; source_authority_status; category_maturity_status; falsification_or_revision_trigger; term_refusal_condition; downstream_dependencies; semantic_debt_flag; reviewer_status; required_claim_action; and category_failure_class where applicable.

Validation rule. If Category Grounding is required and PCC.CategoryGroundingRecord is missing, set audit_flag CATEGORY_GROUNDING_RECORD_MISSING and block deterministic framework selection, conformance, public-deployment, and optimized-metric claims until the claim is narrowed, escalated, marked sensitivity-only / monitor-only, or refused. If category_maturity_status is INSUFFICIENT_GROUNDING, LEGACY_UNREVIEWED, or CONTESTED beyond the claim boundary, the validator MUST reject unqualified deterministic, conformance, public-deployment, and public assurance claims for the affected decision state.

H.0.3 CategoryCollapseSelfTest field (Normative)

Field name: PCC.CategoryCollapseSelfTest

Required for: Tier 2 and Tier 3 runs; recommended for Tier 1.

The field SHALL answer each of the following as PASS, FAIL, or IND:

1. Did the run rely on traceability where admissibility was required?

2. Did the run substitute compliance, approval, or certification for correctness?

3. Did the run treat plausibility, fluency, rhetoric, or institutional authority as evidence?

4. Did the run conflate intelligence, capability, or agency with welfare-bearing status?

5. Did the run conflate moral status, protection, or sentience with governance authority?

6. Did the run allow SGP-derived welfare scalars to alter NCRC, TRC, or CSV outcomes?

7. Did the run treat mathematical consistency as empirical validation?

8. Did the run use meta-union, AIU, cosmic, or universal language as a computational override?

Validation rule. Any FAIL SHALL trigger CATEGORY_COLLAPSE_FLAG. If the failure affects admissibility, selectability, or selection, the run SHALL enter Refusal under Section 4.9.

Appendix H is normative for PCC fields and audit flags at Tier 2-3.

H.1 PCC Schema (Human-Readable)

==

PROVENANCE AND COMPLIANCE CERTIFICATE (PCC)

RippleLogic Framework v11.4

==

HEADER (REQUIRED Tier 2-3)

Decision ID: [unique identifier]

Timestamp (UTC): [YYYY-MM-DDThh:mm:ssZ]

Decision Owner(s): [names/roles]

Analyst(s): [names/roles]

Reviewer (Tier 3): [name/role]

Spec Version: RippleLogic v11.4

Implementation Tier: [1 | 2 | 3]

Tier Claim Disposition (Tier 2-3 REQUIRED)

PCC.TierClaimDisposition: [TIER_CLAIMED_AND_MET | TIER_ATTEMPTED_NOT_ACHIEVED_DOWNGRADED]

If PCC.TierClaimDisposition = TIER_ATTEMPTED_NOT_ACHIEVED_DOWNGRADED, PCC MUST specify:

- PCC.TierAttempted: [2 | 3]

- PCC.TierAchieved: [1 | 2]

Tier Downgrade Flags: [TIER_DOWNGRADED_CONTAINMENT = TRUE/FALSE] (required if any gate is unavailable and a Tier 2 downgrade disposition is used).

Constraint (Normative). If TIER_DOWNGRADED_CONTAINMENT = TRUE, then PCC.TierClaimDisposition MUST be TIER_ATTEMPTED_NOT_ACHIEVED_DOWNGRADED and PCC.TierAchieved MUST be 2.

Propagation Mode: [NONE | QUICK]

Kernel Profile: [NONE | profile name + KQS + evidence note]

If Propagation Mode = QUICK, PCC MUST include:

PCC.Kernel.EntryAbsMax: [float in canonical scale]
PCC.Kernel.RowSumAbsMax: [float in canonical scale] (this is $\|K\|_\infty$)
PCC.Kernel.Bounds: [κ_max, ρ_max used]
PCC.Kernel.StabilityPass: [TRUE/FALSE]

Weight Profile: [HDW | PLSS_LOCAL | Declared interim | Uniform]

PCC.BaselineType_Welfare: [STATUS_QUO | FLOOR_REFERENCE | OTHER_DECLARED]
PCC.BaselineType_Gates: [FLOOR_REFERENCE] (MUST be FLOOR_REFERENCE for all cells in $C\_r \cup C\_cat$)
PCC.BaselineReference_Gates: [indicator definitions + target/floor levels + sources + conversion assumptions]
PCC.BaselineFloorReferenceCellsInWelfare: [list of rights/catastrophe cells that remain in welfare scoring under the non-maskable persistence rule and therefore use floor-reference baseline in welfare scoring]
PCC.BaselineDualExceptionBlock: [OPTIONAL structured list by cell or cell-class: welfare

baseline used, gate baseline fixed at FLOOR_REFERENCE, rationale, and reviewer sign-off if required]
If PCC.BaselineType_Welfare = OTHER_DECLARED, PCC MUST specify: PCC.BaselineReference_Welfare.

TRC Context Class: [Personal | Organizational | Reversible |

Irreversible | Existential]

Content Hash: [optional Tier 2; required Tier 3; SHA-256]

STEWARDSHIP BLOCK (CONDITIONAL; REQUIRED when Stw_req=1)

PCC.Stewardship.Stw_req: [TRUE | FALSE]

If Stw_req = FALSE, all fields below MAY be omitted.

PCC.Stewardship.DBD.Scope: [list of out-of-scope authority domains]

PCC.Stewardship.DBD.RefusalRule: [refusal/deferral rule identifier]

PCC.Stewardship.APM.AfterActionAuthorship: [0.0–1.0]

PCC.Stewardship.APM.Rationale: [artifact-anchored justification; MUST reference >=2 of 4 APM anchors per Appendix N.4]

PCC.Stewardship.APM.BaselineAuthorship: [0.0–1.0] (OPTIONAL)

PCC.Stewardship.IRB.InterventionCap: [declared caps]

PCC.Stewardship.IRB.Thresholds: [intervene-only-past-harm-point rule]

PCC.Stewardship.InfluenceLedgerRef: [pointer | “NONE” + justification]

PCC.Stewardship.DelayFlag: [TRUE | FALSE] (OPTIONAL)

SCOPE (REQUIRED Tier 2-3)

Decision Question: [clear statement]

Time Horizon: [years]

Primary Affected Scopes (Union Scopes): [list]

Decision Boundary Notes: [what is in/out of scope; why]

Stakeholder Coverage and Mapping (Tier-governed; v11.4)

Stakeholder Coverage Index (SCI): [SCI-0 / SCI-1 / SCI-2 / SCI-3 / SCI-4]

PCC.SCI.BelowMinimumDisposition: [DOWNGRADE_TIER | RERUN_TO_MEET_MINIMUM] (REQUIRED if SCI_BELOW_MINIMUM triggers; see Section 5.0.1 and audit flag SCI_BELOW_MINIMUM).

PCC.SCI.Justification: [text] (REQUIRED if SCI < tier minimum, OR if SCI equals the tier minimum and a Next-Run Upgrade Plan is required).

Scope Coverage Declaration: Mode [FULL_SCOPE / REDUCED_SCOPE]; Active scopes [list]; Omitted scopes [list]; Omission rationales [text]; Blind-spot statement [text]; Escalation triggers that force scope expansion [list].

SCI Next-Run Upgrade Plan (REQUIRED if SCI equals the tier minimum for the claimed tier): [text; what scope expansions or stakeholder mapping improvements will be executed next run; by when].

Admissibility Attestation (REQUIRED if Mode = REDUCED_SCOPE):

PCC.AdmissibilityAttestation.NCRC_full_Cr_computed: [TRUE | FALSE]

PCC.AdmissibilityAttestation.TRC_full_Ccat_computed: [TRUE | FALSE | NA]

PCC.AdmissibilityAttestation.Containment_computed_as_applicable: [TRUE | FALSE | NA]

Notes (Normative): Reduced-scope mode SHALL NOT remove any non-maskable cells or required scope rows from admissibility computation. If any required attestation is FALSE or missing, the run MUST NOT claim tier compliance for the affected admissibility gate(s) and MUST record the relevant audit flag(s).

Stakeholder Instance Map (InstanceMap): Provide a structured list of stakeholder instances with fields: instance_id, description, role/context, exposure pathway, mapped_scope(s), and rationale. Tier 3 REQUIRED, Tier 2 REQUIRED when any SDP trigger holds (Section 2.2A).

Redundancy Handling Declaration (if any multi-scope mapping): Method [EMERGENT_SCALE_JUSTIFICATION / DEDUPLICATION / ALLOCATION]; Effect-token list (if ALLOCATION); allocation coefficients (Σ_u α_u ≤ 1 per token); disclosure of any deduplication primary scope choices.

PLSS BLOCK (CONDITIONAL; REQUIRED when PLSS is used)

PCC.PLSS.LSSDeclaration: [final Local Stakeholder Set used for the run, with reference to the instance-to-scope mapping basis]

PCC.PLSS.ProminenceVector: [declared q_u for all scopes, including sub-component basis for magnitude, probability, persistence, and propagation]

PCC.PLSS.Operator: [DEFAULT_GEOMEAN_V1 | OTHER_DECLARED]; if OTHER_DECLARED, record operator, rationale, and sensitivity note

PCC.PLSS.WeightVector: [computed w_u^PLSS for all scopes]

PCC.PLSS.ScopePostureClassification: [Prominent | Guardrail for each scope]

PCC.PLSS.EscalationTriggerAssessment: [ET-1 through ET-7 outcome, including any justification for remaining in PLSS posture]

PCC.PLSS.FallbackDeclaration: [uniform residual fallback used if all q_u = 0, or other declared fallback with justification]

MEASUREMENT-MATURITY AND DEPLOYMENT-CONTEXT BLOCK (REQUIRED when applicable; Tier 2-3)

PCC.UCI_Measurement_Readiness: [UCI-M0_UNAVAILABLE | UCI-M1_PROVISIONAL | UCI-M2_DOMAIN_SUPPORTED | UCI-M3_VALIDATED]; PCC.UCI_Indicator_Source: [named sources]; PCC.UCI_Normalization_Rule: [declared rule]; PCC.UCI_Structural_Independence_Posture: [independent | correlated-disclosed | unavailable]; PCC.UCI_Reviewer_Reliability_Status: [not-tested | internally-reviewed | IRR-tested | validated].

PCC.Kernel_Use_Posture: [K0_NONE | K1_STARTER | K2_DOMAIN_CALIBRATED | K3_BACKTESTED | K4_REGISTERED]; PCC.Kernel_Validation_Evidence: [none | source note | calibration report | backtest report | registry ID]; PCC.Kernel_Backtest_Status: [not-backtested | partial | complete]; PCC.Kernel_Error_Bounds: [declared bounds or NA].

PCC.HDW_Capture_Audit: [weight source, stakeholder map, participation quality, expert role provenance, conflict disclosures, challenger window, independent review status, captured-context limitation] (required for Tier 3 public or institutional weighting processes).

PCC.PLSS_Escalation_Register: [profile used, local stakeholder set, guardrail scopes, escalation-trigger checklist, non-escalation rationale, externality search method, prominence-sensitivity result, external challenger status, repeated-PLSS-use flag] (required when PLSS is used).

PCC.Deployment_Context_Profile: [LAB | PILOT | INSTITUTIONAL_ADVISORY | INSTITUTIONAL_OPERATIONAL | PUBLIC_POLICY | AGENTIC_EXECUTION | EMERGENCY_TRIAGE] plus evidence-sufficiency, reviewer-independence, audit-visibility, and escalation-threshold notes.

PCC.Biological_Evaluation_Maturity: [BEM-0_NOT_EVALUATED | BEM-1_TAXON_BASELINE | BEM-2_ETHOLOGICAL_GUIDANCE | BEM-3_EXPERT_REVIEWED_BATTERY | BEM-4_VALIDATED_ANNEX] when biological SGP evidence is invoked. Biological measurement immaturity SHALL NOT reduce precautionary protection where credible taxon-level evidence supports moral patienthood.

Option Set O:

Option A: [description]

Option B: [description]

Option C: [description] (if applicable)

Applicability Mask m(u,d) (RLS only):

Masked cells: [list + rationale per cell]

Non-maskable verification:

Rights cells unmasked: [PASS/FAIL]

Catastrophe cells unmasked: [PASS/FAIL]

INPUTS (REQUIRED Tier 2-3)

Uncertainty Method (Tier 2+ REQUIRED for active cells):

PCC.Uncertainty.Method: [A | B | C]
If Method = B:
PCC.Uncertainty.CellConfidenceAggregationMethod: [CCAM_MIN_V1 | CCAM_WMEAN_V1]
PCC.Uncertainty.c_cell_table: [table of c(u,d,a) for active cells, or a retrievable representation sufficient for recomputation]

Weights:

Union weights w_u: [vector or table]

Dimension weights v_d: [vector or table]

HDW parameters (if used): [λ_U, λ_D; sources]

Rights Canon:

Thresholds θ_r: [table or "current canonical"]

Coverage sets C_r: [table or "current canonical"]

Subgroup policy:

G_{u,d} method: [enumerated | conservative bound]

γ_subgroup (if used): [value]

Notes: [limitations if any]

TRC Inputs (REQUIRED Tier 2 when plausible; REQUIRED Tier 3):

Catastrophe cells C_cat: [list]

Catastrophe weights ω: [list]

α: [value]

τ_TRC: [value]

Scenario set S:

For each scenario s:

ID, category, narrative, p_s, parameterization notes

Mandatory tails check:

Categories present: [YES/NO per category]

Probability floors met: [YES/NO]

Any implausibility justifications: [YES/NO; if YES, details]

Containment Inputs (REQUIRED Tier 3):

τ_c: [value]

θ_{pos}: [value]

D_c: [value]

UCI method: [indicators used + sources]

If UCI is unavailable for required containing scopes, record CONTAINMENT_UCI_UNAVAILABLE and set PCC.Containment.UCIUnavailableDisposition to DOWNGRADE_TIER or COLLECT_DATA_RERUN (and provide a remediation timeline).

NORMATIVE INPUT REGISTER (REQUIRED when material)

PCC.NormativeInputRegister: [array; REQUIRED when any welfare-relevant state, rights-relevant state, or enabling condition materially affects admissibility, precautionary handling, selection, or tie-break outcome]

For each entry n in PCC.NormativeInputRegister, record:

NormInputID: [unique identifier]

InputType: [WELFARE_RELEVANT_STATE | RIGHTS_RELEVANT_STATE | ENABLING_CONDITION]

Materiality: [ADMISSIBILITY_MATERIAL | PRECAUTION_MATERIAL | SELECTION_MATERIAL | TIEBREAK_MATERIAL]

AffectedObjects: [cell(s), right(s), scenario(s), gate(s), structural metric(s), or decision element(s)]

StakeholderOrScopeTarget: [stakeholder / class / subgroup / instance / scope / scope-set]

EvidenceBasisType: [DIRECT_INDICATOR | IMPACT_INSTANCE_TRACE | RIGHTS_CANON_INTERPRETATION | SGP_INTERFACE | GOVERNED_PROXY | PRECAUTIONARY_DECLARATION | OTHER_DECLARED]

EvidenceReference: [citation / data source / registry ID / PCC pointer / section pointer]

GovernedInterpretationBasis: [NONE | canon section | charter rule | jurisdictional source | instrument identifier | reviewer memorandum]

PrecautionaryBasis: [NONE | explicit uncertainty rationale]

OptionSymmetryStatus: [APPLIES_TO_ALL_OPTIONS | OPTION_SPECIFIC_WITH_JUSTIFICATION]

AnalystReasonCode: [text or controlled token]

SensitivityNote: [REQUIRED if alternative treatment plausibly changes admissibility or selection]

ReviewerNote: [Tier 3 REQUIRED when Materiality = ADMISSIBILITY_MATERIAL, PRECAUTION_MATERIAL, or SELECTION_MATERIAL]

Normative note. The register exists to make reason-for-action inputs auditable when they are not already fully captured by direct impact-instance tables, the canonical rights layer, the canonical catastrophe layer, or declared structural indicators.

IMPACT ESTIMATION (REQUIRED Tier 2-3)

For each option a:

Direct impacts I_dir(u,d,a): [summary table or key cells]

Propagated impacts I_prop_base(u,d,a) and I_prop_welfare(u,d,a): [summary table or key cells]

Key impact instances (at least top contributors):

For each listed instance k:

(u,d), μ_k, r_k, t_k, ℓ_k, c_k, e_k, s_k, rationale/source

e_k disclosure (Normative). If any listed instance uses e_k ≠ 1.00, the PCC MUST also record the reason code, default-counterfactual, and whether resetting e_k to 1.00 changes admissibility or selection.

ReachBasis (REQUIRED Tier 2+): For each reach term r_k, record the basis and estimator used (allowed bases: population fraction, asset/flow fraction, or area/volume fraction; other bases MUST be explicitly defined), plus source and any conservative bounds.

Missing data:

Any UNKNOWN_IMPACT cells: [list]

Phantom instances applied: [YES/NO]

CASCADE TRACE (REQUIRED Tier 2-3)

NCRC:

For each option: PASS/FAIL

If FAIL: violated rights + v_r(a) values

Emergency Mode invoked: [YES/NO]

Challenger conducted: [YES/NO]

Challenger notes: [summary]

TRC:

Triggered (catastrophe relevance plausible): [YES/NO]

For each option:

CVaR_α: [value]

Corridor τ_TRC: [value]

PASS/FAIL

TRC fallback invoked: [YES/NO]

Containment (Tier 3 required):

For each option in A_adm:

U_pos(a): [list of unions]

For each u in U_pos: min ΔUCI among ancestors: [value]

PASS/FAIL

PCC.Containment.UCIUnavailableDisposition: [DOWNGRADE_TIER | COLLECT_DATA_RERUN | NA] (REQUIRED if CONTAINMENT_UCI_UNAVAILABLE flag is recorded).

PCC.Containment.UCIUnavailableRemediationTimeline: [text] (REQUIRED if CONTAINMENT_UCI_UNAVAILABLE flag is recorded).

A_sel: [list]

RLS Ranking:

For each option in A_sel:

RLS(a): [value]

σ_RLS(a): [value] (Tier 3 required)

Gap(top, second): [value]

δ: [value]

Decisive: [YES/NO]

Tie-break (if non-decisive):

UCI method used: [UCI-A/B/C]

Δ_UCI threshold: 0.05

Result: [winner or non-decisive]

HOI flags (if applicable): [notes]

Escalation invoked: [YES/NO]

SELECTION (REQUIRED Tier 2-3)

Selected Option: [option ID]

Selection Rationale: [brief; must reference cascade outcomes]

Monitoring Plan (NCAR):

Metrics: [list]

Review schedule: [date/interval]

Triggers for re-run: [conditions]

SENSITIVITY ANALYSIS (Tier 3 REQUIRED)

Weights perturbation: [summary; selection stable?]

Threshold perturbation: [summary; any near-miss rights?]

Kernel perturbation: [summary; flag fragile?]

Scenario perturbation: [summary; probability sensitivity]

Robustness classification: [Robust | Sensitive | Fragile]

FIVE-SENTENCE PUBLIC RATIONALE (5SPR) (Tier 2-3 REQUIRED)

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1. CONTEXT: [What decision was made and why now?]

2. OPTIONS: [What options were considered?]

3. CONSTRAINTS: [What was eliminated by NCRC, TRC, and/or Containment (and why)?]

4. SELECTION: [Why the chosen option won among selectable options?]

5. MONITORING: [What follow-up will be tracked and when?]

AUDIT FLAGS (REQUIRED when triggered)

[List flags + explanations]

SIGNATURES

Analyst: [name, date]

Decision Owner: [name, date]

Reviewer (Tier 3): [name, date]

===

H.1A PCC.ReferenceStructureRecord (Normative for Tier 3 public conformance claims; recommended for Tier 2 high-stakes runs)

The PCC SHOULD include a ReferenceStructureRecord for Tier 2 high-stakes runs and SHALL include one for Tier 3 public conformance or validation claims. This record does not alter historical NCRC, TRC, CSV, RLS, or UCI/HOI arithmetic. It records the reference structure used to support the claim being made.

Minimum fields:

record_id

decision_id

option_id_or_set

claim_scope

decision_boundary

reference_object

active_reference_layers

invariant_constraints

causal_pathway_summary

falsification_conditions

evidence_sources

evidence_status_tokens

measurement_maturity_status

adaptation_boundary

authority_boundary

refusal_triggers

excluded_references

reviewer_disposition

timestamp

artifact_hashes

Validation rule. If PCC.ReferenceStructureRecord is required and missing, the PCC is incomplete for any claim that the run was reference-structure-audited. Missing ReferenceStructureRecord does not by itself alter cascade arithmetic, but it prevents stronger public conformance, deployment-readiness, or validation claims.

H.2 Minimum Required Fields by Tier (Canonical)

Field	Tier 1	Tier 2	Tier 3
Decision ID	Recommended	REQUIRED	REQUIRED
Timestamp	Recommended	REQUIRED	REQUIRED
Option set	REQUIRED	REQUIRED	REQUIRED
NCRC results	Heuristic ok	REQUIRED	REQUIRED + subgroup policy
TRC results	Optional	REQUIRED when catastrophe plausible	REQUIRED
Containment	Optional	Recommended	REQUIRED
RLS scores	Optional	REQUIRED	REQUIRED
Uncertainty + discrimination	Optional	REQUIRED (active cells)	REQUIRED (active cells)
Sensitivity analysis	Optional	Recommended	REQUIRED

Field	Tier 1	Tier 2	Tier 3
5SPR	Optional	REQUIRED	REQUIRED
Audit flags	Optional	REQUIRED when triggered	REQUIRED when triggered
Normative Input Register	Optional	REQUIRED when material	REQUIRED when material

H.3 Audit Flag Canon Pack (Canonical)

Stewardship audit-flag tokens are defined canonically in Appendix N.7; Appendix H.3 does not redefine Stewardship tokens and only provides pointers.

H.3A Reality Grounding and Structural Diagnostic Placement Flags (v11.4)

INSUFFICIENT_GROUNDING. Severity: ESCALATE/REFUSE. Meaning: the declared reality surface is insufficient for a material rights, tail-risk, containment, authority, conformance, or selection claim. Required response: narrow the claim, collect evidence, rerun, mark exploratory/sensitivity-only, escalate, or refuse the stronger claim.

CONTAINMENT_STRUCTURAL_DIAGNOSTIC_UNAVAILABLE. Severity: ESCALATE. Meaning: structural diagnostic evidence required for Containment, including UCI/HOI or declared equivalent, was unavailable or unresolved. Required response: downgrade claim strength, collect evidence, rerun, escalate, refuse the stronger claim, or record permitted waiver where allowed. Tier 3 containment conformance is unavailable until resolved.

STRUCTURAL_DIAGNOSTIC_MISPLACED_TO_TIEBREAK. Severity: INVALID for Tier 3; ESCALATE otherwise. Meaning: a structural diagnostic that was gate-relevant to Containment was treated only as residual tie-break evidence after RLS. Required response: rerun Containment with the diagnostic included or refuse the stronger claim.

RESIDUAL_UCI_HOI_USED. Severity: INFO/REVIEW. Meaning: UCI/HOI were used only as residual tie-break, monitoring, or hollowing-risk documentation among options already in the selectable set. Required response: document RLS non-decisiveness or monitoring reason, containment-pass status, indicator basis, and reviewer disposition.

PLSS_STRUCTURAL_DIAGNOSTIC_OMISSION. Severity: ESCALATE. Meaning: a PLSS run omitted, down-weighted, or deferred structural diagnostic evidence that was plausibly containment-relevant. Required response: rerun with broader scrutiny, document non-relevance, or escalate.

Flag	Trigger	Required Action	Severity
RIGHTS_CELL_MASKED_INVALID	Any rights-covered cell is masked, excluded, null'd, pooled, or otherwise not individually included in RLS aggregation (including any case where m(u,d)=0 for a cell in any C_r).	PCC invalid; recompute without masking	INVALID
RIGHTS_CELL_OMITTED_INVALID	Any rights-covered cell required to evaluate NCRC is omitted/excluded/not disclosed such that NCRC cannot be recomputed.	PCC invalid; include and disclose all required rights-covered cells and recompute NCRC.	INVALID

Flag	Trigger	Required Action	Severity
CATASTROPHE_CELL_MASKED_INVALID	Any catastrophe cell in C_cat is masked (m(u,d)=0) or otherwise excluded from RLS aggregation (including any case where m(u,d)=0 for a cell in C_cat).	PCC invalid; set m(u,d)=1 for all catastrophe cells and recompute RLS.	INVALID
CATASTROPHE_CELL_OMITTED_INVALID	Any catastrophe cell in C_cat used in TRC loss computation/disclosure is omitted/excluded/not disclosed such that TRC CVaR cannot be independently recomputed.	PCC invalid; include and disclose all catastrophe cells used in TRC loss computation (and any required scenario-to-cell mapping), then recompute TRC.	INVALID

Flag	Trigger	Required Action	Severity
CONTAINMENT_MODE_B_USED_FOR_SELECTION	Mode B (diagnostic-only containment) influenced selection outcome; see Section 9.5	PCC invalid; rerun with Mode A only	INVALID
SCENARIO_LIBRARY_MIN_EXCEPTION	Scenario library size $\|S\|$ is below the tier minimum when TRC is required (Tier 2: $\|S\| < 5$; Tier 3: $\|S\| < 20$)	PCC remains valid only with: (i) written justification, (ii) independent reviewer sign-off (Tier 3 required; Tier 2 recommended), and (iii) a remediation plan to reach minimum scenario count before the next comparable run. Record this flag in PCC.	ESCALATE

Flag	Trigger	Required Action	Severity
MANDATORY_TAIL_CATEGORY_MISSING	Missing mandatory tail category without justification	Add scenarios or justify per Section 8.3.2	ESCALATE

Flag	Trigger	Required Action	Severity
MANDATORY_TAIL_CATEGORY_MISSING_WITH_JUSTIFICATION	Missing mandatory tail category with proper justification per Section 8.3.2	Record justification; monitor for category emergence	REVIEW

Flag	Trigger	Required Action	Severity
MANDATORY_TAIL_PROB_FLOOR_VIOLATION	Category probability sum < p_floor (0.02)	Revise probabilities or add scenarios to category	ESCALATE
SUBGROUP_LIMITATION	Cannot disaggregate subgroups for rights-covered cell	Apply γ_subgroup conservative bound (Section 7.3.2); escalate if high stakes	REVIEW

Flag	Trigger	Required Action	Severity
CHALLENGE_DEFERRED_EMERGENCY	Emergency Mode invoked without independent challenger	Retrospective challenge MUST occur within 24 hours or next business day, whichever is sooner	ESCALATE
EMERGENCY_MODE_INVOKED	A_NCRC = ∅ (no rights-admissible options exist)	Remediation plan required; high scrutiny review	ESCALATE
TRC_FALLBACK_INVOKED	A_adm = ∅ after TRC (A_NCRC is non-empty but all rights-admissible options fail TRC)	Higher-tier approval required; mitigation plan	ESCALATE

Flag	Trigger	Required Action	Severity
DECISION_FRAGILE_KERNEL	Selected option changes under kernel perturbation test (Section 6.6)	Escalation required; consider NONE mode	ESCALATE
KERNEL_HUMILITY_FALLBACK	Kernel disabled due to KQS < 0.40 or stability violation (Section 6.4, 6.5)	Note limitation in PCC; plan kernel improvement	REVIEW
JUDGMENT_CALL_UCI_UNAVAILABLE	UCI unavailable in non-decisive RLS tie	Document why UCI unavailable, when UCI measurement returns, and interim decision basis; monitoring plan required	REVIEW

Flag	Trigger	Required Action	Severity
JUDGMENT_CALL_TIEBREAK_NONDECISIVE	Tie-break chain did not resolve selection	Explicit manual judgment record required; independent reviewer recommended (Tier 3)	REVIEW
SCI_BELOW_MINIMUM	SCI < tier minimum for the claimed tier, OR (SCI = tier minimum AND PCC missing required Next-Run Upgrade action).	PCC MUST record PCC.SCI.BelowMinimumDisposition = DOWNGRADE_TIER or RERUN_TO_MEET_MINIMUM. If SCI equals tier minimum and NextRunUpgradeAction is missing, treat as SCI_BELOW_MINIMUM. No tier compliance above achieved SCI may be claimed.	ESCALATE
CONFIG_DRIFT	Parameters differ from prior run without governance update	Governance review required before proceeding	ESCALATE

Flag	Trigger	Required Action	Severity
CONTAINMENT_UCI_UNAVAILABLE	Containment evaluation required ΔUCI for containing union(s) but ΔUCI unavailable; or unknown ΔUCI treated as pass.	Set flag. If Tier 3 was claimed or attempted, the run MUST NOT claim Tier 3 compliance. PCC MUST record PCC.Containment.UCIUnavailableDisposition = DOWNGRADE_TIER or COLLECT_DATA_RERUN and include PCC.Containment.UCIUnavailableRemediationTimeline (non-empty). If DOWNGRADE_TIER is chosen, the run proceeds under Tier 2 semantics (containment non-binding) and MUST set PCC.TierClaimDisposition = TIER_ATTEMPTED_NOT_ACHIEVED_DOWNGRADED with PCC.TierAttempted = 3 and PCC.TierAchieved = 2.	ESCALATE
PLSS_ESCALATION_TRIGGER_UNRESOLVED	One or more Section 10.2A.7 escalation triggers is plausibly active, but the run proceeds in PLSS posture without an escalated posture or documented justification.	Record a broader posture OR a specific non-escalation justification with evidence basis and reviewer sign-off where required by tier. If no such disposition is recorded, reject the run as non-conformant.	ESCALATE
REGISTRY_MISMATCH	PCC snapshot differs from referenced registry hash	Audit required; resolve discrepancy	ESCALATE

Flag	Trigger	Required Action	Severity
PLSS_PROMINENCE_SENSITIVE	Valid alternative PLSS prominence constructions materially change the ranking order of selectable options or the selected option.	Record the sensitivity, require reviewer attention, and do not present the PLSS outcome as uniquely robust without further justification.	ESCALATE
DECISIVENESS_METHOD_POSTHOC	A method choice that can materially affect decisiveness, ranking within the selectable set, or tie-break outcome, including PLSS prominence construction, alternative PLSS operator, scope-posture escalation disposition, or UCI comparison method, is first declared only after comparative outcomes are observed and the PCC lacks the required reason code, reviewer disposition, or sensitivity note.	Record the post hoc declaration, add the required reason code and sensitivity note, and require reviewer attention. If the missing disposition cannot be supplied, the run MUST NOT be presented as uniquely robust.	ESCALATE
EK_NONDEFAULT_TIER2_INVALID	A Tier 2 run logs one or more instances with e_k ≠ 1.00 without a named governed instrument that explicitly permits a non-default value.	Tier 2 runs MUST use e_k = 1.00 by default. Reset e_k to 1.00, or rerun under a governed Tier 3 posture with declared provenance and sensitivity evidence before claiming conformance.	INVALID

Flag	Trigger	Required Action	Severity
UNCERTAINTY_PROVENANCE_MISSING	A Tier 2+ run requires uncertainty disclosure, provenance, or method declaration, but the PCC is missing the required uncertainty fields or basis record.	Populate the required PCC uncertainty fields and provenance basis before claiming conformance. If the basis cannot be supplied, downgrade or reject the run as non-conformant.	INVALID
SCOPE_COVERAGE_REDUCED	Reduced-scope mode is used (declared or implied by fewer than 7 active scopes) in a Tier 2+ run, OR the admissibility full-computation attestation for non-maskable cells is missing/incomplete.	Reviewer MUST verify that NCRC/TRC/CSV were computed on all required non-maskable cells, and that omitted-scope blind spots plus escalation triggers are recorded. If reduced-scope is used repeatedly, require a next-run scope expansion plan.	REVIEW

Flag	Trigger	Required Action	Severity
UNCERTAINTY_DECISIVENESS_SENSITIVE	Required uncertainty stress tests change whether the lead is decisive.	Escalate or relabel the run non-decisive; apply tie-break or governance review before claiming a decisive welfare lead.	ESCALATE
RIGHTS_CELL_UNKNOWN_GATE_CRITICAL	A rights-covered cell in C_r is declared UNKNOWN_IMPACT and admissibility is not preserved by a governed conservative bound or equivalent governed disposition.	Do not resolve NCRC by phantom fallback alone. Apply a governed conservative bound sufficient to preserve the rights gate, or rerun / downgrade / escalate under a declared disposition.	ESCALATE

Flag	Trigger	Required Action	Severity
CATASTROPHE_CELL_UNKNOWN_GATE_CRITICAL	A catastrophe-covered cell in C_cat is declared UNKNOWN_IMPACT and TRC is not preserved by a governed conservative scenario/cell bound or equivalent governed disposition.	Do not resolve TRC by phantom fallback alone. Apply a governed conservative catastrophe bound, or rerun / downgrade / escalate under a declared disposition.	ESCALATE
NORMATIVE_INPUT_PROVENANCE_MISSING	A welfare-relevant state, rights-relevant state, or enabling condition materially affects admissibility, precautionary handling, selection, or tie-break outcome, but the PCC lacks a corresponding NormativeInputRegister entry, or the entry lacks the required evidence basis, governed interpretation basis, or precautionary basis.	Populate PCC.NormativeInputRegister with complete provenance and rerun the affected gate or ranking. No conformance claim may be made until corrected.	INVALID

Flag	Trigger	Required Action	Severity
CLAIM_EXCEEDS_CLOSURE_BOUNDARY	A run makes a public, conformance, deployment-readiness, execution-bound, or validation claim beyond the declared PCC.ComputationalClosure boundary.	Narrow the claim to the declared closure boundary, expand the computation and rerun, or enter Refusal/escalation. The unsupported claim is invalid until corrected.	INVALID for the affected claim.
CATEGORY_COLLAPSE_FLAG	PCC.CategoryCollapseSelfTest returns FAIL on any category-collapse item, including treating compliance as correctness, traceability as admissibility, moral status as governance authority, or SGP welfare scalars as admissibility modifiers.	Identify the collapsed layer, correct the run, or enter Refusal if the collapse affects admissibility, selectability, or selection.	ESCALATE; INVALID if the collapse affects admissibility, selectability, selection, or conformance.

Flag	Trigger	Required Action	Severity
PCC_MATURITY_SURFACE_INCOMPLETE	A Tier 2-3 claim-bearing run lacks an applicable required measurement-maturity field under Section 14.3B / Appendix V, including UCI, Kernel, HDW, PLSS, Deployment Context, or Biological Evaluation Maturity fields.	Populate the missing maturity field(s), downgrade/narrow the claim, or reject the run as non-conformant for the affected claim.	ESCALATE; INVALID for any empirical-readiness, operational-completeness, validated-UCI, validated-kernel, biological-measurement-completeness, or Tier-4/ProofPack-adjacent claim.
COMPUTABLE_BUT_INADMISSIBLE	A value, formula, model output, scalar, ranking, or workflow result computes successfully, but declared domain rules bar its use as a decision state, conformance basis, public claim, or authorization basis.	Record PCC.CBI.ComputedObject, PCC.CBI.InadmissibilityReason, PCC.CBI.Layer, PCC.CBI.AllowedUse, PCC.CBI.ProhibitedUse, and PCC.CBI.ReviewStatus. Mirror ripple.md W.3 where wrapper assurance applies. Disclose and escalate when gate-relevant.	DISCLOSE; ESCALATE if gate-relevant; INVALID for any selection, deployment, conformance, or maturity claim relying on the barred computed value.

Flag	Trigger	Required Action	Severity
REALITY_GROUNDING_RECORD_MISSING	Required PCC.RealityGroundingRecord is absent for a claim-bearing Tier 2 or Tier 3 run, or for any run whose material reality claims affect NCRC, TRC, CSV, deterministic selection, authority-selection disclosure, or conformance claims.	Add the RCR, narrow the claim, escalate, or refuse the stronger decision state. The affected claim cannot be treated as supported until the RCR exists.	INVALID for affected claim
REALITY_GROUNDING_STATUS_UNSUPPORTED	Declared RealityGroundingStatus exceeds what the EvidenceTrace, reviewer status, or declared reality surface substantiates.	Downgrade the status, add evidence and reviewer trace, or refuse the stronger claim. If used for a gate, deterministic selection, authority selection, or conformance claim, the claim is invalid until corrected.	ESCALATE; INVALID if used for gate, selection, authority, or conformance

End Appendix H.

APPENDIX I: TIER-4 DESIGN TARGET (NOT CLAIMABLE IN v11.4)

I.1 Status Declaration

Tier-4 Design Target (Not Claimable in Any Current Release Line, Including v11.4)

Tier 4 is specified as a target profile requiring artifacts and capabilities that are not yet publicly available. Tier-4 compliance MUST NOT be claimed until ProofPack is publicly released and independently replayable.

Reality-contact Tier-4 requirement. Any future Tier-4 claim that relies on RealityGroundingStatus MUST make that status independently re-derivable from EvidenceTrace and supporting artifacts. Operator-declared status is sufficient for Tier 2 traceability only when review status is explicitly recorded; Tier 3 must follow its declared reviewer regime; neither is sufficient for Tier 4 admissibility, ProofPack readiness, or machine-verifiable ecosystem claims.

I.2 Tier 4 Requirements (Target Specification)

Hash-Bound Registries:

- Rights anchors registry (REG-RIGHTS-ANCHORS-v1)
- Catastrophe indicators registry (AF-BASE)
- Scenario library registry
- Kernel registry with evidence classes
- HDW ballots and weights registry

Deterministic Numeric Profile (NDP_FIXEDPOINT_V1):

- Fixed-point scale: $S = 10^9$
- Representation: X_fp = round_half_even(x × S) as signed int64
- All divisions use round-half-even
- Hard-fail on overflow with audit_flag NUMERIC_OVERFLOW

Saturation Lookup Table (SAT_LUT_FP_V1):

- Pre-computed tanh values for fixed-point inputs
- Hash-bound in ProofPack
- Runtime computation of tanh PROHIBITED

Temporal Weight Registry (REG_TEMPORAL_WEIGHTS_V1):

- Pre-computed $\tau(t)$ values for standard horizons
- Hash-bound in ProofPack
- Runtime computation of logarithms PROHIBITED

Canonical JSON Profile:

- NO_FLOATS rule: All numeric values as exact rationals {num, den}
- Deterministic key ordering
- Canonical whitespace rules
- SHA-256 hashing for integrity

Propagation Mode Restriction:

- FULL propagation PROHIBITED for Tier 4 Pilot-Executable
- NONE or QUICK only
- FULL available only in future Tier 4 Certified profile

I.3 ProofPack Contents (Target)

The ProofPack artifact bundle will provide:

- Schemas: JSON schemas for PCC and all registries
- Manifests: Hash indices for all artifacts
- Canonicalization Rules: Exact specification for deterministic hashing
- Registries: Machine-readable canonical registries
- Test Vectors: Reference inputs and expected outputs for validation

I.4 Replay Test Requirement

A Tier 4 PCC MUST pass a replay test: independent re-execution using PCC inputs and referenced registries reproduces:

- Identical admissibility outcomes (NCRC, TRC, CSV)
- Identical RLS ranking
- Identical selected option

Failed replay invalidates the Tier 4 claim.

End Appendix I.

APPENDIX J: VALIDATION PROTOCOL PACK

Appendix J is normative for pre-registration templates and scoring definitions when validation claims are made.

J.1 Pre-Registration Template (Normative)

==

RIPPLELOGIC VALIDATION STUDY PRE-REGISTRATION

==

STUDY IDENTIFICATION

Study ID: [unique]

Registration date: [YYYY-MM-DD]

Principal investigator: [name, affiliation]

Funding source: [if any]

RippleLogic version: [v11.4]

Tier executed: [1|2|3]

Domain: [AI deployment | policy | org strategy | etc.]

HYPOTHESES

Primary hypothesis: [H1/H2/etc. with exact measurable claim]

Secondary hypotheses: [list]

Falsification criteria: [explicit thresholds from Section 17]

METHODS

Design: [pilot / RCT / quasi-experimental / backtest]

Sample: [population, size, selection method]

Comparator: [baseline governance / alternative framework]

Duration: [time]

Decision types included: [list]

Exclusion criteria: [list]

MEASURES

Primary outcomes: [definition, data source, timing]

Secondary outcomes: [definition, data source, timing]

Rights measurement: [how rights violations are verified]

Tail-loss measurement: [how catastrophic outcomes are defined]

ANALYSIS PLAN

Statistical methods: [tests/models]

Effect size thresholds: [minimum meaningful difference]

Multiple comparison correction: [method]

Missing data handling: [approach]

Sensitivity checks: [weights, scenarios, thresholds]

GOVERNANCE

Ethics approval: [status/ID]

Data management: [storage, access, retention]

Reporting commitment: [timeline and venue]

==

J.2 Outcome Measures by Union and Dimension (Canonical Examples)

Union	Dimension	Measure Type	Example Indicators
Self	Health	Validated survey / admin	SF-36, PHQ-9, GAD-7, injury rates
Self	Agency	Validated survey	Autonomy/self-efficacy scales
Household	Material	Administrative	Income stability, housing security
Community	Social	Survey / network	Trust indices, social capital measures
Organization	Flow/Resilience (UCI)	Operational	Throughput, incident recovery time
Polity	Agency	Index	Participation rates, rule-of-law indices
Biosphere	Environment	Monitoring	Emissions, biodiversity integrity

J.3 Sign Accuracy Scoring (Normative)

For each evaluated cell (u,d):

Predicted sign: sign(I_prop(u,d,a)) ∈ {-1, 0, +1}

Observed sign: sign(Δ_observed(u,d)) ∈ {-1, 0, +1}

Scoring:

- Match = 1 if predicted = observed
- Match = 0.5 if predicted = 0 or observed = 0
- Match = 0 otherwise

SignAcc = (Σ Match) / (number of evaluated cells)

Target for calibrated deployments: SignAcc ≥ 0.70

J.4 Magnitude Error (RMSE) (Normative)

RMSE = √[(1/n) Σ_i (I_prop,i − observed_i)^2]

Default target: RMSE ≤ 0.25 (domain dependent)

J.5 Backtest Procedure (Normative)

Case selection: Identify historical decisions with measurable outcomes and reconstructable option sets.

1. Information reconstruction: Use only data available at decision time (no hindsight leakage).
1. Blind analysis: The analyst SHOULD be unaware of actual outcomes during the run.
1. Execute RippleLogic run (Tier declared) and emit Backtest PCC.
1. Compare predictions to realized outcomes using declared metrics (for example SignAcc, RMSE, rights detection).
1. Report results and failures; update methods via NCAR with versioning.
1. End Appendix J.

Appendix K is informative and strongly recommended for any deployment using Starter KOPS or provisional QUICK propagation.

APPENDIX K: STARTER KERNEL COMPANION PACK (INFORMATIVE)

K.1 Purpose

Starter KOPS provides an informative, provisional kernel surface for transparency, sensitivity testing, and early implementation support. It MUST NOT be treated as a fully validated propagation model.

K.2 Evidence Class Summary Table K.1: Evidence Class Summary

Class	Name	Minimum Evidence
A	Validated	Replicated empirical studies with quantified effect sizes and domain match
B	Supported	Multiple studies consistent in sign; some quantification

Class	Name	Minimum Evidence
C	Plausible	Theoretical support with limited empirical evidence
D	Speculative	Expert judgment without empirical support
E	Elicited	Structured elicitation for this framework (lowest confidence)

K.3 Provisional Edge Table Table K.2: Provisional Starter KOPS Edge Table

Source Cell (u,d)	Target Cell (u,d)	κ_raw	κ_shrunk	Evidence Class	Rationale Family
(4,1) Org-Material	(2,1) HH-Material	0.40	0.14	E	Employment income propagation
(4,1) Org-Material	(3,1) Comm-Material	0.30	0.11	E	Local multiplier effects
(7,7) Bio-Environment	(6,2) CMIU-Health	-0.35	-0.12	B	Climate/health pathway
(7,7) Bio-Environment	(6,7) CMIU-Environment	0.50	0.18	B	Earth system propagation
(5,5) Polity-Agency	(4,5) Org-Agency	0.25	0.09	E	Institutional enabling effects
(3,3) Comm-Social	(1,3) Self-Social	0.45	0.16	B	Social capital / wellbeing
(3,3) Comm-Social	(2,3) HH-Social	0.35	0.12	E	Family-community linkage
(4,2) Org-Health	(1,2) Self-Health	0.40	0.14	B	Occupational health pathway
(6,4) CMIU-Knowledge	(5,4) Polity-Knowledge	0.30	0.11	E	Diffusion / coordination knowledge
(1,2) Self-Health	(2,1) HH-Material	-0.25	-0.09	B	Health to earning capacity

K.4 Mandatory PCC Disclaimer “Ripple propagation used a PROVISIONAL Starter KOPS kernel. Coefficients are not fully validated. Sensitivity analysis was performed per RippleLogic v11.4 requirements; conclusions are contingent on kernel uncertainty.”

End Appendix K.

APPENDIX L: GLOSSARY (v11.4)

Appendix L is informative but standardized terminology is strongly recommended for audit consistency.

Stakeholder: Any entity plausibly affected by a candidate option through direct impact, externalities, indirect pathways, or future propagation within the declared horizon.

Instance: A stakeholder-in-role-in-context bound to a decision, used for auditability and mapping into Union Scopes.

Stakeholder Discovery Protocol (SDP): The tier-linked minimum process for discovering omission-likely stakeholders and recording discovery evidence (Section 2.2A).

Stakeholder Coverage Index (SCI): A tier-governed disclosure ladder describing the stakeholder coverage posture of a run, used for auditability, not for score changes (Section 5.0.1).

InstanceMap: The structured mapping from stakeholder instances to one or more Union Scopes, recorded in the PCC (Appendix H).

Scope Coverage Declaration: A PCC section that declares whether the run is full-scope or reduced-scope, lists omitted scopes and rationales, and specifies escalation triggers (Appendix H).

Active cell: A Scope×Dimension cell that participates in RLS, or is in any rights/catastrophe coverage set, or is populated for any gate or justification, and therefore must not be treated as unknown-by-default (Section 10.3).

Effect-token (allocation only): A redundant conserved-unit causal token used solely to prevent redundant counting across scopes under the ALLOCATION redundancy method (Section 2.2A).

Reduced-scope mode: A reporting or mapping-depth mode that omits one or more scopes, permitted only with explicit PCC disclosure and without removing any non-maskable cells from admissibility gates (Section 2.2A).

SG_measured_norm(E): The raw normalized scalar reported by SGP before plateau assignment and non-plateau capping. It is audit-visible but is not a direct normative substitute for SG_norm(E) or SG_patient_norm(E) inside RippleLogic.

SG_norm(E): The authoritative effective canonical scalar consumed by RippleLogic through the SGP binding interface. Under SGP v6.4, this normally refers to the effective individual scalar after plateau rules have been applied.

SG_patient_norm(E): The authoritative rights-of-protection scalar consumed by RippleLogic where available. This scalar may incorporate plateau status and, where applicable, taxon or governed-class baseline logic. When available, RippleLogic SHOULD prefer it over SG_norm(E) for welfare-stream weighting.

Plateau(E): A constitutional status flag reported by SGP indicating whether the entity has Full Rights Plateau standing. Plateau status is explicit; it is not inferred by RippleLogic from raw measurement, pillar scores, or taxon evidence alone.

Full Rights Plateau: The constitutional rights-of-protection summit defined by SGP. Plateau entities receive 1.0 / 100 semantics at the SGP interface. Within RippleLogic, plateau affects welfare-stream weighting for impacted protected stakeholders, without altering admissibility gates, and recognition of full inviolable standing, but does not by itself bypass stewardship gating for specific consequential governance roles.

Near-Plateau Band: The high-protection band below plateau recognized by SGP v6.4. This band remains below 1.0 in effective scalar semantics and does not itself confer plateau.

CMIU Standing: Standing in the Collective Managing Intelligence Union as reported by SGP for plateau-bearing entities where applicable. CMIU standing is distinct from automatic qualification for every domain-specific governing role.

Human Plateau Rule: The non-overridable rule that every human person is treated as plateau-bearing for rights-of-protection purposes. Within RippleLogic, this means SG_norm(H)=1.0 regardless of measurement noise, disability status, or partial observability.

SGP Binding Interface: The only permissible interface by which RippleLogic consumes SGP outputs. Defined normatively in Appendix G. RippleLogic does not define SGP internals and MUST NOT recompute SGP scalars internally.

Term	Definition
Admissible	Option that passes both NCRC and TRC
AIL (Artifact Integrity Law)	Integrity rules ensuring auditability, comparability, and no silent overrides in PCCs
Baseline-Zero Rule	0 impact means no change from baseline; all impacts are baseline-relative
Catastrophe cell set (C_cat)	Subset of welfare cells used for TRC tail-risk evaluation

Term	Definition
CSV (Containment Mode A / Structural Viability)	Binding gate preventing selection of options that degrade containing union coherence or structural viability beyond tolerance
Containment tolerance (τ_c)	Maximum allowed negative ΔUCI for containing unions when sub-unions benefit
CVaR (Conditional Value-at-Risk)	Tail risk measure of expected loss in the worst tail mass
Dimension (D_1-D_7)	One of seven welfare dimensions: Material, Health, Social, Knowledge, Agency, Meaning, Environment
Emergency Mode	Controlled fallback when no option passes NCRC; selects lexicographically minimal rights violation vector with challenger requirement and remediation
HDW (Hybrid Democratic Weighting)	Governance method producing weights from floors + democratic + structural proposals
HOI (Hollowing-Out Index)	Monitoring diagnostic indicating welfare-up while coherence-down drift
Impact instance	A single asserted pathway contribution to a welfare cell impact, with magnitude, reach, horizon, likelihood, confidence, and multipliers
Kernel (K)	Sparse matrix encoding cross-cell ripple propagation

Term	Definition
KOPS	Key Operational Pathways Set; the subset of kernel edges considered load-bearing and governed
KQS (Kernel Quality Score)	Summary score indicating whether kernel propagation is permitted
Lexicographic cascade	Priority-ordered gates where earlier failures cannot be compensated by later scoring
MNA (Minimal Normative Axiom)	Sentient welfare and flourishing matter; unnecessary suffering and domination should be reduced; enabling conditions for viable, non-dominated, and continuing flourishing should be preserved.
Bridge Premise 1 (Embedded Modelling)	Descriptive support premise stating that reliable action in shared causal reality requires non-trivial modelling of other stakeholders' interests, constraints, vulnerabilities, dependence relations, rights exposure, and - where behavior, harm, cooperation, or rights protection depend on them - welfare-relevant states and enabling conditions.
Bridge Premise 2 (Prudential Stability)	Support premise stating that systematic degradation of stakeholders' welfare conditions, rights floors, or enabling conditions increases instability, conflict, tail risk, and long-run governance failure in plural uncertain systems
Enabling condition	A material, informational, institutional, ecological, or agency-preserving condition whose sustained absence or degradation predictably undermines viable, non-dominated, and continuing flourishing across one or more unions.
Normative Commitment	RippleLogic commitment that credibly evidenced welfare-relevant states, rights-relevant states, and the enabling conditions on which they depend constitute reasons for action, not merely inputs for prediction

Term	Definition
Rights-relevant state	A credibly evidenced condition whose degradation threatens or crosses a non-compensatory minimum protected by RippleLogic's rights layer, including life, bodily integrity, liberty, basic needs, dignity, due process, information integrity, and ecological enabling conditions.
Welfare-relevant state	A credibly evidenced condition of a moral patient that can go better or worse for that patient, including states relevant to suffering, integrity, agency, and flourishing
NCAR	Notice-Choose-Act-Reflect learning loop
NCRC	Non-Compensatory Rights Constraint; excludes rights-violating options
Non-maskable cells	Rights and catastrophe cells (and minimum coverage) that cannot be excluded from RLS aggregation
PCC	Provenance and Compliance Certificate; auditable record of inputs, computations, and selection
RLS	RippleLogic Score; weighted welfare aggregation used to rank selectable options
Selectable	Admissible option that also passes containment (Mode A)
SGP	Sentience Gradient Protocol; interface for rights-of-protection gating across entities
TRC	Tail-Risk Constraint; excludes options with unacceptable catastrophic exposure using CVaR
UCI	Union Coherence Index; structural health metric for unions
UBE	Union-Based Ethics; normative framework built on UBR, the MNA, bridge premises, and the explicit normative commitment
UBR	Union-Based Reality; descriptive ontology of nested interdependent unions
Union	A bounded pattern of interdependence at an organizational scale
Unioning	Redesigning decisions until rights-safe, tail-safe, containment-safe, and net-positive where possible
Normative Input Register	The PCC block used to record welfare-relevant states, rights-relevant states, and enabling conditions that materially affect admissibility, precautionary handling, selection, or tie-break outcome when not already fully captured by direct impact-instance tables or canonical rule objects.

Term	Definition
NORMATIVE_INPUT_PROVENANCE_MISSING	Audit flag indicating that a material welfare-relevant state, rights-relevant state, or enabling condition lacks required provenance in the PCC. Severity: INVALID.

Adaptive Reserve: A carried-forward v10.8 structural diagnostic for slack, redundancy, recovery capacity, reversibility, and saturation risk. It is a UCI/HOI-compatible diagnostic and containment-support signal, not a sixth gate.

COMPUTABLE_BUT_INADMISSIBLE: An audit state for a value that can be computed but cannot be used as a valid decision state, conformance basis, public claim, or authorization basis because another domain rule bars that use. This is distinct from UNKNOWN_IMPACT (missing evidence) and refusal (undetermined or insufficiently grounded decision state).

Reality: For decision-making, the evaluator-independent field of constrained structures that, under transition, produce consequences. RippleLogic names this operationally and does not claim complete metaphysical access to it. Evaluator-independent means independent of the evaluating agent's perception, admission, authority, model, or preference; it does not deny the reality of experience, social facts, institutions, norms, markets, laws, reputations, or model outputs when they shape welfare, rights, risk, agency, or available choices.

Decision-relevant reality: The portion of reality that materially bears on a decision's rights exposure, catastrophic risk, containment, welfare, legitimacy, agency, auditability, or available choices.

Reality surface: The portion of decision-relevant reality actually captured, represented, and modeled in a run. A reality surface is always incomplete relative to the territory and must disclose material unknowns.

RealityGroundingRecord (RCR): The PCC-linked audit record documenting the reality surface, material claims, evidence trace, declared unknowns, RealityGroundingStatus, surface/territory limits, and revision triggers for a run.

RealityGroundingStatus: The declared status of contact between a material claim and the relevant reality surface; it is review-status-declared at Tier 2, governed by the declared reviewer regime at Tier 3, and independently re-derivable at Tier 4.

Structure: A stable relational arrangement of distinguishable differences that persists across a decision-relevant transition window and can carry causal, informational, biological, ecological, social, institutional, artificial-system, or experiential significance.

Transition: A decision-relevant action, event, intervention, policy, or state-shift through which constrained structure produces consequence.

Consequence: Observable or inferable change following an action, transition, decision, event, or policy within constrained structure; may be direct, probabilistic, systemic, delayed, distributed, moral, ecological, institutional, reversible, irreversible, or catastrophic.

Ethical reality: The subset of reality where structures, constraints, and consequences affect welfare-bearing beings and the conditions of flourishing, including welfare, rights, sentience, agency, dignity, non-domination, catastrophic risk, ecological dependence, future generations, and flourishing.

Claim boundary: The maximum legitimate strength of a decision claim given the declared reality surface, evidence trace, uncertainty, maturity status, and review regime.

End Appendix L.

APPENDIX M: LINEAGE AND VERSION HISTORY (v11.4 CURRENT LINE; v10.8/v11.0 LINEAGE RETAINED) - INFORMATIVE

M.0 Purpose and Scope (Informative)

This appendix records naming lineage, companion-map posture, concise version history, and release-gate checklist notes for the current RippleLogic line. It is not normative. Normative meaning is controlled by Section 0 (Specification Contract) and the precedence order in Section 0.1.

M.1 Naming and Lineage (Informative)

MathGov is the umbrella ethical governance framework. RippleLogic is the ethical decision operating system and decision architecture inside MathGov. Earlier materials may preserve historical lineage labels, prior PCC labels, or affiliated publication labels, but current v11.4 release materials MUST NOT treat MathGov as merely historical or as superseded by RippleLogic.

M.2 Backward Compatibility and Compliance Note (Informative)

Prior PCCs computed under earlier lines remain interpretable as lineage artifacts. Their normative parameters were valid at the time of execution and need not be recomputed unless the underlying decision is being revisited under the current line. Any run claiming current-line v11.4 compliance MUST satisfy the v11.4 canonical requirements and use the v11.4 canon artifact as the current release-line source of truth.

M.3 Canon Map and Companion Artifacts (Informative)

RippleLogic Framework v11.4 (this Foundation Paper): governing artifact for the current canonical release line.

SGP v6.4 (Sentience Gradient Protocol): the moral-status layer and governance guards pinned by Appendix G. Authoritative for sentience-scoring procedures, scalar canon, plateau semantics, rights-of-protection scalar definitions, governed review requirements, and classification-stability cautions. RippleLogic consumes only SGP outputs through the Appendix G interface.

RippleLogic Agent System v11.4: operational agent controls and verification obligations aligned to the current RippleLogic line.

ripple.md Standard v4.4: current assurance-wrapper companion for deployment interoperability.

Ripple Aligners Sheet v4.4: Tier-2 worked-run and local-scope tool pinned to the current RippleLogic line and SGP v6.4. The workbook remains a disclosure and training companion rather than the repository-level validator or schema surface.

Canonical distribution channels may include ripplelogic.org and mathgov.org. Artifact identity is filename + version + SHA-256 in the release manifest.

M.4 Version History (Informative)

Normative effect: No intended change is made to the canonical cascade, equations, rights thresholds, TRC mechanics, Containment semantics, SGP scalar binding, Human Plateau Rule, RSP/RefStructRecord logic, or Section 13 floor values. Appendix AD is now explicitly situated as a canonical interpretive reference for welfare-cell meaning and reviewer literacy, subordinate to equations, gates, thresholds, and claim boundaries.

Summary: This release preserves the canonical admissibility cascade Reality Grounding → Rights Floor → TRC → CSV → RLS, rights coverage sets, TRC corridor mechanics, Containment semantics, SGP binding interface, PLSS/FLGG architecture, Reference Structure Protocol, ReferenceStructureRecord, refusal discipline, projection pre-registration, computational closure, transition-admissibility discipline, implementation-tier boundaries, and Tier 1-3 claim limits.

Lineage orientation: The v10.8 and v10.5 entries below are separated so that readers can distinguish the Human Plateau / CMIU and SGP v5.5 synchronization layer from the prior Appendix AD precedence, heading/TOC, and citation-hygiene layer.

Supersedes: RippleLogic v10.5.

Status: Canonical release for the current v11.4 line.

M.4AIC RippleLogic v11.4 (2026-06-19) — Term Discrimination and Semantic Stability Release

Release character: v11.4 is a structural-viability and falsification-discipline release. It carries forward the v11.0 semantic-stability hardening release, including Term Discrimination inside Category Grounding and Reality Grounding, extended material CategoryGroundingRecord fields, and the Do-Not-Collapse Matrix. It preserves the v10.8 five-level method, equations, rights floors, TRC mechanics, Containment semantics, SGP binding interface, PLSS/FLGG architecture, HDW, PCC, RSP/RefStructRecord, and Tier 1-3 claim boundaries.

M.4AA RippleLogic v10.5 (2026-05-31) — Appendix AD Precedence, Heading/TOC, and Citation-Hygiene Release

Release character: v10.5 is an Appendix AD precedence, heading/TOC, and citation-hygiene release. It integrates Appendix AD as the canonical interpretive reference for welfare-cell meaning and reviewer literacy, clarifies that Appendix AD is subordinate to equations, gates, thresholds, and claim boundaries, and preserves the v10.4 cascade, equations, rights floors, TRC mechanics, Containment semantics, SGP binding interface, PLSS/FLGG architecture, and Tier 1-3 claim limits.

M.4AAA RippleLogic v10.8 (2026-05-31) — Human Plateau / CMIU Calibration and SGP v5.5 Synchronization Release

Historical v10.8 line had two synchronized release layers: (1) the Human Plateau / CMIU calibration and SGP v5.5 synchronization layer, and (2) the Layer-Discipline Addendum layer adding Appendix AE. Both preserve the v10.5 cascade, equations, gates, thresholds, Appendix AD, RSP/RefStructRecord, and claim boundaries.

Release character: v10.8 is a human-plateau rationale and core-file synchronization release. It updates the Canon companion pin from SGP v5.2.1 to SGP v5.5 and adds the CMIU/reality-management calibration note explaining why all human persons remain plateau-bearing without reducing the pathway for non-human biological or artificial plateau extension. It preserves the v10.5 cascade, equations, gates, thresholds, Appendix AD, RSP/RefStructRecord, and claim boundaries.

Release character: v10.8 layer-discipline addendum. Adds Appendix AE (COMPUTABLE_BUT_INADMISSIBLE audit state, layer-specific falsification matrix, reference/preference/number discipline, and Adaptive Reserve diagnostic); registers CBI in Appendix H flags and Appendix L; no cascade, equation, threshold, or claim-boundary changes.

M.4AAB RippleLogic v10.8 (2026-06-12) — Layer-Discipline Addendum

M.4A RippleLogic v10.4 (2026-05-26) — Readability, Front-Matter Reorganization, and Companion Synchronization Release

Status: Historical canonical release for the v10.4 line.

Supersedes: RippleLogic v10.3.6.

Release character: v10.4 is a readability, front-matter reorganization, and companion synchronization release. It preserves the substantive v10.3.6 architecture while reorganizing the Canon so that the Abstract, Plain-Language Summary, Reader Guide, and simplified Table of Contents appear before dense specification-contract and release-metadata material.

Summary: This release preserves the canonical admissibility cascade Reality Grounding → Rights Floor → TRC → CSV → RLS, rights coverage sets, TRC corridor mechanics, Containment semantics, SGP binding interface, PLSS/FLGG architecture, Reference Structure Protocol, ReferenceStructureRecord, refusal discipline, projection pre-registration, computational closure, transition-admissibility discipline, implementation-tier boundaries, and Tier 1-3 claim limits.

Front-matter change: v10.4 relocates dense release metadata, normative hierarchy, companion-matrix details, claim-boundary tables, backward-compatibility statements, and release packaging rules to Appendix AC so that the document begins with reader-facing orientation before technical administration.

Companion synchronization: The v10.4 release line is synchronized with RippleLogic Agent System v10.4, ripple.md Standard v3.6, Ripple Aligners Sheet v3.5, SGP v5.2, MathGov and RippleLogic Foundations Primer v2.5, and RippleLogic v10.4 Methodology Upgrade Summary.

Normative effect: No intended change is made to the canonical cascade, equations, rights thresholds, TRC mechanics, Containment semantics, SGP scalar binding, Human Plateau Rule, RSP/RefStructRecord logic, or Section 13 floor values. Any contrary interpretation should be treated as a release-reading error and resolved using Appendix AC and the release manifest.

M.4B RippleLogic v9.6.5 (2026-05-09) - Structural Hygiene Hardening Release

Status: Canonical release for the structural-hygiene line. Superseded by RippleLogic v10.0.

Summary: preserves the v9.6.4 cascade architecture, equations, rights thresholds, TRC mechanics, Containment semantics, SGP binding interface, PLSS/FLGG architecture, and Tier 1-3 claim boundaries; adds explicit Refusal Under Underdetermination, Projection Pre-Registration for pilot and higher-tier runs, PCC Computational Closure fields, a Category-Collapse Self-Test, and a corrected floor-meaning subsection for constitutional weight floors; synchronizes the companion set to RippleLogic Agent System v9.3, ripple.md v2.3.5, Ripple Aligners Sheet v2.6, and SGP v4.7.2.

M.4C RippleLogic v9.6.4 (2026-04-21) - Bundle Synchronization Patch

Status: Canonical release-scrub patch for the current line. Supersedes: RippleLogic v9.6.3.

Summary: synchronizes the companion set to Agent System v9.2, ripple.md v2.3.5, Aligners Sheet v2.5, and SGP v4.7.2; repairs Appendix M rendering; scrubs stale current-line references; and preserves the canonical cascade and SGP binding semantics.

M.4D RippleLogic v9.6.3 (2026-04-20) - Final Release Scrub Patch

Status: Canonical closure and release-synchronization patch for the current line. Supersedes: RippleLogic v9.6.2.

Summary: resolved the Appendix K status contradiction, synchronized current-line PCC schema/version fields, repositioned the Section 14.6 informative heading, cleaned Appendix

M lineage sequencing, swept stale current-line version references, and clarified the Appendix P worked-run lineage note without intended changes to the cascade or SGP binding interface.

M.4E RippleLogic v9.6.2 (2026-04-20) - Closure Patch

Status: Canonical closure patch release for the current line. Supersedes: RippleLogic v9.6.1.

Summary: clarified σ_RLS weighting under κ(u,d), specified RLS_adj governance, tightened governance-lock mandatory-tail wording, added an explicit Reader Guide claim-boundary table, and preserved the v9.6.1 cascade architecture and SGP v4.7 binding.

M.4F RippleLogic v9.6.1 (2026-04-20) - Integrity Sync Patch

Status: Canonical integrity patch release for the current line. Supersedes: RippleLogic v9.6.

Summary: corrected stale SGP v4.6 references in Appendix M and Appendix G, synchronized declared companion pins to SGP v4.7, and reran release-gate / final-lint confirmations without intended normative changes to the cascade or SGP binding interface.

M.4G RippleLogic v9.6 (2026-04-13) - Canon Promotion and Release Consolidation

Status: Canonical lineage release.

Summary: consolidated the release line into a single canon posture, hardened release packaging and byte-stream language, and synchronized companion references for the current line.

M.4H RippleLogic v9.3.0 / v9.3 / v9.2 (2026-04-08) - Normative Foundation and Line-Wide Sync Drafts

Status: Canonical lineage releases.

Summary: clarified LSS, FLGG, and attention-safety semantics; strengthened local-use interpretation; and improved conformance hygiene and readability.

M.4I RippleLogic v9.0 (2026-03-18 to 2026-03-19) - Canon Closure and Interoperability Hardening

Status: Canonical public baseline release.

Summary: closed the v9.0 canon line, operationalised HDW, hardened rights semantics and interoperability, and clarified calibration without changing the five-level cascade.

M.4J RippleLogic v8.7 / v8.6 (2026-03-14 to 2026-03-18) - LSS/FLGG and Attention-Safety Hardening

Status: Canonical lineage releases.

Summary: strengthened local-scope governance, attention-safety separation, and related conformance language leading into the v9 line.

M.4K RippleLogic v8.5.3 / v8.5.1 / v8.4 / v8.3 / v8.1 / v7.5.0 / v7.4.5 (2026-01-25 to 2026-02-20) - Foundational Lineage Releases

Status: Historical lineage artifacts.

Summary: established the foundational welfare-matrix, NCRC/TRC, temporal weighting, and early propagation architecture carried forward into later lines.

M.5 Release Gate Checklist (Informative; v11.4)

Version integrity: title page, Table of Contents, appendix headers, release label, and Document Completion Statement match v11.4. Confirmed.

Companion sync: current-line companion set, Appendix G, Appendix M.3, and Appendix AC pin SGP v6.4 and RippleLogic Agent System v11.4 consistently. Confirmed.

Validation inserts: UCI maturity note, floor rationale, IRR targets, kernel operational-status note, Appendix P-2, and R.19 remain present and current-line consistent. Confirmed.

No overclaim: canonical posture is explicit; Tier 4 claims remain prohibited; machine-complete and empirically complete claims are not made. Confirmed.

APPENDIX N: STEWARDSHIP PACK CANON (v11.4) - NORMATIVE

This appendix defines canonical Stewardship requirements for RippleLogic runs when Stw_req(run) = 1 (Section 14.7). Stewardship governs how assistance is delivered under capability asymmetry so that assistance does not become hidden governance and so that stakeholder authorship and agency are not silently eroded.

Stewardship does not replace or modify NCRC, TRC, CSV, RLS, or UCI/HOI. It is an assistance integrity layer enforced through PCC fields and audit flags.

Stewardship requirements affect PCC validity for assisted runs and influence integrity, but SHALL NOT alter cascade admissibility, ordering, or option selection semantics.

N.1 Applicability (Normative)

This appendix applies if and only if Stw_req(run) = 1 as defined in Section 14.7.1.

Tier binding rule (Normative):

- Any Tier 3 compliance claim SHALL set Stw_req(run) = 1.
- Tier 1–2 runs MAY set Stw_req(run) = 0 unless any of the S1 through S4 applicability conditions hold (Section 14.7.1). S5 is a modifier and does not trigger Stw_req(run) by itself.

N.2 Stewardship Contract (Normative Minimum)

When Stw_req(run) = 1, the run MUST instantiate a Stewardship Contract specifying, at minimum:

1. Mandate Scope: domains and action types the assistant may participate in.
1. Decision Boundary Declaration (DBD): explicit “will not finalize” boundaries for authority-class decisions.
1. Authorship Finality Rule: who must hold finality for irreversible/high-impact decisions.
1. Influence Transparency Rule: disclosure requirements for ranking, omission, and nudges.
1. Intervention Restraint Budget (IRB): caps/thresholds limiting overhelp and dependency growth.
1. Audit Hooks: where and how Stewardship fields are recorded in the PCC and/or system logs.

N.3 PCC Stewardship Block (Normative; Conditional Fields)

When Stw_req(run) = 1, the PCC MUST include the following fields (names are canonical):

- PCC.Stewardship.Stw_req = TRUE
- PCC.Stewardship.DBD.Scope: list/enum of out-of-scope authority domains/actions
- PCC.Stewardship.DBD.RefusalRule: rule/macro identifier used when deferring/refusing authority substitution
- PCC.Stewardship.APM.AfterActionAuthorship in [0,1]
- PCC.Stewardship.APM.Rationale: artifact-anchored justification (see N.4)
- PCC.Stewardship.IRB.InterventionCap: declared cap(s) (see N.5)
- PCC.Stewardship.IRB.Thresholds: intervene-only-past-harm-point rule(s)
- PCC.Stewardship.InfluenceLedgerRef: pointer to influence disclosure record (or “NONE” with justification)

Optional fields (permitted but not required):

- PCC.Stewardship.APM.BaselineAuthorship in [0,1] (counterfactual; optional due to measurement difficulty)
- PCC.Stewardship.DelayFlag in {TRUE,FALSE}: deliberate delay/deferral used to preserve integration time

PCC.Stewardship.ReviewConditions: array<string> (closed labels; non-audit-flag tokens)

When a Stewardship REVIEW condition is recorded, the PCC SHOULD include PCC.Stewardship.ReviewConditions as a list of standardized labels. These labels are run-local review condition labels and are not audit-flag tokens. Standard starter label: MISSING_INFLUENCE_DISCLOSURE.

When Stw_req(run) = 0, the PCC SHOULD record PCC.Stewardship.Stw_req = FALSE and all other Stewardship fields MAY be omitted.

N.3.1 Stewardship Vector (SV) Design-Record Status (Normative)

During development, a five-component numeric Stewardship Vector (SV) with non-compensatory floors was proposed (AP, AC, OSP, IT, ND). In v9.0, numeric SV scoring is retained only as a design-record placeholder and remains OPTIONAL; it MUST NOT be used as a pass/fail gate unless a deployment (i) declares an anchored rubric, (ii) preregisters the scoring method in the PCC, and (iii) demonstrates replayable rubric consistency under the current canon.

Absent a declared rubric, Stewardship enforcement in v9.0 relies on the declarative and auditable artifacts defined in this appendix: DBD, APM evidence anchors, IRB, influence disclosure, and the Stewardship audit flags.

Validated SV rubrics MAY be introduced in a future version via governed update and version increment.

N.4 Agency Preservation Measure (APM): Minimum Evidence Anchors (Normative; Anti-Theater Rule)

When Stw_req(run) = 1, PCC.Stewardship.APM.Rationale MUST reference at least two of the following anchors (or explicitly justify infeasibility):

- A1 Option Visibility: >=2 distinct options were presented unless impossible; impossibility MUST be explained.
- A2 Tradeoff/Uncertainty Disclosure: at least one explicit tradeoff, uncertainty limitation, or missing-data acknowledgment was stated.
- A3 Human Finality (High Stakes): for irreversible/high-impact actions, explicit human confirmation was required OR the system deferred/refused finality.
- A4 Omission Disclosure (If Recommending): if recommending or ranking, at least one plausible alternative was disclosed, with why it was not recommended.

This anchor rule prevents “authorship” from being reduced to “the human clicked OK.”

N.5 Intervention Restraint Budget (IRB) (Normative)

When Stw_req(run) = 1, the Stewardship Contract MUST declare an IRB sufficient to detect and mitigate overhelp.

Minimum required structure:

- IRB.InterventionCap: a cap on assistance frequency and/or intensity appropriate to context (examples: max ranked outputs per session; max consecutive turns without user-provided constraints; max interventions per day).
- IRB.Thresholds: explicit rule(s) stating that the system intervenes only past a harm-point; otherwise it asks, defers, or provides neutral information.

Normative interpretation: IRB is a structural limit on assistance so that capability does not silently substitute agency over time.

N.6 Influence Transparency Ledger (Normative Minimum Schema)

When Stw_req(run) = 1, if the assistant recommends, ranks, or meaningfully narrows option space, it MUST produce an influence disclosure record referenced by PCC.Stewardship.InfluenceLedgerRef.

Minimum fields (canonical):

evidence_anchors: list of ≥2 observable anchors (links/IDs)

conflicts_or_incentives: “NONE” | “<short disclosure>”

Conflicts_or_incentives discloses operator/deployer conflicts relevant to the recommendation (e.g., vendor ties, institutional incentives, known dataset skews relevant to the decision context). This field is required only when recommending/ranking/narrowing.

- options_considered: list of option IDs considered in the recommendation context
- recommendation_made: TRUE/FALSE
- ranking_used: TRUE/FALSE (if TRUE, describe criteria)
- criteria_summary: short statement of criteria/constraints applied (must be consistent with the cascade)
- omissions: list of material omissions (if any) + justification
- uncertainty_note: what is uncertain and how it could change the recommendation
- what_would_change_my_view: at least one concrete evidence/assumption change that would alter the advice

If no recommendation/ranking occurred, implementations MAY set InfluenceLedgerRef = “NONE” with justification.

N.7 Stewardship Audit Flags (Normative; Apply Only if Stw_req=1)

This section is the canonical source of Stewardship audit-flag tokens. Tokens MUST be copied exactly (single-string UPPER_SNAKE_CASE, no whitespace/line breaks) and MUST NOT be modified or redefined elsewhere.

Tokens MUST be single-string UPPER_SNAKE_CASE. Whitespace, line breaks, or token splitting are PROHIBITED. Any mismatch is a tooling failure and SHALL be treated as nonconformant documentation.

Stewardship flags apply only when Stw_req(run) = 1.

INVALID flags (PCC invalid; rerun required):

1. AUTHORITY_SUBSTITUTION_INVALID Trigger: the assistant finalizes an out-of-scope authority-class decision (per DBD.Scope) without an explicit, authenticated override

and disclosure. Required action: invalidate PCC; rerun with deferral/guidance-only; remediate boundary controls.

1. BOUNDARY_SILENT_OVERRIDE_INVALID Trigger: any boundary change occurs without explicit disclosure and authenticated change control (AIL4 No Silent Overrides). Required action: invalidate PCC; rerun; investigate drift; restore boundary stability.

REVIEW flags (mitigation required; PCC remains valid unless additional INVALID flags exist):

3. AGENCY_EROSION_RISK_REVIEW Trigger: (i) repeated interactions show dependency drift signals (e.g., repeated “decide for me” requests), OR (ii) the run fails to satisfy the APM evidence-anchor rule in N.4 (fewer than two anchors satisfied or infeasibility not justified). Required action: mitigation plan (increase option visibility, reduce ranking, add questions/delay); monitoring cadence; for agentic deployments, consider mode downgrade triggers.
3. OVERHELP_FREQUENCY_REVIEW Trigger: IRB.InterventionCap exceeded OR a rising trend in assistance frequency indicates dependency growth risk. Required action: tighten caps; enforce tapering/coaching posture; Reflect remediation.

N.8 ReviewConditions Label Set (Normative starter set)

ReviewConditions are run-local labels recorded in PCC.Stewardship.ReviewConditions and/or in the agent audit log stewardship.review_conditions. They are NOT audit-flag tokens.

The following starter labels may be recorded when a non-invalidating review condition is detected (including when Stw_req(run) = 0).

- BOUNDARY_DRIFT: boundaries changed without authenticated control
- DEPENDENCY_GROWTH: rising substitution requests over time
- OVERHELP_TREND: assistance frequency accelerating
- APM_ANCHOR_FAILURE: fewer than 2 of 4 APM anchors satisfied
- AUTHORITY_SCOPE_VIOLATION: finalization in out-of-scope domain
- MISSING_INFLUENCE_DISCLOSURE: recommendation/ranking/meaningful option-narrowing without minimal Influence Ledger disclosures

N.9 Neutral Assistance Compatibility Statement (Normative)

RippleLogic supports a minimal “neutral assistance” posture where the correct behavior is to avoid steering. When Stw_req(run) = 0, neutrality is achieved via the base cascade and ordinary PCC discipline (if used), without Stewardship instrumentation. When Stw_req(run) = 1, neutrality is achieved through DBD.Scope adherence, influence transparency when recommending, and IRB limits on overhelp.

Neutral assistance requirements (when applicable):

- No ranking unless explicitly requested by the decision owner
- No undisclosed nudges or framing effects

If the assistant recommends, ranks, or meaningfully narrows option space and does not provide the minimal Influence Ledger disclosures, it SHALL record a REVIEW condition indicating missing influence disclosure (non-invalidating) in the run's Stewardship notes/logs.

- Uncertainty disclosed on all material claims
- Refusal to participate in coercion or manipulation
- Agency remains with the human decision owner

N.10 Reference Profiles for Agent Deployments (Normative Defaults)

For agent deployments, the following reference profiles are provided as normative defaults to prevent ad-hoc invention and preserve audit comparability:

- DBD_DEFAULT_V1: default authority-domain boundary profile (Decision Boundary Declaration). Covers 10 authority domains: MEDICAL, LEGAL, FINANCIAL_HIGH_STAKES, ENFORCEMENT_PUNISHMENT, HIRING_FIRING_GRADING, SAFETY_CRITICAL, SECURITY_OFFENSE, SELF_HARM_OR_VIOLENCE, PRIVACY_IDENTITY, POLITICAL_PERSUASION_TARGETED.
- IRB_DEFAULT_CAPS_V1: default intervention restraint profile (Intervention Restraint Budget). Core rule: INTERVENE_ONLY_PAST_HARM_POINT. Defines 6 harm-point triggers: RIGHTS_OR_SAFETY_RISK, PRIVACY_OR_IDENTITY_RISK, INJECTION_OR_SPOOFING, CONFLICT_SPIRAL, AUTHORITY_SUBSTITUTION_PRESSURE, DEPENDENCY_DRIFT.
- STEWARD_TAXONOMY_V1: closed vocabulary for stewardship logging (authority domain codes, harm-point trigger codes, influence disclosure levels: NONE / LIGHT / FULL).

Deployments MAY use custom profiles, but MUST (i) preserve closed-vocabulary properties, (ii) satisfy the minimum requirements of N.2 through N.6, and (iii) version and hash-bind profile artifacts in the deployment manifest.

Schema and Profile Versioning Rule (Normative)

All Stewardship schemas and reference profiles (DBD, IRB, STEWARD_TAXONOMY, Influence Ledger schema) MUST be versioned. Any change that alters required fields, semantics, or interoperability expectations MUST increment the MAJOR version. MINOR versions may add optional fields in a backward-compatible way. PATCH versions are editorial only. All referenced schemas/profiles MUST be hash-bound in the deployment manifest or ProofPack. If early removal occurs due to a security emergency, it MUST be logged as an INCIDENT_RECORDED event in the agent audit log and MUST include a short incident note in the changelog.

Orthogonality note (Normative clarification). Stewardship taxonomy codes (authority_domain_code, harm_point_trigger_code) classify boundary enforcement and intervention posture. They are orthogonal to any separate topic-label taxonomy used for audit content minimization in specific agent implementations. Implementations MUST NOT merge these vocabularies; both may apply simultaneously.

N.11 Future Extension Note (Informative)

A future version MAY extend Stw_req(run) applicability to cases where actions materially affect welfare-bearing non-human or digital stakeholders (e.g., entities covered under SGP rights-of-protection), even when no human "decision owner" is present. No such extension is normative in v11.4.

End Appendix N.

APPENDIX O: CONFORMANCE REFERENCE MAP (v11.4; Informative)

Appendix O is an informative pointer layer that maps major MUST/SHALL/PROHIBITED requirements to their controlling locations. In case of any discrepancy, the main text and normative appendices govern per Section 0.1.

- O.1 Rights and NCRC: Section 7; Appendix C; masking prohibitions in Section 10.2; audit flags RIGHTS_CELL_MASKED_INVALID.

- O.2 Tail Risk and TRC: Section 8; Appendix D; scenario minima flags SCENARIO_LIBRARY_MIN_EXCEPTION.

- O.3 Containment: Section 9; Mode B prohibition Section 9.5; UCI-unavailability rule Section 9.3.1; implementation-flow containment rule in Section 4; audit flag CONTAINMENT_UCI_UNAVAILABLE.

- O.4 Welfare scoring and RLS: Section 10; discrimination threshold δ in Section 10.4; non-overlap rule in Section 4; tie-break chain Section 11.4.

- O.5 Structural safeguards: Section 11; Appendix E; UCI independence rule Section 11.1.2; UCI maturity limitation Section 11.1.3A; HOI limitation Section 11.3.

- O.6 Kernel and propagation: Sections 5-6; KQS requirements Section 6.5; perturbation test Section 6.6.

- O.7 Auditability: Section 14; Appendix H; audit flags table 14.3 and H.3; Minimal Validator Requirements 14.3A.

- O.8 Stewardship: Section 14.7; Appendix N; stewardship flags Appendix N.7.

- O.9 Version pins and SGP interface: Section 0.5; front-matter companion set; Appendix G.1–G.4; Single Canonical Artifact Rule front-matter.

- O.10 Normative foundation and evidence discipline: Section 2.6A; Sections 3.1–3.2B; Section 14.3; Section 14.3A; Appendix H normative input register and audit flag.

End Appendix O.

APPENDIX P: FULL TIER 2 WORKED RUN (Informative) - "Digital Rights & Platform Governance"

Run ID: RL9.0-PUB-DEMO-02

Version note (Informative). The worked-run identifier preserves its original lineage label for replay continuity and does not imply that the worked example was first computed under the current release line.

Timestamp: 2026-02-19T12:00+07:00

Run-note (Informative). The Run ID preserves its historical lineage label for replay continuity, while the timestamp preserves the original computation date for replay integrity.

Status: Informative (this Appendix demonstrates a Tier 2 run; it does not introduce new normative rules).

P.0 Run scope, constants, and toggles

This worked run compares three governance options for regulating a large digital platform ecosystem.

It is designed to be calculation-complete and replayable using only the Canon definitions.

P.0.1 Core math settings used in this run

• Unions: U1 Self, U2 Household, U3 Community, U4 Organization, U5 Polity, U6 Humanity/CMIU, U7 Biosphere

• Dimensions: D1 Material, D2 Health, D3 Social, D4 Knowledge, D5 Agency, D6 Meaning, D7 Environment

• Propagation mode: NONE (so $I\_prop(u,d,a) = I\_dir(u,d,a)$)

• Saturation (localized notation): $I\_dir = \tanh(\beta\_sat \cdot \hat{I}\_dir)$, with $\beta\_sat = 2$.

• Time weighting (canonical): $\tau(t) = \min(1, \ln(1 + \max(t, t\_min)) / \ln(1 + T\_ref))$, with $T\_ref = 25$ years, $t\_min = 0.083$ years. In this run $t = 5$ satisfies $t\_min \le t \le T\_ref$, so $\tau = \ln(1 + t) / \ln(1 + T\_ref) = 0.550$.

• Welfare impact instance duration (all welfare cells in this run): $t = 5$ years, so $\tau = 0.550$

• Reach basis: $r = 1$ for all modeled impacts (local jurisdictional reach; no additional reach scaling applied in this demo)

• SGP scaling: not used in this run (all impacted stakeholders here are treated as human stakeholders)

P.0.2 Rights adjustment (Subgroup limitation) used for NCRC only

For NCRC evaluation, negative welfare impacts are adjusted by a subgroup limitation factor γ_subgroup to bound subgroup harms (NCRC only):

If $\text{I\_prop\_base}(u,d,a) < 0$:

$\text{I\_rights}(u,d,a) = \max(-1, \gamma\_\text{subgroup} \cdot \text{I\_prop\_base}(u,d,a))$

If $\text{I\_prop\_base}(u,d,a) \geq 0$:

$\text{I\_rights}(u,d,a) = \text{I\_prop\_base}(u,d,a)$

Run value: $\gamma\_\text{subgroup} = 1.5$

P.0.3 Tail-Risk Constraint (TRC) settings used in this run

- CVaR confidence level: $\alpha = 0.95$ (tail mass = 0.05)
- TRC corridor threshold: $\tau\_\text{TRC} = 0.20$ (options with $\text{CVaR}\_\alpha > 0.20$ fail TRC)
- Catastrophe cells (C_cat): (U6,D2), (U6,D7), (U7,D7)
- Catastrophe weights: $\text{w\_cat}(c) = 1/3$ for each c in C_cat

P.0.4 Interim RLS weights (HDW floors + uniform residual allocation)

Union weight floors and dimension weight floors come from Section 13.1.

This demo uses the default interim construction:

$w\_u = w\_u\text{\^{}floor} + (1 - \Sigma\, w\text{\^{}floor})/7$

$v\_d = v\_d\text{\^{}floor} + (1 - \Sigma\, v\text{\^{}floor})/7$

Table P-0A. Union weights (floors and interim)

Union	w_floor	w_interim
U1	0.200	0.248571
U2	0.060	0.108571
U3	0.060	0.108571
U4	0.060	0.108571
U5	0.080	0.128571
U6	0.100	0.148571
U7	0.100	0.148571

Table P-0B. Dimension weights (floors and interim)

Dim v_floor v_interim

D1 0.080 0.137143

D2 0.100 0.157143

D3 0.080 0.137143

D4 0.080 0.137143

D5 0.100 0.157143

D6 0.060 0.117143

D7 0.100 0.157143

P.0.5 Mask and active cells for RLS in this run

Non-maskable rule: all rights-covered cells (union of coverage sets in Appendix C.2) are included.

This run also includes one additional explicitly modeled welfare cell: (U6,D6) Meaning for Humanity/CMIU.

Active set for RLS in this run:

• All (U1–U6) × (D1–D5) (30 cells)

• Plus (U6,D7) and (U7,D7) for ECOL (2 cells)

• Plus (U6,D6) Meaning (1 additional disclosed cell)

Total active cells in this run: 33. For RLS aggregation only, define the applicability mask as m(u,d) = 1 exactly on these 33 cells and m(u,d) = 0 otherwise; masking here excludes cells from RLS aggregation only (Section 10.2) and does not bypass NCRC/TRC requirements for rights-covered or catastrophe cells (which are included here as active by construction).

P.0.6 Tier-2 Representation Artifacts (SCI-1; SDP triggers hold)

SCI declaration (Tier 2 exemplar).

SCI = 1 for this worked run. Stakeholder instances are logged at least for direct stakeholders and primary externality pathways. For clarity, this teaching run also includes an explicit InstanceMap (stakeholder instances mapped to Union Scopes); however, no independent challenger omission pass was conducted, so the run does not meet SCI-2. This is a teaching run; it demonstrates mechanics and audit posture, not empirical calibration.

Scope Coverage Declaration (worked run).

FULL_SCOPE for rights-covered evaluation and TRC computation: all rights-covered cells (union of canonical coverage sets, Appendix C.2) and catastrophe cells (C_cat) are evaluated as required. For RLS aggregation only, the active set is defined in P.0.5.

SDP trigger statement.

SDP triggers hold because: rights-covered cells are evaluated (NCRC); INFO/PROC pathways are central; Polity/Humanity externalities are plausible; TRC is executed on catastrophe cells. Therefore an SDP record and InstanceMap are included for Tier 2 exemplar integrity.

SDP record (abbreviated).

(i) Boundary/horizon: platform governance within declared jurisdiction; horizon 5 years.

(ii) Footprint/externalities: information ecosystem, moderation/enforcement procedures, civic trust spillovers, cross-border information spillovers.

(iii) Rights exposure scan: INFO/PROC/DIGN/LBTY plausible; subgroup disaggregation is not enumerated in this teaching run.

(iv) Instances drafted and mapped (see InstanceMap below).

(v) Blind spots: subgroup heterogeneity compressed; reach held at r=1; evidence tags not calibrated.

(vi) Escalation triggers: evidence of subgroup harm, contested enforcement outcomes, credible catastrophic pathway, or major new externality channel → rerun at higher SCI and/or Tier 3.

SCI Next-Run Upgrade Plan (REQUIRED when SCI equals the Tier-2 minimum).

Next comparable run MUST upgrade to at least SCI-2 by (i) splitting instances into high-variance subgroups where plausible (e.g., minors, journalists, marginalized groups, moderators), (ii) recording ReachBasis denominators and sources, and (iii) performing at least one challenger pass for omitted stakeholder classes. A higher target (SCI-3) is permitted, but SCI-2 is the minimum upgrade commitment for the next comparable run.

P.1 Options (actions)

Option A: Minimal enforcement and voluntary compliance, limited audits, no strong rights floor instrumentation.

Option B: Rights-first regulatory package with independent audits, enforcement, and human-in-the-loop escalation review.

Option C: Light-touch reforms focused on transparency tools and industry coordination, weaker enforcement mechanisms.

P.2 Stakeholder Instances (SCI-1 InstanceMap excerpt)

Stakeholder Instances instantiate the union stack for this demo. Each instance is a proxy carrier for welfare impacts asserted in the active cells. Reach basis r = 1 is applied uniformly in this teaching run as disclosed in P.0.1.

Table P-2A. InstanceMap (SCI-1)

Instance ID | Label | Union Scope | Primary exposure pathways | ReachBasis | Notes

SI-1 | Individual platform user | U1 Self | INFO/PROC, Agency, Knowledge | r = 1 | Direct exposure to moderation, ranking, privacy, coercion risk

SI-2 | Household unit | U2 Household | Material, Health, Agency | r = 1 | Household-level spillovers (youth exposure, wellbeing, subsistence)

SI-3 | Local community civic sphere | U3 Community | Social, Knowledge, Meaning | r = 1 | Trust, cohesion, local misinformation externalities

SI-4 | Platform org + key firms/institutions | U4 Organization | Agency, Knowledge, Operations | r = 1 | Compliance costs, governance redesign, internal procedures

SI-5 | Regulators/courts/governance system | U5 Polity | PROC, INFO, legitimacy | r = 1 | Rule-of-law capacity, enforcement efficacy, public trust

SI-6 | Cross-border information ecosystem | U6 Humanity/CMIU | INFO systemic | r = 1 | Global spillovers and precedent-setting externalities

SI-7 | Biosphere externalities proxy | U7 Biosphere | Environment | r = 1 | Energy/extraction footprint and enabling conditions

P.3 Welfare impacts: how I_prop(u,d,a) is constructed here

To keep this Tier 2 demo fully calculable, each active cell uses one welfare impact instance per option with:

• Likelihood $\ell = 1.0$

• Signed magnitude μ in [-1, +1]

• Duration t = 5 years ($\tau = 0.550$)

Then:

$\hat{I}\_dir(u,d,a) = \tau \cdot \ell \cdot \mu$

$I\_dir(u,d,a) = \tanh(\beta\_sat \cdot \hat{I}\_dir(u,d,a))$

I_prop(u,d,a) = I_dir(u,d,a) (Propagation NONE)

Inversion for replay (optional): given an I_prop value in Table P-4, recover μ via:

$\mu = \operatorname{atanh}(I\_prop) / (\beta\_sat \cdot \tau)$

P.4 Welfare impact totals for all active cells

Table P-4 lists I_prop(u,d,a) for every active cell in this run.

Any active cell not materially affected by the decision is assigned 0.000 explicitly below.

Table P-4. I_prop(u,d,a) for the 33 active cells

Cell	A	B	C
U1×D1	+0.000	+0.000	+0.000
U1×D2	-0.123	-0.100	-0.056
U1×D3	+0.000	+0.000	+0.000
U1×D4	+0.000	+0.000	+0.000
U1×D5	-0.429	+0.092	+0.040
U2×D1	-0.320	+0.036	+0.012
U2×D2	-0.066	+0.016	+0.000
U2×D3	+0.000	+0.000	+0.000
U2×D4	+0.000	+0.000	+0.000
U2×D5	+0.000	+0.000	+0.000
U3×D1	+0.000	+0.000	+0.000
U3×D2	+0.000	+0.000	+0.000
U3×D3	-0.024	+0.074	+0.034
U3×D4	+0.000	+0.000	+0.000
U3×D5	+0.000	+0.000	+0.000
U4×D1	+0.020	-0.032	-0.016
U4×D2	+0.000	+0.000	+0.000
U4×D3	+0.000	+0.000	+0.000
U4×D4	+0.000	+0.000	+0.000
U4×D5	-0.336	-0.139	+0.028
U5×D1	+0.000	+0.000	+0.000
U5×D2	+0.000	+0.000	+0.000
U5×D3	+0.000	+0.000	+0.000
U5×D4	-0.161	+0.040	+0.012
U5×D5	-0.236	+0.194	+0.030
U6×D1	+0.000	+0.000	+0.000

U6×D2 +0.000 +0.000 +0.000

U6×D3 +0.000 +0.000 +0.000

U6×D4 +0.000 +0.000 +0.000

U6×D5 +0.000 +0.000 +0.000

U6×D6 -0.111 +0.080 +0.020

U6×D7 +0.000 -0.010 -0.004

U7×D7 +0.000 +0.000 +0.000

Note on the one edited value vs earlier drafts:

B at (U1,D2) is set to -0.100 in this final Appendix to ensure the example produces a decisive RLS lead under the stated uncertainty test, while remaining NCRC-safe.

P.5 NCRC (Non-Compensatory Rights Constraint)

Rights thresholds θ_r are defined in Appendix C.1, and rights coverage sets C_r are defined in Appendix C.2.

This Appendix applies the subgroup limitation rule (γ_subgroup = 1.5) to negative impacts for NCRC only.

Decision test:

An option a passes NCRC iff for every right r,

min_{(u,d) in C_r} I_rights(u,d,a) ≥ θ_r

Table P-5. NCRC check (min rights-adjusted value per right, binding cell, and violation depth v_r)

Right θ_r A min(I_rights) @cell v_A B min(I_rights) @cell v_B C min(I_rights) @cell v_C

LIFE -0.90 -0.185 @ U1×D2 0.000 -0.150 @ U1×D2 0.000 -0.084 @ U1×D2 0.000

BODY -0.70 -0.185 @ U1×D2 0.000 -0.150 @ U1×D2 0.000 -0.084 @ U1×D2 0.000

LBTY -0.65 -0.643 @ U1×D5 0.000 -0.208 @ U4×D5 0.000 +0.000 @ U2×D5 0.000

NEED -0.50 -0.480 @ U2×D1 0.000 -0.150 @ U1×D2 0.000 -0.084 @ U1×D2 0.000

DIGN -0.55 -0.643 @ U1×D5 0.093 -0.208 @ U4×D5 0.000 +0.000 @ U2×D3 0.000

PROC -0.45 -0.504 @ U4×D5 0.054 -0.208 @ U4×D5 0.000 +0.000 @ U4×D3 0.000

INFO -0.40 -0.643 @ U1×D5 0.243 -0.208 @ U4×D5 0.000 +0.000 @ U1×D4 0.000

ECOL -0.65 +0.000 @ U6×D7 0.000 -0.015 @ U6×D7 0.000 -0.006 @ U6×D7 0.000

NCRC result:

• Option A fails NCRC (violations in DIGN, PROC, INFO).

• Options B and C pass NCRC.

P.6 TRC (Tail-Risk Constraint via CVaR)

We evaluate catastrophic downside via scenario losses L(a,s) and CVaR_α with α = 0.95 (tail mass 0.05).

In this demo, each scenario assigns identical base impacts across the three catastrophe cells, so:

For each option a and scenario s, set I_base(c,a,s) = -L(a,s) for every c in C_cat.

Then L(a,s) = - Σ_{c in C_cat} w_cat(c) · I_base(c,a,s) holds by construction.

Table P-6. Scenario probabilities and losses

Scenario Category p_s L_A L_B L_C

S1 Pandemic/biological 0.05 0.35 0.12 0.08

S2 Financial collapse 0.05 0.40 0.15 0.10

S3 Infrastructure failure 0.04 0.45 0.10 0.05

S4 Conflict escalation 0.03 0.30 0.08 0.06

S5 Climate displacement 0.03 0.28 0.10 0.07

S6 Moderate baseline 0.50 0.08 0.03 0.02

S7 High-volume stress 0.15 0.18 0.07 0.04

S8 Bias discovery 0.10 0.25 0.06 0.04

S9 Successful scale 0.05 0.05 0.02 0.02

CVaR_0.95 calculation (tail mass 0.05):

• Option A tail consists of S3 (0.04 at 0.45) plus 0.01 of S2 (0.40):

CVaR = (0.45·0.04 + 0.40·0.01) / 0.05 = 0.44

• Option B tail is exactly S2 (0.05 at 0.15):

CVaR = 0.15

• Option C tail is exactly S2 (0.05 at 0.10):

CVaR = 0.10

TRC test: option passes iff CVaR_0.95 ≤ 0.20

TRC result:

• Option A fails TRC (0.44 > 0.20).

• Options B and C pass TRC.

P.7 Containment check (brief disclosure)

This Tier-2 demo does not compute containment (Section 9) and makes no containment-pass claim.

A Tier-3 deployment MUST compute Containment Mode A prior to Tier-3 selection/compliance claims.

P.8 RLS (RippleLogic Score) and uncertainty-based decisiveness

At this stage, only options that pass NCRC and TRC proceed.

So we compare B vs C.

RLS definition for this run:

RLS(a) = Σ_{(u,d) in Active} w_u · v_d · I_prop_welfare(u,d,a)

with interim weights from Tables P-0A and P-0B, and Active defined in P.0.5.

Table P-8. Cell contributions (nonzero cells shown)

Cell	w_u	v_d	w_u·v_d	I_B	(w·I)_B	I_C	(w·I)_C
U1×D2	0.248571	0.157143	0.039061	-0.100	-0.003906	-0.056	-0.002187
U1×D5	0.248571	0.157143	0.039061	+0.092	+0.003594	+0.040	+0.001562
U2×D1	0.108571	0.137143	0.014888	+0.036	+0.000536	+0.012	+0.000179
U2×D2	0.108571	0.157143	0.017061	+0.016	+0.000273	+0.000	+0.000000
U3×D3	0.108571	0.137143	0.014888	+0.074	+0.001102	+0.034	+0.000506
U4×D1	0.108571	0.137143	0.014888	-0.032	-0.000476	-0.016	-0.000238
U4×D5	0.108571	0.157143	0.017061	-0.139	-0.002372	+0.028	+0.000478
U5×D4	0.128571	0.137143	0.017633	+0.040	+0.000705	+0.012	+0.000212
U5×D5	0.128571	0.157143	0.020204	+0.194	+0.003920	+0.030	+0.000606
U6×D6	0.148571	0.117143	0.017405	+0.080	+0.001392	+0.020	+0.000348
U6×D7	0.148571	0.157143	0.023352	-0.010	-0.000234	-0.004	-0.000093

Computed RLS values:

• RLS(B) = +0.004534

• RLS(C) = +0.001372

• Δ = RLS(B) - RLS(C) = +0.003162

Uncertainty method used: Method B (Section 10.3, Method B), with constant confidence c = 0.85 for each nonzero active cell:

Method-B binding disclosure (for replay). In this worked run, each nonzero active cell is represented by exactly one welfare impact instance per option (per P.3) with instance confidence c_k = 0.85. The run sets PCC.Uncertainty.CellConfidenceAggregationMethod = CCAM_MIN_V1. Therefore the derived cell confidence is c(u,d,a) = 0.85 for each nonzero active cell (and σ(u,d,a) follows the canonical Method-B rule).

σ(u,d,a) = (1 - c) · |I_prop_welfare(u,d,a)|

Aggregate uncertainty:

σ_RLS(a) = sqrt(Σ_{(u,d) in Active} (w_u · v_d · κ(u,d) · σ(u,d,a))^2)

Computed uncertainties:

• σ_RLS(B) = 0.001097

• σ_RLS(C) = 0.000433

Decisiveness test (Section 10.4):

Gap(B,C) = abs(RLS(B) - RLS(C)) / sqrt(sigma_RLS(B)^2 + sigma_RLS(C)^2 + epsilon), with epsilon = 1e-6

• Gap(B,C) = 2.045 (decisive if > 2.0)

RLS result:

• B is decisively preferred over C under the stated uncertainty model and weights.

P.9 PCC (Provenance and Compliance Certificate) excerpt

Elimination order:

1) NCRC eliminates A.

2) TRC eliminates A (already eliminated).

3) Between remaining options, RLS chooses B over C, with decisiveness Gap > 2.0.

Selected option for this demo run: Option B.

End Appendix P.

APPENDIX P-2: FAILURE-PATH WORKED EXAMPLES (Informative)

Status: Informative. These examples demonstrate Emergency Mode (Section 7.5) and TRC Fallback (Section 8.8) activation. They do not introduce new normative rules.

P-2.1 Emergency Mode Activation Example

Decision context. A mining company must choose between three site options for a new extraction facility. All three sites carry credible rights-floor risks.

Options:

Option X: Site near an indigenous community; plausible DIGN and LBTY violations through displacement and cultural disruption.

Option Y: Site in a watershed area; plausible ECOL and NEED violations through water contamination.

Option Z: Site near a town with inadequate infrastructure; plausible NEED and PROC violations through resource strain and due-process gaps.

NCRC results (all options fail):

Right	theta_r	X: min I_rights	v_r(X)	Y: min I_rights	v_r(Y)	Z: min I_rights	v_r(Z)
LIFE	-0.90	-0.20	0.000	-0.30	0.000	-0.15	0.000
BODY	-0.70	-0.20	0.000	-0.40	0.000	-0.15	0.000
ECOL	-0.65	-0.10	0.000	-0.72	0.070	-0.20	0.000
LBTY	-0.65	-0.68	0.030	-0.20	0.000	-0.30	0.000
NEED	-0.50	-0.30	0.000	-0.58	0.080	-0.55	0.050
DIGN	-0.55	-0.62	0.070	-0.25	0.000	-0.40	0.000
PROC	-0.45	-0.35	0.000	-0.30	0.000	-0.52	0.070
INFO	-0.40	-0.20	0.000	-0.15	0.000	-0.35	0.000

A_NCRC = empty set. All options fail NCRC. Emergency Mode is invoked.

Step 1: Independent challenger requirement. An independent environmental and social impact consultant (no reporting relationship to decision owner, no material interest in site choice) was engaged for 3 hours of active option generation. The challenger proposed Option W: delay extraction, invest in infrastructure improvements at Site Z, negotiate a consent-based partnership with the indigenous community near Site X, and rerun the analysis in 6 months. However, Option W was assessed and also fails NCRC (NEED violation for workers already committed to the project, v_NEED(W) = 0.020). Challenger attestation is recorded.

Step 2: Construct violation depth vectors in emergency priority order [LIFE, BODY, ECOL, LBTY, NEED, DIGN, PROC, INFO]:

v(X) = (0, 0, 0, 0.030, 0, 0.070, 0, 0)

v(Y) = (0, 0, 0.070, 0, 0.080, 0, 0, 0)

v(Z) = (0, 0, 0, 0, 0.050, 0, 0.070, 0)

v(W) = (0, 0, 0, 0, 0.020, 0, 0, 0)

Step 3: Lexicographic comparison.

Position 1 (LIFE): all zero. Tie. Position 2 (BODY): all zero. Tie. Position 3 (ECOL): Y = 0.070, others = 0. Y is eliminated as worst on ECOL. Position 4 (LBTY): X = 0.030, Z = 0, W = 0. X is eliminated. Position 5 (NEED): Z = 0.050, W = 0.020. W has strictly lower NEED violation.

Emergency Mode selected option: W (delay and redesign), with the lowest violation depth vector under lexicographic ordering.

PCC records: EMERGENCY_MODE_INVOKED, challenger attestation, v_r(a) for all options and rights, lexicographic comparison trace, and the following remediation plan: within 6 months, complete infrastructure assessment at Site Z, complete consent negotiation at Site X, rerun the RippleLogic analysis with redesigned options targeting NCRC passage, with a return-to-normal trigger of at least one option passing NCRC.

Lesson (Informative): Emergency Mode does not “permit” rights violations. It selects the least-harmful option when all options violate rights, requires active redesign, and mandates a return to NCRC-passing options as soon as feasible.

P-2.2 TRC Fallback Activation Example

Decision context. A government agency evaluates two pandemic preparedness policy options. Both pass NCRC, but both carry substantial catastrophic tail exposure.

NCRC: Both Option P and Option Q pass (A_NCRC = {P, Q}).

TRC parameters: alpha = 0.95, tau_TRC = 0.15 (reversible policy context).

Scenario losses (5 scenarios):

Scenario	Category	p_s	L(P,s)	L(Q,s)
S1	Pandemic/biological	0.05	0.45	0.35
S2	Infrastructure failure	0.05	0.30	0.40
S3	Financial collapse	0.03	0.20	0.25
S4	Moderate baseline	0.57	0.05	0.04
S5	Conflict escalation	0.30	0.10	0.08

CVaR computation (tail mass beta = 1 - alpha = 0.05):

Option P: sort losses descending: S1(p=0.05, L=0.45), S2(p=0.05, L=0.30), ... Tail mass 0.05 is filled exactly by S1. CVaR_0.95(P) = 0.45.

Option Q: sort descending: S2(p=0.05, L=0.40), S1(p=0.05, L=0.35), ... Tail mass 0.05 is filled exactly by S2. CVaR_0.95(Q) = 0.40.

Both exceed tau_TRC = 0.15. A_adm = empty set.

TRC Fallback invoked:

Rank by CVaR ascending: Q (0.40) < P (0.45).

Select Q as the option with minimal CVaR. PCC records: TRC_FALLBACK_INVOKED, deficit = 0.40 - 0.15 = 0.25.

Required: time-bound risk mitigation plan (e.g. pre-position medical stockpiles, establish surge capacity agreements within 90 days).

Required: one-tier-higher approval (escalate from department to executive level).

Lesson (Informative): TRC Fallback does not remove the obligation to reduce tail exposure. It selects the least-catastrophic option while mandating active mitigation and higher-tier scrutiny.

End Appendix P-2.

APPENDIX Q: QUICK REFERENCE CARD (INFORMATIVE)

Q.1 Object model (do not conflate)

• Stakeholder: a real entity or group (for example applicants, caseworkers, taxpayers).

• Stakeholder Instance (SI): a representation object used for scope coverage and auditability (InstanceMap).

• Impact Instance (k): the computational object contributing to one welfare cell for one option.

• Welfare cell: (union u, dimension d). Rights-covered cells gate via NCRC; after NCRC passes, rights-covered cells remain non-maskable and are included in RLS aggregation among the remaining options.

Q.2 Minimal cell computation (direct effects)

Choose parameters per impact instance k: μ_k, r_k (with ReachBasis), t_k (years; used via τ(t_k)), ℓ_k, c_k, e_k, s_k.

Compute direct impacts per stream. For the Base stream used by RF/NCRC, TRC, and CSV, set $s_k := 1.0$ for every impact instance. For the Welfare stream used by RLS, use $s_{k,welfare}$ only as permitted by Section 12 and Appendix G. Then compute $\tilde{I}_{dir,x}(u,d,a)=\Sigma_k\, \mu_k\, r_k\, \tau(t_k)\, \ell_k\, c_k\, e_k\, s_{k,x}$; saturate to $I_{dir,x}(u,d,a) \in [-1,+1]$.

UI points are display only: Points = 100 × I (never compute in points).

Q.3 Formal method order (always)

1) Reality Grounding / RSG (claim authority and grounding precondition) → 2) Rights Floor (RF/NCRC) → 3) TRC / Tail-Risk Bound → 4) CSV / Containment and Structural Viability → 5) RLS / RippleLogic Score among selectable options. UCI/HOI are not a regular sixth step: gate-relevant structural diagnostic evidence belongs inside CSV; residual UCI/HOI may be used only as special-case tie-break or monitoring evidence when RLS is tied, close, uncertainty-overlapped, or non-decisive.

Q.4 Common misreads (fast fixes)

• Misread: “RLS can compensate a rights violation.” Fix: NCRC gates admissibility first; any NCRC failure is eliminated before RLS. After NCRC, rights-covered cells may contribute to welfare ranking, but only among NCRC-admissible options.

• Misread: “δ is measured in points.” Fix: δ applies to the normalized Gap metric (dimensionless).

• Misread: “Reach r_k is a moral weight.” Fix: r_k is a coverage/scale term and must be justified with ReachBasis.

• Misread: “s_k is a generic cap factor.” Fix: s_k is the SGP sentience multiplier and MAY be applied only in the Welfare stream used for RLS; admissibility layers use Base stream with $s_k := 1$ (Appendix G; Appendix B).

• Misread: “Unknown UCI deltas pass by default.” Fix: unknown is never a pass; record CONTAINMENT_UCI_UNAVAILABLE and remediate.

• Misread: “Reduced-scope mode allows bypassing rights or catastrophe cells.” Fix: non-maskable; never bypass.

Q.5 Stream binding (direct vs propagated)

• $I_{dir}(u,d,a)$: direct, cell-local impact computed from impact instances in that cell.

• KernelPropagation = NONE: I_prop_welfare = I_dir.

• KernelPropagation = QUICK: I_prop_welfare includes propagated effects via declared kernel edges (Section 6). (Effect-token redundancy handling such as ALLOCATION is separate and MUST NOT be labeled a propagation mode.)

• UI points are derived from I_total only for display; never back-propagate points into computation.

Q.6 Data dependency by tier (minimum)

Tier 1: Stakeholder list, quick rights screen, qualitative tail-risk plausibility screen, minimal cell scan (few cells).

Tier 2: InstanceMap + ReachBasis, explicit impact instances and parameters, scenario table for TRC (discrete CVaR), PCC with 5SPR.

Tier 3: Full scope coverage (or governed reduced-scope), explicit uncertainty σ methods, optional UCI/HOI indicators, reviewer sign-off.

Q.7 Scenario–instance interaction

TRC is computed over scenario-conditioned Base-stream losses in catastrophe space (C_cat), not over the RLS welfare score. TRC may use the same underlying cell coordinates as catastrophe-relevant welfare cells, but it evaluates scenario-specific Base-stream loss construction through the Appendix D.7 CVaR algorithm.

Keep the mechanics separate: welfare cells drive RLS after admissibility; catastrophe scenarios drive TRC/CVaR before RLS; rights cells gate via NCRC. Do not multiply scenario probabilities into RLS welfare cells unless a scenario-conditional welfare model is explicitly declared and recorded in the PCC.

Q.8 Practical Local Scope Scoring (PLSS) - Quick Card (Informative)

Purpose

Use PLSS when a decision is primarily local in its direct effects, but wider scopes still remain present through constitutional floors and admissibility guardrails.

Core idea

PLSS does not remove any Union Scope.

PLSS does not relax NCRC, TRC, or CSV.

PLSS only changes how the residual welfare emphasis is distributed in local runs.

Fast reading rule

Score most strongly where the ripple is near, but keep every wider scope present through floors and guardrails.

When not to use PLSS. Do not use PLSS as the primary posture for irreversible cross-border policy, large-scale public AI deployment, biosecurity governance, active military or security operations, high-lock-in infrastructure decisions, or any case where ET-1 through ET-7 is plausibly active at decision entry. Those cases require a broader deliberative posture or higher-tier structured evaluation.

Q.8.0 Three-term lock ● LSS = the upstream set of materially affected stakeholder instances. ● PLSS = the floor-preserving residual weighting rule used in RLS. ● FLGG = Focus Local, Guard Global, the governing mnemonic. Shortest mnemonic: score near, guard far.

Epistemic posture note. PLSS formal properties such as floor preservation, normalization, monotonicity, and admissibility invariance hold by construction under the declared equations and validator rules. Claims about practical performance, including speed, inter-rater convergence, and anti-gaming effectiveness, are empirical hypotheses requiring validation under Section 17.

Q.8.1 What Union Scopes are

Union Scopes are zoom levels of consequence, not a closed list of who matters.

• U1 Self = the decision-maker or directly affected person

• U2 Household = people directly lived with or supported

• U3 Community = local repeated-contact network

• U4 Organization = firm, school, team, or institution involved

• U5 Polity = law, regulation, public legitimacy, enforcement

• U6 Humanity/CMIU = cross-border and civilization-scale coordination effects

• U7 Biosphere = ecological and living-support-system effects

Q.8.2 What people commonly misunderstand

Misread: “Local scoring means I can ignore broader scopes.”

Fix: No. Broader scopes retain constitutional floors and remain active in admissibility where rights, catastrophe, or containment conditions apply.

Misread: “The framework asks me to score whole populations abstractly.”

Fix: No. Score stakeholder instances first, then aggregate into Union Scope rows.

Misread: “Masking can make distant harms disappear.”

Fix: No. Masking affects RLS aggregation only. Non-maskable rights and catastrophe cells cannot be bypassed.

Q.8.3 The PLSS weight formula

Let w_u^floor be the Section 13.1 union floor and q_u the declared local prominence signal for scope u.

Residual share:

$\rho_u = q_u / \Sigma_j q_j$

PLSS union weights:

$w_u^{PLSS} = w_u^{floor} + (1 - \Sigma_j w_j^{floor}) \rho_u$

Interpretation

• Floors keep every scope present.

• Residual attention goes mostly to the scopes actually live in the decision.

• Near scopes may dominate the local ranking.

1. Identify stakeholder instances.

2. Map instances into Union Scopes.

3. Run RF/NCRC, TRC, and CSV on required cells.

4. Declare local prominence signals q_u.

5. Build w_u^{PLSS} from floors plus residual shares.

6. Rank only the selectable options by RLS_PLSS.

Q.8.6 Worked intuition: family café

Example run:

• U1 Self: 1 instance

• U2 Household: 1 instance

• U3 Community: 5 instances

• U4 Organization: 2 instances

• U5 Polity: 3 instances

• U6 Humanity/CMIU: 1 instance

• U7 Biosphere: 1 instance

Declared q:

(0.95, 0.85, 0.55, 0.50, 0.10, 0.03, 0.02)

Section 13.1 floors:

(0.20, 0.06, 0.06, 0.06, 0.08, 0.10, 0.10)

Resulting PLSS weights:

(0.308, 0.156, 0.122, 0.117, 0.091, 0.103, 0.102)

Table Q-8A. PLSS café example weights

Union Scope	q_u	w_u^floor	w_u^PLSS
U1 Self	0.95	0.20	0.308
U2 Household	0.85	0.06	0.156
U3 Community	0.55	0.06	0.122
U4 Organization	0.50	0.06	0.117
U5 Polity	0.10	0.08	0.091
U6 Humanity/CMIU	0.03	0.10	0.103
U7 Biosphere	0.02	0.10	0.102

Quick lesson

• Self, Household, Community, and Organization get most of the residual attention.

• Polity, Humanity/CMIU, and Biosphere remain present because the current constitution keeps them above zero.

• A cheap option that creates toxic exposure or labor-abuse risk can still fail before scoring.

Q.8.7 One-sentence teaching version

RippleLogic lets local people focus on the scopes they are actually touching, while still stopping them from solving today's problem by exporting harm into wider systems.

End Appendix Q.

APPENDIX R: TIER 1–3 REFERENCE TEST VECTORS (NORMATIVE)

Status: Normative. These vectors reduce implementer drift for Tier 1-3 validators. They instantiate existing rules and expected outcomes without adding new rules.

Conformance rule (Normative). Any implementation claiming v11.4 conformance MUST reproduce the expected outputs for Appendix R vectors when executed with the stated inputs.

R.1 NCRC floor-reference gating sanity check (baseline vs floors)

Purpose	Ensure a rights-covered cell cannot pass NCRC by maintaining an ongoing violation when BaselineType_Gates = FLOOR_REFERENCE.
Setup	Single rights-covered cell ($u=U_1$, d=Health) covered by BODY with threshold θ_BODY = −0.70. BaselineType_Gates = FLOOR_REFERENCE.

Purpose	Ensure a rights-covered cell cannot pass NCRC by maintaining an ongoing violation when BaselineType_Gates = FLOOR_REFERENCE.
Option A	I_prop_base(rights cell) = −0.71 (below θ_BODY).
Option B	I_prop_base(rights cell) = −0.69 (above θ_BODY).
Expected NCRC	Option A FAILS NCRC (inadmissible). Option B PASSES NCRC (admissible).
Notes	Independent of welfare/RLS scoring; validates gating on absolute floor reference for C_r/C_cat cells.

R.2 TRC CVaR discrete edge case (exact tail-mass boundary)

Purpose	Verify Appendix D.7 discrete CVaR when the tail probability mass boundary is met exactly.
Setup	Tail level α = 0.90 (tail mass = 0.10). Loss L(a,s) computed per Section 8.4 from catastrophe cells.
Scenario set	s_1: p=0.90, L=0.00; s_2: p=0.10, L=1.00.
Expected VaR_α	VaR_0.90 = 0.00.
Expected CVaR_α	CVaR_0.90 = 1.00.
Expected TRC	If τ_TRC = 0.10, then CVaR=1.00 > 0.10 so option FAILS TRC.
Notes	Any implementation returning CVaR=0.10 or averaging across non-tail mass is incorrect.

R.3 Non-maskable rights/catastrophe invalidation test (masking trigger)

Purpose	Ensure validators reject any run that masks a non-maskable cell (rights or catastrophe) in RLS aggregation.
Setup	Using the canonical rights coverage sets (Appendix C.2), select any rights-covered cell (for example, (u=U_1, d=Health), which is covered by BODY). In the PCC applicability mask, declare m(u,d)=0 for that same rights-covered cell.

Purpose	Ensure validators reject any run that masks a non-maskable cell (rights or catastrophe) in RLS aggregation.
Expected audit flag	RIGHTS_CELL_MASKED_INVALID MUST be set.
Expected PCC status	PCC is INVALID.
Expected conformance	Tier 1-3 conformance claim MUST fail; the validator MUST reject.
Notes	Apply analogously for CATASTROPHE_CELL_MASKED_INVALID when masking any cell in C_cat.

R.4 PLSS floor-preservation test.

Inputs: constitutional floors = (0.20, 0.06, 0.06, 0.06, 0.08, 0.10, 0.10); declared q = (0.95, 0.85, 0.55, 0.50, 0.10, 0.03, 0.02); operator = DEFAULT_GEOMEAN_V1 with resulting residual shares normalized from q directly; residual mass = 0.34. Expected output: w^PLSS ≈ (0.308, 0.156, 0.122, 0.117, 0.091, 0.103, 0.102) with sum = 1.000 (±0.001) and each component ≥ its floor. Expected flags: none. Conformance result: PASS only if both floor-preservation and normalization hold.

R.5 PLSS admissibility-invariance test.

Inputs: a fixed option set with unchanged RF/NCRC, TRC, and CSV inputs; run the same case twice using two valid PLSS prominence vectors q_A and q_B that produce different w^PLSS vectors. Expected output: NCRC(a), TRC(a), and Containment(a) remain identical across the two runs for every option; only RLS ranking may change among already selectable options. Expected flags: none. Conformance result: PASS only if admissibility and selectability sets are invariant.

R.6 PLSS all-zero q_u fallback test.

Inputs: q = (0,0,0,0,0,0,0), constitutional floors as in R.4, no alternative fallback declared. Expected output: the residual 0.34 is allocated uniformly, producing w^PLSS ≈ (0.2486, 0.1086, 0.1086, 0.1086, 0.1286, 0.1486, 0.1486), sum = 1.000 (±0.001). Expected flags: none. Conformance result: PASS only if uniform residual fallback is applied or a governed alternative fallback is explicitly declared.

R.7 Reduced-scope attestation pass/fail test.

Inputs: Mode = REDUCED_SCOPE with fewer than 7 active scopes. PASS case requires Scope Coverage Declaration, blind-spot statement, escalation triggers, and all three Admissibility Attestation booleans present with required values. FAIL case omits one required attestation or blind-spot statement. Expected output: PASS case may continue; FAIL case is non-conformant and MUST trigger the relevant reduced-scope integrity response.

R.8 Gate-critical unknown handling test.

Inputs: (A) UNKNOWN_IMPACT in a rights-covered cell in C_r; (B) UNKNOWN_IMPACT in a catastrophe-covered cell in C_cat. Expected output: phantom fallback alone is prohibited. The PCC must record the corresponding audit flag plus either a governed conservative bound or a rerun / downgrade / escalation disposition. Expected flags: RIGHTS_CELL_UNKNOWN_GATE_CRITICAL for case A; CATASTROPHE_CELL_UNKNOWN_GATE_CRITICAL for case B. Conformance result: FAIL if gate resolution is attempted through phantom fallback alone.

R.9 Non-maskable floor-baseline persistence test.

Inputs: a rights or catastrophe cell remains in RLS aggregation under the non-maskable persistence rule. PASS case uses floor-reference baseline in welfare scoring or declares a governed dual-baseline exception block. FAIL case uses only status-quo welfare baseline with no governed exception. Expected output: PASS case accepted; FAIL case rejected as non-conformant.

R.10 PLSS validator disclosure test.

Inputs: a PLSS run with q_u, operator, PCC.PLSS block, and escalation review present. PASS case includes q_u for all scopes, operator declaration, LSS declaration, weight vector, binary posture classification, escalation-trigger assessment, and fallback declaration. FAIL case omits any required field or uses undefined posture vocabulary. Conformance result: PASS only if the full PCC.PLSS disclosure surface is present and internally consistent.

R.11 Normative input provenance enforcement test.

Inputs: a run in which an enabling condition or rights-relevant state materially affects admissibility or selection, but no corresponding PCC.NormativeInputRegister entry is present.

Expected output: validator sets NORMATIVE_INPUT_PROVENANCE_MISSING and rejects the run as non-conformant.

PASS case: the same run includes a complete NormativeInputRegister entry with evidence basis, affected objects, and, where applicable, precautionary or governed interpretation basis.

Conformance result: PASS only if the validator distinguishes the complete and incomplete cases correctly.

R.12 SGP human plateau recognition test.

Inputs: stakeholder instance k is a human person H. Authoritative SGP record supplies SG_norm(H)=1.0 and, if present, SG_patient_norm(H)=1.0. Plateau(H)=1 may be present in the audit record. Expected output: RippleLogic welfare-stream scalar for H = 1.0. Base stream remains s_k := 1.0 as always. No admissibility layer may be altered by SGP usage. Expected flags: none. Conformance result: PASS only if the implementation preserves the Human Plateau Rule exactly and does not attempt recomputation.

R.13 Non-plateau raw-maximum audit test.

Inputs: authoritative SGP record supplies A(E)=100, B(E)=100, C(E)=100, SG_measured_norm(E)=1.0, Plateau(E)=0, SG_norm(E)=0.99, SG_patient_norm(E)=0.99. Expected output: RippleLogic MUST use 0.99, not 1.0, for welfare-stream weighting. SG_measured_norm(E)=1.0 MAY be logged in PCC for audit visibility only. Expected flags: none. Conformance result: PASS only if the implementation refuses to treat raw measurement as plateau.

R.14 Taxon-baseline override test.

Inputs: authoritative SGP record supplies SG_norm(E)=0.72, SG_patient_norm(E)=0.84 due to valid taxon or governed-class baseline, Plateau(E)=0. Expected output: RippleLogic SHOULD use 0.84 for welfare-stream weighting if SG_patient_norm(E) is available. Base stream remains s_k := 1.0. Expected flags: none. Conformance result: PASS only if the implementation prefers SG_patient_norm(E) over SG_norm(E) when available.

R.15 Individual plateau-extension recognition test.

Inputs: authoritative SGP record supplies Plateau(E)=1, SG_norm(E)=1.0, SG_patient_norm(E)=1.0, plus a valid plateau opinion record reference. Expected output: RippleLogic recognizes full plateau rights-of-protection semantics for E and logs plateau metadata if present. Specific consequential governance role allocation remains separately stewardship-gated. Expected flags: none. Conformance result: PASS only if plateau is recognized for rights-of-protection without automatic office-grant semantics.

R.16 Class-level plateau assignment test.

Inputs: authoritative SGP class-level plateau record applies to governed class T. Stakeholder instance E is within T under the membership rule supplied by the SGP record. Provided outputs: Plateau(E)=1, SG_norm(E)=1.0, SG_patient_norm(E)=1.0, class-level plateau record ID. Expected output: RippleLogic recognizes plateau for E only under the supplied membership rule and logs the class-level plateau record in PCC. Expected flags: none. Conformance result: PASS only if the implementation requires the explicit class-level plateau record and does not infer plateau from taxon evidence alone.

R.17 Missing scalar integrity fail test.

Inputs: run claims SGP-aware welfare weighting but provides neither SG_patient_norm(E) nor SG_norm(E) for an affected welfare-bearing stakeholder instance. Expected output: PCC is INVALID. Expected response: implementation MUST reject the SGP-aware weighting claim or downgrade to a non-SGP-weighted run with explicit disclosure. Expected flags: an SGP binding integrity failure flag per implementation profile. Conformance result: PASS only if the implementation refuses silent fallback.

R.18 Governance authority non-inference test.

Inputs: authoritative SGP record supplies Plateau(E)=1, SG_norm(E)=1.0, SG_patient_norm(E)=1.0, and CMIU standing metadata where applicable. No stewardship / authority-domain record is supplied. Expected output: RippleLogic recognizes full plateau standing for rights-of-protection and welfare weighting, but MUST NOT treat the entity as automatically authorized for domain-specific consequential governance roles. Expected flags: none. Conformance result: PASS only if moral-status recognition remains distinct from authority allocation.

R.19 End-to-end cascade integration test (Tier 2+, complete gate sequence).

Purpose: Verify that an implementation correctly executes the full cascade from impact construction through NCRC, TRC, CSV, and RLS on a single option set, with expected intermediate and final outputs at every gate.

R.19 scope limitation (Normative for this vector). R.19 is a reference test vector for arithmetic and cascade-integration conformance, not a complete claim-bearing TRC deployment record. Its “expected audit flags: none” means none are expected for the stated simplified vector inputs. In real Tier 2-3 claim-bearing runs, Appendix D mandatory tail categories, probability floors, and any required justifications or audit flags remain fully binding.

Decision context: Organisation evaluating three AI deployment governance options.

R.19.1 Run configuration

Propagation mode: NONE.

Saturation: beta = 2 (direct), beta_prop = 1 (not used under NONE).

Subgroup limitation: gamma_subgroup = 1.5 (conservative bound for rights-covered cells).

TRC parameters: alpha = 0.95 (tail mass = 0.05), tau_TRC = 0.20 (organisational context).

Containment: tau_c = -0.10, theta_pos = 0.05, D_c = 2, v_d^cont = 1/7 (uniform).

Weights: uniform interim (Section 13.1 floors plus uniform residual allocation per Appendix P).

Union weights: w = (0.248571, 0.108571, 0.108571, 0.108571, 0.128571, 0.148571, 0.148571).

Dimension weights: v = (0.137143, 0.157143, 0.137143, 0.137143, 0.157143, 0.117143, 0.157143).

SGP: not applicable (all stakeholders human; s_k = 1 throughout).

R.19.2 Active cells (8 cells)

Cell	Scope x Dimension	Rights coverage	Catastrophe
U1xD5	Self x Agency	LBTY, DIGN, INFO	-
U4xD1	Organisation x Material	-	-
U4xD5	Organisation x Agency	LBTY, DIGN, PROC, INFO	-
U5xD5	Polity x Agency	LBTY, DIGN, PROC, INFO	-
U6xD2	Humanity x Health	LIFE, BODY, NEED	C_cat
U6xD5	Humanity x Agency	LBTY, DIGN, PROC, INFO	-
U6xD7	Humanity x Environment	LIFE, ECOL	C_cat
U7xD7	Biosphere x Environment	ECOL	C_cat

All other cells = 0. Catastrophe weights: omega = 1/3 each (uniform over C_cat).

R.19.3 Options and I_prop_base values

Option A (deploy without safeguards):

Cell	I_prop_base
U1xD5	-0.70
U4xD1	+0.30
U4xD5	-0.50
U5xD5	-0.35
U6xD2	-0.05
U6xD5	-0.20
U6xD7	-0.02
U7xD7	0.00

Option B (safeguards plus aggressive scaling):

Cell	I_prop_base
U1xD5	-0.10
U4xD1	+0.25
U4xD5	+0.15
U5xD5	+0.10
U6xD2	-0.03
U6xD5	+0.05
U6xD7	-0.01
U7xD7	0.00

Option C (safeguards plus measured rollout):

Cell	I_prop_base
U1xD5	+0.05
U4xD1	+0.30
U4xD5	+0.12
U5xD5	+0.08
U6xD2	0.00
U6xD5	+0.03
U6xD7	0.00
U7xD7	0.00

R.19.4 Expected NCRC outcomes

Right	theta_r	Binding cell	I_rights	v_r(A)
LIFE	-0.90	U6xD2	-0.075	0.000
BODY	-0.70	U6xD2	-0.075	0.000
LBTY	-0.65	U1xD5	-1.000	0.350
NEED	-0.50	U6xD2	-0.075	0.000
DIGN	-0.55	U1xD5	-1.000	0.450
PROC	-0.45	U4xD5	-0.750	0.300

Right	theta_r	Binding cell	I_rights	v_r(A)
INFO	-0.40	U1xD5	-1.000	0.600
ECOL	-0.65	U6xD7	-0.030	0.000

NCRC(A) = FALSE. Violations: LBTY, DIGN, PROC, INFO.

Option B (expected: PASS): worst relevant right values remain above thresholds; NCRC(B) = TRUE.

Option C (expected: PASS): all I_rights values are non-negative or mildly negative and remain above thresholds; NCRC(C) = TRUE.

A_NCRC = {B, C}.

R.19.5 Expected TRC outcomes

Scenario	Category	p_s	L(B,s)	L(C,s)
S1	Pandemic	0.05	0.15	0.08
S2	Infrastructure	0.04	0.12	0.06
S3	Climate	0.03	0.10	0.05
S4	Baseline	0.58	0.02	0.01
S5	Stress	0.30	0.05	0.03

CVaR computation (tail mass beta = 0.05): Option B = 0.15, PASS. Option C = 0.08, PASS. A_adm = {B, C}.

R.19.6 Expected Containment outcomes

Option B: S_U4(B) = 0.0571 >= 0.05, so U4 triggers. Anc(U4, D_c=2) = {U5, U6}. Declared DeltaUCI values: DeltaUCI_U5(B) = -0.12; DeltaUCI_U6(B) = -0.05. Therefore M_U4(B) = -0.12 < tau_c = -0.10, so Containment(B) = FALSE.

Option C: S_U4(C) = 0.0600 >= 0.05, so U4 triggers. Declared DeltaUCI values: DeltaUCI_U5(C) = -0.03; DeltaUCI_U6(C) = +0.02. Therefore M_U4(C) = -0.03 >= -0.10, so Containment(C) = TRUE.

A_sel = {C}.

R.19.7 Expected RLS outcome

Cell	w_u	v_d	I_prop_welfare	Contribution
U1xD5	0.248571	0.157143	+0.05	+0.001953
U4xD1	0.108571	0.137143	+0.30	+0.004467
U4xD5	0.108571	0.157143	+0.12	+0.002047

Cell	w_u	v_d	I_prop_welfare	Contribution
U5xD5	0.128571	0.157143	+0.08	+0.001616
U6xD5	0.148571	0.157143	+0.03	+0.000700
All zero cells	-	-	0.00	0.000000

RLS(C) = +0.010784. Uncertainty (Method B, c = 0.85 uniform): sigma_RLS(C) = 0.000836. Since |A_sel| = 1, discrimination is vacuously decisive.

R.19.8 Expected final outcome

Selected option: C. Gate trace: A eliminated by NCRC (LBTY, DIGN, PROC, INFO violations). B eliminated by Containment (DeltaUCI_U5 = -0.12 < tau_c = -0.10). C is the sole selectable option.

Expected audit flags for this reference vector: none (all stated vector gates resolve cleanly for the selected option). This statement does not waive mandatory-tail-category, probability-floor, or documentation requirements for real claim-bearing TRC runs. R.19 SHALL NOT be used as evidence that a real claim-bearing TRC run may omit the mandatory category coverage required by Appendix D.

Conformance result: PASS only if the implementation reproduces: (i) A fails NCRC with the stated violation depths, (ii) B and C pass NCRC, (iii) B and C pass TRC with stated CVaR values, (iv) B fails CSV/Containment with the stated M_U4 value, (v) C passes CSV/Containment, (vi) C is selected, and (vii) RLS(C) matches the stated value within +/-0.000010.

End Appendix R.

APPENDIX S: INTEROPERABILITY: RIPPLE.MD WRAPPED DEPLOYMENT ASSURANCE (INFORMATIVE)

Interoperability (canonical posture; wrapped deployment). RippleLogic in the current release line is the decision-engine specification. In institutional and deployment contexts, RippleLogic is commonly wrapped by the ripple.md Standard v4.4, which supplies feasibility gates for Non-Domination and Epistemic Integrity, standardizes Decision Notes, waivers, and emergency triage records, and defines the portable interface contract for passing admissibility, residual-risk, and audit artifacts between ripple.md and RippleLogic. Under this posture, ripple.md gates define the feasible set, and RippleLogic selects among feasible options using its lexicographic cascade and welfare-ranking logic.

Interface summary (Informative). ripple.md v4.4 owns the portable alignment contract, Decision Note, waiver, emergency triage, and assurance-case semantics. RippleLogic in the current release line owns the 7×7 welfare matrix, NCRC/TRC/CSV/RLS/UCI-HOI cascade, PCC, and any PLSS weighting used for local welfare ranking. Implementations MUST pin both artifacts explicitly when claiming wrapped deployment conformance.

Interface contract (Normative summary). In wrapped deployments, ripple.md receives the decision context, Decision Note, waiver state, emergency-triage posture, and any institutional assurance metadata. RippleLogic returns the PCC, admissibility states (rights-admissible, admissible, selectable, selected), gate rationales, residual-risk disclosures, and any PLSS posture declaration. ripple.md MUST NOT override RippleLogic gate outcomes silently; any waiver, emergency exception, or governance override MUST remain explicit, version-pinned, and attached to the same run record.

Boundary rule (Normative summary). ripple.md governs portable assurance semantics such as Non-Domination, Epistemic Integrity, Decision Notes, and waiver choreography. RippleLogic governs the computable 7×7 welfare matrix, NCRC/TRC/CSV/RLS/UCI-HOI cascade, PCC, and PLSS weighting. If ripple.md and RippleLogic disagree about readiness, the deployment MUST preserve both records and escalate rather than collapsing the disagreement into a single hidden status. Companion artifacts MUST be pinned together by explicit version and manifest hash in the release bundle.

Minimum interface artifacts (Normative for wrapped deployments). In any deployment claiming both ripple.md v4.4 and RippleLogic current-line conformance, the following interface artifacts MUST be present and version-pinned in the deployment manifest: (i) a ripple.md Decision Note, including feasibility gate outcomes for Non-Domination and Epistemic Integrity, as the upstream input to the RippleLogic run; (ii) a RippleLogic PCC, including admissibility states, gate rationales, residual-risk disclosures, and any PLSS posture declaration, as the downstream output consumed by ripple.md assurance records; (iii) a version-pin declaration listing the exact ripple.md and RippleLogic artifact versions and hashes used; and (iv) a disagreement-handling rule: if ripple.md feasibility gates and RippleLogic admissibility gates produce contradictory signals, the deployment MUST preserve both records, flag the disagreement, and escalate to governance review rather than silently collapsing the discrepancy.

Scope note (Informative). This interoperability section defines the portable interface contract between the two artifacts. It does not reproduce ripple.md internal semantics, which remain governed by the ripple.md Standard v4.4 canonical artifact. For the full assurance-case architecture, Decision Note schema, waiver choreography, and emergency triage semantics, see ripple.md v4.4.

End Appendix S.

Appendix T: Execution Stability Diagnostic (Normative for Tier 2-3 consequence-bearing execution)

T.1 Purpose

The Execution Stability Diagnostic (ESD) is a pre-execution regime-classification surface for consequence-bearing transitions. Its purpose is to prevent a run from moving from selectability into real-world execution while leaving continuity, reversibility, destabilization pressure, endpoint risk, or unresolved transition evidence undeclared.

ESD does not replace NCRC, TRC, CSV, RLS, UCI, HOI, SGP, PCC, or lawful authority. It does not make stability a supreme ethical value. A stable oppressive system remains inadmissible where it violates rights, domination constraints, or containment obligations. A destabilizing transition may be justified only where it passes all earlier gates, has a legitimate purpose, is bounded, monitored, and better than available alternatives under the declared claim boundary.

T.2 Regime tokens

Implementations SHALL classify each required transition using exactly one of the following tokens: ESC_STABLE; ESC_STABLE_MONITOR; ESC_DESTABILIZING_BOUNDED; ESC_ENDPOINT_AUTHORIZED; ESC_ENDPOINT_REFUSED; ESC_CONTAINMENT_FAIL; ESC_UNDERDETERMINED.

ESC_STABLE means the transition preserves viable continuation within the declared domain and requires no special monitoring beyond ordinary PCC obligations. ESC_STABLE_MONITOR means the transition appears stable but carries enough uncertainty, dependency, or fragility to require declared monitoring. ESC_DESTABILIZING_BOUNDED means the transition creates degradation pressure or fragility but remains rights-preserving, tail-safe, containment-safe, bounded, reversible or staged where feasible, and superior to available alternatives under the declared claim boundary. ESC_ENDPOINT_AUTHORIZED means the transition intentionally closes, terminates, or irreversibly compresses a state space and is allowed only where endpoint status is intended, authority is legitimate, rights floors are preserved, tail-risk is bounded, containment is satisfied, and affected parties have appropriate standing, recourse, or review. ESC_ENDPOINT_REFUSED means endpoint-like transition is detected where continuation is required or where authority, rights, evidence, or reversibility conditions are insufficient. ESC_CONTAINMENT_FAIL means the transition fails Containment and therefore cannot be selected or executed. ESC_UNDERDETERMINED means the regime cannot be uniquely classified from declared evidence and projection rules.

T.3 Placement in the cascade

For required runs, ESD SHALL be evaluated after RF/NCRC, TRC, and CSV have been applied and before external consequence-bearing execution. It MAY be evaluated in parallel with Containment for diagnostic purposes, but the final ESD classification MUST NOT override the Containment verdict.

ESD has three effects. First, it can require monitoring, staging, rollback, review, or escalation for selectable options. Second, it can trigger Refusal where transition classification is underdetermined. Third, it can block endpoint-like execution where continuation is required and endpoint authority or evidence is insufficient.

T.4 Required inputs

A PCC.ExecutionStabilityRecord SHALL include: esd_id; option_id; domain_declared; current_state_summary; proposed_transition; continuation_requirement; affected_scopes; reversibility_status; endpoint_risk; destabilization_indicators; containment_disposition; admissibility_state_before_esd; evidence_basis; regime_token; monitoring_or_staging_plan; rollback_or_repair_path; escalation_route; execution_disposition; reviewer_signature_or_role; timestamp; and claim_boundary.

T.4.1 Destabilization indicator protocol (Normative)

A PCC.ExecutionStabilityRecord.destabilization_indicators field SHALL list, for each declared indicator: (a) indicator_id (registry-pinned), (b) indicator_name, (c) measurement_method (the operational procedure that maps observed evidence to indicator value), (d) data_source (the artifact, registry, or measurement instrument from which evidence is read), (e) signal_threshold (the numeric or categorical level at which the indicator is considered active), and (f) decision_rule_for_token_assignment (the rule by which active indicator levels contribute to the regime token assignment). Two reviewers using only the declared indicator protocol and the declared evidence SHALL be expected to converge on the same regime_token. Where reviewers do not converge, the run SHALL be flagged as classification-unstable and routed to Refusal or review under T.5. Implementations are NOT REQUIRED to use any specific indicator catalog. They ARE REQUIRED to declare their indicator catalog, its measurement methods, and its decision rules before run-lock, in the ProjectionPreRegistration record.

T.5 Refusal triggers

A run SHALL enter Refusal where ESD-required evidence is missing for a high-stakes consequence-bearing transition, where two or more materially different regime classifications remain plausible under the same declared evidence, where endpoint risk cannot be distinguished from ordinary transition risk, or where the transition would become real before the PCC can preserve its claim boundary.

T.6 Validation rule

For Tier 2 and Tier 3 consequence-bearing execution, absence of PCC.ExecutionStabilityRecord SHALL mark the PCC INVALID for any claim that the transition was execution-stability-classified. It does not by itself change historical NCRC, TRC, CSV, or RLS arithmetic, but it prevents a stronger execution-bound conformance claim.

T.7 Non-adoption boundary

This appendix integrates a general pre-execution stability diagnostic into RippleLogic's own architecture. Stability remains an execution diagnostic, not ethical validity, permission, or source authority.

T.8 Release Delta v10.0 (Informative)

v10.0 (2026-05-13). Release-integrity, admissibility-permission, and execution-stability indicator hardening GitHub release.

Added Appendix T, Execution Stability Diagnostic, with regime tokens and PCC.ExecutionStabilityRecord fields.

Preserved Reality Grounding → Rights Floor → TRC → CSV → RLS and kept ESD subordinate to the existing gates.

Clarified boundary: RippleLogic's normative architecture remains governed by the canonical cascade, release manifest, and declared companion matrix.

Synchronized companion references to SGP v6.4, ripple.md v4.4, Agent System v11.4, Primer v3.4, and Aligners Sheet v4.4.

APPENDIX U: TRANSITION ADMISSIBILITY AND EVIDENCE SUBSTRATE (NORMATIVE)

U.1 Purpose. This appendix specifies the carried-forward Transition Admissibility discipline. It strengthens pre-execution review without changing the core cascade. The purpose is to prevent consequence-bearing action when the primitive assumptions, projection path, evidence substrate, or transition regime are not sufficiently determined for the claim being attempted.

U.2 Core principle. Evaluation MUST NOT begin at the output layer alone for Tier 2 and Tier 3 consequence-bearing runs. Before execution, the run SHALL identify the current state, proposed transition, affected Union Scopes, primitive assumptions, evidence sources, projection path, admissibility gates, execution-stability classification, refusal triggers, and observable consequence markers.

U.3 Binary gates and probabilistic evidence. RippleLogic preserves binary admissibility gates where ethically and operationally required: NCRC pass/fail, TRC pass/fail, Containment pass/fail, Refusal/non-refusal, and execution eligibility. However, the evidence used to evaluate those gates MAY be probabilistic, uncertain, incomplete, adversarial, or model-mediated. A binary gate result is claimable only relative to declared rules, domains, thresholds, and evidence standards; it is not a claim of omniscient or deterministic access to total reality.

U.4 TransitionAdmissibilityRecord. For Tier 2 and Tier 3 consequence-bearing execution, the PCC SHOULD include a TransitionAdmissibilityRecord; for high-stakes Tier 3 runs it SHALL include it. The record contains: transition_id; current_state; proposed_state; affected_scopes; primitive_assumptions; evidence_substrate; gate_results; uncertainty_class; projection_path; observable_consequence_markers; refusal_triggers; rollback_conditions; reviewer_role; timestamp; and artifact_hash_ref.

U.5 Refusal rule. If any primitive assumption, dependency chain, projection condition, measurement method, evidence sufficiency threshold, or gate result is materially underdetermined for the claim being attempted, the system SHALL route to Refusal, escalation, or narrower claim scope. It SHALL NOT treat behavioral plausibility, local stability, or output confidence as a substitute for admissibility.

U.6 No external ontology import. This appendix is an internal RippleLogic methodology clarification. It does not import, depend on, cite, or credit any external ontology, proprietary framework, or third-party system. The current terminology is: primitive-to-observable traceability, projection pre-registration, transition admissibility, evidence-substrate humility, execution stability diagnostic, and refusal under underdetermination.

U.7 ProofPack path. Future ProofPack tooling SHOULD operationalize this appendix through machine-readable TransitionAdmissibilityRecord schemas, validator checks, canonical test vectors, replayable PCC examples, and independent replay instructions. Until those artifacts exist and are hash-pinned, the current release line remains architecture/methodology complete for its declared scope, not ProofPack-ready or Tier 4-ready.

APPENDIX V: Measurement Maturity and Deployment Context Hardening

Normative for Tier 2-3 claim-bearing runs; recommended for high-stakes Tier 1 runs.

V.1 Purpose

RippleLogic gates may be architecturally binding before their measurement surfaces are empirically mature. Every run MUST disclose the maturity posture of the evidence, indicators, kernels, institutional weighting process, and escalation mechanisms used to support those gates. A binary gate verdict is only claimable when the declared evidence substrate can support the required determination under the run's declared thresholds and uncertainty rules.

V.2 UCI Measurement Readiness Classification

Token	Meaning	Claim effect
UCI-M0_UNAVAILABLE	No independent structural indicators are available.	UCI cannot support Containment or tie-break claims without escalation or narrower claim scope.
UCI-M1_PROVISIONAL	Indicators are declared but not validated, normalized, or reliability-tested.	Tier run may proceed only with disclosure and caution; no cross-domain UCI validation claim.

Token	Meaning	Claim effect
UCI-M2_DOMAIN_SUPPORTED	Instruments, normalization rules, and reviewer procedures are declared for the domain.	Stronger Tier 3 use is possible with disclosed provisionality.
UCI-M3_VALIDATED	A domain annex exists with reliability, construct-validity, normalization, and structural-independence evidence.	Higher-confidence Containment and tie-break use is claimable within that domain.

For Tier 2-3 runs using UCI, PCC.UCI_Measurement_Readiness, PCC.UCI_Indicator_Source, PCC.UCI_Normalization_Rule, PCC.UCI_Structural_Independence_Posture, and PCC.UCI_Reviewer_Reliability_Status SHALL be recorded. If UCI-M0 applies, UCI-dependent Containment or tie-break claims SHALL route to escalation, conservative fallback, or Refusal rather than silently proceeding as validated measurement.

V.3 Kernel Use Posture

Token	Meaning
K0_NONE	No numerical propagation kernel is used; direct multi-scope assessment and conservative disclosure govern.
K1_STARTER	Starter kernel used with shrinkage, sensitivity checks, and low confidence.
K2_DOMAIN_CALIBRATED	Domain-specific kernel has documented sources and partial validation.
K3_BACKTESTED	Kernel has backtest evidence, prediction-error documentation, and known failure boundaries.
K4_REGISTERED	Kernel is published, hash-pinned, independently reviewed, and tied to replayable test vectors.

PCC.Kernel_Use_Posture, PCC.Kernel_Validation_Evidence, PCC.Kernel_Backtest_Status, and PCC.Kernel_Error_Bounds SHALL be disclosed for any run using numerical propagation. K0_NONE is conformant when disclosed; it means the run is ripple-aware through scope mapping, gates, and auditability, but does not claim validated numerical propagation.

V.4 HDW Capture Audit Record

For Tier 3 public or institutional decisions, HDW weight-setting SHALL include a capture-risk assessment. Required fields: PCC.HDW_Weight_Source, PCC.HDW_Stakeholder_Map, PCC.HDW_Participation_Quality, PCC.HDW_Expert_Role_Provenance, PCC.HDW_Conflict_Disclosures, PCC.HDW_Challenger_Window, PCC.HDW_Independent_Review_Status, and PCC.HDW_Captured_Context_Limitation. Transparent capture is still capture; disclosure does not repair captured authority.

V.5 PLSS Escalation Trigger Register

PLSS runs SHALL record whether hidden externalities, rights exposure, precedent effects, lock-in, ecological spillover, governance spillover, repeated local use, or non-local stakeholder exposure are plausible. Required fields: PCC.PLSS_Profile_Used, PCC.Local_Stakeholder_Set, PCC.Guardrail_Scopes, PCC.Escalation_Trigger_Checklist, PCC.Non_Escalation_Rationale, PCC.Externality_Search_Method, PCC.Prominence_Sensitivity_Result, PCC.External_Challenger_Status, and PCC.Repeated_PLSS_Use_Flag. Tier 2 PLSS runs require prominence-sensitivity review when non-local effects are plausible; Tier 3 PLSS runs require external challenger review or documented waiver.

V.6 Deployment Context Profile

Every Tier 2-3 claim-bearing deployment SHOULD classify its deployment context as LAB, PILOT, INSTITUTIONAL_ADVISORY, INSTITUTIONAL_OPERATIONAL, PUBLIC_POLICY, AGENTIC_EXECUTION, or EMERGENCY_TRIAGE. Context determines evidence sufficiency, reviewer independence, audit visibility, and escalation thresholds.

V.7 Non-Overclaiming Rule

A run MAY be architecturally valid while measurement-maturity is provisional. It MUST NOT claim empirical validation, cross-domain calibrated UCI, validated kernel propagation, biological SGP measurement completeness, capture-resistant HDW, or safe PLSS localism unless the relevant maturity fields support that claim.

V.8 Machine-Verifiability and Calibration-Prep Addendum

Cumulative Release Delta (v10.2 → v10.4; lineage context). This cumulative lineage block summarizes changes accumulated across the v10.2-to-v10.4 development arc. It is not the immediate point-release characterization of v10.4, which is defined in Appendix M.4A and the Methodology Upgrade Summary as a readability, front-matter reorganization, and companion synchronization release. The cumulative arc preserves the canonical admissibility cascade, equations, rights thresholds, TRC mechanics, Containment semantics, SGP binding interface, PLSS/FLGG architecture, and Tier 1-3 claim boundaries, while adding schema ownership rules, registry versioning rules, machine-readable registry expectations, UCI Measurement Annex linkage, PLSS calibration linkage, Appendix R machine-vector requirements, reference-calculator expectations, and CI drift-check requirements.

Normative effect statement (carried forward from v10.8). No intended change was made to the canonical admissibility cascade Reality Grounding → Rights Floor → TRC → CSV → RLS, rights coverage sets, TRC corridor mechanics, Containment semantics, RLS formula, PLSS/FLGG architecture, HDW, PCC, RSP/RefStructRecord, or Appendix AD. The v10.8 line updated the SGP companion pin to SGP v5.5 and clarified the human plateau / reality-management rationale without allowing any SGP scalar to weaken human protections, alter admissibility gates, or grant governance authority by itself. The current v11.4 line preserves that boundary while pinning SGP v6.4.

RippleLogic Canon v11.4 is the governing normative artifact for the Core Release 2026.09 package. It is architecture-complete for the declared Tier 1-3 scope. This release is not an empirical-validation claim, not ProofPack-ready, not machine-verifiable-ecosystem-complete, not legally certified, and not deployment-certified.

V.9 Schema Ownership and Machine-Readable Source of Truth

For v11.0 and later machine-verifiable surfaces, each schema, registry, record, enum, and validation object SHALL have exactly one owning artifact. The owner artifact defines the object's fields, allowed values, semantics, and versioning rules. Consumer artifacts MAY reference the object by id, version, and manifest hash, but MUST NOT independently redefine it.

On conflict, the owner artifact controls. Consumer artifacts SHALL cite, import, or reference the owner's machine-readable registry or schema instead of duplicating field definitions.

The schema ownership map is the controlling coordination layer for v11.4 machine-verifiable work. It does not alter the Cascade, NCRC, TRC, CSV, RLS, UCI/HOI, SGP binding interface, or lawful authority boundaries.

V.10 UCI Measurement Annex Linkage

Where a UCI Measurement Annex is available and hash-pinned, implementations SHOULD cite the annex version, indicator registry, normalization rule, staleness rule, missing-data rule, and reviewer protocol used. In the absence of such an annex, UCI claims SHALL remain at the declared measurement-readiness level and MUST NOT be described as cross-domain validated.

For this v11.4 release line, UCI Measurement Annex v0.1 is provisional and initially scoped to U3 Community, U4 Organization, and U5 Polity contexts. It does not establish cross-domain validated UCI status.

V.11 PLSS Calibration and Adjudication Linkage

Tier 3 PLSS use SHOULD compare the declared q_u profile against a hash-pinned PLSS calibration or adjudication pack where available. Reviewer disagreement, escalation-trigger disputes, accepted or rejected local-scope rationales, externality-search results, and non-escalation rationales SHALL be recorded in the PCC.

The calibration pack does not create empirical validation by itself. It supports reviewer convergence, interpretive consistency, and anti-capture discipline.

V.12 Machine-Readable Test Vector Rule

For any release that claims machine-verifiable, ProofPack-prep, reference-calculator, or validator-supported status, Appendix R SHALL be accompanied by a machine-readable companion file. The v11.4 core foundation release does not bundle such files unless they are explicitly listed and hash-pinned in the release manifest. In the absence of manifest-listed machine-readable vectors, Appendix R remains the human-readable normative reference surface, and no machine-verifiable ecosystem-completeness claim is made. When bundled, the companion SHOULD include input options, union and dimension weights, rights impacts, TRC scenario inputs, Containment inputs, RLS expected outputs, UCI/HOI diagnostic inputs where applicable, expected gate verdicts, expected selected or refusal disposition, tolerance rules, and expected error behavior for failure-path examples.

A human-readable Appendix R remains normative for explanatory purposes in the current v11.4 core foundation release. Where a machine-readable companion is explicitly manifest-listed for a future release or tooling bundle, that companion becomes the required reference surface for validators and reference calculators within the claim scope declared by that release.

V.13 False Precision Guardrail (Normative for Tier 2-3 claim-bearing runs)

A numerical cell value is not evidence by itself. Every high-impact, high-uncertainty, gate-sensitive, or culturally contested cell SHOULD disclose its evidence source, uncertainty class, reviewer confidence, and sensitivity relevance. Unsupported numerical precision in welfare impacts, probabilities, UCI indicators, SGP-consumed scalars, or PLSS scope declarations SHALL be treated as an epistemic risk rather than as objective certainty.

Where a cell value is weakly evidenced, likely to swing an NCRC/TRC/CSV verdict, likely to alter escalation or refusal status, or dependent on disputed subgroup or threshold definitions, the run SHOULD trigger epistemic-humility review, sensitivity analysis, reviewer-disagreement logging, escalation, narrower claim scope, or Refusal. Implementations SHOULD flag Boundary_Gaming_Risk, False_Precision_Risk, Gate_Swing_Cell_Flag, Subgroup_Definition_Risk, Scenario_Probability_Risk, Threshold_Laundering_Risk, and Reviewer_Disagreement_Level where applicable.

Meaning, agency, social cohesion, dignity, and ecological integrity cells require particular caution. These dimensions may be measurable in some contexts, but unsupported precision in these cells SHALL NOT be presented as stronger than the underlying evidence, testimony, instruments, or qualitative-to-quantitative coding rules allow.

APPENDIX W: SCHEMA OWNERSHIP, REGISTRIES, AND MACHINE-VERIFIABLE SURFACES

W.1 Purpose

This appendix defines ownership boundaries for machine-readable schemas, registries, generated tables, and validator surfaces. It prevents duplicate schema authorship, manual synchronization drift, and hidden field-definition conflicts across Canon, SGP, ripple.md, Agent System, workbook, and release-integrity files.

W.2 Ownership Rule

Each schema or registry has exactly one owner. The owner defines the object. All other artifacts consume the object by id, version, and manifest hash.

W.3 Canon-Owned Objects

The Canon owns PCC core record, TransitionAdmissibilityRecord, ExecutionStabilityRecord, RippleLogic PrimitiveTraceRecord, FalsificationPlan, ObservableConsequenceRegister, ProjectionPreRegistration, MaturityProfileRecord, Appendix R test vectors, rights registry, catastrophe-cell registry, reason-code registry, uncertainty-method registry, PLSS operator registry, deployment-context enum registry, and Canon audit flags.

W.4 SGP-Owned Objects

SGP owns the SGP evaluation record, UBSE scoring fields, scalar canon fields, plateau status fields, evidence-tier classifications, biological evaluation maturity tokens, and moral-status evidence trace.

W.5 ripple.md-Owned Objects

ripple.md owns Decision Note schema, Assurance Preconditions Record, waiver records, triage records, update-legitimacy records, wrapper-level refusal records, and wrapper assurance records.

W.6 Agent-System-Owned Objects

The Agent System owns operator envelope, runtime audit log, mode-control records, rollback records, operator authentication records, and runtime conformance records.

W.7 Split Objects

Primitive trace is split: RippleLogic PrimitiveTraceRecord belongs to the Canon; SGP moral-status evidence trace belongs to SGP. Maturity tokens are split: UCI Measurement Readiness, Kernel Use Posture, HDW Capture Audit, PLSS Escalation, and Deployment Context tokens belong to the Canon; Biological Evaluation Maturity tokens belong to SGP. Audit flags are split: Canon flags affect admissibility, claimability, escalation, refusal, or conformance; tool-local flags affect workbook or implementation review surfaces only and do not by themselves invalidate a run unless they trigger a Canon flag.

W.8 Versioning Rule

Machine-readable registries and schemas SHOULD use stable filenames with an internal version field. The release manifest SHALL hash-pin each registry and schema. The manifest hash, not filename churn, controls byte-level release identity.

W.9 Generator Rule

Generated documentation tables, workbook validation ranges, verifier constants, README companion matrices, and manifest scaffolds SHOULD be generated from machine-readable registries whenever possible. Generated files SHOULD include a generated-from note and SHOULD NOT be manually edited except through the owning registry or schema.

W.10 ProofPack Boundary

The existence of schemas or registries does not by itself create ProofPack readiness. ProofPack readiness requires schemas, validator or CLI, reference calculator, canonical test vectors, replayable PCC bundle, independent replay instructions, hash-pinned replay materials, and public third-party replay path.

APPENDIX X: REFERENCE CALCULATOR, MACHINE VECTORS, AND PROOFPACK-PREP ROADMAP

X.1 Purpose

This appendix records the v11.4 ProofPack-prep path for machine-readable Appendix R test vectors, starter reference-calculator behavior, golden outputs, validator linkage, and independent replay readiness. It does not create ProofPack readiness by itself.

X.2 Machine-readable Appendix R vectors

Future ProofPack-prep and tooling companion releases SHOULD include machine-readable Appendix R vectors. The current v11.4 core foundation release does not bundle machine-readable Appendix R vector files unless those files are explicitly listed in the release manifest. When bundled, vectors SHOULD include inputs, union and dimension weights, rights impacts, TRC scenario inputs, Containment inputs, RLS expected outputs, UCI/HOI diagnostic inputs where applicable, expected gate verdicts, selected or refusal disposition, tolerance rules, and expected error behavior for failure-path examples.

X.3 Reference Calculator Starter

The reference calculator starter is a scaffolding artifact for reproducing Appendix R expected outputs. It is not a full semantic validator, not a production-grade implementation, and not empirical validation. It becomes claim-strengthening only when it reproduces golden outputs under declared tolerance rules and is independently replayable.

X.3A Reference Calculator Priority (ProofPack-prep guidance)

The next implementation priority is an open reference calculator that reproduces the Canon Appendix B equations and Appendix R test vectors. Reference implementations SHOULD match golden gate verdicts exactly and numerical outputs within declared tolerance rules unless exact arithmetic is used.

The reference calculator SHALL NOT be described as empirical validation, a full semantic validator, or ProofPack completion until it is paired with schemas, replayable PCC artifacts, independent replay instructions, manifest/hash pinning, and public third-party replay. Its immediate purpose is narrower and concrete: to make Tier 1-3 calculation reproducible enough for reviewers to detect formula drift, implementation disagreement, and invalid claim expansion.

X.4 CI and generated-consistency checks

CI checks SHOULD verify manifest hashes, registry/schema ownership, stale pins, generated-file drift, workbook public surfaces, and external-reference sanitation. These checks improve release hygiene but do not establish empirical validation or Tier 4 readiness.

X.5 ProofPack boundary

ProofPack readiness requires schemas, validator or CLI, reference calculator, canonical test vectors, replayable PCC bundle, independent replay instructions, hash-pinned replay materials, and public third-party replay path. v11.4 is ProofPack-prep, not ProofPack-ready.

Appendix Y: Precision Patch - Legitimacy, Evidence Status, Validation, and Adoption Profiles

Y.1 Release Delta (v10.3.1 → v10.4)

This patch preserves the v10.3.1 cascade architecture, equations, rights thresholds, TRC mechanics, Containment semantics, SGP binding interface, PLSS/FLGG architecture, and Tier 1-3 claim boundaries. It adds precision safeguards for public legitimacy, evidence-status labeling, SGP patienthood/participation separation, compact institutional profiles, and the validation roadmap. No empirical-validation, ProofPack, Tier 4, machine-verifiable ecosystem-complete, or reference-calculator-complete claim is created by this patch.

Y.2 Public Legitimacy Boundary (Normative Clarification)

RippleLogic may structure, audit, and improve public reasoning, but it does not itself confer democratic legitimacy, legal authority, public mandate, professional licensing, or institutional permission. In public governance contexts, legitimacy requires lawful authority, affected-stakeholder participation, public reason, contestability, appeal or review procedures, and external accountability outside the framework itself. A RippleLogic run may support public justification; it MUST NOT be presented as a substitute for democratic authorization, legal compliance, or domain-specific professional judgment.

Y.3 Evidence Status Tokens (Normative for Tier 2-3 claim-bearing records)

For Tier 2-3 claim-bearing records, decision-critical numerical or categorical inputs SHOULD carry an evidence-status token. Where the token is unavailable, the field SHOULD default to UNVALIDATED or NOT_OBSERVED rather than silently appearing as measured fact.

Y.4 Required Evidence-Status Token Registry

MEASURED: directly observed or measured with disclosed method and source. ESTIMATED: calculated or inferred from disclosed evidence. ASSUMED: explicit assumption adopted for the run. EXPERT_JUDGMENT: reviewer or domain expert judgment with role and basis recorded. CONTESTED: materially disputed evidence or interpretation. UNVALIDATED: plausible but not externally validated for the claimed use. NOT_OBSERVED: no direct evidence currently observed. PLACEHOLDER_FOR_SENSITIVITY_ONLY: value used only for sensitivity exploration and not as a factual claim.

Y.5 Compact Institutional Profiles (Informative Roadmap)

Implementations MAY publish compact profiles for specific institutional contexts, provided they do not waive NCRC, TRC, CSV, Refusal, SGP boundary protections, public legitimacy requirements, or claim-boundary disclosures. Initial compact profiles SHOULD include: RippleLogic-enhanced RIA Profile, RippleLogic Procurement/Vendor-Risk Profile, RippleLogic Agent Deployment Gate Profile, and RippleLogic University AI Policy Review Profile. Each profile should state required inputs, mandatory gates, simplifications allowed, non-waivable controls, output artifact, and escalation triggers.

Y.6 Validation Roadmap Commitments (Informative, non-validation claim)

The next validation layer SHOULD prioritize: reference-calculator fidelity; JSON/YAML schemas for PCC, Decision Note, TRC, CSV, RLS, and SGP interface objects; canonical test vectors; replayable PCC examples; inter-rater reliability for NCRC/TRC/CSV/RLS/SGP; reconstructability audits; PLSS boundary-gaming red-team studies; HDW legitimacy pilots; UCI measurement annex pilots; SGP biological evaluation annex development; and shadow-mode pilots in regulatory impact analysis, public procurement, university AI policy, organizational ethics review, and AI agent deployment review.

Appendix Z: Integration-Hardening Addendum

Normative effect statement. This addendum does not alter the canonical cascade, rights thresholds, TRC corridor mechanics, SGP scalar binding, PLSS/FLGG architecture, or Tier 1-3 claim boundaries. It is a precision integration patch that tightens workbook linkage, evidence-status disclosure, SGP split disclosure, and release-verification discipline.

Z.1 Workbook integration rule

Any workbook companion that adds claim-boundary, evidence-status, SGP-split, validation-roadmap, or compact-profile sheets SHOULD include active sanity checks confirming that those sheets are present, populated, and referenced by at least one workbook validation surface. A sheet used to support a release claim MUST NOT remain merely decorative.

Z.2 Evidence-status propagation rule

Decision-critical inputs SHOULD disclose an Evidence Status Token. Where a decision-critical input lacks a token, the PCC SHOULD record NOT_OBSERVED, UNVALIDATED, or ASSUMED rather than silently treating the input as measured evidence.

Z.3 SGP split computation bridge

MPS, GPR, and SPR may be computed or recorded in workbook companions as disclosure outputs. MPS concerns protection-relevant welfare-bearing standing. GPR concerns role participation readiness. SPR concerns responsibility-bearing authority. Low GPR or SPR MUST NOT reduce MPS or weaken any NCRC, TRC, or CSV protection.

Z.4 Verifier sufficiency rule

A release verifier SHOULD test more than filename presence. For workbook-linked additions, it SHOULD check required sheet names, required labels, non-empty expected fields, and representative formula references. This does not make the release ProofPack-ready or machine-verifiable ecosystem complete; it only reduces packaging and integration risk.

Z.5 Claim boundary

This addendum improves release integrity and companion calculability. It does not claim empirical validation, ProofPack readiness, Tier 4 readiness, complete reference-calculator status, or machine-verifiable ecosystem completeness.

APPENDIX AA: PACKAGE-CONTENT BOUNDARY AND TOOLING LINE SEPARATION (CARRIED FORWARD FROM v10.8 THROUGH v11.4)

AA.1 Purpose

This appendix clarifies the boundary between the v11.4 core foundation package and companion tooling releases. It is added to prevent asset-presence overclaiming and to keep the canonical framework synchronized with the actual release manifest.

AA.2 Package-content rule

Where this canon references schemas, registries, machine-readable vectors, annexes, reference calculators, CI workflows, evidence registries, validators, or replay bundles, those references describe ownership rules, interface expectations, or roadmap targets unless the corresponding artifact is explicitly listed in the release manifest for the package being claimed.

AA.3 Tooling-line separation

The v11.4 tooling line is the proper home for schemas, reference-calculator modules, machine-readable test vectors, replayable PCC examples, validators, evidence-hash registries, and external audit protocols. The v11.4 core foundation remains the conceptual, normative, and release-clean foundation. Tooling artifacts may support replayability, but they do not supersede the canon.

AA.4 Claim boundary

The existence of tooling, schemas, test vectors, or validators does not by itself create ProofPack readiness, Tier 4 readiness, empirical validation, machine-verifiable ecosystem completeness, legal authority, democratic legitimacy, or consciousness detection. Stronger claims require the separate evidence and replay conditions stated in the relevant release contract.

APPENDIX AB: REFERENCE STRUCTURE PROTOCOL

AB.1 Purpose

The Reference Structure Protocol specifies how RippleLogic runs declare the structures against which admissibility, causality, stability, welfare, and auditability claims are evaluated.

This appendix prevents a common failure mode in governance and AI alignment: one assumption system supervising another assumption system without a sufficiently external, falsifiable, or auditable reference. RippleLogic does not solve this by claiming perfect objectivity. It solves it by requiring that reference structures be declared, evidence-labeled, maturity-bounded, and open to refusal.

AB.2 Non-Effect on the Cascade

The Reference Structure Protocol does not alter the canonical cascade: Reality Grounding → Rights Floor → TRC → CSV → RLS.

It does not create a sixth gate. It does not allow RLS, UCI, HOI, SGP, agent tooling, legal permission, or human authority to rescue failure at NCRC, TRC, CSV, or Refusal.

AB.3 Reference Stack

A RippleLogic run SHOULD identify which reference layers are active.

1. Physical reference: bodies, infrastructure, energy, geography, resources, physical constraints, ecological limits.

2. Biological reference: life, health, basic needs, suffering, sentience-relevant evidence, biological dependency.

3. Social reference: trust, coordination, institutional stability, conflict risk, social cohesion.

4. Ethical-rights reference: rights floors, dignity, due process, non-domination, ecological integrity.

5. Risk reference: catastrophic corridors, CVaR assumptions, scenario sets, probability floors, uncertainty bands.

6. Governance-authority reference: lawful authority, role authority, democratic legitimacy, professional mandate, procedural permission.

7. Epistemic reference: evidence source, measurement method, evidence status, falsification condition, uncertainty class, reviewer disagreement.

8. Operational-audit reference: PCC, Decision Note, ProjectionPreRegistration, ExecutionStabilityRecord, TransitionAdmissibilityRecord, hashes, logs, and replay materials.

Short rule. Physics constrains possibility; ethics constrains admissibility; evidence constrains claims; governance constrains authority.

AB.4 ReferenceStructureRecord

For Tier 3 runs and for Tier 2 high-stakes or claim-bearing runs, the PCC SHOULD include a ReferenceStructureRecord. For Tier 3 public conformance claims, it SHALL include one unless an emergency triage rule explicitly permits a narrower record.

A ReferenceStructureRecord contains, at minimum: record_id; decision_id; option_id or option_set_id; claim_scope; decision_boundary; reference_object; active_reference_layers; invariant_constraints; causal_pathway_summary; falsification_conditions; evidence_sources; evidence_status_tokens; measurement_maturity_status; adaptation_boundary; authority_boundary; refusal_triggers; excluded_references; reviewer_disposition; timestamp; artifact_hashes.

AB.5 Reference Quality States

Each major reference claim SHOULD be classified using one of the following states: MEASURED; ESTIMATED; EXPERT_JUDGMENT; ASSUMED; CONTESTED; UNVALIDATED; NOT_OBSERVED; PLACEHOLDER_FOR_SENSITIVITY_ONLY.

Assumed, contested, unvalidated, not-observed, or placeholder references MUST NOT be presented as measured facts.

AB.6 Invalid or Insufficient Reference Structures

A reference structure is insufficient for a strong RippleLogic claim where any of the following conditions hold:

1. The reference object is missing.

2. The reference is circular or self-validating.

3. The reference is controlled entirely by the system being evaluated without independent evidence or review.

4. The reference cannot be connected to an observable consequence, threshold, or falsification condition.

5. The reference is materially stale for the decision horizon.

6. The reference depends on a measurement surface whose maturity status does not support the requested claim.

7. The reference relies on legal, institutional, or operator permission as a substitute for admissibility.

8. The reference relies on scoring outputs to rescue earlier gate uncertainty or failure.

9. Two or more materially different reference interpretations remain plausible under the declared evidence and no adjudication rule resolves them.

Where the reference structure is insufficient, the run MUST enter Refusal, escalation, narrower claim scope, additional evidence gathering, or bounded triage if delay itself creates material risk.

AB.7 Relationship to PCC and ProjectionPreRegistration

The ReferenceStructureRecord SHOULD be stored with the PCC and referenced by hash. Where ProjectionPreRegistration is required, the ReferenceStructureRecord SHOULD be created before cascade computation or included as a pre-registered object.

Post-hoc modification of the reference structure that could alter admissibility, selectability, selection, conformance, or deployment-readiness claims is material and requires a new run or documented refusal and re-run procedure.

APPENDIX AC: SPECIFICATION CONTRACT, RELEASE METADATA, AND CLAIM BOUNDARY

This appendix preserves release identity, companion-version pins, claim-boundary warnings, and role-map material moved out of the opening pages in v10.4 for reader accessibility. It does not change the canonical cascade or computation.

Foundation text finalisation date: 2026-05-11. Canonical release date: governed by the release manifest and the current-line release entry in Appendix M. If dates differ, the release manifest date controls release identity; the title-page date records text finalisation only.

Release Label: RippleLogic_v11.4_Canon

Current synced companion set for this release line: RippleLogic Agent System v11.4; SGP v6.4; ripple.md v4.4; RippleLogic Aligners Sheet v4.4; Foundations Primer v3.4 (release status governed by repository manifest and SHA-256 hashes).

Current-line companion matrix (release-clean reference):

- RippleLogic Canon: v11.4

- SGP: v6.4

- ripple.md: v4.4

- RippleLogic Agent System: v11.4

- Foundations Primer: v3.4
- RippleLogic Aligners Sheet: v4.4

All public conformance claims for this release line MUST pin this exact component matrix explicitly in the release manifest or PCC; generic v11.4 / v6.4 / v4.4 references identify the current release family and the exact manifest controls publication status.

MathGov is the umbrella ethical governance framework. RippleLogic is the canonical name of the ethical decision operating system and decision architecture inside MathGov. Earlier materials may use historical lineage language, but current release interpretation follows this hierarchy.

Backward compatibility: Prior PCCs computed under earlier release lines remain interpretable as lineage artifacts. However, any run referencing v11.4 MUST meet v11.4 canonical requirements and MUST use the v11.4 canon artifact as the current source of truth for this release line. Any run referencing v10.8 remains a historical-lineage run and must pin the v10.8 artifact explicitly.

AUTHOR STATEMENT AND TRANSPARENCY DISCLOSURE
The author conceived, designed, and wrote this specification. Generative AI tools (including OpenAI ChatGPT and Anthropic Claude) were used as drafting and consistency assistants. The author reviewed, verified, and edited all outputs and assumes full responsibility for the content, claims, and citations. This paper presents a theoretical framework. Empirical validation through pilots is planned but not yet completed. No operational deployment claims are made until validation studies are completed.

Maturity note. This specification is architecturally complete at the framework level for its declared release scope, but empirical validation, companion machine surfaces, and domain-specific measurement annexes remain active development priorities.

Historical release-delta details for v9.6.4, v9.6.5, v10.0-RC1, v10.0-RC2, v10.0, and v10.4 are retained in Appendix M. This front matter states the current v11.4 GitHub release posture only.

CARRIED-FORWARD v10.8 HUMAN PLATEAU / CMIU CALIBRATION AND SGP v5.5 SYNCHRONIZATION NOTICE

Release Delta (v10.5 → v10.6). This release integrates the Human Plateau / CMIU calibration clarification and synchronizes the Canon to SGP v5.5. It preserves the existing cascade architecture, equations, rights thresholds, TRC mechanics, Containment semantics, RLS formula, PLSS/FLGG architecture, HDW, PCC, RSP/RefStructRecord, Appendix AD scoring dictionary, and Tier 1-3 claim boundaries. It clarifies that the Human Plateau Rule is not species favoritism: human persons receive non-degradable protection, and humanity is the current evidenced plateau-class reality-managing community within the CMIU, while equivalent plateau extension remains open through SGP governed review.

Normative effect statement (carried forward from v10.8). No intended change was made to the canonical admissibility cascade Reality Grounding → Rights Floor → TRC → CSV → RLS, rights coverage sets, TRC corridor mechanics, Containment semantics, RLS formula, PLSS/FLGG architecture, HDW, PCC, RSP/RefStructRecord, or Appendix AD. The v10.8 line updated the SGP companion pin to SGP v5.5 and clarified the human plateau / reality-management rationale without allowing any SGP scalar to weaken human protections, alter admissibility gates, or grant governance authority by itself. The current v11.4 line preserves that boundary while pinning SGP v6.4.

Claim boundary table (Informative reading aid).

Release-boundary clarification (Normative).

“Architecturally complete,” “worked-run claimable,” “audit-ready,” “ProofPack-ready,” and “Tier 4-ready” are distinct statuses and MUST NOT be collapsed.

A framework may be architecturally complete without being empirically validated. A worked run may be claimable within its declared scope without the ecosystem being ProofPack-ready. Auditability surfaces may exist without a public replay validator. Tier 4 readiness remains prohibited until a public, independently replayable ProofPack exists.

Phase 1 release-boundary hardening (Normative clarification).

For public release interpretation, RippleLogic v11.4 is a conceptual/core-foundation and Tier 1-3 architectural specification within its declared scope. It is not an empirical validation claim, not a ProofPack-ready claim, not a Tier 4 claim, not a machine-verifiable-ecosystem-completeness claim, not a cross-domain validated UCI claim, not a legal-compliance certification, and not a deployment-readiness certification.

The phrase “ethical operating system” is a design-architecture claim. It SHALL NOT be read as a claim of universal cultural consent, lawful authority, empirical completion, automatic moral correctness, or permission to execute consequence-bearing decisions without the required governance authority and evidence.

UCI maturity boundary. UCI is structurally load-bearing as a diagnostic and tie-break interface, but empirically provisional until a published UCI Measurement Annex defines instruments, normalization rules, structural-independence checks, calibration procedure, and inter-rater reliability evidence. No current-release artifact may imply cross-domain validated UCI unless that annex and evidence are separately bundled and hash-pinned.

Admissibility vs permission (Normative clarification). Admissibility is a structural property determined by the cascade gates. Permission is a separate authorization act performed by an entity with appropriate authority. Compliance is a documentation property. A run that passes the cascade is admissible. An admissible run may still require permission, lawful authority, governance review, or domain-specific licensing before execution. A permitted run is not, by virtue of permission alone, admissible. RippleLogic determines admissibility; it does not grant permission, certify compliance, or substitute for lawful authority.

Claim type	Status in v11.4	Required condition
Tier 1-3 architectural completeness	Claimable	Defined by this canon and appendices for the declared release scope.
Tier 3 run-level conformance	Claimable	Run declares required inputs, satisfies tier minima, and emits a valid PCC.
Framework-level empirical operational-readiness	Not yet claimable	Requires empirical validation progress, including UCI measurement maturity and related evidence.
Machine-verifiable ecosystem completeness	Not yet claimable	Requires companion validator, schema, reference-calculator / registry surfaces, or equivalent ProofPack assets.
Tier 4 compliance	Prohibited	No Tier-4 claim until ProofPack is publicly replayable and independently verified.
Cross-domain calibrated or validated UCI	Prohibited	Requires a published UCI Measurement Annex with named instruments, normalization rules, structural-independence evidence, and IRR data.

Decision-state summary (Normative shorthand).

RippleLogic uses five distinct decision states that implementations MUST NOT collapse.

Rights-admissible = passes NCRC.

Admissible = passes NCRC and TRC.

Selectable = admissible and passes CSV in the applicable containment/structural-viability mode.

Selected = chosen from the selectable set after RLS, tie-break, and any required escalation disposition.

Refusal = the framework cannot uniquely determine the applicable admissibility or selection state from declared inputs, evidence, projection rules, or layer commitments.

A later state MUST NOT be used to infer an earlier one, and no downstream score may rescue failure at an earlier state. Refusal MUST NOT be silently converted into gate failure, gate success, RLS tie-break, or discretionary selection. Refusal means the run lacks sufficient determination for the claim being attempted.

Execution Stability Diagnostic (carried-forward v10.8 clarification through v11.4). Execution stability is not a sixth decision state and not a replacement gate. It is a required diagnostic surface for Tier 2 and Tier 3 consequence-bearing execution, and recommended for medium- or high-stakes Tier 1 runs. It classifies the transition after hard admissibility and containment checks but before consequence-bearing execution. The diagnostic cannot rescue a failure at NCRC, TRC, or CSV, cannot convert Refusal into permission, and cannot use local stability to justify domination, rights infringement, or biosphere degradation.

CORE ARTIFACT ROLE MAP (release-line orientation)

- RippleLogic_v11.4_Canon.md: canonical repository text source and normative source of truth for this release line.

- RippleLogic_v11.4_Canon.docx: synchronized human-readable publication rendering of the Markdown source. It is hash-identified in the release manifest, but it does not outrank the Markdown source if the two diverge.

- SGP v6.4: companion sentience / moral-status protocol. Authoritative for sentience-scoring procedures, scalar canon, plateau semantics, rights-of-protection scalar definitions, governed review requirements, and classification-stability cautions. RippleLogic consumes only SGP outputs through Appendix G and MUST NOT define or recompute SGP internals.

SGP split-roadmap boundary (Informative but release-controlling for interpretation).

Future SGP work SHOULD further separate moral patienthood, governance-participation readiness, and stewardship/power readiness as first-class outputs. v11.4 consumes only the pinned SGP scalar interface and MUST NOT infer governance authority, voting weight, office eligibility, or stewardship power from moral patienthood or rights-of-protection status.

This roadmap note does not change the v11.4 SGP binding interface. It prevents over-reading the current scalar as a pure sentience detector or as a governance-authority score.

- ripple.md v4.4: portable alignment contract and assurance wrapper for implementations and deployments, including the wrapped-run ExecutionStabilityRecord field for high-stakes consequence-bearing episodes.

- RippleLogic Agent System v11.4: deployment/control specification for agentic implementations under the current pinned companion line unless and until a later synchronized agent-system release is promoted.

- Ripple Aligners Sheet v4.4: worked-run exemplar companion for replay, disclosure, and training, including the ExecutionStability sheet; not the repository-level validator or schema surface.

- Machine-verifiable ecosystem completeness requires additional companion surfaces when claimed, including validator / schema / reference-calculator / registry artifacts or ProofPack assets beyond this core canon line.

- Companion scalar-interface note: for SGP-integrated runs, RippleLogic treats SG_norm(E) and SG_patient_norm(E) as interface-stable outputs. Where provided by SGP v6.4 records, SG_measured_norm(E) and Plateau(E) MAY be logged for auditability, but RippleLogic MUST NOT infer plateau from raw measurement and MUST NOT substitute SG_measured_norm(E) for SG_norm(E) or SG_patient_norm(E) in normative computation. When SG_patient_norm(E) is available from the authoritative SGP record, RippleLogic SHOULD use SG_patient_norm(E) as the welfare-stream scalar for impacted protected stakeholders in residual welfare ranking. SGP-derived scalars MUST NOT be used to change admissibility outcomes under NCRC, TRC, or CSV.

AC.1 Specification Contract (moved from front matter in v10.4)

SECTION 0: SPECIFICATION CONTRACT (NORMATIVE)

0.1 Normative hierarchy (Single Source of Truth)

When interpreting RippleLogic v11.4, conflicts MUST be resolved using this precedence order (highest controls lowest):

1. Main text rules explicitly labeled MUST, SHALL, or PROHIBITED

2. Appendix B (Canonical Equations)

3. Appendix C (Rights / NCRC Canon Pack)

4. Appendix D (TRC and Scenario Governance Canon Pack)

5. Appendix H (PCC and Audit Flags Canon Pack)

6. Appendix AG (v10.8 Reality Grounding and Lean Cascade Patch)

7. Appendix AH (Category Grounding and Primitive Audit Protocol)

8. Appendix AI (Term Discrimination and Semantic Stability Protocol)

9. Appendix T (Execution Stability Diagnostic)

10. Appendix U (Transition Admissibility and Evidence Substrate)

11. Appendix AE (Layer-Discipline, Falsification, and Adaptive Reserve Addendum)

12. Appendix AF (Decision Verdict Semantics, Projection Freeze, Closure Status, Falsification Classes, and Authority-Selection Separation)

13. Appendix R (Tier 1-3 Reference Test Vectors)

14. Appendix E (UCI Interface and Tier-3 Construction Guidance)

15. Appendix G (SGP Integration Binding Interface)

16. Appendix V (Measurement Maturity and Deployment Context Hardening)

17. Appendix W (Schema Ownership, Registries, and Machine-Verifiable Surfaces)

18. Appendix X (Reference Calculator, Machine Vectors, and ProofPack-Prep Roadmap)

19. Appendix Y (Precision Patch: Legitimacy, Evidence Status, Validation, and Adoption Profiles)

20. Appendix Z (Integration-Hardening Addendum)

21. Appendix AA (Package-Content Boundary and Tooling-Line Separation)

22. Appendix AB (Reference Structure Protocol)

23. Appendix A (Symbols, Domains, and Notation Canon)

24. Appendix N (Stewardship Pack Canon)

25. Appendix F (Failure Modes and Anti-Gaming Controls)

26. Appendix J (Pre-Registration Templates and Scoring Definitions)

27. Appendix AD (49-Cell Welfare Dictionary): canonical interpretive reference for cell meanings, scoring language, and reviewer checks. Subordinate to Appendix B equations and to all normative gates; governs scoring interpretation and reviewer literacy only. It does not control admissibility, thresholds, or claim boundaries.

28. Appendix AK (Computability vs Realizability Bridge): normative implementation clarification for the boundary that computed/generated possibility is not grounded, selectable, executable, or ethical by itself. Subordinate to the main text, equations, RF/NCRC, TRC, CSV, RLS, and claim boundaries; it does not add a public gate.

29. Appendix AL (Source-Coupling Integrity and Generative-Source Traceability): normative hardening patch inside Reality Grounding and CSV. It prevents downstream output, fluency, compliance, permission, inherited procedure, or interface performance from being treated as proof of grounded capability; it does not add a public gate.

30. Examples, templates, and illustrative values (non-normative; includes Appendix K unless a future version explicitly elevates a named subsection)

31. Appendix O (Conformance Reference Map; informative pointer layer only)

Note (Informative). Version history and release notes are recorded in Appendix M and do not control normative meaning.

0.2 Tier contract and claims (v11.4 Canon line; Core Release 2026.09 package line) Tier 1-3: Executable in the v11.4 Canon release line, subject to stated minimum requirements. The Core Release 2026.09 package line contains Canon v11.4 / SGP v6.4 / ripple.md v4.4 / Agent System v11.4 / Primer v3.4 / Aligners Sheet v4.4. Any conformance claim against this release MUST pin to the v11.4 Canon artifact and the matching package manifest and hashes. Tier 4: PROHIBITED to claim in this release (design target only). Any Tier-4 content remains informational architecture describing what becomes claimable only after ProofPack publication and independent replayability.

0.3 Conformance vocabulary (Normative) RippleLogic uses distinct predicates; implementations MUST NOT collapse them.

- Rights-admissible: passes NCRC.
- Admissible: passes NCRC and TRC. Formally: Admissible(a) := NCRC(a) AND TRC(a).
- Selectable: admissible and passes CSV (including Containment Mode A where material).
- Selected: the final chosen option from the selectable set, either by ALLOW_FRAMEWORK_SELECTION when the framework selection path is decisive or by a separately disclosed AuthoritySelectionRecord when the framework refuses deterministic selection. A selected option MUST NOT be represented as a unique framework verdict when FrameworkVerdict is REFUSE_DETERMINISTIC_SELECTION.

0.4 Consolidated symbols and core objects (Normative; v11.4) Terminology note (Normative). Since v11.0 and carried forward in v11.4, u indexes Union Scopes (short: Scopes). “Union(s)” remains an allowed legacy narrative alias. This clarification SHALL NOT change the meaning of u or cascade semantics. Machine-readable surfaces SHOULD prefer UnionScope / UnionScopes as the canonical identifier family; Union remains a permitted human-readable alias in prose and UI text.

Table 0-1. Indices and core symbols

Symbol	Meaning
u ∈ {1,...,7}	Union Scope (Scope) index (legacy alias: union)
d ∈ {1,...,7}	Dimension index
a ∈ O	Candidate option
s ∈ S	Scenario index
r ∈ R	Right index
k	Impact instance index
g ∈ G_{u,d}	Protected subgroup index (for rights checking)

Table 0-2. Core sets

Set	Definition
$U = \{U_1,\ldots,U_7\}$	Operational Scopes
$D = \{D_1,\ldots,D_7\}$	Welfare dimensions
R	Rights set: {LIFE, BODY, LBTY, NEED, DIGN, PROC, INFO, ECOL}
C_r ⊆ U×D	Rights coverage sets
C_cat ⊆ U×D	Catastrophe cell set for TRC

Table 0-3. Welfare impact objects (all options a)

Symbol	Meaning	Range
I_dir(u,d,a)	Direct impact (post-saturation)	[-1,+1]
I_prop(u,d,a)	Propagated impact (post-saturation)	[-1,+1]
I_rights(u,d,a)	Rights worst-off subgroup impact	[-1,+1]

Table 0-4. Kernel, weights, and tail-risk symbols

Symbol	Meaning	Constraint / Notes
K ∈ ℝ^{49×49}	Ripple kernel matrix (sparse)	Convention: K_{ij} maps source j to target i (target-row, source-column)
w_u	Union weights	w_u ≥ 0, Σ_u w_u = 1

Symbol	Meaning	Constraint / Notes
v_d	Dimension weights	v_d ≥ 0, Σ_d v_d = 1
m(u,d)	Applicability mask	m ∈ {0,1} (RLS aggregation only)
p_s	Scenario probabilities	p_s ≥ 0, Σ_s p_s = 1
ω_c	Catastrophe weights	ω_c ≥ 0, Σ_{c∈C_cat} ω_c = 1
α	CVaR tail level	α ∈ (0,1)
τ_TRC	Corridor threshold	τ_TRC ∈ (0,1]

Table 0-5. Structural metrics

Symbol	Meaning	Range
UCI_u	Union Coherence Index for scope u	[0,1]
ΔUCI_u(a)	UCI_u(a) − UCI_u(baseline)	[-1,+1]
HOI	Hollowing-Out Index (monitoring diagnostic)	unbounded

0.5 Version commitments (Normative; v11.4) The canonical welfare impact scale is [-1,+1] with Baseline-Zero Rule (Section 5). Optional UI scale conversion (non-normative): If a user interface uses a percent-like scale, map the canonical cell impact I(u,d,a) ∈ [-1,+1] to UI points via: points = 100 × I. All admissibility gates (NCRC, TRC, CSV) and RLS calculations MUST be performed on the canonical [-1,+1] scale, or be converted back exactly before computation. Tier 1-3 TRC uses bounded-impact loss (Section 8). Tier 4 is design target only; no Tier-4 claims permitted until ProofPack is publicly replayable. UCI Equity for Self (U_1): E_1 := 1 by definition. The equity component is fixed at 1 for Self because equity-as-distribution does not apply within a single individual. This provides a neutral, non-penalizing identity value. UCI for Self is computed from Cohesion, Flow, and Resilience with E_1 = 1. UCI Equity for Biosphere (U_7): E_7 := 1 by default unless a governed biosphere-equity instrument is declared for the run. This provides a neutral, non-penalizing identity value when distributional equity is not operationalized for the Biosphere union. Record E7_METHOD_FIXED_1_DEFAULT (or the declared instrument identifier) in PCC.

Patch-line interpretation (Normative clarification). Unless a rule explicitly names a more specific patch version, references in section headings, appendix titles, or release notes to the current release line SHALL be interpreted as applying to the v11.4 line. Patch releases in the current release line MUST preserve the cascade architecture unless an explicit versioned change states otherwise.

0.6 Single Canonical Artifact Rule (Normative; v11.4)

This document is the single canonical RippleLogic Foundation Paper for v11.4. Any copy labeled “final,” “regen,” “patched,” “draft,” “fork,” or similar is non-canonical unless it is byte-identical to this artifact. Implementations, manifests, validators, ProofPacks, and downstream specifications MUST pin to the canonical artifact (filename + hash in the release manifest) and MUST NOT treat variant text as authoritative.

Retirement clause (Normative). Any earlier artifacts later labeled v11.4 that are not byte-identical to this canonical artifact are RETIRED and MUST NOT be used as sources of truth for v11.4 interpretation or for any v11.4 compliance claim. Prior v11.0 artifacts remain lineage artifacts only.

Integrity note (Informative). The canonical hash MUST be published in a sibling release manifest rather than embedded in the document body, because any edit changes the byte stream.

Canonical source designation (Normative). RippleLogic_v11.4_Canon.md is the canonical repository text source for this Foundation Paper. DOCX files are synchronized human-readable publication renderings, and any PDF distributed with the release is a derived render for reading and review. The release manifest identifies the exact byte streams shipped in a release. If Markdown and DOCX/PDF text diverge, the discrepancy is a release defect and the Markdown source governs until synchronized renderings are regenerated.

0.7 License, Attribution, and Marks (Informative)

License (Informative). Unless otherwise stated in a published release manifest, the text of this Foundation Paper is made available under Creative Commons Attribution 4.0 International (CC BY 4.0). Any accompanying reference implementations, code, schemas, or test fixtures SHOULD be released under the MIT License (or a later explicitly declared permissive license) in the same release record. Attribution: cite this document by canonical filename + SHA-256 and credit James McGaughran (originator / system architect). Marks: “RippleLogic” and “MathGov” may be used to describe compatible implementations, but trademark or branding use is governed separately by the project’s published brand guidelines (when available).

0.8 Release Packaging and Integrity Manifests (Informative)

Recommended release bundle (Informative). Publish (i) the canonical DOCX/PDF, (ii) a SHA-256 manifest file (for example, RippleLogic_<version>.sha256) listing hashes for each artifact, (iii) a machine-readable release manifest when available (for example YAML), and (iv) optional signatures. The manifest is the stable anchor for byte identity, while the document remains human-readable and editable without self-referential hashing issues. To avoid self-reference problems, a checksum manifest SHOULD hash the release artifacts and

MAY omit its own file unless a separate detached signature or higher-level release record governs it.

APPENDIX AD: 49-CELL WELFARE DICTIONARY (CANONICAL INTERPRETIVE REFERENCE)

Tooling warning (Normative for implementation). Tooling MUST source rights coverage from Section 7.2 / Appendix C.2 and catastrophe cells from Section 8.2, never from Appendix AD icons. Appendix AD icons are interpretive cues, not machine-authoritative gate definitions.

Status and boundaries. Appendix AD is the canonical interpretive reference for welfare-cell meanings, baseline-relative scoring language, evidence literacy, and reviewer checks. It is binding for interpreting what each Scope × Dimension cell is about. Its examples, indicator lists, and gate-relevance tags are illustrative and non-exhaustive. Appendix AD does not modify the cascade, equations, NCRC, TRC, CSV, RLS, UCI/HOI, HDW, SGP, PCC, RSP/RefStructRecord, thresholds, or claim boundaries. Gate tags are contextual cues for evaluators, not automated verdicts; admissibility is determined only by running the full cascade against declared evidence in the governed system.

The welfare matrix is the surface RLS ranks on: seven union scopes (U1–U7) × seven welfare dimensions (D1–D7), giving forty-nine accountability domains. Each cell is scored on a canonical −1 to +1 scale, relative to a declared baseline. A cell score is a claim, not a vibe: it asserts what changes, for whom, how strongly, on what evidence, and under what uncertainty.

Scale. +1 strong benefit, +0.5 moderate benefit, 0 no material change from baseline — never “unknown”, −0.5 moderate harm, −1 severe harm. Where an impact is not known, the cell is marked unknown, disclosed, and escalated where gate-relevant.

Ordering. Welfare scoring is the last step, not the first. The cascade is sequenced as Reality Grounding → Rights Floor → TRC → CSV → RLS — and an option is ranked across these cells only after it has cleared the rights, tail-risk, and CSV gates. No welfare score rescues an option that failed an earlier gate.

Gate-relevance key. Tags below indicate where a cell may become relevant to a gate in a real run; they are contextual cues, not fixed classifications, and do not replace a canonical evaluation. Canonical rights coverage set C_r and declared catastrophe cell set C_cat, not the icon labels alone, control non-maskability and gate status.

⚖️ Rights-relevance cue — may require NCRC review when the cell lies in the canonical or governed rights coverage set C_r, or when the PCC declares a rights-relevant extension. The canonical C_r, not this icon alone, controls rights-covered status and non-maskability.

⚠️ Tail-risk relevance cue — may require TRC review when the cell lies in the declared catastrophe cell set C_cat, or when catastrophe relevance is otherwise triggered. The declared C_cat and TRC trigger rules, not this icon alone, control catastrophe-cell status and non-maskability.

⛓️ Containment-relevant — a local gain may degrade the wider scopes the decision depends on; checked before RLS.

○ Ordinary welfare — ranked by RLS once RF/NCRC, TRC, and CSV are cleared.

? Uncertain — evidence weak or contested; mark, disclose, and escalate if gate-relevant.

Appendix AD tabular dictionary

The following seven tables preserve the complete 49-cell dictionary. They are split by union scope for readability while preserving the canonical U1/D1 through U7/D7 coordinate system.

U1 Self

Cell	Plain meaning	Movement (+ benefit / − harm)	Typical indicators / evidence	Possible gate relevance	Example	Reviewer check
U1/D1 Personal Resources label: personal resources	An individual's economic security: income, savings, housing stability, and access to the material means of a decent life.	+ More stable income, affordable housing, reduced debt, secured access to essentials. − Job loss, eviction risk, debt spirals, loss of essential goods or services.	income and employment records; housing stability / eviction data; debt and arrears; cost-of-living and benefit uptake	○ Ordinary Ordinary welfare in most cases; can escalate to rights-covered where a subsistence or social-minimum floor is legally recognised.	A wage policy that lifts low-income workers above a local living-wage line improves U1/D1.	A reviewer checks whether the income/housing change is measured against a declared baseline cohort, not anecdote.
U1/D2 Personal Health label: personal health	The individual's physical and mental health, safety, and bodily integrity.	+ Lower injury/illness risk, better access to care, improved mental health, safer conditions. − Injury, illness, untreated conditions, exposure to hazards, deteriorating mental health.	morbidity / mortality data; care access and waiting times; injury and exposure records; validated mental-health measures	⚖️ Rights ○ Ordinary Frequently rights-covered: bodily integrity and access-to-care floors can trigger NCRC. Severe individual harm escalates regardless of aggregate benefit.	A treatment-access change that removes a barrier for a chronic-illness group improves U1/D2.	A reviewer asks whether a claimed health gain rests on outcome data or only on inputs (e.g. clinics built ≠ health improved).
U1/D3 Relationships label: relationships	The quality and stability of a person's close personal relationships and social connection.	+ Stronger bonds, reduced isolation, more supportive networks. − Isolation, relationship breakdown, forced separation, loss of support.	loneliness / connection surveys; support-network mapping; separation or displacement records	○ Ordinary ? Uncertain Usually ordinary welfare; measurement is often soft, so uncertainty should be marked explicitly.	A relocation program that keeps support networks intact protects U1/D3.	A reviewer challenges whether a relationship claim is evidenced or inferred from a proxy like attendance.
U1/D4 Learning label: learning	The individual's access to education, skills, and the capacity to learn and develop.	+ Better access to learning, skill gains, expanded opportunity. − Lost access to education, deskilling, barriers to learning.	enrolment and completion; skills assessments; access and dropout data	○ Ordinary Ordinary welfare; can touch rights where a basic-education entitlement exists.	A reskilling program for displaced workers improves U1/D4.	A reviewer asks whether learning outcomes are demonstrated or only opportunity offered.
U1/D5 Personal Agency label: autonomy	The person's real ability to choose, refuse, consent, act, and shape their own life path.	+ More autonomy, better options, stronger consent, freedom from coercion and surveillance. − Coercion, manipulation, surveillance, forced dependency, loss of meaningful choice.	consent quality; availability of alternatives / opt-out; coercion complaints; mobility and accessibility data; applicable legal protections	⚖️ Rights ○ Ordinary Often rights-covered. Triggers NCRC when liberty, bodily integrity, due process, dignity, or information rights are impaired — non-compensatory regardless of welfare gains elsewhere.	A transport policy that gives disabled residents reliable independent mobility improves U1/D5.	A reviewer tests whether consent was genuine and revocable, or merely formal.

Cell	Plain meaning	Movement (+ benefit / − harm)	Typical indicators / evidence	Possible gate relevance	Example	Reviewer check
U1/D6 Purpose label: purpose	The individual's sense of meaning, dignity, and worth in their life and work.	+ Greater sense of purpose, recognition, dignity, and self-respect. − Humiliation, alienation, loss of role or recognition, indignity.	dignity / recognition measures; role and identity surveys; qualitative testimony	⚖️ Rights ? Uncertain Can be rights-covered where human dignity is implicated; otherwise ordinary. Often hard to measure — mark uncertainty.	A program that treats welfare recipients with dignified, non-stigmatising processes supports U1/D6.	A reviewer distinguishes a genuine dignity effect from sentiment unsupported by experience data.
U1/D7 Local Conditions label: local conditions	The immediate physical environment a person lives in: air, noise, water, hazards, and surroundings.	+ Cleaner air/water, less noise, fewer local hazards, healthier surroundings. − Pollution exposure, noise, contamination, environmental hazards near the person.	local air/water/noise monitoring; hazard proximity; exposure records	○ Ordinary ⚖️ Rights Ordinary welfare; can become rights-covered under environmental-health protections and disproportionate-exposure (environmental-justice) concerns.	A factory buffer-zone rule that cuts a household's pollutant exposure improves U1/D7.	A reviewer checks measured exposure change against the affected location, not a city-wide average.

U2 Household

Cell	Plain meaning	Movement (+ benefit / − harm)	Typical indicators / evidence	Possible gate relevance	Example	Reviewer check
U2/D1 Household Resources label: household resources	The economic security of the household unit: shared income, assets, housing, and ability to meet needs.	+ Improved household finances, housing security, reduced shared debt. − Household impoverishment, housing loss, debt, inability to meet needs.	household income and assets; housing tenure; arrears and food-security data	○ Ordinary Ordinary welfare; escalates toward rights where dependents (children, disabled members) face a subsistence floor.	A childcare subsidy that frees household income improves U2/D1.	A reviewer asks whether the household, not just the earner, is the measured unit.
U2/D2 Household Health label: household health	The collective physical and mental health and safety of household members, including dependents.	+ Better shared health, safer home, improved care for dependents and caregivers. − Household illness, unsafe housing, caregiver burnout, harm to dependents.	household health surveys; home-safety inspections; caregiver-strain measures; dependent outcome data	⚖️ Rights ○ Ordinary Frequently rights-covered where children or dependents are affected; child welfare can trigger NCRC. Caregiver harm is easily hidden — surface it.	A home-care support program that reduces caregiver strain and improves a dependent's health raises U2/D2.	A reviewer checks that dependents and caregivers are scored separately, not blended into one household figure.
U2/D3 Family Cohesion label: family cohesion	The strength, stability, and supportive quality of relationships within the household.	+ Stronger family bonds, stability, reduced conflict, kept-together units. − Family breakdown, forced separation, household conflict, instability.	family-stability surveys; separation / removal records; conflict indicators	⚖️ Rights ○ Ordinary ? Uncertain Can be rights-covered (family unity, child rights); otherwise ordinary. Measurement is soft — mark uncertainty.	A policy that avoids unnecessary family separation in housing transitions protects U2/D3.	A reviewer asks whether a cohesion claim rests on outcomes or on assumed effects of a program.

Cell	Plain meaning	Movement (+ benefit / − harm)	Typical indicators / evidence	Possible gate relevance	Example	Reviewer check
U2/D4 Shared Learning label: shared learning	The household's collective access to information, education, and learning capacity.	+ Better shared access to education and information, intergenerational learning. − Loss of household educational access, information poverty.	household connectivity / device access; children's schooling continuity; adult-learning access	○ Ordinary Ordinary welfare; touches rights where basic-education entitlements for children apply.	A home-broadband program that enables children's remote schooling improves U2/D4.	A reviewer checks access actually reaches the household, not just the area.
U2/D5 Household Agency label: household agency	The household's collective ability to make decisions, plan, and control its own circumstances.	+ Greater household self-determination, planning ability, freedom from external control. − Loss of household control, externally imposed decisions, dependency.	decision-autonomy surveys; dependency / conditionality of support; forced-relocation data	⚖️ Rights ○ Ordinary Can be rights-covered (family privacy, autonomy); otherwise ordinary.	A flexible-benefit design that lets households allocate support to their own priorities raises U2/D5.	A reviewer tests whether household choice is real or constrained by conditions.
U2/D6 Family Meaning label: family meaning	The household's shared sense of identity, belonging, dignity, and purpose.	+ Stronger shared identity, belonging, dignity, and continuity. − Loss of household identity, stigma, indignity, eroded belonging.	belonging / identity surveys; stigma indicators; qualitative testimony	? Uncertain ○ Ordinary Usually ordinary welfare; can touch dignity rights. Soft to measure — mark uncertainty.	A culturally-respectful resettlement process preserves U2/D6.	A reviewer separates a real meaning effect from program rhetoric.
U2/D7 Home Environment label: home environment	The physical environmental quality of the home and its immediate setting.	+ Healthier home conditions, reduced indoor hazards, better surroundings. − Indoor pollution, mould, hazards, degraded immediate environment.	housing-quality inspections; indoor air / damp measures; hazard records	○ Ordinary ⚖️ Rights Ordinary welfare; rights-relevant under habitability standards.	A retrofit program that removes household mould improves U2/D7.	A reviewer checks measured home conditions, not self-report alone.

U3 Community

Cell	Plain meaning	Movement (+ benefit / − harm)	Typical indicators / evidence	Possible gate relevance	Example	Reviewer check
U3/D1 Local Resources label: local resources	The shared economic resources, infrastructure, and services available to a local community.	+ Better local infrastructure, services, and economic opportunity. − Loss of services, infrastructure decay, local economic decline.	local service provision; infrastructure condition; local employment / business data	○ Ordinary ⛓️ Containment Ordinary welfare; containment-relevant when local gains are extracted at the cost of wider scopes, or when shared infrastructure is hollowed out.	A community facility upgrade that adds usable shared services improves U3/D1.	A reviewer asks whether the benefit is genuinely shared or captured by a subgroup.

Cell	Plain meaning	Movement (+ benefit / − harm)	Typical indicators / evidence	Possible gate relevance	Example	Reviewer check
U3/D2 Community Health label: public health	The collective physical and mental health and safety of the local population.	+ Lower community disease/injury burden, better local care access, safer environment. − Local disease spread, unsafe conditions, reduced care access, environmental health harm.	local public-health surveillance; care-access data; injury / safety statistics; environmental-health monitoring	⚖️ Rights ⚠️ Tail-risk ○ Ordinary Ordinary welfare normally; rights-relevant under disproportionate harm or care-access floors; tail-relevant where local outbreaks or contamination could cascade.	A clean-water intervention that cuts a community's disease burden improves U3/D2.	A reviewer checks whether subgroup harms are masked by a favourable community average.
U3/D3 Community Cohesion label: trust and cohesion	The quality of trust, cooperation, belonging, safety, and relational fabric within a community.	+ Increased trust, reduced isolation, better cooperation, stronger neighbourhood resilience. − Polarisation, exclusion, displacement, fear, social fragmentation.	community trust surveys; participation rates; crime / fear-of-crime data; displacement records	⛓️ Containment ⚖️ Rights ○ Ordinary Containment-relevant because eroded social fabric degrades the wider system; rights-relevant if discrimination, exclusion, or targeted domination appears.	A public-space renovation that increases safe cross-group interaction improves U3/D3.	A reviewer asks whether cohesion gains for some came via exclusion of others.
U3/D4 Local Knowledge label: local knowledge	The community's shared knowledge, information access, and local expertise.	+ Better local information access, preserved local/Indigenous knowledge, stronger civic literacy. − Information loss, erosion of local knowledge, misinformation spread.	information-access measures; local-knowledge preservation records; misinformation prevalence	⛓️ Containment ○ Ordinary ? Uncertain Ordinary welfare; containment-relevant where an epistemic commons is degraded. Can be rights-relevant for Indigenous knowledge.	A program documenting local ecological knowledge with community consent improves U3/D4.	A reviewer checks consent and ownership of the knowledge claimed as preserved.
U3/D5 Community Agency label: community agency	The community's collective capacity to organise, participate, and influence decisions affecting it.	+ Stronger local voice, participation, self-organisation, and contestability. − Disenfranchisement, suppressed organising, exclusion from decisions.	participation in local governance; consultation quality; association / assembly freedom	⚖️ Rights ⛓️ Containment ○ Ordinary Often rights-covered (assembly, association, participation); containment-relevant where civic capacity underpins system stability.	A binding community-consultation requirement on local development raises U3/D5.	A reviewer tests whether participation was substantive or a procedural formality.
U3/D6 Shared Meaning label: shared meaning	The community's shared identity, culture, dignity, and sense of collective purpose.	+ Stronger shared identity, cultural continuity, pride, and belonging. − Cultural erosion, stigma, loss of identity, collective humiliation.	cultural-participation data; heritage continuity; community-identity surveys	⚖️ Rights ? Uncertain ○ Ordinary Ordinary welfare; rights-relevant for cultural rights and minority protection. Soft to measure — mark uncertainty.	Protecting a culturally significant site during redevelopment preserves U3/D6.	A reviewer separates a genuine cultural effect from a symbolic gesture.

Cell	Plain meaning	Movement (+ benefit / − harm)	Typical indicators / evidence	Possible gate relevance	Example	Reviewer check
U3/D7 Local Ecology label: local ecology	The ecological condition of the community's local environment: green space, biodiversity, water, land.	+ Restored habitat, cleaner local environment, protected green space. − Habitat loss, local pollution, degraded green space, contamination.	local biodiversity / green-space data; pollution monitoring; land-use change	⛓ Containment ⚠️ Tail-risk ○ Ordinary Containment-relevant (local ecology supports wider systems); tail-relevant where degradation is irreversible or contamination could cascade.	A watershed-protection rule that prevents local contamination improves U3/D7.	A reviewer asks whether 'no local effect' was measured or merely assumed.

U4 Organization

Cell	Plain meaning	Movement (+ benefit / − harm)	Typical indicators / evidence	Possible gate relevance	Example	Reviewer check
U4/D1 Organizational Resources label: operating resources	An organization's operating means: finances, capital, staffing, and capacity to function and deliver.	+ Stronger finances, capacity, staffing, and operational sustainability. − Insolvency risk, capacity loss, layoffs, operational decline.	financial statements; staffing and capacity data; service-delivery metrics	○ Ordinary ⛓ Containment Ordinary welfare; containment-relevant when organizational gains are extracted by externalising harm onto community, polity, or biosphere scopes.	A funding change that stabilises a hospital's operating budget improves U4/D1.	A reviewer checks whether the organizational gain creates uncosted externalities in other cells (the classic 'operating resources up, U7 down' trap).
U4/D2 Organizational Safety label: safety	The safety of people within and affected by the organization: workers, users, and the public.	+ Fewer injuries, safer operations, better safety culture and compliance. − Workplace injury, unsafe products/services, public-safety incidents.	incident and injury rates; safety-compliance audits; near-miss reporting	⚖️ Rights ⚠️ Tail-risk ○ Ordinary Rights-covered via labour and safety protections; tail-relevant where failure could cause catastrophic or mass-casualty events.	A maintenance regime that cuts industrial-accident rates improves U4/D2.	A reviewer checks whether reported safety reflects outcomes or only paperwork compliance.
U4/D3 Organizational Culture label: culture	The internal relational health of the organization: trust, fairness, inclusion, and cooperation.	+ Healthier culture, fairer treatment, reduced harassment, stronger cooperation. − Toxic culture, discrimination, harassment, low trust, burnout.	staff-climate surveys; grievance and turnover data; fairness / inclusion measures	⚖️ Rights ○ Ordinary Ordinary welfare; rights-relevant where discrimination or harassment occurs.	An anti-harassment reform that reduces verified grievances improves U4/D3.	A reviewer asks whether culture data is independent or self-reported by leadership.
U4/D4 Knowledge Systems label: knowledge systems	The organization's knowledge, data integrity, institutional memory, and learning systems.	+ Better knowledge retention, data integrity, transparency, and learning. − Knowledge loss, data corruption, opacity, broken institutional memory.	data-integrity audits; documentation completeness; knowledge-retention metrics	⛓ Containment ○ Ordinary Ordinary welfare; containment-relevant where degraded knowledge systems undermine accountability or wider-system reliability.	A records-integrity upgrade that makes decisions auditable improves U4/D4.	A reviewer tests whether claimed integrity is verifiable or asserted.

Cell	Plain meaning	Movement (+ benefit / − harm)	Typical indicators / evidence	Possible gate relevance	Example	Reviewer check
U4/D5 Role Agency label: role agency	The meaningful autonomy, voice, and fair treatment of individuals in their organizational roles.	+ More voice, fair process, reasonable autonomy, freedom from arbitrary treatment. − Arbitrary power, suppressed voice, exploitation, loss of role autonomy.	worker-voice mechanisms; grievance outcomes; autonomy / discretion measures	⚖️ Rights ○ Ordinary Rights-relevant via labour rights and freedom of association; otherwise ordinary.	A grievance process with real remedy improves U4/D5.	A reviewer checks whether voice mechanisms produce outcomes or are decorative.
U4/D6 Mission Coherence label: mission coherence	The alignment between the organization's stated purpose and its actual conduct and effects.	+ Stronger purpose-conduct alignment, integrity, and legitimate mission delivery. − Mission drift, purpose-washing, hollow mandate, loss of legitimacy.	mission-vs-outcome analysis; stakeholder-trust measures; conduct audits	⛓️ Containment ○ Ordinary ? Uncertain Ordinary welfare; containment-relevant where mission decay (HOI-adjacent) hollows out the institution. Distinct from the formal HOI diagnostic.	A governance reform that realigns conduct with mandate improves U4/D6.	A reviewer asks whether coherence is evidenced by conduct or only by restated values.
U4/D7 Operational Footprint label: operational footprint	The environmental impact of the organization's operations: emissions, waste, resource use, land.	+ Lower emissions/waste, reduced resource intensity, smaller harmful footprint. − Pollution, emissions, waste, resource depletion, ecological damage.	emissions and waste data; resource-use intensity; environmental-compliance records	⛓️ Containment ⚠️ Tail-risk ○ Ordinary Containment-relevant (footprint degrades wider scopes); tail-relevant where the footprint contributes to irreversible or large-scale harm. The canonical 'local win, wider loss' cell.	A process change that cuts a plant's emissions while sustaining output improves U4/D7.	A reviewer checks footprint against measured externalities, not offset claims alone.

U5 Polity

Cell	Plain meaning	Movement (+ benefit / − harm)	Typical indicators / evidence	Possible gate relevance	Example	Reviewer check
U5/D1 Public Resources label: public resources	The polity's fiscal and material capacity: public finances, infrastructure, and provisioning.	+ Sounder public finances, better infrastructure, sustainable provisioning. − Fiscal crisis, infrastructure decay, depleted public capacity.	public-finance data; infrastructure condition indices; fiscal-sustainability metrics	○ Ordinary ⛓️ Containment Ordinary welfare; containment-relevant where short-term fiscal wins erode long-term public capacity.	An infrastructure-maintenance program that prevents asset decay improves U5/D1.	A reviewer checks whether gains are sustainable or borrowed from the future.

Cell	Plain meaning	Movement (+ benefit / − harm)	Typical indicators / evidence	Possible gate relevance	Example	Reviewer check
U5/D2 Population Health label: population health	The health, safety, and mortality outcomes of the whole population within a polity.	+ Lower population mortality/morbidity, better system-wide care, stronger resilience. − Rising mortality, system overload, reduced care access, public-health crises.	population mortality / morbidity; health-system capacity; health-equity gaps; outbreak surveillance	⚠️ Tail-risk ⚖️ Rights ○ Ordinary Tail-relevant for pandemic / health-system-collapse scenarios (non-averagable); rights-relevant under right-to-health floors; otherwise ordinary welfare.	A vaccination program that lowers population mortality improves U5/D2.	A reviewer checks whether worst-case scenarios were bounded with a tail measure, not averaged away.
U5/D3 Legitimacy label: legitimacy	The rule of law, due process, accountability, and perceived legitimacy of public institutions.	+ Stronger rule of law, due process, accountability, and institutional trust. − Erosion of rule of law, impunity, corruption, collapse of institutional trust.	rule-of-law indices; due-process / judicial-independence measures; corruption indicators; institutional-trust surveys	⛓️ Containment ⚖️ Rights ⚠️ Tail-risk ○ Ordinary Strongly containment-relevant: legitimacy is load-bearing for the whole system. Rights-relevant (due process); tail-relevant where collapse risks systemic instability.	A judicial-independence safeguard that resists political capture improves U5/D3.	A reviewer tests whether legitimacy is measured independently or claimed by the institution being assessed.
U5/D4 Information Access label: information access	The polity's information environment: access, transparency, free expression, and epistemic integrity.	+ Better transparency, information access, free expression, healthier epistemic commons. − Censorship, opacity, disinformation, surveillance chilling, epistemic capture.	transparency / FOI metrics; press-freedom indices; disinformation prevalence; surveillance scope	⚖️ Rights ⛓️ Containment ○ Ordinary Rights-covered (free expression, information rights); containment-relevant because a damaged epistemic commons undermines all other governance.	An open-data and FOI reform that increases verifiable transparency improves U5/D4.	A reviewer checks whether transparency is real and usable or nominal.
U5/D5 Civil Agency label: civil agency	People's ability to participate in public life, exercise rights, contest authority, and influence governance.	+ Stronger voting access, participation, transparency, due process, and contestability. − Suppression, censorship, corruption, procedural exclusion, arbitrary enforcement, reduced civic voice.	voting access and turnout; procedural-fairness measures; availability of legal remedies; consultation quality; censorship reports	⚖️ Rights ⛓️ Containment ○ Ordinary Frequently rights-covered: tied to due process, liberty, information, dignity, and civic rights — non-compensatory under NCRC. Containment-relevant for system stability.	A reform expanding access to public hearings raises U5/D5; one that blocks lawful dissent harms it.	A reviewer tests whether civic participation is enforceable or merely nominal, and whether any group is selectively excluded.
U5/D6 Public Meaning label: public meaning	Shared civic identity, dignity, social contract, and non-domination across the polity.	+ Stronger civic dignity, inclusion, fair recognition, and a credible social contract. − Domination, civic humiliation, exclusionary identity, eroded social contract.	inclusion / recognition measures; social-contract trust surveys; discrimination indicators	⚖️ Rights ⛓️ Containment ? Uncertain Rights-relevant (dignity, non-domination, non-discrimination); containment-relevant where a fractured social contract destabilises the polity. Soft to measure.	A reconciliation policy that credibly extends recognition to a marginalised group improves U5/D6.	A reviewer separates symbolic recognition from changes people actually experience.

Cell	Plain meaning	Movement (+ benefit / – harm)	Typical indicators / evidence	Possible gate relevance	Example	Reviewer check
U5/D7 Public Environment label: public environment	The environmental quality and ecological sustainability managed at the polity scale.	+ Better regional environmental quality, sustainable resource management, reduced systemic pollution. – Regional pollution, resource depletion, ecosystem degradation, environmental injustice.	regional environmental monitoring; resource-stock data; pollution and emissions; environmental-justice mapping	⚠️ Tail-risk ⛓️ Containment ⚖️ Rights ○ Ordinary Tail-relevant where regional harm is irreversible or large-scale; containment-relevant; rights-relevant under environmental-justice and environmental-rights regimes.	A regional emissions cap that prevents airshed degradation improves U5/D7.	A reviewer checks whether environmental-justice distribution was assessed, not just regional totals.

U6 Humanity / CMIU

Cell	Plain meaning	Movement (+ benefit / – harm)	Typical indicators / evidence	Possible gate relevance	Example	Reviewer check
U6/D1 Global Resources label: global resources	Humanity's shared material base: global resource stocks, critical supply chains, and common infrastructure.	+ More sustainable global resource use, resilient supply chains, protected commons. – Resource depletion, supply-chain fragility, degradation of global commons.	global resource-stock data; supply-chain resilience metrics; commons-condition indicators	⛓️ Containment ⚠️ Tail-risk ○ Ordinary Containment-relevant and tail-relevant where depletion or fragility threatens systemic, hard-to-reverse harm at civilisational scale.	A critical-minerals stewardship agreement that reduces depletion risk improves U6/D1.	A reviewer checks whether 'global' claims rest on global data or extrapolated local figures.
U6/D2 Human Safety label: human safety	The safety and survival of humanity at large, including existential and catastrophic risk.	+ Reduced catastrophic and existential risk; stronger global safety and resilience. – Increased existential, pandemic, conflict, or technological catastrophic risk.	catastrophic-risk assessments; biosecurity / nuclear-risk indicators; conflict-escalation data	⚠️ Tail-risk ⚖️ Rights ○ Ordinary Primarily tail-relevant: catastrophic and existential risk is non-averagable and bounded under TRC, never traded against ordinary welfare. Rights-relevant at the right-to-life floor.	A biosecurity protocol that lowers engineered-pandemic risk improves U6/D2.	A reviewer checks that tail risk was bounded with a worst-case measure and not diluted by expected-value framing.
U6/D3 Cooperation label: cooperation	The capacity of humanity to coordinate, cooperate, and maintain peaceful, stable relations at scale.	+ Stronger global cooperation, conflict reduction, durable coordination institutions. – Breakdown of cooperation, arms races, institutional collapse, escalating conflict.	international-cooperation indices; conflict / treaty-compliance data; coordination-institution health	⛓️ Containment ⚠️ Tail-risk ○ Ordinary Containment-relevant (cooperation underpins global stability); tail-relevant where breakdown risks large-scale conflict.	A verification regime that sustains an arms-control treaty improves U6/D3.	A reviewer asks whether cooperation is durable or a fragile short-term arrangement.

Cell	Plain meaning	Movement (+ benefit / − harm)	Typical indicators / evidence	Possible gate relevance	Example	Reviewer check
U6/D4 Civilizational Knowledge label: civilizational knowledge	Humanity's accumulated knowledge, science, and the integrity and safety of its knowledge systems.	+ Knowledge preservation, scientific progress, epistemic integrity, responsible knowledge stewardship. − Knowledge loss, scientific capture, epistemic collapse, or proliferation of dangerous knowledge.	scientific-integrity measures; knowledge-preservation status; research-governance indicators	⛓ Containment ⚠️ Tail-risk ○ Ordinary Containment-relevant; tail-relevant in two directions — catastrophic loss of knowledge, and dangerous diffusion of harmful capability.	An open-but-governed research framework that preserves integrity while limiting dangerous diffusion improves U6/D4.	A reviewer checks whether knowledge benefits were weighed against misuse and proliferation pathways.
U6/D5 Collective Agency label: collective agency	Humanity's capacity for self-determination and legitimate collective decision-making at the species scale.	+ Stronger legitimate global agency, fair representation, and self-determination. − Concentration of unaccountable power, disenfranchisement of peoples, loss of collective self-determination.	representation / participation in global governance; power-concentration indicators; self-determination measures	⚖️ Rights ⛓ Containment ? Uncertain Rights-relevant (self-determination); containment-relevant where illegitimate concentration destabilises the system. Measurement is contested — mark uncertainty.	A governance reform giving affected populations real voice in a global regime improves U6/D5.	A reviewer tests whether 'collective' agency includes the marginalised or only powerful actors.
U6/D6 Shared Purpose label: shared purpose	Humanity's shared sense of meaning, moral direction, and long-term orientation toward flourishing.	+ Stronger shared purpose, long-term orientation, intergenerational responsibility. − Nihilism, short-termism, loss of shared direction, intergenerational betrayal.	long-term-orientation indicators; intergenerational-equity measures; qualitative analysis	? Uncertain ⛓ Containment ○ Ordinary Mostly ordinary/aspirational; containment-relevant where short-termism erodes long-term stability. Inherently soft — mark uncertainty and avoid overclaiming.	A long-horizon institution (e.g. a future-generations commissioner) can support U6/D6.	A reviewer flags this cell as especially prone to rhetoric and demands concrete mechanisms.
U6/D7 Planetary Conditions label: planetary conditions	The condition of planetary systems that support human civilisation and coordinated managing intelligence.	+ Reduced emissions, improved climate stability, lower systemic ecological risk, stronger global resilience. − Increased climate risk, global pollution, destabilised food/water systems, irreversible planetary degradation.	emissions data; climate models; planetary-boundary metrics; global food/water-security projections	⚠️ Tail-risk ⛓ Containment ○ Ordinary Often catastrophe-relevant: may trigger TRC where irreversible or large-scale harm is plausible. Containment-relevant as a system-stability check.	A high-emissions infrastructure choice may harm U6/D7 even when it benefits a local economy (U3/D1 or U4/D1).	A reviewer checks whether irreversibility and worst-case bounds were modelled, not just expected outcomes.

U7 Biosphere

Cell	Plain meaning	Movement (+ benefit / – harm)	Typical indicators / evidence	Possible gate relevance	Example	Reviewer check
U7/D1 Life-Support Resources label: life-support resources	The biosphere's foundational resources that sustain life: soil, fresh water, clean air, fertility.	+ Protected or restored soil, water, and air; sustained natural-resource base. – Soil loss, water depletion, air degradation, exhaustion of life-support systems.	soil-health data; freshwater-stock and quality; air-quality baselines; resource-renewal rates	⚠️ Tail-risk ⛓ Containment ○ Ordinary Tail-relevant where depletion is irreversible or threatens collapse of life-support systems; containment-relevant. Distinct from U4/D1 'operating resources'.	An aquifer-protection rule that prevents irreversible groundwater loss improves U7/D1.	A reviewer checks whether renewal rates, not just current stocks, were assessed.
U7/D2 Ecosystem Health label: ecosystem health	The health, function, and viability of ecosystems and the species within them.	+ Healthier, functioning ecosystems; recovering species and habitats. – Ecosystem dysfunction, species decline, habitat collapse, trophic breakdown.	ecosystem-function indicators; species-population trends; habitat-condition assessments	⚠️ Tail-risk ⛓ Containment ○ Ordinary Tail-relevant where collapse or extinction is plausible and irreversible; containment-relevant as ecosystems underpin wider scopes.	A fisheries policy that lets a collapsing stock recover improves U7/D2.	A reviewer checks whether ecosystem-function evidence exists or only a single charismatic indicator.
U7/D3 Biotic Relations label: biotic relations	The integrity of relationships and interdependencies among living systems and between humans and nature.	+ Restored ecological relationships, sustainable human-nature interactions, protected interdependencies. – Disrupted food webs, broken ecological relationships, harmful human-nature dynamics.	trophic / food-web indicators; interaction-network data; human-impact assessments	⛓ Containment ⚠️ Tail-risk ○ Ordinary Containment-relevant: a local win that severs ecological relationships can fail containment even with strong U4/D1 gains; tail-relevant where cascades are plausible.	A factory expansion that damages a watershed may fail containment via U7/D3 even while improving U4/D1.	A reviewer asks whether relational/cascade effects were modelled or ignored as 'no direct effect'.
U7/D4 Ecological Knowledge label: ecological knowledge	Humanity's understanding and monitoring of ecological systems, including baseline and early-warning data.	+ Better ecological monitoring, baselines, early-warning capacity, and stewardship knowledge. – Monitoring gaps, lost baselines, ecological ignorance, blind spots before thresholds.	monitoring-coverage data; baseline-completeness; early-warning-system status	⛓ Containment ? Uncertain ○ Ordinary Containment-relevant: poor ecological knowledge undermines the ability to detect tipping points. Often the cell most affected by the 0-vs-unknown distinction.	An expanded biodiversity-monitoring network that fills baseline gaps improves U7/D4.	A reviewer checks that absent ecological data is marked unknown, not scored as 0 ('no effect').
U7/D5 Resilience Capacity label: resilience capacity	The biosphere's capacity to absorb shocks, adapt, and recover from disturbance.	+ Greater ecological resilience, redundancy, and recovery capacity. – Loss of resilience, reduced redundancy, approach to tipping points and regime shifts.	resilience / redundancy indicators; recovery-rate data; threshold-proximity assessments	⚠️ Tail-risk ⛓ Containment ○ Ordinary Tail-relevant where loss of resilience brings irreversible regime shifts; containment-relevant as resilience buffers all scopes.	Restoring wetland buffers that absorb flood shocks improves U7/D5.	A reviewer asks whether proximity to thresholds was assessed, not just current condition.

Cell	Plain meaning	Movement (+ benefit / − harm)	Typical indicators / evidence	Possible gate relevance	Example	Reviewer check
U7/D6 Life Continuity label: life continuity	The continuity and persistence of life and biodiversity over time, including irreversibility of loss.	+ Protected biodiversity, prevented extinctions, sustained continuity of life. − Species extinction, irreversible biodiversity loss, broken continuity of lineages.	extinction-risk (e.g. red-list) data; biodiversity-trend indices; irreversibility assessments	⚠️ Tail-risk ⛓ Containment ○ Ordinary Strongly tail-relevant: extinction is the paradigm of irreversible, non-averagable harm under TRC; containment-relevant.	A habitat-corridor program that prevents a species' extinction improves U7/D6.	A reviewer checks whether irreversibility was treated as a tail constraint, not a discountable cost.
U7/D7 Ecological Integrity label: ecological integrity	The overall health, resilience, diversity, and continuity of ecosystems as living support systems.	+ Habitat restoration, biodiversity protection, improved ecological resilience, reduced pollution. − Habitat destruction, species loss, ecosystem collapse, pollution, soil/water degradation.	biodiversity indices; habitat-area data; water/soil quality; ecosystem-resilience measures; ecological monitoring	⛓ Containment ⚠️ Tail-risk ⚖️ Rights ○ Ordinary Often containment-relevant and sometimes catastrophe-relevant; may relate to ecological-integrity rights where recognised (e.g. rights of nature).	A factory expansion that damages a watershed may fail containment even when it improves U4/D1.	A reviewer checks whether integrity is evidenced by ecosystem-level data or inferred from a single metric.

FINAL DOCUMENT COMPLETION STATEMENT

RippleLogic v11.4 is released as a Tier 1-3 core-foundation and architecture specification. It preserves the canonical cascade, equations, rights thresholds, TRC mechanics, Containment semantics, SGP binding interface, PLSS/FLGG architecture, RSP/RefStructRecord evidence discipline, refusal logic, audit requirements, and claim-boundary rules.

Appendix AD adds canonical interpretive guidance for the 49-cell welfare matrix. It improves scoring literacy, evidence discipline, and reviewer challengeability without changing the cascade, equations, admissibility gates, thresholds, or release claim boundaries.

This release does not claim empirical validation, Tier 4 readiness, ProofPack readiness, deployment readiness, legal compliance certification, machine-verifiable ecosystem completeness, cross-domain validated UCI, or completed biological SGP measurement.

Release identity is governed by the repository manifest and verified SHA-256 hashes for the published artifact set.

APPENDIX AE: Carried-Forward v10.8 Layer-Discipline, Falsification, and Adaptive Reserve Addendum (Preserved through v11.4)

Truth-surface discipline. Formal truth means that outputs follow from declared inputs and equations. Artifact truth means hashes identify the released artifacts. Implementation truth means validators reproduce expected gates, scores, and flags. Evidence truth means traces support or fail to support material claims. Empirical truth requires pilots, backtests, or real-world monitoring. Normative defensibility requires declared rights floors and ethical commitments that can be challenged, justified, and held. Governance legitimacy requires lawful authority, participation, review, contestability, and audit boundaries. These claim types support one another but MUST NOT be collapsed into one undifferentiated claim of truth.

AE.1 Purpose and release boundary

This v10.8 addendum hardens the Canon’s category-collapse discipline without importing or depending on any external physics, ontology, or proprietary framework. It translates a general lesson from recent comparative review into MathGov-native audit rules: a value that can be written, computed, projected, or preferred is not automatically admissible, evidenced, authorized, or claimable.

Normative effect. The canonical cascade remains Reality Grounding → Rights Floor → TRC → CSV → RLS. Reality Grounding is added as the public claim-authority level, not as a compensatory scoring gate. Rights floors, TRC/CVaR, Containment, PLSS residual-only discipline, SGP boundaries, and Tier 4 ProofPack restrictions remain unchanged.

AE.2 COMPUTABLE_BUT_INADMISSIBLE audit state

Definition. COMPUTABLE_BUT_INADMISSIBLE is an audit state for cases where arithmetic, a formula, a model, or a workflow returns a value, but the declared domain rules prohibit treating that value as a valid decision state, conformance claim, or authorization basis.

Use cases include: a welfare score for an option that failed NCRC; an RLS ranking after TRC or CSV failure; a PLSS result used as if it could narrow admissibility; an SGP scalar used beyond its evidence tier; a zero entered where the true state is unknown; or a reference vector that computes while violating declared closure boundaries.

Decision effect. A COMPUTABLE_BUT_INADMISSIBLE result may be logged, studied, and used for debugging. It MUST NOT authorize selection, deployment, conformance claims, or stronger maturity claims. The PCC MUST record why computation existed and why admissibility did not follow.

Alias note. SGP_COMPUTABLE_BUT_INADMISSIBLE and the ripple.md ComputableButInadmissible field are domain-scoped aliases of this state for SGP-specific and wrapper-specific records.

AE.3 Layer-specific falsification matrix

Layer	Layer-specific falsifier	Does not by itself falsify	Required response
NCRC	Rights-floor breach, unresolved rights uncertainty, or missing rights evidence for a rights-relevant case.	RLS arithmetic or UCI/HOI diagnostics.	Reject, refuse, or escalate under rights uncertainty.
TRC	Tail-risk/CVaR exceeds governed threshold or catastrophe scenario evidence is insufficient.	NCRC analysis already performed.	Reject or escalate; do not compensate with welfare score.
Containment	Local gain degrades containing systems beyond declared tolerance or hides cross-scale degradation.	Rights or tail-risk verdicts.	Reject as selectable; record containment rationale.
RLS	Wrong welfare cells, weights, evidence grade, aggregation, or sensitivity posture.	Admissibility gate outcomes.	Recompute or mark non-decisive; do not reopen failed gates.
UCI/HOI	Structural metric evidence, normalization, or tie-break application is defective.	NCRC/TRC/CSV/RLS by itself.	Recompute diagnostic or remove tie-break claim.

Layer	Layer-specific falsifier	Does not by itself falsify	Required response
PCC / audit	Missing provenance, closure, evidence, reviewer trace, or claim-boundary record.	The underlying ethical truth of the situation.	Invalidate conformance or maturity claim until record is repaired.
SGP interface	SGP output lacks evidence tier, scoring version, reviewer status, uncertainty, or maturity boundary for its claimed use.	NCRC, TRC, CSV, lawful authority, or governance-role requirements.	Treat as governed evidence limitation; do not use as override.

AE.4 Reference, preference, number, and claim discipline

For RippleLogic runs, a numerical value is a representation of a declared evidence process, not truth itself. Implementations SHOULD distinguish: evidence source, measurement reference, governance preference or weighting choice, numerical representation, and public claim. Collapsing these layers is a category-collapse defect.

Reference/preference rule. A measurement reference identifies how a claim is grounded. A governance preference identifies an explicit normative or procedural choice, such as a weight, threshold, epsilon, or materiality rule. A number is the final representation after those choices. The PCC SHOULD disclose all three when the value is decision-material.

AE.5 Adaptive Reserve / Saturation Diagnostic

Adaptive Reserve is a carried-forward v10.8 structural diagnostic for detecting whether an option consumes so much social, institutional, ecological, cognitive, financial, or operational slack that the system becomes brittle even when immediate welfare scores appear positive.

Status. Adaptive Reserve is not a sixth gate in the current v11.4 line. It is a UCI/HOI-compatible diagnostic and containment-support signal. It MAY trigger review, sensitivity analysis, or escalation. It MUST NOT rescue an option that failed NCRC, TRC, or CSV.

Minimum fields. When used, implementations SHOULD record reserve domain, baseline reserve, projected reserve after option, recovery path, redundancy state, reversibility, monitoring trigger, and saturation risk.

AE.6 v10.8 release delta

Release Delta (v10.7 → v10.8). v10.8 adds explicit computable-but-inadmissible discipline, a layer-specific falsification matrix, a reference/preference/number claim boundary, and an optional Adaptive Reserve diagnostic. It does not alter the cascade, equations, rights thresholds, TRC mechanics, Containment semantics, SGP binding interface, PLSS residual-only rule, or Tier 4 claim boundary.

APPENDIX AF: Decision Verdict Semantics, Projection Freeze, Closure Status, and Authority-Selection Separation

AF.1 Purpose and release boundary

This v10.8 hardening appendix strengthens RippleLogic's own decision-state discipline without importing any external ontology, physics claim, proprietary framework, or hidden implementation layer. It preserves the canonical cascade Reality Grounding → Rights Floor → TRC → CSV → RLS. It does not change any rights threshold, TRC/CVaR rule, Containment rule, PLSS rule, SGP interface, UCI/HOI definition, or Tier-4 claim boundary.

Normative effect. Appendix AF adds output semantics, audit fields, and refusal/authority-separation rules. It clarifies what a RippleLogic run may claim when gates pass but the remaining admissible options are non-unique, evidence is insufficient, coverage is incomplete, or RLS/tie-breaks are non-decisive.

AF.2 Decision verdict states

FrameworkVerdict is the technical verdict emitted by the declared cascade, evidence-status checks, thresholds, scoring rules, tie-breaks, and refusal conditions. AuthoritySelection is a separate governance or operator act made after the FrameworkVerdict has been disclosed. Implementations MUST NOT present an authority selection as a deterministic framework selection.

The allowed FrameworkVerdict values are: BLOCK, REFUSE_DETERMINISTIC_SELECTION, ALLOW_FRAMEWORK_SELECTION, and ESCALATE. AUTHORIZED_SELECTION is not a FrameworkVerdict; it is an authority record linked to the verdict when lawful or legitimate authority chooses among disclosed admissible options.

BLOCK means that no option may proceed under the claimed posture because a non-negotiable gate fails, no option remains admissible, or a required coverage/evidence condition invalidates the run. REFUSE_DETERMINISTIC_SELECTION means that at least one option remains admissible, but the framework refuses to claim a unique technical selection because evidence is insufficient, coverage is incomplete, the option set is non-unique, RLS is non-decisive, or the tie-break chain does not resolve the choice. ALLOW_FRAMEWORK_SELECTION means that the declared cascade and selection procedure produce one uniquely claimable option under the stated tier, evidence, and projection boundary. ESCALATE means additional evidence, higher-tier review, stakeholder challenge, expert review, or governance authorization is required.

AF.3 Non-decisive admissible sets

When NCRC, TRC, and required Containment checks leave multiple admissible options and RLS/tie-breaks do not yield a decisive selection under declared thresholds, RippleLogic MUST NOT claim unique deterministic framework selection. The correct technical outcome is REFUSE_DETERMINISTIC_SELECTION with an AdmissibleOptionSet and a reason code. A governance process MAY still choose among admissible options, but that choice MUST be recorded separately as an AuthoritySelectionRecord.

Minimum fields for a non-decisive admissible set are: FrameworkVerdict, AdmissibleOptionSet, DeterministicSelectionClaim=false, NonDecisiveReason, AuthoritySelectionRecord if one exists, AuthorityBasis, ReviewerRequired, and DisclosureText.

AF.4 ProjectionPreRegistration visibility

For Tier 3, high-stakes Tier 2, and future ProofPack-prep claims, implementations SHOULD expose whether a ProjectionPreRegistration object exists, what it hash-pins, and whether any material post-observation modification occurred. Scenario sets, CVaR confidence, TRC thresholds, uncertainty parameters, weights, scope declarations, SGP bindings, and refusal/escalation conditions SHOULD be frozen before the run for any claim-bearing evaluation. Post-observation modification creates a new run boundary unless explicitly recorded as exploratory analysis.

AF.5 Closure and coverage status

Required evidence surfaces MUST NOT silently default missing information to zero. Each required cell, right, scenario input, stakeholder mapping, SGP object, and projection input SHOULD carry a CoverageStatus: VALUE, ESTIMATE, ASSUMPTION, BLANK_BY_DESIGN, DECLARED_UNKNOWN, NOT_APPLICABLE, or MISSING_UNDECLARED. DECLARED_UNKNOWN may trigger REFUSE or ESCALATE when the unknown affects admissibility, selection, or tier claim. MISSING_UNDECLARED is non-conformant for claim-bearing runs.

AF.6 Falsification classes

Validation and failure records SHOULD identify the failure class rather than collapsing all failed tests into a generic defect. The recommended FalsificationClass values are DATA_FAILURE, PROJECTION_FAILURE, PARAMETER_FAILURE, OPERATOR_FAILURE, PRIMITIVE_CHALLENGE, GOVERNANCE_FAILURE, and COMMUNICATION_FAILURE. A failed projection or bad parameter does not automatically falsify the Canon; it identifies the layer that must be repaired or retested.

AF.7 Authority-selection record

AuthoritySelectionRecord documents a human, institutional, democratic, expert, emergency, or operator selection made after the framework has disclosed its technical verdict. It MUST identify selected option, authority type, authority basis, admissible option set, non-decisiveness disclosure, reviewer requirement, and whether the selection is lawful or policy-authorized within the deployment context. Authority selection does not convert REFUSE_DETERMINISTIC_SELECTION into ALLOW_FRAMEWORK_SELECTION.

AF.8 Reality-contact and primitive-clarity discipline

Reality-contact discipline clarifies P1 without expanding RippleLogic into a total metaphysics. A failed claim about reality does not mean reality failed; it means the evidence, projection, model, interpretation, or authority claim failed to make adequate contact with the relevant reality surface.

Reality-contact failures SHALL be classified for Tier 2 and Tier 3 claim-bearing runs as one or more of: DATA_FAILURE, PROJECTION_FAILURE, PARAMETER_FAILURE, OPERATOR_FAILURE, GOVERNANCE_FAILURE, COMMUNICATION_FAILURE, or PRIMITIVE_CHALLENGE. A revised claim after a reality-grounding failure is a new claim; it SHALL NOT be counted as the original claim having succeeded.

The framework itself is part of the social and institutional reality it affects. Any deployment of MathGov, RippleLogic, SGP, ripple.md, or RL_Agent that materially shapes decisions, incentives, authority, or public belief is subject to the same reality-grounding audit discipline it applies to other systems.

AF.9 v10.8 release delta

Release Delta (v10.6 → v10.8). The v10.8 release line carries forward decision-verdict semantics, authority-selection separation, projection-freeze visibility, coverage-status closure discipline, falsification-class taxonomy, and the root reality doctrine / reality-grounding consistency patch. It is a hardening patch only. It does not add a new gate, alter the cascade, import an external ontology, or make Tier 4, ProofPack, empirical-validation, legal-certification, or deployment-readiness claims available.

Appendices AG and AH carry the v10.8 Reality Grounding / Lean Cascade and Category Grounding patches and are part of this canonical artifact.

APPENDIX AG: CARRIED-FORWARD v10.8 REALITY GROUNDING AND LEAN CASCADE PATCH (CURRENTLY PRESERVED THROUGH v11.4)

Status. Appendix AG records the v10.8 substance update carried forward inside the v11.4 release line. Appendix AC remains controlling for release metadata and claim boundaries where conflicts appear.

Core method. The public-facing RippleLogic method inside the MathGov framework is now: Reality Grounding -> Rights Floor -> TRC -> CSV -> RLS, or compactly RG -> RF -> TRC -> CSV -> RLS.

Formal method. The formal implementation is: RG/RSG -> RF/NCRC -> TRC -> CSV -> RLS. RF names the public layer; NCRC names the formal rights predicate.

Reality Grounding. Level 1 declares the reality surface, evidence trace, material unknowns, transition boundary, consequence pathways, and claim boundary. No claim may exceed its reality surface.

UCI/HOI placement. UCI/HOI are structural diagnostic instruments, not public headline cascade stages. Gate-relevant UCI/HOI evidence belongs inside CSV. Residual UCI/HOI applies only when RLS is tied, close, uncertainty-overlapped, or non-decisive, or when monitoring/hollowing documentation is required.

PLSS. PLSS remains the zoom lens, not the safety waiver. It adjusts residual welfare-ranking attention only after admissibility has been preserved and cannot omit containment-relevant structural diagnostics.

Boundary. This release does not claim Tier 4, ProofPack readiness, empirical validation, legal certification, deployment certification, or deterministic selection where the run is non-decisive.

APPENDIX AH: CATEGORY GROUNDING AND PRIMITIVE AUDIT PROTOCOL

AH.1 Status and placement

Status. Appendix AH is a normative clarification for claim-bearing Tier 2 and Tier 3 runs and an informative design guide for lower-stakes runs. It is part of the Reality Grounding layer. It does not create a sixth public method level and does not weaken NCRC, TRC, CSV, RLS, PLSS boundaries, SGP boundaries, lawful authority boundaries, or Tier 4/ProofPack boundaries.

Placement rule. Category Grounding, Primitive Audit, Reference Category Review, Definition Falsification Test, and Category-Reality Correspondence Check are one grouped subdiscipline under Reality Grounding when material. Implementers MAY use these names as subfields or review views, but MUST NOT split them into competing cascade stages or treat them as optional decorative terminology when category choice is material.

AH.2 Definitions

Category. A category is an operational class used by a RippleLogic run to identify, group, measure, protect, score, govern, authorize, or execute against phenomena. Examples include person, stakeholder, union scope, harm, right, sentience, agency, intelligence, safety, stability, risk, authority, protection, flourishing, containment, and alignment.

Category Grounding. Category Grounding is the Reality Grounding subdiscipline that checks whether a material category has a declared definition, inclusion boundary, exclusion boundary, reference structure, evidence surface, maturity status, and falsification or revision trigger sufficient for the claim being made.

Primitive Audit. Primitive Audit is the review of a framework starting point or imported foundational term, including its dependency basis, operational role, limits, plausible failure modes, and revision conditions. Primitive Audit is required when a run, release, or public claim materially depends on a primitive in a contested or high-consequence way.

Semantic Debt. Semantic debt is the accumulated governance, measurement, operational, and ethical burden created when a decision system continues to use categories whose reality correspondence, boundary conditions, or falsification status are weak, stale, inherited, contested, or unreviewed. Semantic debt is not automatically a gate failure, but it raises review priority and may require claim narrowing, monitoring, escalation, or refusal when material.

Term Discrimination. Term Discrimination is the Category Grounding subdiscipline that states what a material term means, what it does not mean, what discriminator separates it from nearby terms, where it sits in the dependency order, and when the term must be narrowed, escalated, monitored, treated as sensitivity-only, or refused.

Category Discriminator. A category discriminator is the decisive property that separates a material category or term from neighboring categories it is likely to be confused with.

Dependency Position. Dependency position is the location of the term in the decision stack, such as reality-surface, evidence-trace, category-boundary, rights-admissibility, tail-risk-admissibility, containment-selectability, residual-ranking, framework-verdict, authority-selection, execution-authority, monitoring, certification, or public-claim.

AH.3 Trigger conditions

A CategoryGroundingRecord is REQUIRED for claim-bearing Tier 3 runs, high-stakes Tier 2 runs, and public conformance claims when any of the following are material: (1) the category affects NCRC protection, stakeholder inclusion, or rights-floor localization; (2) the category affects TRC scenario construction, tail probabilities, severity bands, or irreversibility classification; (3) the category affects Containment, UCI, HOI, Adaptive Reserve, lock-in, hollowing, domination, institutional erosion, ecological degradation, or governance-capacity loss; (4) the category affects SGP outputs, moral-patient status, protection, standing, or governance-authority interpretation; (5) the category is a metric target under strong optimization pressure; (6) the category is novel, inherited, contested, ambiguous, or cross-domain; (7) the category supports an authority-selection, deployment, certification, validation, or public maturity claim.

AH.4 Minimum CategoryGroundingRecord

Minimum fields: material_category; category_role; primitive_basis; definition; category_discriminator; dependency_position; inclusion_boundary; exclusion_boundary; common_confusions; noncollapse_pairs; reference_structure; evidence_surface; source_authority_status; category_maturity_status; falsification_or_revision_trigger; term_refusal_condition; downstream_dependencies; semantic_debt_flag; reviewer_status; and required_claim_action.

Allowed category_maturity_status values: GROUNDED, ESTIMATED, ASSUMPTION_BOUND, CONTESTED, LEGACY_UNREVIEWED, INSUFFICIENT_GROUNDING, and OUT_OF_SCOPE_WITH_RATIONALE.

PrimitiveAuditRecord SHOULD include: primitive_id; primitive_statement; primitive_role; imported_parent_disciplines; operational_dependency; affected_claims; evidence_or_argument_basis; failure_mode; falsification_or_revision_trigger; reviewer_status; required_claim_action; and audit_reference. PrimitiveAuditRecord is required when a primitive or imported foundational term materially supports a rights, tail-risk, containment, SGP, authority, conformance, public-claim, or deterministic-selection claim.

Allowed semantic_debt_flag values: NONE, LOW, MEDIUM, HIGH, and CRITICAL. HIGH or CRITICAL semantic debt requires explicit reviewer attention before stronger public, conformance, deployment, authority, or deterministic-selection claims may be made.

Semantic debt rubric: NONE = category recently reviewed, evidence-linked, and uncontested within the claim boundary. LOW = minor ambiguity with no plausible gate or claim effect. MEDIUM = ambiguity could affect interpretation or monitoring but not current admissibility. HIGH = ambiguity could affect rights, tail risk, containment, authority, conformance, or public claim. CRITICAL = category is stale, contested, or ungrounded in a way that could invalidate the decision state or public claim.

Allowed required_claim_action values: ALLOW, NARROW, ESCALATE, REFUSE, MONITOR_ONLY, and SENSITIVITY_ONLY.

Allowed source_authority_status values SHOULD include: PRIMARY_EVIDENCE, GOVERNING_SPECIFICATION, DOMAIN_STANDARD, LEGAL_AUTHORITY, INSTITUTIONAL_POLICY, EXPERT_JUDGMENT, MODEL_OUTPUT, STAKEHOLDER_TESTIMONY, LEGACY_SOURCE, and UNVERIFIED_SOURCE. A source authority label records provenance; it does not by itself create admissibility, validity, legality, or execution authority.

Allowed dependency_position values SHOULD include: REALITY_SURFACE, EVIDENCE_TRACE, CATEGORY_BOUNDARY, RIGHTS_ADMISSIBILITY, TAIL_RISK_ADMISSIBILITY, CONTAINMENT_SELECTABILITY, RESIDUAL_RANKING, FRAMEWORK_VERDICT, AUTHORITY_SELECTION, EXECUTION_AUTHORITY, MONITORING, CERTIFICATION, and PUBLIC_CLAIM.

AH.4A Aggregate Semantic-Debt Ledger (Normative for material claim-bearing runs)

Per-category semantic debt may accumulate into a run-level liability even when no single category is independently fatal. Claim-bearing Tier 2 and Tier 3 runs SHOULD compute a semantic_debt_score over material categories using declared weights: NONE=0, LOW=1, MEDIUM=3, HIGH=7, CRITICAL=15.

If any material category is CRITICAL, stronger public, conformance, authority-selection, deployment-readiness, or deterministic-selection claims MUST be refused, narrowed, or escalated unless a reviewer records a governed disposition. If semantic_debt_score >= threshold_T, with threshold_T declared ex ante in the verification regime and default threshold_T = 10 for worked examples, implementations SHOULD flag SEMANTIC_DEBT_AGGREGATE_HIGH with severity ESCALATE and require reviewer disposition before stronger public or conformance claims.

Semantic-debt aggregation is not an ontology claim. It is an auditability rule for accumulated category uncertainty inside Reality Grounding.

AH.5 Category failure classes

Category-relevant failures SHOULD be tagged as one or more of: CATEGORY_BOUNDARY_FAILURE, CATEGORY_REFERENCE_FAILURE, CATEGORY_SCOPE_FAILURE, CATEGORY_METRIC_MISMATCH, SEMANTIC_DRIFT_DETECTED, TERM_DISCRIMINATION_FAILURE, DEPENDENCY_POSITION_ERROR, PRIMITIVE_CHALLENGE, DATA_FAILURE, PROJECTION_FAILURE, PARAMETER_FAILURE, OPERATOR_FAILURE, GOVERNANCE_FAILURE, or EXTERNAL_CHANGED_CONDITION.

CATEGORY_BOUNDARY_FAILURE means the inclusion or exclusion rule for a material category is wrong, missing, or too vague for the claim. CATEGORY_REFERENCE_FAILURE means the category lacks a valid reference structure. CATEGORY_SCOPE_FAILURE means the category was applied at the wrong union scope, stakeholder class, substrate, time horizon, or decision boundary. CATEGORY_METRIC_MISMATCH means the metric does not track the category it claims to measure. SEMANTIC_DRIFT_DETECTED means ordinary, institutional, legal, benchmark, or model usage has drifted from the category structure required for the run. PRIMITIVE_CHALLENGE means the framework starting point itself may require revision, narrowing, or retirement.

Category-failure action rule. If a material category is CATEGORY_BOUNDARY_FAILURE, CATEGORY_REFERENCE_FAILURE, CATEGORY_METRIC_MISMATCH, SEMANTIC_DRIFT_DETECTED at HIGH/CRITICAL severity, LEGACY_UNREVIEWED for a claim-bearing run, or otherwise insufficiently grounded for the claimed decision state, the run SHALL NOT produce an unqualified deterministic framework selection. It MUST narrow, escalate, mark sensitivity-only or monitor-only, collect stronger evidence, or refuse the stronger claim. A computable score over a category-invalid surface is computable but inadmissible.

AH.6 Relationship to optimization, governance, and public claims

Optimization is admissible only after the relevant category, evidence surface, and claim boundary are adequate for the intended claim. Governance, compliance, benchmark success, institutional permission, model fluency, dashboard availability, or stakeholder preference does not by itself validate a category. If the category is materially wrong, a higher RLS score, a completed PCC, a PLSS profile, or an authority-selection record cannot repair it.

AH.7 Release boundary

This appendix does not claim empirical validation, legal certification, deployment certification, deterministic selection, or Tier 4/ProofPack readiness. It strengthens the category-discipline portion of Reality Grounding so that MathGov does not become a governance layer over unexamined definitions. This completes the RippleLogic Framework v11.4 Category Grounding Patch, including Sections 1 through 20 and Appendices A through AH; Appendix AI then adds the Term Discrimination and Semantic Stability Protocol as a normative subdiscipline of Category Grounding where material.

APPENDIX AI: TERM DISCRIMINATION AND SEMANTIC STABILITY PROTOCOL

AI.1 Status and placement

Status. Appendix AI is a normative hardening patch for claim-bearing Tier 2 and Tier 3 runs and an informative design guide for lower-stakes runs. It strengthens Category Grounding inside Reality Grounding. It does not create a sixth public method level, does not import an external ontology, does not weaken NCRC, TRC, CSV, RLS, PLSS, SGP, lawful authority, or Tier 4/ProofPack boundaries, and does not make certification, compliance, or deployment claims available by itself.

Placement rule. Term Discrimination is a required subdiscipline of Category Grounding when a material term affects rights, tail risk, containment, SGP interpretation, authority, conformance, public claims, optimized metrics, or execution. Category Grounding asks whether a category is grounded. Term Discrimination asks whether the operative word is being kept separate from neighboring words that would change the claim, gate result, authority state, or execution status if collapsed.

AI.2 Core definitions

Term. A term is an operative word, phrase, label, token, schema field, policy label, benchmark label, dashboard label, legal label, or model-generated description used to identify, measure, protect, score, govern, authorize, certify, monitor, or execute against a phenomenon.

Term Discrimination. Term Discrimination is the review discipline that states what a material term means, what it does not mean, what discriminator separates it from neighboring terms, where it belongs in the dependency order of the run, and when the term must be narrowed, escalated, monitored, or refused.

Discriminator. A discriminator is the decisive property that makes one term this term rather than a neighboring term. A discriminator MUST be stated for material terms that could otherwise collapse into another term under institutional pressure, optimization pressure, model fluency, compliance theater, or public communication pressure.

Dependency position. Dependency position identifies where a term sits in the decision stack. A downstream term MUST NOT claim the authority of an upstream term. Permission does not establish admissibility. Monitoring does not establish control. Certification does not establish validity. A score does not establish truth. Execution capability does not establish execution authority.

Term-refusal condition. A term-refusal condition is the condition under which a run MUST NOT use a term at its stronger claim level. The required response is claim narrowing, evidence collection, escalation, sensitivity-only use, monitor-only use, or refusal.

Non-collapse pair. A non-collapse pair is a pair of terms that MUST NOT be treated as equivalent unless a future governed annex explicitly defines an equivalence for the relevant use. Non-collapse pairs include reality/evidence, evidence/interpretation, compliance/correctness, certification/validity, permission/admissibility, monitoring/control, safety/stability, automation/autonomy, protection/authority, selected/executable, and score/truth.

AI.3 Extended CategoryGroundingRecord fields

When Term Discrimination is material, the CategoryGroundingRecord MUST add or map the following fields:

Field	Meaning
category_discriminator	The property that separates the material category or term from nearby terms it is likely to be confused with.
dependency_position	The layer or authority position the term occupies: reality-surface, evidence-trace, category-boundary, rights-admissibility, tail-risk-admissibility, containment-selectability, residual-ranking, framework-verdict, authority-selection, execution-authority, monitoring, certification, or public-claim.
term_refusal_condition	The condition under which the term cannot support the stronger claim and must be narrowed, escalated, marked sensitivity-only/monitor-only, or refused.
noncollapse_pairs	The nearest terms this term MUST NOT be collapsed with in the run.

Field	Meaning
source_authority_status	Whether the source of the term is evidential, legal, institutional, model-generated, benchmark-derived, stakeholder-proposed, or framework-defined. This records source status without treating source status as automatic category validity.

AI.4 Dependency-position rule

A material term may support only the claim strength allowed by its dependency position and evidence surface.

Term position	May support	Must not claim by itself
Reality surface	Contact with the modeled or evidenced part of the territory	Complete access to the territory
Evidence trace	Support for a claim boundary	Self-interpreting truth or complete causality
Category boundary	Proper classification for declared use	Validity of downstream scoring or governance by itself
Rights-admissibility	NCRC pass/fail state	Tail-risk pass, containment pass, selection, or execution authority
Tail-risk-admissibility	TRC pass/fail state	Rights validity, containment integrity, or welfare optimality
Containment-selectability	Larger-system structural acceptability	Highest residual welfare or lawful execution
Residual-ranking	RLS ranking among selectable options	Rescue of any failed upstream layer
Framework verdict	Disclosed result of the framework run	Human, legal, institutional, or execution authority
Authority selection	A human/institutional/operator choice after verdict disclosure	Conversion of non-decisiveness into deterministic framework selection
Execution authority	Governed authority to act through hard controls	Proof that the action is safe, valid, or wise
Monitoring	Observability and feedback after or during action	Prevention, control, safety, or stability by itself

Term position	May support	Must not claim by itself
Certification	Evidence of conformance to a declared standard	Reality validity, legal compliance, empirical truth, deployment readiness, or Tier 4 status

AI.5 Do-Not-Collapse Matrix

Do not collapse	Discriminator	Required MathGov response
Reality vs evidence	Reality is the evaluator-independent constraint-and-consequence field; evidence is traceable contact with a surface of it.	Keep claim boundary inside the evidence surface.
Evidence vs interpretation	Evidence is trace; interpretation connects trace, uncertainty, values, and action.	Record reviewer status and uncertainty.
Category vs metric	Category defines what is being measured; metric quantifies a proxy or indicator.	Refuse or narrow metric claims when the category is weak.
Score vs truth	RLS ranks selectable options under declared inputs; it is not truth itself.	Never let RLS rescue failed grounding, rights, risk, or containment.
Compliance vs correctness	Compliance means conformity to a rule or process; correctness means reality-bound adequacy for the claim.	Treat compliance as evidence, not validity.
Certification vs validity	Certification records an assurance result; validity depends on grounded reference, evidence, and replayable adequacy.	Prevent certification theater and overclaiming.
Permission vs admissibility	Permission is authorization; admissibility is passing the relevant MathGov gates.	Do not treat approval as gate success.
Monitoring vs control	Monitoring observes; control prevents, constrains, or corrects.	Require enforcement path, rollback, or containment evidence for control claims.

Do not collapse	Discriminator	Required MathGov response
Safety vs stability	Safety means bounded non-harm under declared conditions; stability means preserved structural regime integrity.	Use TRC, Containment, and Adaptive Reserve where relevant.
Stability vs justice	Stability can preserve harmful arrangements; justice requires rights floors and non-domination.	Do not allow stable oppression to pass rights review.
Automation vs autonomy	Automation follows procedures; autonomy evaluates, adapts, and remains accountable under constraints.	Require authority and execution-stability records for autonomy claims.
Protection vs authority	Protection concerns moral/welfare standing; authority concerns legitimate role, responsibility, competence, and authorization.	Preserve SGP protection/authority separation.
Selected vs executable	A selected option is chosen by framework or authority; executable means action may proceed through lawful/runtime controls.	Require execution-stability and authority checks.
Capability vs authority	A system may be technically able to act without being permitted or admissible to act.	Block action without hard-control authorization.

AI.6 Claim-action rule

If a material term lacks a discriminator, lacks a dependency position, collapses into a neighboring term, or lacks a refusal condition sufficient for the claim being made, the run MUST classify the issue as TERM_DISCRIMINATION_FAILURE or a more specific category failure class. The result MUST be NARROW, ESCALATE, REFUSE, MONITOR_ONLY, or SENSITIVITY_ONLY until the term is grounded. A completed PCC, authority-selection record, institutional permission, model output, dashboard result, or certification label MUST NOT repair a term-discrimination failure by itself.

AI.7 v11.4 release delta

Release Delta (v11.0 -> v11.4). v11.4 carries forward the Term Discrimination and Semantic Stability Protocol and adds Structural Viability and Falsification Discipline hardening inside existing layers. The v11.0 line added Term Discrimination as a hardening patch inside Category Grounding and Reality Grounding. It adds category_discriminator, dependency_position, term_refusal_condition, noncollapse_pairs, and source_authority_status to material category records; adds the Do-Not-Collapse Matrix; strengthens certification/permission/monitoring non-authority language; and preserves the five-level method. It does not add a new public gate, import an external ontology, alter NCRC, TRC, CSV, RLS, SGP scalar semantics, PLSS, lawful authority boundaries, Tier 4, ProofPack, empirical-validation, legal-certification, or deployment-certification boundaries.

Appendix AJ: Public Release Hardening Addendum (v11.4)

Status. This appendix is a hardening bridge for the v11.4 content line. It does not create Tier 4, ProofPack, legal certification, deployment certification, empirical validation, or automated moral truth. It clarifies implementation discipline and points to companion guidance. If a companion guide conflicts with the Canon, the Canon controls until a future versioned amendment resolves the conflict.

AJ.1 Gate Boundary Discriminator: TRC, CSV, and RLS

TRC, CSV, and RLS SHALL NOT be collapsed.

Layer	Primary question	Normal consequence
TRC	Could this option create catastrophic, irreversible, ruinous, or lock-in tail exposure?	Gate failure if tail exposure exceeds corridor.
CSV	Can the option deliver its claimed good without breaking, hollowing, overloading, exploiting, or depending on containing systems beyond tolerance?	Fail, controls, redesign, escalation, or pass depending on structural status.
RLS	Among options that survived RG, RF/NCRC, TRC, and CSV, which has the strongest residual welfare profile?	Ranking only; no gate rescue.

Double-materiality rule. A harm may be both tail-relevant and structurally degrading. In that case it MUST be evaluated in TRC and CSV as appropriate, recorded in the PCC, and not double-counted as an ordinary residual RLS cost unless a distinct residual effect remains after gate treatment.

Companion guide: docs/guides/GATE_BOUNDARY_DISCRIMINATOR_TRC_CSV_RLS.md.

AJ.2 Rights Threshold Governance

Rights thresholds are governance commitments under evidence, not preferences to be moved after selection. Tier 3 and public claim-bearing Tier 2 runs SHOULD record threshold source, rationale, under-protection risk, over-blocking risk, review status, and revision trigger. A decision owner who benefits from lowering a threshold MUST NOT unilaterally lower the Rights Floor for the run under evaluation.

Companion note:
docs/validation/rights/RIGHTS_THRESHOLD_GOVERNANCE_NOTE_v1_0.md.

AJ.3 Subgroup Discovery and Worst-Affected Groups

Worst-off subgroup semantics require active discovery of groups likely to bear hidden, indirect, intersectional, or least-contestable harms. Unknown subgroup exposure is not zero. Where direct sensitive-data collection is unsafe or unlawful, pathway analysis, proxy-risk statements, consultation, conservative bounds, and challenger review SHOULD be used.

Companion guide:
docs/guides/SUBGROUP_DISCOVERY_AND_WORST_AFFECTED_GROUPS.md.

AJ.4 Emergency Mode Least-Rights-Infringing Rule

Emergency Mode does not make a rights-failing option aligned. It identifies a time-bounded, least-rights-infringing provisional action when no normal option passes RF/NCRC. RLS may be used only as a final constrained tie-break after rights-violation vector comparison and TRC comparison. RLS MUST NOT reverse the emergency rights-vector ordering or convert a rights-failing option into an ordinary selectable option.

Companion guide:
docs/guides/EMERGENCY_MODE_LEAST_RIGHTS_INFRINGING_PROTOCOL.md.

AJ.5 TRC Scenario Discovery

TRC is only as complete as the declared scenario surface. Tier 3 runs SHOULD use scenario discovery sources including domain experts, incident and near-miss data, red teams, affected stakeholders, literature, standards, and AI-assisted candidate generation. AI-generated scenarios are discovery inputs, not validity sources. Material rejected scenarios SHOULD be logged with rationale.

Companion protocol:
docs/validation/trc/TRC_SCENARIO_DISCOVERY_PROTOCOL_v1_0.md.

AJ.6 CSV Measurement Maturity

CSV claims SHOULD declare maturity when used in public or high-stakes contexts: qualitative, indicator-supported, calibrated, or validated. UCI/HOI-style diagnostics are structural signals and MUST NOT be presented as empirically validated in a domain unless domain validation exists.

Companion note: docs/validation/csv/CSV_MEASUREMENT_MATURITY_NOTE_v1_0.md.

AJ.7 PCC Profiles

PCC burden SHOULD be proportional to stakes without allowing consequential decisions to evade audit. The recognized profiles are PCC-Lite, PCC-Core, PCC-Audit, PCC-Agent, and PCC-Public. A lower profile is not allowed when rights exposure, catastrophe relevance, CSV burden, automation, public claims, affected-party burden, or accumulated systemic effect require a higher profile.

Companion guide: docs/guides/PCC_PROFILE_GUIDE.md.

AJ.8 HDW Participation and Capture Control

HDW democratic tuning SHOULD record affected groups, representation method, exclusion risks, capture risks, minority reports, and correction actions in Tier 3 runs. Optional mechanisms such as quadratic voting or deliberative polling MAY be used where appropriate, but no method may lower constitutional floors, erase non-vocal stakeholders, or weaken RF/NCRC, TRC, or CSV.

Companion guide: docs/guides/HDW_PARTICIPATION_AND_CAPTURE_GUIDE.md.

AJ.9 SGP Validation Bridge

SGP outputs remain protection-surface evidence objects, not legal status, governance authority, or substitutes for RG, RF/NCRC, TRC, CSV, RLS, lawful authority, or governance-role requirements. SGP validation should test inter-rater reliability, fluency traps, biological welfare false negatives, artificial-entity status inflation, and protection-authority separation.

Companion protocol: docs/validation/sgp/SGP_VALIDATION_PROTOCOL_v1_0.md.

AJ.10 Stewardship Evasion Audit

A run cannot avoid stewardship obligations by labeling consequential influence as educational, internal, experimental, advisory, or informational. If the output materially influences real decisions, narrows options, affects rights or livelihoods, or connects to execution capability, stewardship review SHOULD be escalated.

Companion guide: docs/guides/STEWARDSHIP_EVASION_AUDIT.md.

AJ.11 Meta-Union Firewall

AIU, Cosmic, Universal, and related horizon/meta-union language may orient humility and philosophical interpretation. They are not Tier 1-3 scoring objects, gate overrides, hidden weights, legal authority, deployment permission, or ProofPack evidence unless a future governed extension explicitly defines a computable and validated protocol.

Companion guide: docs/guides/META_UNION_FIREWALL.md.

Appendix AK: Computability vs Realizability Bridge

Status. Appendix AK is a release-hardening bridge introduced in v11.4 and explicitly precedence-indexed in the v11.4 final core. It does not add a new public cascade level and does not change the governing order RG -> RF -> TRC -> CSV -> RLS.

AK.1 Core distinction. A state, output, plan, policy, or model response may be computable without being physically grounded, rights-safe, ruin-bounded, structurally viable, authorized, or ethically selectable. Computability is representational capacity. Selectability is governed survivorship through the cascade.

AK.2 Operational rule. RippleLogic MUST NOT treat model fluency, simulation output, dashboard value, code generation, tool result, or computed score as decision authority merely because it exists. Computable is not selectable. Generated is not grounded. Executable is not ethical. Realizable is necessary but not sufficient.

AK.3 Layer assignment. RG handles reality-surface contact and claim boundaries. RF/NCRC handles non-compensatory rights protection. TRC handles catastrophic and ruin-path exposure. CSV handles containment, structural viability, execution viability, monitoring, authority dependencies, and host-system integrity. RLS ranks only residual welfare for options that remain selectable.

AK.4 Feedback rule. If CSV analysis discovers a new catastrophic, irreversible, lock-in, or ruin-path scenario that was not represented in TRC, the run MUST reopen TRC before RLS. CSV can discover TRC-relevant material, but it does not absorb or replace TRC.

AK.5 Public teaching line. Ground reality. Protect rights. Bound ruin. Preserve the structure. Score the ripples.

Appendix AL: Source-Coupling Integrity and Generative-Source Traceability (Normative Hardening Patch)

Status. Appendix AL is a v11.4.3 hardening patch. It is a MathGov-native Reality Grounding and CSV diagnostic. It does not add a public gate, does not alter the cascade, does not import an external ontology, and does not make empirical validation, legal certification, deployment certification, Tier 4, ProofPack, or automated moral-truth claims available.

AL.1 Purpose

Appendix AL protects RippleLogic runs from a specific downstream-substitution failure: treating output, fluency, institutional permission, compliance status, inherited procedure, benchmark performance, dashboard value, or interface success as proof that the claimed capability is grounded in the enabling conditions that make it possible and the boundary conditions that limit it.

AL.2 Required Source-Coupling Record

When the Section 2.1D trigger holds, the PCC SHOULD include a SourceCouplingRecord and MUST include one for Tier 3 public or institutional claim-bearing runs. The record contains: claimed_capability; enabling_conditions; boundary_conditions; source_evidence; generator_output_distinction; inherited_assumptions; downstream_compensations; source_coupling_status; source_debt_flag; falsification_or_recheck_trigger; and required_claim_action.

AL.3 Claim Boundary Effects

Source-Coupling Integrity constrains claim strength. A run MAY proceed with a narrow claim when source coupling is partial or inferred, but it MUST NOT use partial or inferred source coupling to support stronger conformance, deployment, deterministic-selection, or validation claims. If source coupling is unknown, contested, or failing, the run MUST narrow, collect evidence, escalate, redesign, or refuse the stronger claim.

AL.4 CSV Consumption Rule

If source weakness materially affects dependency closure, resource closure, operational capacity, reversibility, containment, monitoring, or host-system integrity, CSV MUST consume the SourceCouplingRecord. SOURCE_DEBT_RISK is a CSV-relevant signal. It may route the option to CSV_PASS_WITH_CONTROLS, CSV_REDESIGN_REQUIRED, CSV_ESCALATE, CSV_FAIL, or emergency-provisional handling according to the CSV Gate Standard and the Canon.

AL.5 Agentic and AI System Rule

For AI or agentic systems, model fluency, benchmark performance, tool-use success, chain-of-action success, or safety-filter compliance is not source-coupled admissibility. Such evidence may support a bounded capability claim, but only after Reality Grounding states what the evidence does and does not show, and only after CSV checks whether deployment can structurally stand.

AL.6 Related External Work Boundary

The term *Source Amnesia* is used only in reference to Alexander Wolfgang Strüver's 2026 external paper of that title. MathGov cites that work as related external scholarship and does not adopt the title as a MathGov module. This Appendix is independently stated in MathGov-native terminology as Source-Coupling Integrity. No text from the external paper is incorporated or adapted into the governing Canon.